

GoodWe Modbus Protocol Hybrid

For Energy Storage Inverters.

Updated on February 11th, 2026



CHANGE RECORDS

Ver.	Date	Modification	Prepared by	Approved by
1	20221231		Morries	Eric
1.1	20231016	Add registers	Cindy	Eric
1.2	20240202	Add registers	Cindy	Eric
1.3	20240916	Add registers		
1.4	20251210	Add registers	Even	Cindy
1.5	20260211	Add registers	Even	Cindy

1. PROTOCOL DESCRIPTION

This is a map document of standard MODBUS RTU protocol for only GoodWe energy storage inverters compatible with HV

Inverter Address: Can be assigned from 1~247. 247 is factory default assignment.

Communication baud rate: The default baud rate is 9600 bps

Error Code Returned From Inverter Device:

02H: Register address fault or overflow of read register number
 03H: Data error
 04H: Built-in verification code error

CRC Verification:

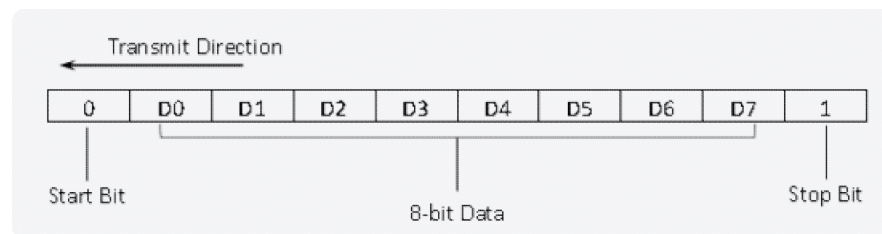
CRC Verification formula : $X^{16}+X^{12}+X^5+1$
 CRC Verification code refer to No.10 Chapter.

Function code:

03H:Reading
 06H: writing single register
 10H:writing multiple registers

Byte Format:

Every byte consists of 1 start bit, 8-bit binary code and 1 stop bit, 10 bit in total. The byte transmit sequence is described as below. D0 is the lowest bit of data and D7 is the highest



Communication Data Format:

Data is transmitted as word or double word format.

Data Type	Amount of Register	Amount of Byte	Description
Byte Data	1	1	
Integer Data	1	2	Return at one time, from high bit to low bits
Long integer	2	4	Return in two parts, from high bit to low bits
Floating Point Data			

2. Data Frame Format**2.1 Read Register (Function Code: 03H)****2.1.1 Data Frame Format from Host PC**

Data NO	Content	Sample	Description
1	Inverter Address	1	Communication address(1-247, Default 0XF7)
2	03H	03H	Function code
3	High byte of first register	00H	Address of register 0001H
4	Low byte of first register	01H	
5	Amount. of High bit Register	00H	Amount of register 02H
6	Amount. of Low bit Register	02H	
7	CRC16 Verification (high bit)	95H	CRC Code of verification
8	CRC16 Verification (low bit)	CBH	

2.1.2 Data Frame Format from Inverter (If Data Reading Successfully)

Data NO	Content	Description
1	Inverter Address	Communication address(1-247, default 0xF7)
2	03H	Function code
3	Amount of byte of data (2N)	Amount of byte of data
4	High byte of data of first register	High byte of first register
5	Low byte of data of first register	Low byte of first register
...	...	...
2N+2	High byte of data of the Nth register	High byte of the Nth register
2N+3	Low byte of data of the Nth register	Low byte of the Nth register
2N+4	High byte of CRC16 verification code	High byte of CRC verification code
2N+5	Low byte of CRC16 verification code	Low byte of CRC verification code

2.1.3 Data Frame Format from Inverter (Register Addr. or register number is wrong)

Data NO	Content	Description
1	Inverter Address	Communication Address (1-247)
2	83H	Function code
3	02H	Error Code
4	High byte of CRC16 verification code	CRC verification code
5	Low byte of CRC16 verification code	CRC verification code

2.2Set/Writing Register (Function code: 10H)

2.2.1 Data Frame Format from AP

Sr.	CODE	Sample	Description
1	Inverter Address	0xF7	Communication Address (1-247, default 0xF7)
2	10H	10H	Function Code
3	High byte of data of first register	00H	Address of register: 0000H
4	Low byte of data of first register	00H	
5	High byte of amount of registers	00H	Amount of registers: 01H
6	Low byte of amount of registers	01H	
7	Amount of bytes (N)	02H	No. of Register Bytes 02H
8	High byte of data	0AH	Data: 0AF0H
9	Low byte of data	F0H	
10	High byte of CRC16 verification code	A0H	CRC verification
11	Low byte of CRC16 verification code	B4H	

2.2.2Data Frame Format from Inverter (when OK)

Sr.	CODE	SAMPLE	EXPLANATION
1	Device Addr.	0xF7	Device communication address (1-247)
2	10H	10H	Function Code
3	High Bit of Start Register Addr.	00H	Register Address 0000H
4	Low Bit of Start Register Addr.	00H	
5	High Bit of Register No.	00H	Number of Register 01H
6	Low Bit of Register No.	01H	
7	CRC16 Verification (high bit)	01H	CRC Verification
8	CRC16 Verification (low bit)	C9H	

2.2.3 Data Frame Format from Inverter (when data is faulty)

Sr.	CODE	EXPLANATION
1	Device Address	Device communication address (1-247)
2	90H	Function Code
3	03H	Error Code
4	CRC16 Verification (high bit)	CRC Verification
5	CRC16 Verification (low bit)	

2.2.4 Data Frame Format from Inverter (when address or amount of register is faulty)

Sr.	CODE	EXPLANATION
1	Device Address	Device communication address (1-247)
2	90H	Function Code
3	02H	Error Code
4	CRC16 Verification (high bit)	CRC Verification
5	CRC16 Verification (low bit)	

2.3 Writing single register (function code 06H)

2.3.1 Frame Format From Host Computer

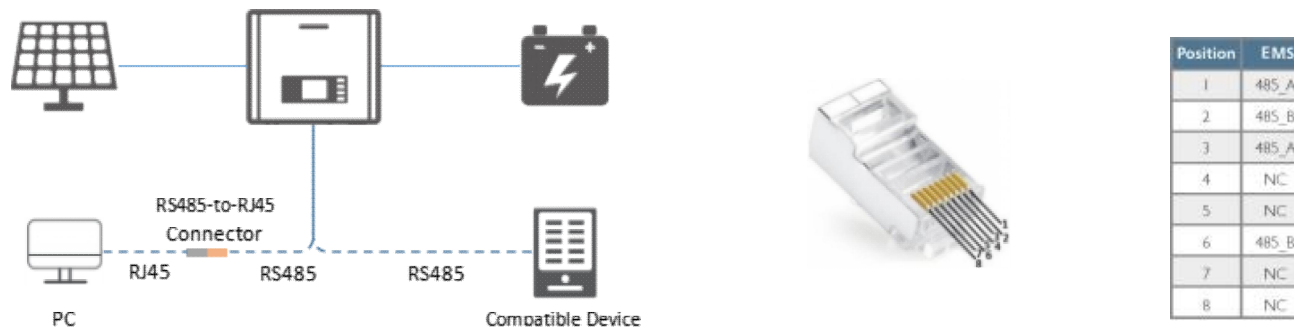
Sr.	CODE	SAMPLE	EXPLANATION
1	Device Addr.	1	Device communication address (1-247)
2	06H	06H	Function Code
3	High Bit of Start Register Addr.	00H	Register Address 0000H
4	Low Bit of Start Register Addr.	00H	
5	High Bit of Data	0AH	Data 0AF0H
6	Low Bit of Data	F0H	
7	CRC16 Verification (high bit)	8FH	CRC Verification
8	CRC16 Verification (low bit)	2EH	

2.3.2 Frame Format Return from Device (data writing successfully)

Sr.	CODE	SAMPLE	EXPLANATION
1	Device Addr.	1	Device communication address (1-247)
2	06H	06H	Function Code
3	High Bit of Start Register Addr.	00H	Register Address 0000H
4	Low Bit of Start Register Addr.	00H	
5	High Bit of Data	0AH	Data 0AF0H
6	Low Bit of Data	F0H	
7	CRC16 Verification (high bit)	8FH	CRC Verification
8	CRC16 Verification (low bit)	2EH	

3. System Wiring Instruction

This is the basic wiring and instructions before starting reading interactive log between GoodWe energy storage inverter and the



Solar inverter must be powered up by DC or AC power before it can communicate successfully to the compatible device.

	#Address	English Name	Chinese Name	#R/W	#Type	#Size	#SF	#Units	Range	Flash Save	Note(English)	Note(Chinese)
<i>Pack cell voltage</i>												
1	3001	Rack Cell 1 Voltage	Rack单体电池1电	RO	U16	1	1	mV				
2	3002	Rack Cell 2 Voltage	Rack单体电池2电	RO	U16	1	1	mV				
3	3003	Rack Cell 3 Voltage	Rack单体电池3电	RO	U16	1	1	mV				
4	3004	Rack Cell 4 Voltage	Rack单体电池4电	RO	U16	1	1	mV				
5	3005	Rack Cell 5 Voltage	Rack单体电池5电	RO	U16	1	1	mV				
6	3006	Rack Cell 6 Voltage	Rack单体电池6电	RO	U16	1	1	mV				
7	3007	Rack Cell 7 Voltage	Rack单体电池7电	RO	U16	1	1	mV				
8	3008	Rack Cell 8 Voltage	Rack单体电池8电	RO	U16	1	1	mV				
9	3009	Rack Cell 9 Voltage	Rack单体电池9电	RO	U16	1	1	mV				
10	3010	Rack Cell 10 Voltage	Rack单体电池10电 压	RO	U16	1	1	mV				
11	3011	Rack Cell 11 Voltage	Rack单体电池11电 压	RO	U16	1	1	mV				
12	3012	Rack Cell 12 Voltage	Rack单体电池12电 压	RO	U16	1	1	mV				
13	3013	Rack Cell 13 Voltage	Rack单体电池13电 压	RO	U16	1	1	mV				
14	3014	Rack Cell 14 Voltage	Rack单体电池14电 压	RO	U16	1	1	mV				
15	3015	Rack Cell 15 Voltage	Rack单体电池15电 压	RO	U16	1	1	mV				
16	3016	Rack Cell 16 Voltage	Rack单体电池16电 压	RO	U16	1	1	mV				
17	3017	Rack Cell 17 Voltage	Rack单体电池17电 压	RO	U16	1	1	mV				
18	3018	Rack Cell 18 Voltage	Rack单体电池18电 压	RO	U16	1	1	mV				
19	3019	Rack Cell 19 Voltage	Rack单体电池19电 压	RO	U16	1	1	mV				
20	3020	Rack Cell 20 Voltage	Rack单体电池20电 压	RO	U16	1	1	mV				
21	3021	Rack Cell 21 Voltage	Rack单体电池21电 压	RO	U16	1	1	mV				
22	3022	Rack Cell 22 Voltage	Rack单体电池22电 压	RO	U16	1	1	mV				

23	3023	Rack Cell 23 Voltage	Rack单体电池23电压	RO	U16	1	1	mV				
24	3024	Rack Cell 24 Voltage	Rack单体电池24电压	RO	U16	1	1	mV				
25	3025	Rack Cell 25 Voltage	Rack单体电池25电压	RO	U16	1	1	mV				
26	3026	Rack Cell 26 Voltage	Rack单体电池26电压	RO	U16	1	1	mV				
27	3027	Rack Cell 27 Voltage	Rack单体电池27电压	RO	U16	1	1	mV				
28	3028	Rack Cell 28 Voltage	Rack单体电池28电压	RO	U16	1	1	mV				
29	3029	Rack Cell 29 Voltage	Rack单体电池29电压	RO	U16	1	1	mV				
30	3030	Rack Cell 30 Voltage	Rack单体电池30电压	RO	U16	1	1	mV				
31	3031	Rack Cell 31 Voltage	Rack单体电池31电压	RO	U16	1	1	mV				
32	3032	Rack Cell 32 Voltage	Rack单体电池32电压	RO	U16	1	1	mV				
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40	3040	Rack Cell 40 Voltage	Rack单体电池40电压	RO	U16	1	1	mV				

41	3041	Rack Cell 41 Voltage	Rack单体电池41电压	RO	U16	1	1	mV				
42	3042	Rack Cell 42 Voltage	Rack单体电池42电压	RO	U16	1	1	mV				
43	3043	Rack Cell 43 Voltage	Rack单体电池43电压	RO	U16	1	1	mV				
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45	3045	Rack Cell 45 Voltage	Rack单体电池45电压	RO	U16	1	1	mV				
46	3046	Rack Cell 46 Voltage	Rack单体电池46电压	RO	U16	1	1	mV				
47	3047	Rack Cell 47 Voltage	Rack单体电池47电压	RO	U16	1	1	mV				
48	3048	Rack Cell 48 Voltage	Rack单体电池48电压	RO	U16	1	1	mV				
49	3049	Rack Cell 49 Voltage	Rack单体电池49电压	RO	U16	1	1	mV				
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113	3113	Rack Cell 113 Voltage	Rack单体电池113 电压	RO	U16	1	1	mV				
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126	3126	Rack Cell 126 Voltage	Rack单体电池126 电压	RO	U16	1	1	mV				
127	3127	Rack Cell 127 Voltage	Rack单体电池127 电压	RO	U16	1	1	mV				
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130	3130	Rack Cell 130 Voltage	Rack单体电池130 电压	RO	U16	1	1	mV				

131	3131	Rack Cell 131 Voltage	Rack单体电池131 电压	RO	U16	1	1	mV				
132	3132	Rack Cell 132 Voltage	Rack单体电池132 电压	RO	U16	1	1	mV				
133	3133	Rack Cell 133 Voltage	Rack单体电池133 电压	RO	U16	1	1	mV				
134	3134	Rack Cell 134 Voltage	Rack单体电池134 电压	RO	U16	1	1	mV				
135	3135	Rack Cell 135 Voltage	Rack单体电池135 电压	RO	U16	1	1	mV				
136	3136	Rack Cell 136 Voltage	Rack单体电池136 电压	RO	U16	1	1	mV				
137	3137	Rack Cell 137 Voltage	Rack单体电池137 电压	RO	U16	1	1	mV				
138	3138	Rack Cell 138 Voltage	Rack单体电池138 电压	RO	U16	1	1	mV				
139	3139	Rack Cell 139 Voltage	Rack单体电池139 电压	RO	U16	1	1	mV				
140	3140	Rack Cell 140 Voltage	Rack单体电池140 电压	RO	U16	1	1	mV				
141	3141	Rack Cell 141 Voltage	Rack单体电池141 电压	RO	U16	1	1	mV				
142	3142	Rack Cell 142 Voltage	Rack单体电池142 电压	RO	U16	1	1	mV				
143	3143	Rack Cell 143 Voltage	Rack单体电池143 电压	RO	U16	1	1	mV				
144	3144	Rack Cell 144 Voltage	Rack单体电池144 电压	RO	U16	1	1	mV				
145	3145	Rack Cell 145 Voltage	Rack单体电池145 电压	RO	U16	1	1	mV				
146	3146	Rack Cell 146 Voltage	Rack单体电池146 电压	RO	U16	1	1	mV				
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148	3148	Rack Cell 148 Voltage	Rack单体电池148 电压	RO	U16	1	1	mV				

149	3149	Rack Cell 149 Voltage	Rack单体电池149 电压	RO	U16	1	1	mV				
150	3150	Rack Cell 150 Voltage	Rack单体电池150 电压	RO	U16	1	1	mV				
151	3151	Rack Cell 151 Voltage	Rack单体电池151 电压	RO	U16	1	1	mV				
152	3152	Rack Cell 152 Voltage	Rack单体电池152 电压	RO	U16	1	1	mV				
153	3153	Rack Cell 153 Voltage	Rack单体电池153 电压	RO	U16	1	1	mV				
154	3154	Rack Cell 154 Voltage	Rack单体电池154 电压	RO	U16	1	1	mV				
155	3155	Rack Cell 155 Voltage	Rack单体电池155 电压	RO	U16	1	1	mV				
156	3156	Rack Cell 156 Voltage	Rack单体电池156 电压	RO	U16	1	1	mV				
157	3157	Rack Cell 157 Voltage	Rack单体电池157 电压	RO	U16	1	1	mV				
158	3158	Rack Cell 158 Voltage	Rack单体电池158 电压	RO	U16	1	1	mV				
159	3159	Rack Cell 159 Voltage	Rack单体电池159 电压	RO	U16	1	1	mV				
160	3160	Rack Cell 160 Voltage	Rack单体电池160 电压	RO	U16	1	1	mV				
161	3161	Rack Cell 161 Voltage	Rack单体电池161 电压	RO	U16	1	1	mV				
162	3162	Rack Cell 162 Voltage	Rack单体电池162 电压	RO	U16	1	1	mV				
163	3163	Rack Cell 163 Voltage	Rack单体电池163 电压	RO	U16	1	1	mV				
164	3164	Rack Cell 164 Voltage	Rack单体电池164 电压	RO	U16	1	1	mV				
165	3165	Rack Cell 165 Voltage	Rack单体电池165 电压	RO	U16	1	1	mV				
166	3166	Rack Cell 166 Voltage	Rack单体电池166 电压	RO	U16	1	1	mV				

167	3167	Rack Cell 167 Voltage	Rack单体电池167 电压	RO	U16	1	1	mV				
168	3168	Rack Cell 168 Voltage	Rack单体电池168 电压	RO	U16	1	1	mV				
169	3169	Rack Cell 169 Voltage	Rack单体电池169 电压	RO	U16	1	1	mV				
170	3170	Rack Cell 170 Voltage	Rack单体电池170 电压	RO	U16	1	1	mV				
171	3171	Rack Cell 171 Voltage	Rack单体电池171 电压	RO	U16	1	1	mV				
172	3172	Rack Cell 172 Voltage	Rack单体电池172 电压	RO	U16	1	1	mV				
173	3173	Rack Cell 173 Voltage	Rack单体电池173 电压	RO	U16	1	1	mV				
174	3174	Rack Cell 174 Voltage	Rack单体电池174 电压	RO	U16	1	1	mV				
175	3175	Rack Cell 175 Voltage	Rack单体电池175 电压	RO	U16	1	1	mV				
176	3176	Rack Cell 176 Voltage	Rack单体电池176 电压	RO	U16	1	1	mV				
177	3177	Rack Cell 177 Voltage	Rack单体电池177 电压	RO	U16	1	1	mV				
178	3178	Rack Cell 178 Voltage	Rack单体电池178 电压	RO	U16	1	1	mV				
179	3179	Rack Cell 179 Voltage	Rack单体电池179 电压	RO	U16	1	1	mV				
180	3180	Rack Cell 180 Voltage	Rack单体电池180 电压	RO	U16	1	1	mV				
181	3181	Rack Cell 181 Voltage	Rack单体电池181 电压	RO	U16	1	1	mV				
182	3182	Rack Cell 182 Voltage	Rack单体电池182 电压	RO	U16	1	1	mV				
183	3183	Rack Cell 183 Voltage	Rack单体电池183 电压	RO	U16	1	1	mV				
184	3184	Rack Cell 184 Voltage	Rack单体电池184 电压	RO	U16	1	1	mV				

185	3185	Rack Cell 185 Voltage	Rack单体电池185 电压	RO	U16	1	1	mV				
186	3186	Rack Cell 186 Voltage	Rack单体电池186 电压	RO	U16	1	1	mV				
187	3187	Rack Cell 187 Voltage	Rack单体电池187 电压	RO	U16	1	1	mV				
188	3188	Rack Cell 188 Voltage	Rack单体电池188 电压	RO	U16	1	1	mV				
189	3189	Rack Cell 189 Voltage	Rack单体电池189 电压	RO	U16	1	1	mV				
190	3190	Rack Cell 190 Voltage	Rack单体电池190 电压	RO	U16	1	1	mV				
191	3191	Rack Cell 191 Voltage	Rack单体电池191 电压	RO	U16	1	1	mV				
192	3192	Rack Cell 192 Voltage	Rack单体电池192 电压	RO	U16	1	1	mV				
193	3193	Rack Cell 193 Voltage	Rack单体电池193 电压	RO	U16	1	1	mV				
194	3194	Rack Cell 194 Voltage	Rack单体电池194 电压	RO	U16	1	1	mV				
195	3195	Rack Cell 195 Voltage	Rack单体电池195 电压	RO	U16	1	1	mV				
196	3196	Rack Cell 196 Voltage	Rack单体电池196 电压	RO	U16	1	1	mV				
197	3197	Rack Cell 197 Voltage	Rack单体电池197 电压	RO	U16	1	1	mV				
198	3198	Rack Cell 198 Voltage	Rack单体电池198 电压	RO	U16	1	1	mV				
199	3199	Rack Cell 199 Voltage	Rack单体电池199 电压	RO	U16	1	1	mV				
200	3200	Rack Cell 200 Voltage	Rack单体电池200 电压	RO	U16	1	1	mV				
201	3201	Rack Cell 201 Voltage	Rack单体电池201 电压	RO	U16	1	1	mV				
202	3202	Rack Cell 202 Voltage	Rack单体电池202 电压	RO	U16	1	1	mV				

203	3203	Rack Cell 203 Voltage	Rack单体电池203 电压	RO	U16	1	1	mV				
204	3204	Rack Cell 204 Voltage	Rack单体电池204 电压	RO	U16	1	1	mV				
205	3205	Rack Cell 205 Voltage	Rack单体电池205 电压	RO	U16	1	1	mV				
206	3206	Rack Cell 206 Voltage	Rack单体电池206 电压	RO	U16	1	1	mV				
207	3207	Rack Cell 207 Voltage	Rack单体电池207 电压	RO	U16	1	1	mV				
208	3208	Rack Cell 208 Voltage	Rack单体电池208 电压	RO	U16	1	1	mV				
209	3209	Rack Cell 209 Voltage	Rack单体电池209 电压	RO	U16	1	1	mV				
210	3210	Rack Cell 210 Voltage	Rack单体电池210 电压	RO	U16	1	1	mV				
211	3211	Rack Cell 211 Voltage	Rack单体电池211 电压	RO	U16	1	1	mV				
212	3212	Rack Cell 212 Voltage	Rack单体电池212 电压	RO	U16	1	1	mV				
213	3213	Rack Cell 213 Voltage	Rack单体电池213 电压	RO	U16	1	1	mV				
214	3214	Rack Cell 214 Voltage	Rack单体电池214 电压	RO	U16	1	1	mV				
215	3215	Rack Cell 215 Voltage	Rack单体电池215 电压	RO	U16	1	1	mV				
216	3216	Rack Cell 216 Voltage	Rack单体电池216 电压	RO	U16	1	1	mV				
217	3217	Rack Cell 217 Voltage	Rack单体电池217 电压	RO	U16	1	1	mV				
218	3218	Rack Cell 218 Voltage	Rack单体电池218 电压	RO	U16	1	1	mV				
219	3219	Rack Cell 219 Voltage	Rack单体电池219 电压	RO	U16	1	1	mV				
220	3220	Rack Cell 220 Voltage	Rack单体电池220 电压	RO	U16	1	1	mV				

221	3221	Rack Cell 221 Voltage	Rack单体电池221 电压	RO	U16	1	1	mV				
222	3222	Rack Cell 222 Voltage	Rack单体电池222 电压	RO	U16	1	1	mV				
223	3223	Rack Cell 223 Voltage	Rack单体电池223 电压	RO	U16	1	1	mV				
224	3224	Rack Cell 224 Voltage	Rack单体电池224 电压	RO	U16	1	1	mV				
225	3225	Rack Cell 225 Voltage	Rack单体电池225 电压	RO	U16	1	1	mV				
226	3226	Rack Cell 226 Voltage	Rack单体电池226 电压	RO	U16	1	1	mV				
227	3227	Rack Cell 227 Voltage	Rack单体电池227 电压	RO	U16	1	1	mV				
228	3228	Rack Cell 228 Voltage	Rack单体电池228 电压	RO	U16	1	1	mV				
229	3229	Rack Cell 229 Voltage	Rack单体电池229 电压	RO	U16	1	1	mV				
230	3230	Rack Cell 230 Voltage	Rack单体电池230 电压	RO	U16	1	1	mV				
231	3231	Rack Cell 231 Voltage	Rack单体电池231 电压	RO	U16	1	1	mV				
232	3232	Rack Cell 232 Voltage	Rack单体电池232 电压	RO	U16	1	1	mV				
233	3233	Rack Cell 233 Voltage	Rack单体电池233 电压	RO	U16	1	1	mV				
234	3234	Rack Cell 234 Voltage	Rack单体电池234 电压	RO	U16	1	1	mV				
235	3235	Rack Cell 235 Voltage	Rack单体电池235 电压	RO	U16	1	1	mV				
236	3236	Rack Cell 236 Voltage	Rack单体电池236 电压	RO	U16	1	1	mV				
237	3237	Rack Cell 237 Voltage	Rack单体电池237 电压	RO	U16	1	1	mV				
238	3238	Rack Cell 238 Voltage	Rack单体电池238 电压	RO	U16	1	1	mV				

239	3239	Rack Cell 239 Voltage	Rack单体电池239 电压	RO	U16	1	1	mV				
240	3240	Rack Cell 240 Voltage	Rack单体电池240 电压	RO	U16	1	1	mV				
241	3241	Rack Cell 241 Voltage	Rack单体电池241 电压	RO	U16	1	1	mV				
242	3242	Rack Cell 242 Voltage	Rack单体电池242 电压	RO	U16	1	1	mV				
243	3243	Rack Cell 243 Voltage	Rack单体电池243 电压	RO	U16	1	1	mV				
244	3244	Rack Cell 244 Voltage	Rack单体电池244 电压	RO	U16	1	1	mV				
245	3245	Rack Cell 245 Voltage	Rack单体电池245 电压	RO	U16	1	1	mV				
246	3246	Rack Cell 246 Voltage	Rack单体电池246 电压	RO	U16	1	1	mV				
247	3247	Rack Cell 247 Voltage	Rack单体电池247 电压	RO	U16	1	1	mV				
248	3248	Rack Cell 248 Voltage	Rack单体电池248 电压	RO	U16	1	1	mV				
249	3249	Rack Cell 249 Voltage	Rack单体电池249 电压	RO	U16	1	1	mV				
250	3250	Rack Cell 250 Voltage	Rack单体电池250 电压	RO	U16	1	1	mV				
251	3251	Rack Cell 251 Voltage	Rack单体电池251 电压	RO	U16	1	1	mV				
252	3252	Rack Cell 252 Voltage	Rack单体电池252 电压	RO	U16	1	1	mV				
253	3253	Rack Cell 253 Voltage	Rack单体电池253 电压	RO	U16	1	1	mV				
254	3254	Rack Cell 254 Voltage	Rack单体电池254 电压	RO	U16	1	1	mV				
255	3255	Rack Cell 255 Voltage	Rack单体电池255 电压	RO	U16	1	1	mV				
256	3256	Rack Cell 256 Voltage	Rack单体电池256 电压	RO	U16	1	1	mV				

257	3257	Rack Cell 257 Voltage	Rack单体电池257电压	RO	U16	1	1	mV				
258	3258	Rack Cell 258 Voltage	Rack单体电池258电压	RO	U16	1	1	mV				
259	3259	Rack Cell 259 Voltage	Rack单体电池259电压	RO	U16	1	1	mV				
260	3260	Rack Cell 260 Voltage	Rack单体电池260电压	RO	U16	1	1	mV				
Pack cell temperature												
1	3501	Rack Cell 1 Temperature	Rack单体电池1温度	RO	S16	1	10	°C				
2	3502	Rack Cell 2 Temperature	Rack单体电池2温度	RO	S16	1	10	°C				
3	3503	Rack Cell 3 Temperature	Rack单体电池3温度	RO	S16	1	10	°C				
4	3504	Rack Cell 4 Temperature	Rack单体电池4温度	RO	S16	1	10	°C				
5	3505	Rack Cell 5 Temperature	Rack单体电池5温度	RO	S16	1	10	°C				
6	3506	Rack Cell 6 Temperature	Rack单体电池6温度	RO	S16	1	10	°C				
7	3507	Rack Cell 7 Temperature	Rack单体电池7温度	RO	S16	1	10	°C				
8	3508	Rack Cell 8 Temperature	Rack单体电池8温度	RO	S16	1	10	°C				
9	3509	Rack Cell 9 Temperature	Rack单体电池9温度	RO	S16	1	10	°C				
10	3510	Rack Cell 10 Temperature	Rack单体电池10温度	RO	S16	1	10	°C				
11	3511	Rack Cell 11 Temperature	Rack单体电池11温度	RO	S16	1	10	°C				
12	3512	Rack Cell 12 Temperature	Rack单体电池12温度	RO	S16	1	10	°C				
13	3513	Rack Cell 13 Temperature	Rack单体电池13温度	RO	S16	1	10	°C				
14	3514	Rack Cell 14 Temperature	Rack单体电池14温度	RO	S16	1	10	°C				
15	3515	Rack Cell 15 Temperature	Rack单体电池15温度	RO	S16	1	10	°C				
16	3516	Rack Cell 16 Temperature	Rack单体电池16温度	RO	S16	1	10	°C				
17	3517	Rack Cell 17 Temperature	Rack单体电池17温度	RO	S16	1	10	°C				

18	3518	Rack Cell 18 Temperature	Rack单体电池18温度	RO	S16	1	10	°C				
19	3519	Rack Cell 19 Temperature	Rack单体电池19温度	RO	S16	1	10	°C				
20	3520	Rack Cell 20 Temperature	Rack单体电池20温度	RO	S16	1	10	°C				
21	3521	Rack Cell 21 Temperature	Rack单体电池21温度	RO	S16	1	10	°C				
22	3522	Rack Cell 22 Temperature	Rack单体电池22温度	RO	S16	1	10	°C				
23	3523	Rack Cell 23 Temperature	Rack单体电池23温度	RO	S16	1	10	°C				
24	3524	Rack Cell 24 Temperature	Rack单体电池24温度	RO	S16	1	10	°C				
25	3525	Rack Cell 25 Temperature	Rack单体电池25温度	RO	S16	1	10	°C				
26	3526	Rack Cell 26 Temperature	Rack单体电池26温度	RO	S16	1	10	°C				
27	3527	Rack Cell 27 Temperature	Rack单体电池27温度	RO	S16	1	10	°C				
28	3528	Rack Cell 28 Temperature	Rack单体电池28温度	RO	S16	1	10	°C				
29	3529	Rack Cell 29 Temperature	Rack单体电池29温度	RO	S16	1	10	°C				
30	3530	Rack Cell 30 Temperature	Rack单体电池30温度	RO	S16	1	10	°C				
31	3531	Rack Cell 31 Temperature	Rack单体电池31温度	RO	S16	1	10	°C				
32	3532	Rack Cell 32 Temperature	Rack单体电池32温度	RO	S16	1	10	°C				
33	3533	Rack Cell 33 Temperature	Rack单体电池33温度	RO	S16	1	10	°C				
34	3534	Rack Cell 34 Temperature	Rack单体电池34温度	RO	S16	1	10	°C				
35	3535	Rack Cell 35 Temperature	Rack单体电池35温度	RO	S16	1	10	°C				

36	3536	Rack Cell 36 Temperature	Rack单体电池36温度	RO	S16	1	10	°C				
37	3537	Rack Cell 37 Temperature	Rack单体电池37温度	RO	S16	1	10	°C				
38	3538	Rack Cell 38 Temperature	Rack单体电池38温度	RO	S16	1	10	°C				
39	3539	Rack Cell 39 Temperature	Rack单体电池39温度	RO	S16	1	10	°C				
40	3540	Rack Cell 40 Temperature	Rack单体电池40温度	RO	S16	1	10	°C				
41	3541	Rack Cell 41 Temperature	Rack单体电池41温度	RO	S16	1	10	°C				
42	3542	Rack Cell 42 Temperature	Rack单体电池42温度	RO	S16	1	10	°C				
43	3543	Rack Cell 43 Temperature	Rack单体电池43温度	RO	S16	1	10	°C				
44	3544	Rack Cell 44 Temperature	Rack单体电池44温度	RO	S16	1	10	°C				
45	3545	Rack Cell 45 Temperature	Rack单体电池45温度	RO	S16	1	10	°C				
46	3546	Rack Cell 46 Temperature	Rack单体电池46温度	RO	S16	1	10	°C				
47	3547	Rack Cell 47 Temperature	Rack单体电池47温度	RO	S16	1	10	°C				
48	3548	Rack Cell 48 Temperature	Rack单体电池48温度	RO	S16	1	10	°C				
49	3549	Rack Cell 49 Temperature	Rack单体电池49温度	RO	S16	1	10	°C				
50	3550	Rack Cell 50 Temperature	Rack单体电池50温度	RO	S16	1	10	°C				
51	3551	Rack Cell 51 Temperature	Rack单体电池51温度	RO	S16	1	10	°C				
52	3552	Rack Cell 52 Temperature	Rack单体电池52温度	RO	S16	1	10	°C				
53	3553	Rack Cell 53 Temperature	Rack单体电池53温度	RO	S16	1	10	°C				

54	3554	Rack Cell 54 Temperature	Rack单体电池54温度	RO	S16	1	10	°C				
55	3555	Rack Cell 55 Temperature	Rack单体电池55温度	RO	S16	1	10	°C				
56	3556	Rack Cell 56 Temperature	Rack单体电池56温度	RO	S16	1	10	°C				
57	3557	Rack Cell 57 Temperature	Rack单体电池57温度	RO	S16	1	10	°C				
58	3558	Rack Cell 58 Temperature	Rack单体电池58温度	RO	S16	1	10	°C				
59	3559	Rack Cell 59 Temperature	Rack单体电池59温度	RO	S16	1	10	°C				
60	3560	Rack Cell 60 Temperature	Rack单体电池60温度	RO	S16	1	10	°C				
61	3561	Rack Cell 61 Temperature	Rack单体电池61温度	RO	S16	1	10	°C				
62	3562	Rack Cell 62 Temperature	Rack单体电池62温度	RO	S16	1	10	°C				
63	3563	Rack Cell 63 Temperature	Rack单体电池63温度	RO	S16	1	10	°C				
64	3564	Rack Cell 64 Temperature	Rack单体电池64温度	RO	S16	1	10	°C				
65	3565	Rack Cell 65 Temperature	Rack单体电池65温度	RO	S16	1	10	°C				
66	3566	Rack Cell 66 Temperature	Rack单体电池66温度	RO	S16	1	10	°C				
67	3567	Rack Cell 67 Temperature	Rack单体电池67温度	RO	S16	1	10	°C				
68	3568	Rack Cell 68 Temperature	Rack单体电池68温度	RO	S16	1	10	°C				
69	3569	Rack Cell 69 Temperature	Rack单体电池69温度	RO	S16	1	10	°C				
70	3570	Rack Cell 70 Temperature	Rack单体电池70温度	RO	S16	1	10	°C				
71	3571	Rack Cell 71 Temperature	Rack单体电池71温度	RO	S16	1	10	°C				

72	3572	Rack Cell 72 Temperature	Rack单体电池72温度	RO	S16	1	10	°C				
73	3573	Rack Cell 73 Temperature	Rack单体电池73温度	RO	S16	1	10	°C				
74	3574	Rack Cell 74 Temperature	Rack单体电池74温度	RO	S16	1	10	°C				
75	3575	Rack Cell 75 Temperature	Rack单体电池75温度	RO	S16	1	10	°C				
76	3576	Rack Cell 76 Temperature	Rack单体电池76温度	RO	S16	1	10	°C				
77	3577	Rack Cell 77 Temperature	Rack单体电池77温度	RO	S16	1	10	°C				
78	3578	Rack Cell 78 Temperature	Rack单体电池78温度	RO	S16	1	10	°C				
79	3579	Rack Cell 79 Temperature	Rack单体电池79温度	RO	S16	1	10	°C				
80	3580	Rack Cell 80 Temperature	Rack单体电池80温度	RO	S16	1	10	°C				
81	3581	Rack Cell 81 Temperature	Rack单体电池81温度	RO	S16	1	10	°C				
82	3582	Rack Cell 82 Temperature	Rack单体电池82温度	RO	S16	1	10	°C				
83	3583	Rack Cell 83 Temperature	Rack单体电池83温度	RO	S16	1	10	°C				
84	3584	Rack Cell 84 Temperature	Rack单体电池84温度	RO	S16	1	10	°C				
85	3585	Rack Cell 85 Temperature	Rack单体电池85温度	RO	S16	1	10	°C				
86	3586	Rack Cell 86 Temperature	Rack单体电池86温度	RO	S16	1	10	°C				
87	3587	Rack Cell 87 Temperature	Rack单体电池87温度	RO	S16	1	10	°C				
88	3588	Rack Cell 88 Temperature	Rack单体电池88温度	RO	S16	1	10	°C				
89	3589	Rack Cell 89 Temperature	Rack单体电池89温度	RO	S16	1	10	°C				

90	3590	Rack Cell 90 Temperature	Rack单体电池90温度	RO	S16	1	10	°C				
91	3591	Rack Cell 91 Temperature	Rack单体电池91温度	RO	S16	1	10	°C				
92	3592	Rack Cell 92 Temperature	Rack单体电池92温度	RO	S16	1	10	°C				
93	3593	Rack Cell 93 Temperature	Rack单体电池93温度	RO	S16	1	10	°C				
94	3594	Rack Cell 94 Temperature	Rack单体电池94温度	RO	S16	1	10	°C				
95	3595	Rack Cell 95 Temperature	Rack单体电池95温度	RO	S16	1	10	°C				
96	3596	Rack Cell 96 Temperature	Rack单体电池96温度	RO	S16	1	10	°C				
97	3597	Rack Cell 97 Temperature	Rack单体电池97温度	RO	S16	1	10	°C				
98	3598	Rack Cell 98 Temperature	Rack单体电池98温度	RO	S16	1	10	°C				
99	3599	Rack Cell 99 Temperature	Rack单体电池99温度	RO	S16	1	10	°C				
100	3600	Rack Cell 100 Temperature	Rack单体电池100温度	RO	S16	1	10	°C				
101	3601	Rack Cell 101 Temperature	Rack单体电池101温度	RO	S16	1	10	°C				
102	3602	Rack Cell 102 Temperature	Rack单体电池102温度	RO	S16	1	10	°C				
103	3603	Rack Cell 103 Temperature	Rack单体电池103温度	RO	S16	1	10	°C				
104	3604	Rack Cell 104 Temperature	Rack单体电池104温度	RO	S16	1	10	°C				
105	3605	Rack Cell 105 Temperature	Rack单体电池105温度	RO	S16	1	10	°C				
106	3606	Rack Cell 106 Temperature	Rack单体电池106温度	RO	S16	1	10	°C				
107	3607	Rack Cell 107 Temperature	Rack单体电池107温度	RO	S16	1	10	°C				

108	3608	Rack Cell 108 Temperature	Rack单体电池108 温度	RO	S16	1	10	°C				
109	3609	Rack Cell 109 Temperature	Rack单体电池109 温度	RO	S16	1	10	°C				
110	3610	Rack Cell 110 Temperature	Rack单体电池110 温度	RO	S16	1	10	°C				
111	3611	Rack Cell 111 Temperature	Rack单体电池111 温度	RO	S16	1	10	°C				
112	3612	Rack Cell 112 Temperature	Rack单体电池112 温度	RO	S16	1	10	°C				
113	3613	Rack Cell 113 Temperature	Rack单体电池113 温度	RO	S16	1	10	°C				
114	3614	Rack Cell 114 Temperature	Rack单体电池114 温度	RO	S16	1	10	°C				
115	3615	Rack Cell 115 Temperature	Rack单体电池115 温度	RO	S16	1	10	°C				
116	3616	Rack Cell 116 Temperature	Rack单体电池116 温度	RO	S16	1	10	°C				
117	3617	Rack Cell 117 Temperature	Rack单体电池117 温度	RO	S16	1	10	°C				
118	3618	Rack Cell 118 Temperature	Rack单体电池118 温度	RO	S16	1	10	°C				
119	3619	Rack Cell 119 Temperature	Rack单体电池119 温度	RO	S16	1	10	°C				
120	3620	Rack Cell 120 Temperature	Rack单体电池120 温度	RO	S16	1	10	°C				
121	3621	Rack Cell 121 Temperature	Rack单体电池121 温度	RO	S16	1	10	°C				
122	3622	Rack Cell 122 Temperature	Rack单体电池122 温度	RO	S16	1	10	°C				
123	3623	Rack Cell 123 Temperature	Rack单体电池123 温度	RO	S16	1	10	°C				
124	3624	Rack Cell 124 Temperature	Rack单体电池124 温度	RO	S16	1	10	°C				
125	3625	Rack Cell 125 Temperature	Rack单体电池125 温度	RO	S16	1	10	°C				

126	3626	Rack Cell 126 Temperature	Rack单体电池126 温度	RO	S16	1	10	°C				
127	3627	Rack Cell 127 Temperature	Rack单体电池127 温度	RO	S16	1	10	°C				
128	3628	Rack Cell 128 Temperature	Rack单体电池128 温度	RO	S16	1	10	°C				
129	3629	Rack Cell 129 Temperature	Rack单体电池129 温度	RO	S16	1	10	°C				
130	3630	Rack Cell 130 Temperature	Rack单体电池130 温度	RO	S16	1	10	°C				
131	3631	Rack Cell 131 Temperature	Rack单体电池131 温度	RO	S16	1	10	°C				
132	3632	Rack Cell 132 Temperature	Rack单体电池132 温度	RO	S16	1	10	°C				
133	3633	Rack Cell 133 Temperature	Rack单体电池133 温度	RO	S16	1	10	°C				
134	3634	Rack Cell 134 Temperature	Rack单体电池134 温度	RO	S16	1	10	°C				
135	3635	Rack Cell 135 Temperature	Rack单体电池135 温度	RO	S16	1	10	°C				
136	3636	Rack Cell 136 Temperature	Rack单体电池136 温度	RO	S16	1	10	°C				
137	3637	Rack Cell 137 Temperature	Rack单体电池137 温度	RO	S16	1	10	°C				
138	3638	Rack Cell 138 Temperature	Rack单体电池138 温度	RO	S16	1	10	°C				
139	3639	Rack Cell 139 Temperature	Rack单体电池139 温度	RO	S16	1	10	°C				
140	3640	Rack Cell 140 Temperature	Rack单体电池140 温度	RO	S16	1	10	°C				
141	3641	Rack Cell 141 Temperature	Rack单体电池141 温度	RO	S16	1	10	°C				
142	3642	Rack Cell 142 Temperature	Rack单体电池142 温度	RO	S16	1	10	°C				
143	3643	Rack Cell 143 Temperature	Rack单体电池143 温度	RO	S16	1	10	°C				

144	3644	Rack Cell 144 Temperature	Rack单体电池144 温度	RO	S16	1	10	°C				
145	3645	Rack Cell 145 Temperature	Rack单体电池145 温度	RO	S16	1	10	°C				
146	3646	Rack Cell 146 Temperature	Rack单体电池146 温度	RO	S16	1	10	°C				
147	3647	Rack Cell 147 Temperature	Rack单体电池147 温度	RO	S16	1	10	°C				
148	3648	Rack Cell 148 Temperature	Rack单体电池148 温度	RO	S16	1	10	°C				
149	3649	Rack Cell 149 Temperature	Rack单体电池149 温度	RO	S16	1	10	°C				
150	3650	Rack Cell 150 Temperature	Rack单体电池150 温度	RO	S16	1	10	°C				

	#Address	English Name	Chinese Name	#R/W	#Type	#Size	#SF	#Units	Range	Flash Sav	Note(English)	Note(Chinese)
<i>Parallel System</i>												
1	10400	Inverter Quantity	并机系统系统标记/ 并机系统机器数量	RO	U16	1	1	N/A			1 : Single inverter system > 1 : Parallel system	设置的机器数量是1， 即单机系统；机器数 量大于1即并机系统
2	10401	Firmware Version_ARM	ARM软件版本号	RO	U16	1	1	N/A			Refer to 35020	参考寄存器 35020
3	10402	Firmware Version_DSP_Master	主DSP软件版本号	RO	U16	1	1	N/A			Refer to 35016	参考寄存器35016
4	10403	Firmware	副DSP软件版本号	RO	U16	1	1	N/A			Refer to 35017	参考寄存器35017
5	10404	Online Quantity	机器在线数量	RO	U16	1	1	N/A				
6	10405	APP Mode	EMS 工作模式	RO	U16	1	1	N/A			Refer to app mode	参考寄存器 47000
7	10406	Safety Country	系统安规	RO	U16	1	1	N/A			Refer to 45244	参考寄存器 45244
8	10407	Work Mode	并机工作模式	RO	U16	1	1	N/A			workmode : waiting/Ongrid/battery mode/check/fault Refer to 35187	工作模式 : waiting/Ongrid/battery mode/check/fault 参考 35187
9	10408	Meter comm status	电表通信状态	RO	U16	1	1	N/A			Refer to 36003	参考寄存器 36003
10	10409	BackUp Enable	backup功能开关状 态	RO	U16	1	1	N/A			Refer to 45252	参考寄存器 45252
11	10410	Controller Status Code	控制状态码	RO	U16	1	1	N/A			This address is used for host control of DRED, RCR, one-click shutdown, etc. It is a multiplexed address. When DRED is enabled, it serves as the DRED status code; when RCR is enabled, it serves as the RCR status code.	用于主机控制 DRED,RCR,一键关断 等， 本地址为复用地址当 DRED使能时为DRED 状态码，RCR使能时 为RCR状态码

12	10411	Reserved	预留	RO	U16	1	1	N/A				
13	10412	PV Total Power	PV总功率	RO	U32	2	1	W			Master DSP add up all PV power	由主dsp累加各路PV , arm直接读取
14	10414	Battery Total power	电池总功率	RO	S32	2	1	W			Refer to 35182-35183	参考寄存器 35182-35183
15	10416	Total Back-Up Load Power	Backup总功率	RO	S32	2	1	W			Refer to 35169-35170	参考寄存器 35169-35170
16	10418	Meter Power	电表功率	RO	S32	2	1	W			Refer to 36025-36026	参考寄存器 36025-36026
17	10420	Total Inverter Power	逆变总功率	RO	S32	2	1	W			Refer to 35137-35138	参考寄存器 35137-35138
18	10422	R Phase Inverter Power	R相逆变功率	RO	S32	2	1	W			Refer to 35124-35125	参考寄存器 35124-35125
19	10424	S Phase Inverter Power	S相逆变功率	RO	S32	2	1	W			Refer to 35129-35130	参考寄存器 35129-35130
20	10426	T Phase Inverter Power	T相逆变功率	RO	S32	2	1	W			Refer to 35134-35135	参考寄存器 35134-35135
21	10428	Backup Active Power-R	Back-Up负载端R相功率	RO	S32	2	1	W			Refer to 35149-35150	参考寄存器 35149-35150
22	10430	Backup Active Power-S	Back-Up负载端S相功率	RO	S32	2	1	W			Refer to 35155-35156	参考寄存器 35155-35156
23	10432	Backup Active Power-T	Back-Up负载端T相功率	RO	S32	2	1	W			Refer to 35161-35162	参考寄存器 35161-35162
24	10434	Meter Frequence	MeterFrequence	RO	U16	1	100	HZ			Refer to 36014	参考寄存器 36014
25	10435	Battery Mode	充放电模式	RO	U16	1	1	N/A			Refer to 35184	参考寄存器 35184
26	10436	Battery Voltage 1	电池电压1	RO	U16	1	10	V			Refer to 35180	参考寄存器 35180
27	10437	Battery Voltage 2	电池电压2	RO	U16	1	10	V			Refer to 35262	参考寄存器 35262
28	10438	Total Inverter Power	机器额定总功率	RO	U32	2	1	W			Refer to 35001	参考寄存器35001
29	10440	Generator operating mode	发电机当前状态	RO	U16	1	1	N/A	0: 市电模式:1:发电机模式		refer to 35369	参考寄存器35368
30	10441	Generator R Phase Power	发电机R相功率	RO	S32	2	1	W			Generator Total R Phase Power In Parallel	并机展示发电机R相总功率
31	10443	Generator R Phase Power	发电机S相功率	RO	S32	2	1	W			Generator Total S Phase Power In Parallel	并机展示发电机S相总功率
32	10445	Generator R Phase Power	发电机T相功率	RO	S32	2	1	W			Generator Total T Phase Power In Parallel	并机展示发电机T相总功率

33	10447	Generator Total Power	发电机总功率	RO	S32	2	1	W			Generator Total Power In Parallel	并机展示发电机总功率
34	10449	Total Generator Energy	发电机输出电能	RO	U32	2	1	kwh			Total value of Generator electric energy sold by the system meter	并机展示发电机输出电能
35	10451	DSP workmode	DSP工作模式	RO	U16	1	1	N/A			Dsp workmode in parallel	并机展示DSP工作模
36	10452	BMS1 Max DisCharge Power	BMS1 电池最大放电功率	RO	U32	2	1	W			Max discharge power in parallel	并机展示电池最大放电功率
37	10454	BMS1 Max Charge Power	BMS1 电池最大充电功率	RO	U32	2	1	W			Max charge power in parallel	并机展示电池最大充电功率
38	10456	EMS Power Mode	能量管理模式	RO	U16	1	N/A	N/A			refer to 47511	参考47511(数采并机使用)
39	10457	ECO Mode Enable	经济模式使能	RO	U16	1	N/A	N/A			refer to 47612	参考47612(数采并机使用)
40	10458	App Work Mode Index	APP 工作模式索引	RO	U16	1	N/A	N/A			refer to 47000	参考47000(数采并机使用)
41	10459	BMS2 Max DisCharge Power	BMS2电池最大放电功率	RO	U32	2	1	W			Max discharge power in parallel	并机展示电池最大放电功率
42	10461	BMS2 Max Charge Power	BMS2 电池最大充电功率	RO	U32	2	1	W			Max charge power in parallel	并机展示电池最大充电功率
1	10470	Meter Check Value	CT 检测结果	RO	U16	1	1	N/A			Refer to 47001	参考寄存器47001，可以使用同一变量
2	10471	Meter Connect Check Flag	CT检测使能状态	RO	U16	1	1	N/A			Refer to 47002	参考47002，可使用同一变量，10439只读，47002读写
3	10472	SOC	每台SOC	RO	U16	1	1	N/A			The cloud board calculates the average value based on the SOC of each machine and acquires the register	云板根据每台机器SOC计算平均值，获取每台寄存器37007

[illegible]

Evcharger												
1	10600	CP Voltage	CP电压	RO	U16	1	10	V				
2	10601	Leak Current	漏电流	RO	U16	1	1	mA				
3	10602	CP Status	CP状态	RO	U16	1	NA	NA			0 disconnected 1 connected 2 charging 3 fault	0 连接断开 1 已连接 2 充电中 3 故障
4	10603	Solar Power for Charge	太阳能充电	RO	U16	1	1	W				
5	10604	Battery Power for Charge	电池充电	RO	U16	1	1	W				
6	10605	Grid Power for Charge	电网充电	RO	U16	1	1	W				
7	10606	Current Charge Energy	当前充电量	RO	U16	1	10	kwh				
8	10607	Current Charge Time	当前充电时间	RO	U16	1	1	min				
9	10608	Charge Current	充电电流	RO	U16	1	10	A				
10	10609	EV Charger Status	电动汽车充电器状态	RO	U16	1	N/A	N/A			0 Initial state 1 Fault state 2 Standby state, no machine failure, waiting for external conditions to be met. 3 Running state 4 Shutdown (no fault, user-initiated shutdown) 5 Reservation status 6 Self-testing 7 Starting 8 The machine is ready to power on, but it still needs some time to really power on.	0 初始状态 1 故障状态 2 待机状态，机器无故障，等待外部条件满足 3 运行状态 4 停机（没有故障，用户主动关机） 5 预约状态 6 自检中 7 启动中 8 机器做好开机准备，但还需要等待一定时间才能真正开机

11	10610	EV Charger Error Message	电动汽车充电器故障信息	RO	U32	2	N/A	N/A				
12	10612	Charging Mode	充电模式	RO	U16	1	N/A	N/A				
13	10613	Max Charge Current	最大充电电流	RO	U16	1	1	A				
14	10614	Current Charge Power	当前充电电流	RO	U16	1	1	W				
15	10615	Total Charge Energy	总充电量	RO	U32	2	10	kwh				
16	10617	Total Charge Time	总充电时间	RO	U32	2	10	H				
17	10619	EV Ouput Voltage	EV端口电压	RO	U16	1	10	V				
18	10620	AC Input Voltage	AC输入电压	RO	U16	1	10	V				
19	10621	EV Communication lost flag	EV通讯丢失位	RO	U16	1	N/A	N/A			0: Com not loss 1: Com loss	0 : 通讯未丢失 1 : 通讯丢失
1	10669	GIT Version	软件版本	RO	U16	1	N/A	N/A				
2	10670	Software Version	软件版本	RO	U16	1	N/A	N/A				
3	10671	EV charger SN	电动车充电器序列	RO	STR	8	N/A	N/A				
4	10679	EV Charger Model Name	电动车充电器模型名称	RO	STR	16	N/A	N/A				
Evcharger Fault												
1	10700	Total Pages	总页数	RO	U16	1	N/A	N/A				
2	10701	Current Page	当前页	RO	U16	1	N/A	N/A			Auto-increment for each read Read page can be set via 20321 to prevent read failure	每读一次自动递增 可通过20321设置读取页，防止读取失败
3	10702	Error Time	故障时间	RO	U32	2	N/A	N/A			unixtime	unixtime

4	10704	Error Code H	高位故障代码	RO	U16	1	N/A	N/A			0 Over Temperature 1 Overload 2 Utility over-voltage 3 Utility undervoltage 4 Utility overcurrent 5 Utility overfrequency 6 Utility under-frequency 7 User shutdown 8 Emergency shutdown 9 Low temperature 10 Leakage current 11 low res ground fault 12 Bluetooth connection failure 13 lock rocker blocked 14 relay abnormal 15 Ground fault	0 过温 1 过载 2 市电过压 3 市电欠压 4 市电过流 5 市电过频 6 市电欠频 7 用户关机 8 紧急停机 9 温度过低 10 漏电流 11 低电阻接地故障 12 蓝牙连接故障 13 lock摇臂遇阻 14 relay异常 15 接地故障
5	10705	Error Code L	低位故障代码	RO	U16	1	N/A	N/A			0 Inverted phase 1 Abnormal leakage current circuit 2 Attitude detection 3 Bluetooth module matching fault 4 Range module matching fault 5 can communication abnormality 6 Meter communication failure 7 cp abnormality,cp below 2V 8 B-gun over-temperature 9 B-gun low temperature	0 反相 1 漏电流电路异常 2 姿态检测 3 蓝牙模块匹配故障 4 测距模块匹配故障 5 can 通讯异常 6 电表通信故障 7 cp异常 cp低于2V 8 B枪过温 9 B枪低温
6	10706	Error Information	故障信息	RO	U32	2	N/A	N/A				

7	10708	Error Code Count	故障代码计数	RO	U16	1	N/A	N/A				
8	10709	Check Sum	检测汇总	RO	U16	1	N/A	N/A				
9	10710	...	...	RO		48	N/A	N/A			Only 8 records are passed per page, one record occupies 8	每页只传8条记录，一条记录占8个寄存器
10	10758	Error Time	故障时间	RO	U32	2	N/A	N/A			unixtime	unixtime
11	10760	Error Code H	高位故障代码	RO	U16	1	N/A	N/A				
12	10761	Error Code L	低位故障代码	RO	U16	1	N/A	N/A				
13	10762	Error Information	故障信息	RO	U32	2	N/A	N/A				
14	10764	Error Code Count	故障代码计数	RO	U16	1	N/A	N/A				
15	10765	Check Sum	检测汇总	RO	U16	1	N/A	N/A				
16	10766	Last Page	最后一页	RO	U16	1	N/A	N/A				
Evcharger Charging												
1	10800	Total Pages	总页数	RO	U16	1	N/A	N/A				
2	10801	Current Page	当前页数	RO	U16	1	N/A	N/A			Auto-increment for each read Read page can be set via 20321 to prevent read failure	每读一次自动递增 可通过20322设置读取页，防止读取失败
3	10802	Operation Time	运行时间	RO	U32	2	N/A	N/A			unixtime	unixtime
4	10804	Charging Mode	充电模式	RO	U16	1	N/A	N/A				

[illegible]

ABD												
1	10900	Grid Voltage RS	RS相电网电压	RO	U16	1	10	V				
2	10901	Grid Voltage R	R相电网电压	RO	U16	1	10	V				
3	10902	Grid Voltage S	S相电网电压	RO	U16	1	10	V				
4	10903	Grid Frequency	电网频率	RO	U16	1	100	Hz				
5	10904	Relay Status	延时状态	RO	U16	1	N/A	N/A				
6	10905	Error Message	故障信息	RO	U32	2	N/A	N/A			bit0 +12V detect fail bit1 relay fail bit2 Utility over-voltage bit3 Utility undervoltage bit4 Utility overfrequency bit5 Utility under-frequency	bit0 +12V异常 bit1 relay故障 bit2 市电过压 bit3 市电欠压 bit4 市电过频 bit5 市电欠频 bit6 通信丢失
7	10907	Inverter Voltage RS	RS相逆变电压	RO	U16	1	10	V				
8	10908	Inverter Voltage R	R相逆变电压	RO	U16	1	10	V				
9	10909	Inverter Voltage S	S相逆变电压	RO	U16	1	10	V				
10	10910	Inverter Frequency	逆变频率	RO	U16	1	100	Hz				
11	10911	Power Voltage	功率电压	RO	U16	1	N/A	V				
12	10912	Box Com Loss	Box通讯丢失位	RO	U16	1	N/A	N/A			0 : Com not loss 1 : Com loss	0 : 通讯未丢失 1 : 通讯丢失
13	10913	Stop Button Flag	停止按钮标志	RO	U16	1	N/A	N/A				
1	10980	Software Verion	软件版本	RO	U16	1	N/A	N/A				
2	10981	ABD SN	ABD序列号	RO	STR	8	N/A	N/A				
3	10989	ABD Model Name	ABD模型名称	RO	STR	16	N/A	N/A				
4	11005	git version	小版本	RO	U16	1	N/A	N/A				

485 parallel												
1	11280	485 Parallel Baud	485并机接口波特	RO	U32	2	N/A	N/A		Y		
2	11282	485 Parallel Addr	并机地址	RW	U16	1	N/A	N/A		Y		
3	11283	485 Parallel Role	节点类型	RW	U16	1	N/A	N/A		Y	0x00: Slave machine 0x01: Master machine 0x02: Paralleled Unit(For Automatic Networking) 0x03: Single Unit Forming Three-Phase (For Automatic Networking) 0x04: Multiple Units Forming Three-Phase (For Automatic Networking) 0xFFFF: Standalone	0x00:从机 0x01:主机 0x02:并机 (自动组网用) 0x03:单机组三相(自动组网用) 0x04:多机组三相(自动组网用) 0xFFFF:单机(默认值)
4	11284	485 Parallel State	组网状态	RO	U16	1	N/A	N/A		Y	0x00: Network setup failed 0x01: Network setup successful 0x02: Networking in progress (For automatic	0x00:组网失败 0x01:组网成功 0x02 : 组网中(自动组网用)
7	11287	Parallel operation anti-reverse current protection	并机防逆流保护使能开关	RW	U16	1	N/A	N/A			0 : close , 1 : open	0 : 关闭 , 1 : 打开
8	11288	RSVD	RSVD	RW	U16	1	N/A	N/A				
EvCharger Setting												
1	20300	Start Time_1	开始时间_1	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
2	20301	End Time_1	结束时间_1	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
3	20302	Work Week_1	工作星期_1	RW	U16	1	N/A	N/A		Y		
4	20303	Parameter1_1	参数1_1	RW	U16	1	N/A	N/A		Y		
5	20304	Parameter1_2	参数1_2	RW	U16	1	N/A	N/A		Y		
6	20305	Start Time_2	开始时间_2	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		

7	20306	End Time_2	结束时间_2	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
8	20307	Work Week_2	工作星期_2	RW	U16	1	N/A	N/A		Y		
9	20308	Parameter2_1	参数2_1	RW	U16	1	N/A	N/A		Y		
10	20309	Parameter2_2	参数2_2	RW	U16	1	N/A	N/A		Y		
11	20310	Start Time_3	开始时间_3	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
12	20311	End Time_3	结束时间_3	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
13	20312	Work Week_3	工作星期_3	RW	U16	1	N/A	N/A		Y		
14	20313	Parameter3_1	参数3_1	RW	U16	1	N/A	N/A		Y		
15	20314	Parameter3_2	参数3_2	RW	U16	1	N/A	N/A		Y		
16	20315	Start Time_4	开始时间_4	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
17	20316	End Time_4	结束时间_4	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
18	20317	Work Week_4	工作星期_4	RW	U16	1	N/A	N/A		Y		
19	20318	Parameter4_1	参数4_1	RW	U16	1	N/A	N/A		Y		
20	20319	Parameter4_2	参数4_2	RW	U16	1	N/A	N/A		Y		
21	20320	Charging Mode	充电模式	RW	U16	1	N/A	N/A		Y	0:Plug and Play 1:Appoint charging 2:PV charge only 3:Check 1 and 2 for default plug-and-charge	0:即插即充 1:预约充电 2:仅PV充电 3:复选1和2 默认即插即充

22	20321	Error Page Set	故障页设置	RW	U16	1	N/A	N/A		N		
23	20322	Log Page Set	日志页设置	RW	U16	1	N/A	N/A		N		
24	20323	Max Charge Current	最大充电电流	RW	U16	1	1	A		Y		
25	20324	Real-Time Clock_Year Month	实时时钟_年月	RW	U16	1	1	N/A		Y	High Byte Year/Low Byte Month:13-99/1-12	高字节年/低字节 月:13-99/1-12
26	20325	Real-Time Clock_Day Hour	实时时钟_日时	RW	U16	1	1	N/A		Y	High Byte Day/Low Byte Hour:1-31/0-23	高字节日/低字节时:1- 31/0-23
27	20326	Real-Time Clock_Minute Second	实时时钟_分秒	RW	U16	1	1	N/A		Y	High Byte minute/Low Byte Second:0-59/0-59	高字节分，低字节 秒:0-59/0-59
33	20332	Off Grid Charge Enable	离网充电使能	RW	U16	1	1	N/A		Y	0: Disabled(default) 1: Enabled	0:失能 1:使能 默认0
ARM功能设置参数2 26000~26399												
1	26000	rsvd	预留	RW	U16	1	1	N/A	[0,1]	Y		
2	26001	rsvd	预留	RW	U16	1	1	Kwh	[0,	Y		
3	26002	rsvd	预留	RW	U16	1	1	Kwh	[0,	Y		
4	26003	rsvd	预留	RW	U16	1	1	%	[0, 100]	Y		
5	26004	Buzzer swtich	蜂鸣器开关	RW	U16	1	N/A	N/A	[0, 1]	Y	0-close; 1-open;	0-关闭;1-开启;
7	26006	JA12 enable Switch	JA12使能开关	RW	U16	1	1	N/A	[0, 1]	Y	0-close; 1-open;	0-关闭;1-开启;
8	26007	Certified capacity	认证容量	RW	U16	1	1	Kwh	[0,	Y		
9	26008	BMS1 system capacity	BMS1电池容量	RW	U16	1	1	Kwh	[0,	Y		
10	26009	JA12 BMS1 limit SOC	JA12限制BMS1 SOC	RW	U16	1	1	%	[0, 100]	Y	Send the quotient of the setting value of parameter 26007 divided by the setting value of parameter 26008 to the device	26007的设置值除以 26008的设置值下发设 备
11	26010	BMS2 Battery capacity	BMS2电池容量	RW	U16	1	1	Kwh	[0,	Y	Platform-side Usage	平台端使用
12	26011	JA12 BMS2 limit SOC	JA12限制BMS2 SOC	RW	U16	1	1	%	[0, 100]	Y		
ARM BMS&EMS管理设置参数2 26400~26799												
1	26400	DOD Charge Power permillage	DOD充电功率千分 比	RW	U16	1	1	‰	[10,1000]	Y		

	#Address	English Name	Chinese Name	#R/W	#Type	#Size	#SF	#Units	Range	Flash Save	Note(English)	Note(Chinese)
32000 - 32099 (Error Message)												
1	32000	Utility Fault 1	电网端故障信息1	RO	U16	1	1					
2	32001	Utility Fault 2	电网端故障信息2	RO	U16	1	1				NA	NA
3	32002	System Fault 1	系统故障信息1	RO	U16	1	1				NA	NA
4	32003	System Fault 2	系统故障信息2	RO	U16	1	1				NA	NA
5	32004	Device Fault 1	设备故障信息1	RO	U16	1	1				NA	NA
6	32005	Device Fault 2	设备故障信息2	RO	U16	1	1				NA	NA
7	32006	DC Fault 1	DC侧故障信息	RO	U16	1	1				NA	NA
8	32007	PV Fault 1	PV侧故障信息1	RO	U16	1	1				NA	NA
9	32008	PV Fault 2	PV侧故障信息2	RO	U16	1	1				NA	NA
10	32009	BAT Fault 1	Bat侧故障信息	RO	U16	1	1				NA	NA
11	32010	BAT Fault 2	Bat侧故障信息	RO	U16	1	1				NA	NA
12	32011	BAT Fault 3	Bat侧故障信息	RO	U16	1	1				NA	NA
17	32016	Alarm 1	告警信息1	RO	U16	1	1				NA	NA
18	32017	Alarm 2	告警信息2	RO	U16	1	1				NA	NA
22	32021	System Fault 3	系统故障信息3	RO	U16	1	1					
23	32022	Alarm 3	告警信息3	RO	U16	1	1					
1	32026	wSolar Fault Summary	wSolar故障信息汇总	RO	U32	2	1				NA	NA
2	32028	wSolar Alarm Summary	wSolar警告信息汇总	RO	U32	2	1				NA	NA
3	32030	Grid Voltage Protection Subcode	电网电压保护子码	RO	U16	1	1				NA	NA
4	32031	Grid Frequency Protection Subcode	电网频率保护子码	RO	U16	1	1				NA	NA

5	32032	Internal Com Module	内部通讯模块子码	RO	U16	1	1				NA	NA
6	32033	Sensor Abnormal Subcode	传感器异常子码	RO	U16	1	1				NA	NA
7	32034	Relay Abnormal	继电器异常子码	RO	U16	1	1				NA	NA
8	32035	Internal Fun Fault Subcode	内部风扇故障子码	RO	U16	1	1				NA	NA
9	32036	External Fun Fault Subcode	外部风扇故障子码	RO	U16	1	1				NA	NA
10	32037	Temperature Fault Subcode	温度故障子码	RO	U16	1	1				NA	NA
11	32038	PV IGBT Short Circuit Fault Subcode	PV IGBT 短路故障子码	RO	U16	1	1				NA	NA
12	32039	PV IGBT Open Circuit Fault Subcode	PV IGBT开路故障子码	RO	U16	1	1				NA	NA
13	32040	PV HCT Fault Subcode	PV HCT故障子码	RO	U16	1	1				NA	NA
14	32041	PV Over Voltage	PV输入过压子码	RO	U16	1	1				NA	NA
15	32042	PV Continuous Hardware Overcurrent Subcode	PV硬件持续过流子码	RO	U16	1	1				NA	NA
16	32043	PV Continuous Software Overcurrent Subcode	PV软件持续过流子码	RO	U16	1	1				NA	NA

17	32044	FlyCap Software Overvoltage	飞跨电压软件 过压子码	RO	U16	1	1				NA	NA
18	32045	FlyCap Hardware Overvoltage Subcode	飞跨电压硬件 过压子码	RO	U16	1	1				NA	NA
19	32046	FlyCap Undervoltage Subcode	飞跨电压欠压 子码	RO	U16	1	1				NA	NA
20	32047	FlyCap Precharge Abnormal Subcode	飞跨电容预充 失败子码	RO	U16	1	1				NA	NA
21	32048	FlyCap Precharge Limit Subcode	飞跨电容无法 预充子码	RO	U16	1	1				NA	NA
22	32049	String Overcurrent Fault Subcode Low	组串过流故障 子码低位	RO	U16	1	1				NA	NA
23	32050	String Overcurrent Fault Subcode High	组串过流故障 子码高位	RO	U16	1	1				NA	NA
24	32051	String Reversed Fault Subcode Low	组串反接故障 子码低位	RO	U16	1	1				NA	NA
25	32052	String Reversed Fault Subcode High	组串反接故障 子码高位	RO	U16	1	1				NA	NA
26	32053	Fuse Fault Subcode Low	熔丝故障子码 低位	RO	U16	1	1				NA	NA

27	32054	Fuse Fault Subcode High	熔丝故障子码 高位	RO	U16	1	1				NA	NA
28	32055	PV Hardware Single Overcurrent Subcode	PV硬件单次过 流子码	RO	U16	1	1				NA	NA
29	32056	PV Software Single Overcurrent Subcode	PV软件单次过 流子码	RO	U16	1	1				NA	NA
30	32057	PV Low Energy Subcode	PV能量低子码	RO	U16	1	1				NA	NA
31	32058	Model Recognition Fault Subcode	机种识别故障 子码	RO	U16	1	1				NA	NA
32	32059	External Comm Failure	外部通讯断链	RO	U16	1	1				NA	NA
33	32060	Trip Switch Trip Warning Subcode	脱扣开关跳脱 警告子码	RO	U16	1	1				NA	NA
34	32061	History PV IGBT Short Circuit Warning Subcode	历史PV IGBT 短路警告子码	RO	U16	1	1				NA	NA
35	32062	History PV Reverse Warning Subcode	历史PV反接警 告子码	RO	U32	1	1				NA	NA
37	32064	Flash Read And Write Warning	Flash读写警告 子码	RO	U16	1	1				NA	NA

38	32065	Mos Persistent Overvoltage Subcode	Mos持续过压子 码	RO	U16	1	1				NA	NA
39	32066	Mos Single Overvoltage Subcode	Mos单次过压子 码	RO	U16	1	1				NA	NA
40	32067	Parallel IO Fault Subcode	并机IO故障子 码	RO	U16	1	1				NA	NA
41	32068	SVG Pre- charge failure	SVG预充失败 子码	RO	U16	1	1				NA	NA
42	32069	PID Fault Subcode	PID故障子码	RO	U16	1	1				NA	NA
43	32070	String Mismatch	组串失配子码	RO	U16	1	1				NA	NA
44	32071	PID/SVG Boost Module Fault Subcode	PID/SVG升压 模块故障子码	RO	U16	1	1				NA	NA
45	32072	PCS Voltage And Current Protection Subcode	PCS电压电流 保护子码	RO	U16	1	1				NA	NA
46	32073	AFCI Fault Subcode	AFCI故障子码	RO	U32	1	1				NA	NA
48	32075	Generator Fault Subcode	发电机故障子 码	RO	U16	1	1				NA	NA
49	32076	String Terminal Overtemperatu	组串端子温度 高子码1	RO	U16	1	1				NA	NA
50	32077	String Terminal Overtemperatu	组串端子温度 高子码2	RO	U16	1	1				NA	NA

51	32078	String Terminal Overtemperatu	组串端子温度高子码3	RO	U16	1	1				NA	NA
52	32079	String Reverse Fault Subcode3	组串反接故障子码3	RO	U16	1	1				NA	NA
53	32080	Fuse Fault Subcode3	熔丝故障子码3	RO	U16	1	1				NA	NA
54	32081	String Overcurrent Fault Subcode3	组串过流故障子码3	RO	U16	1	1				NA	NA
55	32082	String Reversed Warning	历史PV反接警告3子码	RO	U16	1	1				NA	NA
56	32083	AFCI Fault Subcode3	AFCI故障子码3	RO	U16	1	1				NA	NA
57	32084	Relay Fault SubCode2	继电器异常子码2	RO	U16	1	1				NA	NA
58	32085	Ref Abnormal SubCode	基准电压异常子码	RO	U16	1	1				NA	NA
32100 - 32119 (Operation Message)												
1	32100	Device Status	设备状态	RO	U16	1	1				NA	NA
2	32101	Running Mark	运行标志1	RO	U16	1	1				NA	NA
3	32102	Running Mark	运行标志2	RO	U16	1	1				NA	NA
4	32103	Running Mark	运行标志3	RO	U16	1	1				NA	NA
5	32104	Running Mark	运行标志4	RO	U16	1	1				NA	NA
6	32105	Running Mark	运行标志5	RO	U16	1	1				NA	NA
7	32106	Running Mark	运行标志6	RO	U16	1	1				NA	NA
8	32107	Running Mark	运行标志7	RO	U16	1	1				NA	NA
9	32108	Running Mark	运行标志8	RO	U16	1	1				NA	NA
10	32109	Switch Status 1	功能开关状态1	RO	U16	1	1				NA	NA
11	32110	Switch Status 2	功能开关状态2	RO	U16	1	1				NA	NA
12	32111	Switch Status 3	功能开关状态3	RO	U16	1	1				NA	NA

13	32112	Switch Status 4	功能开关状态4	RO	U16	1	1				NA	NA
14	32113	Switch Status 5	功能开关状态5	RO	U16	1	1				NA	NA
15	32114	Switch Status 6	功能开关状态6	RO	U16	1	1				NA	NA
32200- 32219(ARM Fault Message)												
1	32200	External Device Protection Fault	外部设备保护故障	RO	U16	1	1				bit0 : Arc Fault(Subcode) , bit1 : DRED0 bit2 : External Shutdown Signal bit3 : Meter NG bit4 : Meter Abnormal	bit0 : 拉弧故障 (子码) , bit1 : DRED0 bit2 : 外部关断指令 , bit3 : 硬防逆流电表NG bit4 : 异常电表
2	32201	Flash Fault	flash故障	RO	U16	1	1				bit0 : Generation data storage failure, Parameter info, BAT info	bit0 : 发电量参数存储失败,参数信息, 电池参数
3	32202	Reserved	预留	RO	U16	1	1					
4	32203	Com Fault 1	通讯故障1	RO	U16	1	1				dsp , box , ev , commodule , ble , meter ,	dsp , box , ev , commodule , ble , meter ,
5	32204	Com Fault 2	通讯故障2	RO	U16	1	1					
6	32205	Burning Abnormal	烧录异常	RO	U16	1	1				dsp , meter , afci , box , ev	dsp , meter , afci , box , ev
7	32206	Soft Reset	软复位	RO	U16	1	1				Watchdog , can com failure, function safe	看门狗, can通讯失败, 功能安全
8	32207	Reserved 1	预留1	RO	U16	1	1					
9	32208	BMS1 Fault 1	BMS1故障1	RO	U16	1	1				Refer to Subcode list	详见子码列表
10	32209	BMS1 Fault 2	BMS1故障2	RO	U16	1	1				Refer to Subcode list	详见子码列表

11	32210	BMS2 Fault 1	BMS2故障1	RO	U16	1	1				Refer to Subcode list	详见子码列表
12	32211	BMS2 Fault 2	BMS2故障2	RO	U16	1	1				Refer to Subcode list	详见子码列表
13	32212	LCD Fault	LCD故障状态	RO	U16	1	1				bit0 : 0 comm OK ; 1 comm NG	bit0:0为通信正常, 1为通信故障
14	32213	Cabinet Parallel Fault	并柜故障	RO	U16	1	1				bit0 : Error parallel setting bit1 : Master setup error bit2 : Slaver setup error bit3 : Parallel communication failure bit4 : Master shutdown setting	bit0 : 并机设置错误 bit1 : 主机设置错误 bit2 : 从机设置错误 bit3 : 并机通讯故障 bit4 : 主机设置停机
15	32214	BMS1 Fault 3	BMS1故障3	RO	U16	1	1				Refer to Subcode List	详见子码列表
16	32215	BMS2 Fault 3	BMS2故障3	RO	U16	1	1				Refer to Subcode List	详见子码列表
17	32216	Contactor fault	接触器故障	RO	U16	1	1				Refer to Subcode List	详见子码列表
18	32217	Parallel ARM fault	并机ARM故障	RO	U16	1	1				Refer to the wsolar list	详见wsolar列表
32220- 32439(ARM Fault Message)												
1	32220	BAT1 Voltage Protection Subcode	电池1电压保护子码	RO	U16	1	1					

2	32221	BAT1 Current Protection Subcode	电池1电流保护子码	RO	U16	1	1					
3	32222	BAT1 Temperature Protection Subcode	电池1温度保护子码	RO	U16	1	1					
4	32223	BAT1 Unbalance Protection	电池1不平衡保护子码	RO	U16	1	1					
5	32224	BAT1 ISO Protection/Coll ecting Thread Subcode	电池1绝缘电阻保护/采集线子码	RO	U16	1	1					
6	32225	BAT1 Other Protection Subcode	电池1其他保护子码	RO	U16	1	1					
7	32226	AC System Protection Subcode	空调系统保护子码	RO	U16	1	1					
8	32227	BAT2 Voltage Protection Subcode	电池2电压保护子码	RO	U16	1	1					
9	32228	BAT2 Current Protection Subcode	电池2电流保护子码	RO	U16	1	1					
10	32229	BAT2 Temperature Protection Subcode	电池2温度保护子码	RO	U16	1	1					
11	32230	BAT2 Unbalance Protection	电池2不平衡保护子码	RO	U16	1	1					

12	32231	BAT2 ISO Protection/Collecting Thread Subcode	电池2绝缘电阻/采集线保护子码	RO	U16	1	1					
13	32232	BAT2 Other Protection Subcode	电池2其他保护子码	RO	U16	1	1					
14	32233	Reserved	预留	RO	U16	1	1					
15	32234	Arc Fault Subcode	拉弧故障子码	RO	U16	1	1				bit0-bit3 respectively express four	bit0-bit3 分别表示四路故障
16	32235	Function Safety Fault Subcode	功能安全故障子码	RO	U16	1	1					
17	32236	Burning Abnormal Failure	烧录异常失败子码	RO	U16	1	1					
18	32237	DCDC1 Fault Subcode	DCDC1故障子码	RO	U16	1	1					
19	32238	DCDC1 Alarm Subcode	DCDC1告警子码	RO	U16	1	1					
20	32239	DCDC2 Fault Subcode	DCDC2故障子码	RO	U16	1	1					
21	32240	DCDC2 Alarm Subcode	DCDC2告警子码	RO	U16	1	1					
22	32241	BMS1ClusterCirculatingCurrentProrSubCode	第一路BMS簇间环流保护子码	RO	U16	1	1					
23	32242	BMS1 Other Fault3 SubCode(LES)	BMS1其他故障3子码(大储类)	RO	U16	1	1					
24	32243	BMS1Cooling Warning1SubC	第一路BMS液冷_1告警子码	RO	U16	1	1					

25	32244	BMS1CoolingWarning2SubC	第一路BMS液冷_2告警子码	RO	U16	1	1					
26	32245	BMS1CoolingFaultSubCode	第一路BMS液冷故障子码	RO	U16	1	1					
27	32246	BMS1Apollo_CabinetInternalEquipmentFaultSubCode	第一路BMS阿波罗柜内设备故障子码	RO	U16	1	1					
28	32247	BMS2ClusterCirculatingCurrentProrSubCode	第二路BMS簇间环流保护子码	RO	U16	1	1					
29	32248	BMS2 Other Fault3 SubCode(LES)	BMS2其他故障3子码(大储类)	RO	U16	1	1					
30	32249	BMS2CoolingWarning1SubC	第二路BMS液冷_1告警子码	RO	U16	1	1					
31	32250	BMS2CoolingWarning2SubC	第二路BMS液冷_2告警子码	RO	U16	1	1					
32	32251	BMS2CoolingFaultSubCode	第二路BMS液冷故障子码	RO	U16	1	1					
33	32252	BMS2Apollo_CabinetInternalEquipmentFaultSubCode	第二路BMS阿波罗柜内设备故障子码	RO	U16	1	1					
34	32253	AirConditionerFaultSubCode	空调故障子码	RO	U32	2	1					
35	32255	FireFightingFaultSubCode	消防故障子码	RO	U32	2	1					
36	32257	BMS1 Internal Fault	BMS1内部故障子码	RO	U16	1	1					
36	32258	BMS2 Internal Fault	BMS2内部故障子码	RO	U16	1	1					

37	32259	BMS1 Other Fault2 SubCode(RES)	BMS1其他故障 2子码(户储类)	RO	U16	1	1					
38	32260	BMS2 Other Fault2 SubCode(RES)	BMS2其他故障 2子码(户储类)	RO	U16	1	1					
39	32261	BMS1Cooling Warning3SubC	第一路BMS液 冷_3告警子码	RO	U16	1	1					
40	32262	BMS1Cooling Warning4SubC	第一路BMS液 冷_4告警子码	RO	U16	1	1					
41	32263	BMS1Cooling Warning5SubC	第一路BMS液 冷_5告警子码	RO	U16	1	1					
42	32264	BMS2Cooling Warning3SubC	第二路BMS液 冷_3告警子码	RO	U16	1	1					
43	32265	BMS2Cooling Warning4SubC	第二路BMS液 冷_4告警子码	RO	U16	1	1					
44	32266	BMS2Cooling Warning5SubC	第二路BMS液 冷_5告警子码	RO	U16	1	1					
45	32267	BMS1 Other Fault 4 SubCode(LES)	BMS1其他故障 4子码(大储类)	RO	U16	1	1					
46	32268	BMS2 Other Fault 4 SubCode(LES)	BMS2其他故障 4子码(大储类)	RO	U16	1	1					
47	32269	BMS1 Internal Heating Fault	BMS1内部加热 故障子码	RO	U16	1	1					
48	32270	BMS2 Internal Heating Fault	BMS2内部加热 故障子码	RO	U16	1	1					
49	32271	Contactor	接触器故障1	RO	U16	1	1					
50	32272	Contactor	接触器故障2	RO	U16	1	1					

51	32273	BMS1 Other Fault 5 SubCode(RES)	BMS1其他故障 5子码(户储类)	RO	U16	1	1					
52	32274	BMS2 Other Fault 5 SubCode(RES)	BMS2其他故障 5子码(户储类)	RO	U16	1	1					
54	32275	BMS1 Other Fault 6 SubCode(LES)	BMS1其他故障 6子码(大储类)	RO	U16	1	1					
55	32276	BMS2 Other Fault 6 SubCode(LES)	BMS2其他故障 6子码(大储类)	RO	U16	1	1					
32240- 32509(ARM Fault Message)												
1	32479	BMS1 cluster1~2 error status	BMS1 1~2簇故 障状态	RO	U16	1	1				Refer to Subcode List	详见子码列表
2	32480	BMS1 cluster3~4 error status	BMS1 3~4 簇故 障状态	RO	U16	1	1				Refer to Subcode List	详见子码列表
3	32481	BMS1 cluster5~6 error status	BMS1 5~6簇故 障状态	RO	U16	1	1				Refer to Subcode List	详见子码列表
4	32482	BMS1 cluster7~8 error status	BMS1 7~8 簇故 障状态	RO	U16	1	1				Refer to Subcode List	详见子码列表
5	32483	BMS1 cluster9~10 error status	BMS1 9~10簇 故障状态	RO	U16	1	1				Refer to Subcode List	详见子码列表
6	32484	BMS1 cluster11~12 error status	BMS1 11~12簇 故障状态	RO	U16	1	1				Refer to Subcode List	详见子码列表

7	32485	BMS1 cluster13~14 error status	BMS1 13~14 簇故障状态	RO	U16	1	1				Refer to Subcode List	详见子码列表
8	32486	BMS1 cluster15~16 error status	BMS1 15~16簇 故障状态	RO	U16	1	1				Refer to Subcode List	详见子码列表
9	32487	BMS1 cluster17~18 error status	BMS1 17~18簇 故障状态	RO	U16	1	1				Refer to Subcode List	详见子码列表
10	32488	BMS1 cluster19~20 error status	BMS1 19~20簇 故障状态	RO	U16	1	1				Refer to Subcode List	详见子码列表
11	32489	BMS1 cluster21~22 error status	BMS1 21~22 簇故障状态	RO	U16	1	1				Refer to Subcode List	详见子码列表
12	32490	BMS1 cluster23~24 error status	BMS1 23~24簇 故障状态	RO	U16	1	1				Refer to Subcode List	详见子码列表
13	32491	BMS1 cluster25~26 error status	BMS1 25~26簇 故障状态	RO	U16	1	1				Refer to Subcode List	详见子码列表
14	32492	BMS1 cluster27~28 error status	BMS1 27~28簇 故障状态	RO	U16	1	1				Refer to Subcode List	详见子码列表
15	32493	BMS1 cluster29~30 error status	BMS1 29~30簇 故障状态	RO	U16	1	1				Refer to Subcode List	详见子码列表
16	32494	BMS2 cluster1~2 error status	BMS2 1~2簇故 障状态	RO	U16	1	1				Refer to Subcode List	详见子码列表
17	32495	BMS2 cluster3~4 error status	BMS2 3~4簇故 障状态	RO	U16	1	1				Refer to Subcode List	详见子码列表

[illegible]

PV Setting												
1	32950	MPPT Number	MPPT 路数	RO	U16	1	1	NA	[0 ,0]		Example : SDT G3 three-phase inverter Model name: DTG DTI DTK 8- 15K: 1MPPT1PV 17-27K: 2MPPT3PV 30K: 2MPPT4PV Monitoring platform needs model name and power range to	譬如 : SDT G3 三相机 三码: DTG DTI DTK 8-15K: 1MPPT1PV 17-27K: 2MPPT3PV 30K: 2MPPT4PV 监控平台判断路串 需要每个机型的 三码+功率端 来判 断 该机型MPPT总
2	32951	MPPT1 String Number	MPPT1 组串路 数	RO	U16	1	1	NA	[0 ,0]		String numbers for this MPPT	该MPPT支持的组 串路数
3	32952	MPPT2 String Number	MPPT2 组串路 数	RO	U16	1	1	NA	[0 ,0]		String numbers for this MPPT	该MPPT支持的组 串路数
4	32953	MPPT3 String Number	MPPT3 组串路 数	RO	U16	1	1	NA	[0 ,0]		String numbers for this MPPT	该MPPT支持的组 串路数
5	32954	MPPT4 String Number	MPPT4 组串路 数	RO	U16	1	1	NA	[0 ,0]		String numbers for this MPPT	该MPPT支持的组 串路数
6	32955	MPPT5 String Number	MPPT5 组串路 数	RO	U16	1	1	NA	[0 ,0]		String numbers for this MPPT	该MPPT支持的组 串路数

7	32956	MPPT6 String Number	MPPT6 组串路数	RO	U16	1	1	NA	[0 ,0]		String numbers for this MPPT	该MPPT支持的组串路数
8	32957	MPPT7 String Number	MPPT7 组串路数	RO	U16	1	1	NA	[0 ,0]		String numbers for this MPPT	该MPPT支持的组串路数
9	32958	MPPT8 String Number	MPPT8 组串路数	RO	U16	1	1	NA	[0 ,0]		String numbers for this MPPT	该MPPT支持的组串路数
10	32959	MPPT9 String Number	MPPT9 组串路数	RO	U16	1	1	NA	[0 ,0]		String numbers for this MPPT	该MPPT支持的组串路数
11	32960	MPPT10 String Number	MPPT10 组串路数	RO	U16	1	1	NA	[0 ,0]		String numbers for this MPPT	该MPPT支持的组串路数
12	32961	MPPT11 String Number	MPPT11 组串路数	RO	U16	1	1	NA	[0 ,0]		String numbers for this MPPT	该MPPT支持的组串路数
13	32962	MPPT12 String Number	MPPT12 组串路数	RO	U16	1	1	NA	[0 ,0]		String numbers for this MPPT	该MPPT支持的组串路数
14	32963	MPPT13 String Number	MPPT13 组串路数	RO	U16	1	1	NA	[0 ,0]		String numbers for this MPPT	该MPPT支持的组串路数
15	32964	MPPT14 String Number	MPPT14 组串路数	RO	U16	1	1	NA	[0 ,0]		String numbers for this MPPT	该MPPT支持的组串路数
16	32965	MPPT15 String Number	MPPT15 组串路数	RO	U16	1	1	NA	[0 ,0]		String numbers for this MPPT	该MPPT支持的组串路数

	#Address	English Name	Chinese Name	#R/W	#Type	#Size	#SF	#Units	Range	Flash Save	Note(English)	Note(Chinese)
ETC/BTC Message												
17	33524	Peakshaving Mode Start	需量管理模式启动时间_1	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		hh:mm
18	33525	Peakshaving Mode End Time_1	需量管理模式结束时间_1	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
19	33526	Peakshaving Mode Start	需量管理模式启动时间_2	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		hh:mm
20	33527	Peakshaving Mode End Time_2	需量管理模式结束时间_2	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
21	33528	Peakshaving Mode Start	需量管理模式启动时间_3	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		hh:mm
22	33529	Peakshaving Mode End Time_3	需量管理模式结束时间_3	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
23	33530	Peakshaving Mode Start	需量管理模式启动时间_4	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		hh:mm
24	33531	Peakshaving Mode End Time_4	需量管理模式结束时间_4	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
27	33534	Backup Mode Start Time_1	备用模式启动时间_1	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		hh:mm
28	33535	Backup Mode End Time_1	备用模式结束时间_1	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
29	33536	Backup Mode Start Time_2	备用模式启动时间_2	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		hh:mm
30	33537	Backup Mode End Time_2	备用模式结束时间_2	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
31	33538	Backup Mode Start Time_3	备用模式启动时间_3	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		hh:mm
32	33539	Backup Mode End Time_3	备用模式结束时间_3	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		

[illegible]

D	#Address	English Name	Chinese Name	#R/W	#Type	#Size	#SF	#Units	Range	Flash Save	Note(English)	Note(Chinese)
DSP												
1	34900	Smart device 1 status information	智能设备1状态信息	RO	U16	1	1	N/A		N	bit0 - Whether the current position has detected an intelligent device; bit1 to bit2 - Type of the intelligent device; (electric meter / socket) bit3 - Online status of the intelligent device; bit4 - Working status of the intelligent	bit0-当前位置是否搜索到智能设备； bit1~bit2-智能设备类型；(电表/插座) bit3-智能设备在线状态； bit4-智能设备工作状态；
3	34901	Smart device 1SN	智能设备1SN	RO	STR	7	1	N/A		N		
4	34908	Smart device 1 reserved	智能设备1预留	RO	U32	2	1	N/A		N		
5	34910	Smart device 2 status information	智能设备2状态信息	RO	U16	1	1	N/A		N	bit0 - Whether the current position has detected an intelligent device; bit1 to bit2 - Type of the intelligent device; (electric meter / socket) bit3 - Online status of the intelligent device; bit4 - Working status of the intelligent	bit0-当前位置是否搜索到智能设备； bit1~bit2-智能设备类型；(电表/插座) bit3-智能设备在线状态； bit4-智能设备工作状态；
6	34911	Smart device 2SN	智能设备2SN	RO	STR	7	1	N/A		N		
7	34918	Smart device 2 reserved	智能设备2预留	RO	U32	2	1	N/A		N		

8	34920	Smart device 3 status information	智能设备3状态信息	RO	U16	1	1	N/A		N	bit0 - Whether the current position has detected an intelligent device; bit1 to bit2 - Type of the intelligent device; (electric meter / socket) bit3 - Online status of the intelligent device; bit4 - Working status of the intelligent	bit0-当前位置是否搜索到智能设备； bit1~bit2-智能设备类型；(电表/插座) bit3-智能设备在线状态； bit4-智能设备工作状态；
9	34921	Smart device 3SN	智能设备3SN	RO	STR	7	1	N/A		N		
10	34928	Smart device 3 reserved	智能设备3预留	RO	U32	2	1	N/A		N		
11	34930	Smart device 4 status information	智能设备4状态信息	RO	U16	1	1	N/A		N	bit0 - Whether the current position has detected an intelligent device; bit1 to bit2 - Type of the intelligent device; (electric meter / socket) bit3 - Online status of the intelligent device; bit4 - Working status of the intelligent	bit0-当前位置是否搜索到智能设备； bit1~bit2-智能设备类型；(电表/插座) bit3-智能设备在线状态； bit4-智能设备工作状态；
12	34931	Smart device 4SN	智能设备4SN	RO	STR	7	1	N/A		N		
13	34938	Smart device 4 reserved	智能设备4预留	RO	U32	2	1	N/A		N		

[illegible]

Device Info												
2	35001	Rate Power	额定功率	RO	U16	1	1	N/A			Inverter rated power	逆变器额定功率
4	35003	Inverter SN	逆变器序列号	RO	STR	8	1	N/A			ASCII code,16 bytes.Read together, include OEM	ASCII码，16个字节。一起读，包括OEM产品。
5	35011	Model Name	模型名称	RO	STR	5	1	N/A			ASCII code, 10 bytes	ASCII码，10个字节
6	35016	Firmware Version_DSP_Master	主控DSP固件版本	RO	U16	1	N/A	N/A				
7	35017	Firmware Version_DSP_Slave	从属DSP固件版本	RO	U16	1	N/A	N/A				
8	35018	Beta Version_DSP_Master	主控DSP的SVN版本号	RO	U16	1	N/A	N/A				
9	35019	Firmware Version_ARM	ARM固件版本	RO	U16	1	N/A	N/A				
10	35020	Beta Version_ARM	ARM测试版(小版本号)	RO	U16	1	N/A	N/A				
11	35021	DSP Internal Firmware Version	DSP内部固件版本	RO	STR	6	N/A	N/A			Example '04004-13-S01'	例如'04004-13-S01'
12	35027	ARM Internal Firmware Version	ARM内部固件版本	RO	STR	6	N/A	N/A			Example '02034-04-S01'	例如'02034-04-S01'
15	35035	Firmware Version_DSP_DCDC	DCDC_DSP固件版本	RO	U16	1	N/A	N/A			Only For BTC.DCDC module firmware version	仅适用于BTC.DCDC模块固件版本
16	35036	Beta Version_DCDC	DCDC测试版	RO	U16	1	N/A	N/A			Only For BTC.DCDC module beta version	仅适用于BTC.DCDC模块测试版
17	35037	Firmware Version_DSP_MPPT	MPPT_DSP固件版本	RO	U16	1	N/A	N/A			Only For BTC.MPPT module firmware version	仅适用于BTC.MPPT模块固件版本

[illegible]

Inverter Operation Data												
1	35100	Real-Time Clock_Year Month	实时时钟_年月	RO	U16	1	1	N/A			High Byte Year/Low Byte Month:13-99/1-	高字节年/低字节月:13- 99/1-12
2	35101	Real-Time Clock_Day Hour	实时时钟_日时	RO	U16	1	1	N/A			High Byte Day/Low Byte Hour:1-31/0-23	高字节日/低字节时:1- 31/0-23
3	35102	Real-Time Clock_Minute Second	实时时钟_分秒	RO	U16	1	1	N/A			High Byte minute/Low Byte	高字节分, 低字节秒:0- 59/0-59
4	35103	PV1 Voltage	PV电压1	RO	U16	1	10	V				
5	35104	PV1 Current	PV电流1	RO	U16	1	10	A				
6	35105	PV1 Power	PV功率1	RO	U32	2	1	W			200ms Average	200ms平均值
7	35107	PV2 Voltage	PV电压2	RO	U16	1	10	V				
8	35108	PV2 Current	PV电流2	RO	U16	1	10	A				
9	35109	PV2 Power	PV功率2	RO	U32	2	1	W			200ms Average	200ms平均值
10	35111	PV3 Voltage	PV电压3	RO	U16	1	10	V				
11	35112	PV3 Current	PV电流3	RO	U16	1	10	A				
12	35113	PV3 Power	PV功率3	RO	U32	2	1	W			200ms Average	200ms平均值
13	35115	PV4 Voltage	PV电压4	RO	U16	1	10	V				
14	35116	PV4 Current	PV电流4	RO	U16	1	10	A				
15	35117	PV4 Power	PV功率4	RO	U32	2	1	W			200ms Average	200ms平均值
16	35119	PV Mode	PV模式	RO	U32	2	N/A	N/A			To check each MPPT mode,Table 8-3 & 8-4	检测每一个MPPT的模 式, 见表8-3&8-4
17	35121	R Phase Inverter Voltage	R相逆变电压	RO	U16	1	10	V				
18	35122	R Phase Inverter Current	R相逆变电流	RO	U16	1	10	A				
19	35123	R Phase Inverter	R相逆变频率	RO	U16	1	100	Hz				
20	35124	R Phase Inverter Power	R相逆变功率	RO	S32	2	1	W			Inverter Power 200ms Average	逆变功率200ms平均值
21	35126	S Phase Inverter Voltage	S相逆变电压	RO	U16	1	10	V				
22	35127	S Phase Inverter Current	S相逆变电流	RO	U16	1	10	A				
23	35128	S Phase Inverter Frequency	S相逆变频率	RO	U16	1	100	Hz				
24	35129	S Phase Inverter Power	S相逆变功率	RO	S32	2	1	W			Inverter Power 200ms Average	逆变功率200ms平均值
25	35131	T Phase Inverter Voltage	T相逆变电压	RO	U16	1	10	V				

26	35132	T Phase Inverter Current	T相逆变电流	RO	U16	1	10	A				
27	35133	T Phase Inverter Frequency	T相逆变频率	RO	U16	1	100	Hz				
28	35134	T Phase Inverter Power	T相逆变功率	RO	S32	2	1	W			Inverter Power 200ms Average	逆变功率200ms平均值
29	35136	Grid Mode	电网模式	RO	U16	1		N/A			Grid connection status,Refer to Table 8-10	电网连接状态，见表8-10
30	35137	Total Inverter Power	逆变器总功率	RO	S32	2	1	W			200ms Average	200ms平均值
31	35139	AC Active Power	AC有功功率	RO	S32	2	1	W			Total Active on Grid Power Of Inverter. (If meter connection ok, it is meter power.If meter connection fail, it is inverter on-grid port power)	逆变器并网端口的总有功功率。(如果电表连接正常，则为电表功率。如果电表连接失败，则为逆变器并网端口功率。)200ms平均值
32	35141	AC Reactive Power	AC无功功率	RO	S32	2	1	Var			Total Reactive Power Of Inverter	逆变器的总无功功率
33	35143	AC Apparent Power	AC视在功率	RO	S32	2	1	VA			Total Apparent Power Of Inverter	逆变器的总视在功率
34	35145	R Phase Load Voltage on Back-Up	Back-Up负载端 R相电压	RO	U16	1	10	V			Use R phase data for 1-phase inverter	使用R相数据的单相逆变器
35	35146	R Phase Load Current of Back-Up	Back-Up负载端 R相电流	RO	U16	1	10	A				
36	35147	R phase Load Frequency of Back-Up	Back-Up负载端 R相频率	RO	U16	1	100	Hz				
37	35148	Load Mode_R	R相负载模式	RO	U16	1	N/A	N/A			Off means there is No voltage of Backup port. Also used for 1- p inverter.refer to Table 8-11	关闭则Back-up端口没有电压。且用于单相逆变器。见表8-11

38	35149	R Phase Load Power of Back-Up	Back-Up负载端R相功率	RO	S32	2	1	W			200ms Average	200ms平均值
39	35151	S Phase Load Voltage of Back-Up	Back-Up负载端S相电压	RO	U16	1	10	V				
40	35152	S Phase Load Current of Back-Up	Back-Up负载端S相电流	RO	U16	1	10	A				
41	35153	S Phase Load Frequency of Back-Up	Back-Up负载端S相频率	RO	U16	1	100	Hz				
42	35154	Load Mode_S	S相负载模式	RO	U16	1	N/A	N/A			not for 1-phase inverter, refer to Table 8-11	不用于单相逆变器，见表8-11
43	35155	S Phase Load Power of Back-Up	Back-Up负载端S相功率	RO	S32	2	1	W			200ms Average	200ms平均值
44	35157	T Phase Load Voltage of Back-Up	Back-Up负载端T相电压	RO	U16	1	10	V				
45	35158	T Phase Load Current of Back-Up	Back-Up负载端T相电流	RO	U16	1	10	A				
46	35159	T Phase Load Frequency of Back-Up	Back-Up负载端T相频率	RO	U16	1	100	Hz				
47	35160	Load Mode_T	T相负载模式	RO	U16	1	N/A	N/A			not for 1-phase inverter, refer to Table 8-11	不用于单相逆变器，见表8-11
48	35161	T Phase Load Power of Back-Up	Back-Up负载端T相功率	RO	S32	2	1	W			200ms Average	200ms平均值
49	35163	R Phase On-Grid Load Power	负载端R相功率	RO	S32	2	1	W			backup not included	不包括Back-up负载，200ms平均值
50	35165	S Phase On-Grid Load	负载端S相功率	RO	S32	2	1	W				
51	35167	T Phase On-Grid Load	负载端T相功率	RO	S32	2	1	W				

52	35169	Total Back-Up Load Power	Back-up端负载总功率	RO	S32	2	1	W				
53	35171	Total Load Power	负载总功率	RO	S32	2	1	W			Total Power of on-grid load(backup not included)	并网负载的总功率(不包括Back-up负载)
54	35173	Ups Load Percent	Ups负载百分比	RO	U16	1	100	%			BackupLoad Power/Rated power	Backup负载功率/额定功率
55	35174	Air Temperature	空气温度	RO	S16	1	10	°C			Inverter Internal Temperature	逆变器内部温度
56	35175	Model Temperature	模块温度	RO	S16	1	10	°C			Inverter Unit Temperature	逆变器单元温度
57	35176	Hitsink Temperature	散热器温度	RO	S16	1	10	°C			Inverter Heat Sink Temperature	逆变器散热器温度
58	35177	Function Bit Value	比特值函数	RO	U16	1	N/A	N/A			Refer to Table 8-37	
59	35178	BUS Voltage	总线电压	RO	U16	1	10	V				
60	35179	NBUS Voltage	NBUS总线电压	RO	U16	1	10	V				
61	35180	Battery1 Voltage	电池组1电压	RO	U16	1	10	V			If BMS communication successfully, it is the voltage BMS send to	如果BMS通信成功, 则电压由BMS发送给逆变器。
62	35181	Battery1 Current	电池组1电流	RO	S16	1	10	A				
63	35182	Battery1 Power	电池组1功率	RO	S32	2	1	W			200ms Average	200ms平均值
64	35184	Battery1 Mode	电池组1模式	RO	U16	1	N/A	N/A			The charging and discharging status of the battery,Refer toTable 8-9	电池的充放电状态, 见表8-9
65	35185	Warning Code	警告代码	RO	U16	1	N/A	N/A				
66	35186	Safety Country	安规国家	RO	U16	1	N/A	N/A			Refer to Table 8-21	见表8-21
67	35187	Work Mode	工作模式	RO	U16	1	N/A	N/A			Refer to Table 8-1	见表8-1
69	35189	Error Message	故障信息	RO	U32	2	N/A	N/A			Refer to Table 8-12	见表8-12
70	35191	PV Energy-Total	光伏发电总量	RO	U32	2	10	1KW.Hr			The total PV production energy from installation date	自安装之日起光伏发电总量

71	35193	PV Energy-Day	当日光伏发电量	RO	U32	2	10	1KW.Hr			Total PV production energy of the day	当日光伏发电量
73	35197	Hour-Total	总时长	RO	U32	2	1	H			Accumulated operation hours from installation date	自安装之日起累积的工作时数
74	35199	Energy-Day-Sell	当日卖电量	RO	U16	1	10	1KW.Hr			The accumulated exporting energy to grid of the day	当天逆变的输出电量
76	35202	Energy-Day-Buy	当日买电量	RO	U16	1	10	1KW.Hr			The accumulated energy imported from grid of the day	当天逆变的输入电量
77	35203	Energy-Total-Load	负载消耗总电量	RO	U32	2	10	1KW.Hr			From the installation date, not include backup load	自安装之日起，不包括backup负载
78	35205	Energy-Load-Day	当日负载消耗电量	RO	U16	1	10	1KW.Hr			Accumulated load consumption energy of the day, not include backup load	每天负载消耗的电量，不包括backup负载
79	35206	Energy-Battery Charge	电池充电总量	RO	U32	2	10	1KW.Hr			From the installation date, not include backup load	自安装之日起，不包括backup负载
80	35208	Energy-Charge-Day	当日电池充电	RO	U16	1	10	1KW.Hr			Not from BMS	不来自BMS
81	35209	Energy-Battery Discharg	电池放电总量	RO	U32	2	10	1KW.Hr			From the installation date Not from BMS	自安装之日起，不来自BMS
82	35211	Energy-Discharg-Day	当日电池放电	RO	U16	1	10	1KW.Hr			Not from BMS	不来自BMS
83	35212	Battery Strings	电池节数	RO	U16	1	N/A	N/A			number of battery modules	电池模块的数量
84	35213	CPLD Warning Code	CPLD警告代码	RO	U16	1	N/A	N/A			only for after-sales,Refer to Table 8-17	仅用于售后，见表8-17
88	35218	Diagnosis Status High	高位诊断状态	RO	U32	2	N/A	N/A			Refer to Table 8-13	见表8-13
89	35220	Diagnosis Status Low	低位诊断状态	RO	U32	2	N/A	N/A			Refer to Table 8-14	见表8-14

93	35225	EH Battery Function Active	EH电池功能激活	RO	U16	1	N/A	N/A			0: failure 1: Activated 2: Deactivated 3: Locked only for HER	0:失效 1:已激活 2:未 激活 3:锁止 仅用于HER
94	35226	ARC Self Check Status	拉弧自检测状态	RO	U16	1	N/A	N/A			0: failure 1: Activated 2: Deactivated 3: Locked only for inverters with AFCI protection	0:未检测 1:正常 2:检 测中 3:失败 仅用于带AFCI保护的 逆变器
95	35227	Hitsink Temperature2	散热器温度2	RO	S16	1	10	°C			Inverter Heat Sink Temperature	逆变器散热器温度2
96	35228	R Phase Load Voltage on Back-Up	Back-Up2负载 端R相电压	RO	U16	1	10	V			Use R phase data for 1-phase inverter	使用R相数据的单相逆 变器
97	35229	R Phase Load Current of Back-Up	Back-Up2负载 端R相电流	RO	U16	1	10	A				
98	35230	R phase Load Frequency of Back-Up	Back-Up2负载 端R相频率	RO	U16	1	100	Hz				
99	35231	Load Mode_R	Back-Up2R相 负载模式	RO	U16	1	N/A	N/A			Off means there is No voltage of Backup port. Also used for 1- p inverter.refer to Table 8-11	关闭则Back-up端口没 有电压。且用于单相逆 变器。见表8-11
100	35232	R Phase Load Power of Back-Up	Back-Up2负载 端R相功率	RO	S32	2	1	W			200ms Average	200ms平均值
101	35234	S Phase Load Voltage of Back-Up	Back-Up2负载 端S相电压	RO	U16	1	10	V				
102	35235	S Phase Load Current of Back-Up	Back-Up2负载 端S相电流	RO	U16	1	10	A				
103	35236	S Phase Load Frequency of Back-Up	Back-Up2负载 端S相频率	RO	U16	1	100	Hz				
104	35237	Load Mode_S	Back-Up2S相 负载模式	RO	U16	1	N/A	N/A			not for 1-phase inverter,refer to Table 8-11	不用于单相逆变器，见 表8-11
105	35238	S Phase Load Power of Back-Up	Back-Up2负载 端S相功率	RO	S32	2	1	W			200ms Average	200ms平均值

106	35240	T Phase Load Voltage of Back-Up	Back-Up2负载端T相电压	RO	U16	1	10	V				
107	35241	T Phase Load Current of Back-Up	Back-Up2负载端T相电流	RO	U16	1	10	A				
108	35242	T Phase Load Frequency of Back-Up	Back-Up2负载端T相频率	RO	U16	1	100	Hz				
109	35243	Load Mode_T	Back-Up2T相负载模式	RO	U16	1	N/A	N/A			not for 1-phase inverter, refer to Table 8-11	不用于单相逆变器，见表8-11
110	35244	T Phase Load Power of Back-Up	Back-Up负载端T相功率	RO	S32	2	1	W			200ms Average	200ms平均值
111	35246	Total Back-Up Load Power	Back-up2端负载总功率	RO	S32	2	1	W				
112	35248	DSP Com Compleet Flag	DSP通讯完成标志位	RO	U16	1	1	N/A			bit0: Communication Fail bit1: Communication Success	bit0 : 未通讯上 bit1 : 通讯成功

113	35249	DSP Work Mode	控制芯片的工作模式	RO	U16	1	N/A	N/A			bit0 to bit3: Master control state 0 indicates wait; 1 indicates normal; 2 indicates offline; 3 indicates fault; 4 indicates programming; 5 indicates check) bit4 to bit7: Slave control 1 state (defined the same as master control) bit8 to bit11: Slave control 2 state (defined the same as master control) bit12 to bit15: Slave control 3 state (defined the same as	bit0~bit3：主控状态0表示wait；1表示normal；2表示离网；3表示fault；4表示烧录；5表示check) bit4~bit7：副控1状态(定义同主控) bit8~bit11：副控2状态(定义同主控) bit12~bit15：副控3状态(定义同主控)
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Inverter Operation Data												
1	35250	Safety Detailed Error Message	安全详细故障信息	RO	U64	4	1	N/A			Detail information of grid failure,Refer to Table 8-30	电网相关详细故障告警，见表8-30
2	35254	Inverter Detailed Error Message	逆变器详细故障信息	RO	U64	4	1	N/A			Detail information of inverter failure,Refer to Table 8-31	逆变器保护相关详细故障报警，见表8-31
3	35258	Inverter Detailed Status Message	逆变器详细状态信息	RO	U64	4	1	N/A			Refer to Table 8-32	见表8-32
9	35268	Max Grid Frequency within 1minute	每分钟最大电网频率	RO	U16	1	100	Hz				
10	35269	Min Grid Frequency within 1minute	每分钟最小电网频率	RO	U16	1	100	Hz				
11	35270	Max Grid Voltage within 1minute	每分钟R相电网最大电压	RO	U16	1	10	V				
12	35271	Min Grid Voltage within 1minute	每分钟R相电网最小电压	RO	U16	1	10	V				
13	35272	Max Grid Voltage within 1minute	每分钟S相电网最大电压	RO	U16	1	10	V				
14	35273	Min Grid Voltage within 1minute	每分钟S相电网最小电压	RO	U16	1	10	V				
15	35274	Max Grid Voltage within 1minute	每分钟T相电网最大电压	RO	U16	1	10	V				
16	35275	Min Grid Voltage within 1minute	每分钟T相电网最小电压	RO	U16	1	10	V				
17	35276	Max Backup Power within 1minute	每分钟R相Backup负载端最大频率	RO	U32	2	1	W				
18	35278	Max Backup Power within 1minute	每分钟S相Backup负载端最大频率	RO	U32	2	1	W				

19	35280	Backup Power within 1minute	每分钟T相Backup负载端最大频率	RO	U32	2	1	W				
20	35282	Backup Power within 1minute	每分钟Backup负载端最大功	RO	U32	2	1	W				
21	35284	Grid Hvrt Event Times	电网高压穿越次数	RO	U16	1	1	N/A			clear after restart	重启后清零
22	35285	Grid Lvrt Event Times	电网低压穿越次数	RO	U16	1	1	N/A				
26	35292	Offline To Online Delay Second	离网转并网等待时间	RO	U16	1	1	N/A				
27	35293	Grid Hvrt Status	电网高压穿越状态	RO	U16	1	1	N/A				
28	35294	Grid Lvrt Status	电网低压穿越状态	RO	U16	1	1	N/A				
29	35295	Safety National Category Flag	安规国家分类标志	RO	U16	1	1	N/A			1:Australian 2:North American	1:澳洲类 2:北美类
32	35298	Battery1 Sample Voltage	电池组1采样电压	RO	U16	1	10	V				
33	35299	Battery2 Sample Voltage	电池组2采样电压	RO	U16	1	10	V				
35	35301	PV Total Power	总PV功率	RO	U32	2	1	W				
36	35303	PV channel	PV路数	RO	U16	1	1	N/A				
37	35304	PV5 Voltage	PV电压5	RO	U16	1	10	V			String Voltage	和并网保持一致，代表组串电压
38	35305	PV5 Current	PV电流5	RO	U16	1	10	A			String Current	和并网保持一致，代表组串电流
39	35306	PV6 Voltage	PV电压6	RO	U16	1	10	V				
40	35307	PV6 Current	PV电流6	RO	U16	1	10	A				
41	35308	PV7 Voltage	PV电压7	RO	U16	1	10	V				
42	35309	PV7 Current	PV电流7	RO	U16	1	10	A				

43	35310	PV8 Voltage	PV电压8	RO	U16	1	10	V				
44	35311	PV8 Current	PV电流8	RO	U16	1	10	A				
45	35312	PV9 Voltage	PV电压9	RO	U16	1	10	V				
46	35313	PV9 Current	PV电流9	RO	U16	1	10	A				
47	35314	PV10 Voltage	PV电压10	RO	U16	1	10	V				
48	35315	PV10 Current	PV电流10	RO	U16	1	10	A				
49	35316	PV11 Voltage	PV电压11	RO	U16	1	10	V				
50	35317	PV11 Current	PV电流11	RO	U16	1	10	A				
51	35318	PV12 Voltage	PV电压12	RO	U16	1	10	V				
52	35319	PV12 Current	PV电流12	RO	U16	1	10	A				
43	35320	PV13 Voltage	PV电压13	RO	U16	1	10	V				
44	35321	PV13 Current	PV电流13	RO	U16	1	10	A				
45	35322	PV14 Voltage	PV电压14	RO	U16	1	10	V				
46	35323	PV14 Current	PV电流14	RO	U16	1	10	A				
47	35324	PV15 Voltage	PV电压15	RO	U16	1	10	V				
48	35325	PV15 Current	PV电流15	RO	U16	1	10	A				
49	35326	PV16 Voltage	PV电压16	RO	U16	1	10	V				
50	35327	PV16 Current	PV电流16	RO	U16	1	10	A				
51	35328	Warning Message	32bit警告信息	RO	U32	2	N/A	N/A			Warning information, bit mode, different from 35185. See the fault table for specific	警告信息，位模式，区别于35185，具体故障详见不同机型的故障表
52	35330	Grid10minAvgVoltR	电网R相10min平均电压	RO	U16	1	10	V				
53	35331	Grid10minAvgVoltS	电网S相10min平均电压	RO	U16	1	10	V				
54	35332	Grid10minAvgVoltT	电网T相10min平均电压	RO	U16	1	10	V				

55	35333	Error Message Extend	32bit故障信息扩展	RO	U32	2	N/A	N/A			Error information, bit mode, extend from 35189. See the fault table for specific	故障信息，位模式，扩展于35189，具体故障详见不同机型的故障表
57	35335	Warning Message Extend	32bit警告信息扩展	RO	U32	2	N/A	N/A			Warningr information, bit mode, extend from 35328. See the fault table for specific	警告信息，位模式，扩展于35328，具体故障详见不同机型的故障表
59	35337	MPPT Power 1	MPPT1功率	RO	U16	1	1	W			Pv_boost1 total power, not string	为Pv_boost1路总功率，而非组串功率
60	35338	MPPT Power 2	MPPT2功率	RO	U16	1	1	W			Pv_boost2 total power, not string	为Pv_boost2路总功率，而非组串功率
61	35339	MPPT Power 3	MPPT3功率	RO	U16	1	1	W			Pv_boost3 total power, not string	为Pv_boost3路总功率，而非组串功率
62	35340	MPPT Power 4	MPPT4功率	RO	U16	1	1	W			Pv_boost4 total power, not string	为Pv_boost4路总功率，而非组串功率
63	35341	MPPT Power 5	MPPT5功率	RO	U16	1	1	W			Pv_boost5 total power, not string	为Pv_boost5路总功率，而非组串功率
64	35342	MPPT Power 6	MPPT6功率	RO	U16	1	1	W			Pv_boost6 total power, not string	为Pv_boost6路总功率，而非组串功率
65	35343	MPPT Power 7	MPPT7功率	RO	U16	1	1	W			Pv_boost7 total power, not string	为Pv_boost7路总功率，而非组串功率
66	35344	MPPT Power 8	MPPT8功率	RO	U16	1	1	W			Pv_boost8 total power, not string	为Pv_boost8路总功率，而非组串功率
67	35345	MPPT Current 1	MPPT1电流	RO	U16	1	10	A				
68	35346	MPPT Current 2	MPPT2电流	RO	U16	1	10	A				
69	35347	MPPT Current 3	MPPT3电流	RO	U16	1	10	A				
70	35348	MPPT Current 4	MPPT4电流	RO	U16	1	10	A				
71	35349	MPPT Current 5	MPPT5电流	RO	U16	1	10	A				
72	35350	MPPT Current 6	MPPT6电流	RO	U16	1	10	A				
73	35351	MPPT Current 7	MPPT7电流	RO	U16	1	10	A				

74	35352	MPPT Current 8	MPPT8电流	RO	U16	1	10	A				
75	35353	Inverter Reactive Power R	逆变R相无功功率	RO	S32	2	1	VAR				
77	35355	Inverter Reactive Power S	逆变S相无功功率	RO	S32	2	1	VAR				
79	35357	Inverter Reactive Power T	逆变T相无功功率	RO	S32	2	1	VAR				
81	35359	Inverter Apparent Power R	逆变R相视在功率	RO	S32	2	1	VA				
83	35361	Inverter Apparent Power S	逆变S相视在功率	RO	S32	2	1	VA				
85	35363	Inverter Apparent Power T	逆变T相视在功率	RO	S32	2	1	VA				
87	35365	ISO Value	ISO检测值	RO	U16	1	10	KΩ				
88	35366	generator cumulative	发电机累计电	RO	U32	2	10	1kw.h				
88	35368	BMS_Temperature	电池温度采集值	RO	S16	1	10	℃			DSP Reads BAT Temperature	DSP读取电池温度
106	35369	Generator operating mode	发电机工作模式	RO	U16	1	1	N/A			0:Grid mode 1:Generator mode	0：市电模式；1：发电机模式
107	35370	Generator R Phase Voltage	发电机R相电压	RO	U16	1	10	V				
108	35371	Generator R Phase Current	发电机R相电流	RO	U16	1	10	A				
109	35372	Generator R Phaser Frequency	发电机R相频率	RO	U16	1	100	Hz				
110	35373	Generator R Phase Power	发电机R相功率	RO	S32	2	1	W				
112	35375	Generator S Phase Voltage	发电机S相电压	RO	U16	1	10	V				
113	35376	Generator S Phase Current	发电机S相电流	RO	U16	1	10	A				
114	35377	Generator S Phase	发电机S相频率	RO	U16	1	100	Hz				
115	35378	Generator S Phase Power	发电机S相功率	RO	S32	2	1	W				
117	35380	Generator T Phase Voltage	发电机T相电压	RO	U16	1	10	V				
118	35381	Generator T Phase Current	发电机T相电流	RO	U16	1	10	A				
119	35382	Generator T Phase	发电机T相频率	RO	U16	1	100	Hz				
120	35383	Generator T Phase Power	发电机T相功率	RO	S32	2	1	W				

122	35386	DSP Work As Meter	DSP代替内置电表标志	RO	U16	1	1	N/A			Use Uniq, read from DSP. 1 indicates that the DSP replaces the built-in electricity	Uniq使用，从DSP读取，1表示DSP代替内置电表
123	35387	Current Derating Flag	生效降载标志位	RO	U16	1	1	N/A	[0, 0]		0x101: 指令降载_有功调度 0x102: 调度降载_防逆流 0x103: 调度降载_数采指令降载 0x104: 调度降载_RCR降载 0x105: 调度降载_风扇降载 0x201: 过温降载_逆变模块过温 0x202: 过温降载_boost模块过温 0x203: 过温降载_腔体过温 0x301: 过压降载_BUS过压 0x401: 安规降载_PU曲线 0x402: 安规降载_过频降载 0x403: 安规降载_德国70%降载 0x404: 安规降载	
124	35388	Current Derating Ref Value	生效降载功率参考值	RO	S16	1	10	%Pn	[0, 0]			

125	35389	Current Reactive Flag	生效无功运行标志位	RO	U16	1	1	N/A	[0, 0]		0x101: 指令_无功调度 0x102: 指令_数采指令调度 0x201: 安规_QU曲线 0x202: 安规_cosphi_P曲线 0x203: 安规_QP曲线 0x301: 夜间SVG_默认值 0x302: 夜间SVG_调度值 0x401: 其他_端电压控制 0x501: VRT降载_无	
126	35390	Current Reactive power Ref Value	生效无功参考值	RO	S16	1	10	%Pn	[0, 0]			
132	35400	Connect State	设备连接状态	RO	U16	1	1	N/A	[0,0]		0: 默认 1: 首次自检; 2: 非首次上电	
133	35401	MicroGrid Power	微网运行功率	RO	S32	2	1	W	[0, 0]		微网运行功率	
134	35403	MicroGrid Energy-Total	微网总累计电量	RO	U32	2	10	1KW.Hr	[0, 0]		微网累计发电量	
135	35405	MicroGrid Energy-Day	微网当日累计电能	RO	U32	2	10	1KW.Hr	[0, 0]		微网当日发电量	
184	35455	PV Terminal Max Temperature	PV端子最大温度	RO	S16	1	10	°C			Inverter Internal PV Terminal Max Temperature	逆变器内部PV端子最大温度
185	35456	AC Terminal Max Temperature	AC端子最大温度	RO	S16	1	10	°C			Inverter Internal AC Terminal Max Temperature	逆变器内部AC端子最大温度
186	35457	BAT Terminal Max Temperature	电池端子最大温度	RO	S16	1	10	°C			Inverter Internal BAT Terminal Max Temperature	逆变器内部电池端子最大温度

New Registers for BTC												
1	35600	Hitsink Temperaturure-DCDC	DCDC散热器温度	RO	S16	1	10	°C			Only For BTC	仅用于BTC系列
2	35601	Hitsink Temperaturure-MPPT	MPPT散热器温度	RO	S16	1	10	°C				
3	35602	Hitsink Temperaturure-STC	STC散热器温度	RO	S16	1	10	°C				
External Data Collection												
1	36000	Communication Mode	通信代码	RO	U16	1	N/A	N/A			01:WIFI 02:GPRS 03:LAN e20 04:a21 WIFI mode of WIFI+LAN 05:a21 LAN mode of WIFI+LAN module	01:WIFI 02:GPRS 03:LAN e20 04:a21 WIFI mode of WIFI+LAN 05:a21 LAN mode of WIFI+LAN module
2	36001	RSSI	接收的信号强度指示	RO	U16	1	N/A	N/A			RSSI of wifi module	RSSI的WIFI模块
4	36003	Meter Connect Status	电表连接状态	RO	U16	1	N/A	N/A			1:connect correctly 2:connect reverse(CT) 4:connect incorrectly 0:not checked For example: 0X0124 means Phase connect incorrectly , Phase T connect reverse, Phase S	1:连接正确 2:连接保 留(CT) 4:连接不正确 8 : CT和 电压采样同时错相 0:未 检测 例如 : 0X0124就是R相 连接不正确, T相连接 保留, S相连接正确

5	36004	Meter Communication	电表通讯状态	RO	U16	1	N/A	N/A			1:OK 0:NG	1:成功 0:失败
6	36005	Meter Active Power R	电表R相有功功率	RO	S16	1	1	W			If ARM Version>9,please refer to 36019~36041,which can be identified by the server	ARM09版后由36019~36041新地址替换，服务器端识别处理
7	36006	Meter Active Power S	电表S相有功功率	RO	S16	1	1	W				
8	36007	Meter Active Power T	电表T相有功功率	RO	S16	1	1	W				
9	36008	Meter Total Active Power	电表总有功功率	RO	S16	1	1	W				
10	36009	Meter Total Reactive Power	电表总无功功率	RO	S16	1	1	W				
11	36010	Meter Power Factor_R	电表R相功率因数	RO	S16	1	1000	N/A				
12	36011	Meter Power Factor_S	电表S相功率因数	RO	S16	1	1000	N/A				
13	36012	Meter Power Factor_T	电表T相功率因数	RO	S16	1	1000	N/A				
14	36013	Meter Power Factor	电表功率因数	RO	S16	1	1000	N/A				
15	36014	Meter Frequence	电表频率	RO	U16	1	100	N/A				
16	36015	Energy-Total-Sell	卖电总量	RO	float	2	N/A	N/A			Total Feed Energy To Grid. Read from	输给电网的总电量。读取自电表
17	36017	Energy-Total-Buy	买电总量	RO	float	2	N/A	N/A			Total Energy From Grid. Read from	来自电网的总电量。读取自电表

18	36019	Meter Active Power R	电表R相有功功率	RO	S32	2	1	W			the address of ARM Version>9	ARM09版后的地址
19	36021	Meter Active Power S	电表S相有功功率	RO	S32	2	1	W				
20	36023	Meter Active Power T	电表T相有功功率	RO	S32	2	1	W				
21	36025	Meter Total Active Power	电表总有功功	RO	S32	2	1	W				
22	36027	Meter Reactive Power R	电表R相无功功率	RO	S32	2	1	W				
23	36029	Meter Reactive Power S	电表S相无功功率	RO	S32	2	1	W				
24	36031	Meter Reactive Power T	电表T相无功功率	RO	S32	2	1	W				
25	36033	Meter Total Reactive Power	电表总无功功	RO	S32	2	1	W				
26	36035	Meter Apparent Power R	电表R相视在功率	RO	S32	2	1	W				
27	36037	Meter Apparent Power S	电表S相视在功率	RO	S32	2	1	W				
28	36039	Meter Apparent Power T	电表T相视在功率	RO	S32	2	1	W				
29	36041	Meter Total Apparent Power	电表总视在功	RO	S32	2	1	W				

30	36043	Meter Type	电表类型	RO	U16	1	1	N/A			Only for GoodWe Smart Meter(0:Single phase 1:3P3W 2:3P4W 3:HomeKit 4:GM1000D)	仅用于固德威智能电表 (0:单相 1: 三相三线 2:三相四线 3:HomeKit 4:GM1000D)
31	36044	Meter Software Version	电表软件版本	RO	U16	1	1	N/A			Only for GoodWe Smart Meter	仅用于固德威智能电表
32	36045	Meter CT2 Active Power	电表CT2有功功率	RO	S32	2	1	W			Only for AC Couple inverter.Detect PV inverter power	仅用于逆变器AC耦合。检测PV逆变器功率
33	36047	CT2-Energy-Total-Sell	CT2总卖电量	RO	U32	2	100	1KW.Hr				
34	36049	CT2-Energy-Total-Buy	CT2总买电量	RO	U32	2	100	1KW.Hr				
35	36051	MTCT2 Status	MTCT2状态	RO	U16	1	1	N/A				
36	36052	Meter Voltage R	R相电表电压	RO	U16	1	10	V				
37	36053	Meter Voltage S	S相电表电压	RO	U16	1	10	V				
38	36054	Meter Voltage T	T相电表电压	RO	U16	1	10	V				
39	36055	Meter Current R	R相电表电流	RO	U16	1	10	A				

40	36056	Meter Current S	S相电表电流	RO	U16	1	10	A				
41	36057	Meter Current T	T相电表电流	RO	U16	1	10	A				
43	36065	ARC fault channel	拉弧故障通道	RO	U16	1	1	N/A				
44	36066	Parallel Communication Status	并机通讯状态	RO	U16	1	1	N/A			Bit0 (Communication status between SEC1000S and inverters.) --0:NG 1:Success Bit1 (Communication status between parallel inverters.) --	Bit0 (在SEC1000S和逆变器之间通讯状态) 0:失败 1:成功 Bit1 (逆变器并机通讯状态) 0:失败 1:成功
45	36067	ARC Software Version	电弧故障分段器软件版本	RO	U16	1	1	N/A			AFCI Version	电弧故障分段器版本
46	36067	ARC Software Version	电弧故障分段器软件版本	RO	U16	1	1	N/A			AFCI Version	电弧故障分段器版本

55	36092	Active Energy Total Sell_R	R相总有功电量 (卖电)	RO	U64	4	100	1KW.Hr			745、753	ARM745、753平台专用
56	36096	Active Energy Total Sell_S	S相总有功电量 (卖电)	RO	U64	4	100	1KW.Hr				
57	36100	Active Energy Total Sell_T	T相总有功电量 (卖电)	RO	U64	4	100	1KW.Hr				
58	36104	Active Energy Total Sell_Total	总有功电量(卖 电)	RO	U64	4	100	1KW.Hr				
59	36108	Active Energy Total Buy_R	R相总有功电量 (买电)	RO	U64	4	100	1KW.Hr				
60	36112	Active Energy Total Buy_S	S相总有功电量 (买电)	RO	U64	4	100	1KW.Hr				
61	36116	Active Energy Total Buy_T	T相总有功电量 (买电)	RO	U64	4	100	1KW.Hr				
62	36120	Active Energy Total Buy_Total	总有功电量(买 电)	RO	U64	4	100	1KW.Hr				
63	36124	Real-Time Clock_Year Month	实时时钟_年月	RO	U16	1	1	N/A			High Byte Year/Low Byte Month:13-99/1-	高字节年/低字节月:13- 99/1-12
64	36125	Real-Time Clock_Day Hour	实时时钟_日时	RO	U16	1	1	N/A			High Byte Day/Low Byte Hour:1-31/0-23	高字节日/低字节时:1- 31/0-23
65	36126	Real-Time Clock_Minute Second	实时时钟_分秒	RO	U16	1	1	N/A			High Byte minute/Low Byte	高字节分, 低字节秒:0- 59/0-59

66	36127	Meter Active Power R	电表R相有功功率	RO	S32	2	1	W			Used for the data of the second channel metering chip	用于第二路计量芯片数据
67	36129	Meter Active Power S	电表S相有功功率	RO	S32	2	1	W				
68	36131	Meter Active Power T	电表T相有功功率	RO	S32	2	1	W				
69	36133	Meter Total Active Power	电表总有功功	RO	S32	2	1	W				
70	36135	Active Energy Total Sell_R	R相总有功电量(卖电)	RO	U64	4	100	1KW.Hr			745、753	ARM745、753平台专用，用于第二路计量数据
71	36139	Active Energy Total Sell_S	S相总有功电量(卖电)	RO	U64	4	100	1KW.Hr				
72	36143	Active Energy Total Sell_T	T相总有功电量(卖电)	RO	U64	4	100	1KW.Hr				
73	36147	Active Energy Total Sell_Total	总有功电量(卖电)	RO	U64	4	100	1KW.Hr				
74	36151	Active Energy Total Buy_R	R相总有功电量(买电)	RO	U64	4	100	1KW.Hr				
75	36155	Active Energy Total Buy_S	S相总有功电量(买电)	RO	U64	4	100	1KW.Hr				
76	36159	Active Energy Total Buy_T	T相总有功电量(买电)	RO	U64	4	100	1KW.Hr				
77	36163	Active Energy Total Buy_Total	总有功电量(买电)	RO	U64	4	100	1KW.Hr				

82	36171	SAPN FeedPower State 1	SAPN防逆流计 划组1状态	RO	U16	1	1	N/A			unexecuted : 0 running : 1 execution is completed : 2	未执行 : 0 执行中 : 1 执行完成 : 2
83	36172	SAPN FeedPower State 2	SAPN防逆流计 划组2状态	RO	U16	1	1	N/A				
84	36173	SAPN FeedPower State 3	SAPN防逆流计 划组3状态	RO	U16	1	1	N/A				
85	36174	SAPN FeedPower State 4	SAPN防逆流计 划组4状态	RO	U16	1	1	N/A				
86	36175	SAPN FeedPower State 5	SAPN防逆流计 划组5状态	RO	U16	1	1	N/A				
87	36176	SAPN FeedPower State 6	SAPN防逆流计 划组6状态	RO	U16	1	1	N/A				
88	36177	SAPN FeedPower State 7	SAPN防逆流计 划组7状态	RO	U16	1	1	N/A				
89	36178	SAPN FeedPower State 8	SAPN防逆流计 划组8状态	RO	U16	1	1	N/A				
90	36179	SAPN FeedPower State 9	SAPN防逆流计 划组9状态	RO	U16	1	1	N/A				
91	36180	SAPN FeedPower State 10	SAPN防逆流计 划组10状态	RO	U16	1	1	N/A				
92	36181	SAPN FeedPower State 11	SAPN防逆流计 划组11状态	RO	U16	1	1	N/A				
93	36182	SAPN FeedPower State 12	SAPN防逆流计 划组12状态	RO	U16	1	1	N/A				
94	36183	SAPN FeedPower State 13	SAPN防逆流计 划组13状态	RO	U16	1	1	N/A				

95	36184	SAPN FeedPower State 14	SAPN防逆流计 划组14状态	RO	U16	1	1	N/A				
96	36185	SAPN FeedPower State 15	SAPN防逆流计 划组15状态	RO	U16	1	1	N/A				
97	36186	SAPN FeedPower State 16	SAPN防逆流计 划组16状态	RO	U16	1	1	N/A				
98	36187	SAPN FeedPower State 17	SAPN防逆流计 划组17状态	RO	U16	1	1	N/A				
99	36188	SAPN FeedPower State 18	SAPN防逆流计 划组18状态	RO	U16	1	1	N/A				
100	36189	SAPN FeedPower State 19	SAPN防逆流计 划组19状态	RO	U16	1	1	N/A				
101	36190	SAPN FeedPower State 20	SAPN防逆流计 划组20状态	RO	U16	1	1	N/A				
102	36191	SAPN FeedPower State 21	SAPN防逆流计 划组21状态	RO	U16	1	1	N/A				
103	36192	SAPN FeedPower State 22	SAPN防逆流计 划组22状态	RO	U16	1	1	N/A				
104	36193	SAPN FeedPower State 23	SAPN防逆流计 划组23状态	RO	U16	1	1	N/A				
105	36194	SAPN FeedPower State 24	SAPN防逆流计 划组24状态	RO	U16	1	1	N/A				
106	36195	SAPN FeedPower Limit	SAPN实时防逆 流设定值	RO	S32	2	1	W				

107	36197	Generator operating mode	发电机工作模式	RW	U16	1	1	N/A			0:Grid mode 1: Generator mode	0 : 市电模式 ; 1 : 发电机模式
108	36198	Communication Module software version	通讯模块软件版本	RO	STR	10	1	N/A		N		
108	36208	Inverter scheduling mode display	逆变器调度模式展示	RO	U16	1	1	N/A		N	1 : 自发自用 2 : 充电-光伏 3 : 放电-光伏 4 : 导入 5 : 导出 6 : 节约 7 : 离网 8 : 电池待机 9 : 购买电力 10 : 出售电力 11 : 充电 12 : 放电100 : 备用模式 101 : 经济模式——充电 102 : 经济模式——放电 103 : 延时充电——卖电超峰值 104 : 延时充电——卖电未超峰值 105 : 峰值削减——在特定时间段内 106 : 峰值削减——在特定时间段之外	1 : 自发自用 2 : Charge-PV 3 : Discharge-PV 4 : Import 5 : Export 6 : Conserve 7 : Offgrid 8 : Battery standby 9 : Buy Power 10 : Sell Power 11 : Charge 12 : Discharge 100 : 备用模式 101 : 经济模式——充电 102 : 经济模式——放电 103 : 延时充电——卖电超峰值 104 : 延时充电——卖电未超峰值 105 : peakshaving——时间段内 106 :

111	36210	Negative Electric Price Plan State	负电价计划状态	RO	U16	1	1	N/A	[0,4]	N	0 : Negative Electric Price Plan Disable 1 : Sell Electric Plan Enable 2 : Buy price and Sell price is all Positive 3 : No New Negative Electric Price Plan 4 : Buy Electric Plan Enable	0 : 未开启负电价计划功能 1 : 卖电计划生效 2 : 买卖电价格都为正 3 : 未收到更新的负电价计划 4 : 买电计划生效
BMS Operation Data												
1	37000	DRM Status	DRM状态	RO	U16	1	N/A	N/A			DRED only for Australia	仅用于澳洲的命令响应设备
3	37002	BMS Status	BMS状态	RO	U16	1	N/A	N/A				
4	37003	BMS Package Temperature	BMS电池包温	RO	U16	1	10	°C				
5	37004	Max BMS Charge Current	BMS最大充电电流	RO	U16	1	1	A				
6	37005	Max BMS Discharge Current	BMS最大放电电流	RO	U16	1	1	A				
7	37006	BMS Error Code Low	BMS低位故障代码	RO	U16	1	N/A	N/A				
8	37007	SOC	电池剩余电量百分比	RO	U16	1	1	%			Capacity of BAT1	第1组电池容量
9	37008	BMS SOH	电池健康度	RO	U16	1	1	%			BMS SOH	BMS电池健康度

10	37009	BMS Battery Strings	BMS电池节数	RO	U16	1	N/A	N/A				
11	37010	BMS Warning Code Low	BMS低位警告代码	RO	U16	1	N/A	N/A				
12	37011	Battery Protocol	电池协议	RO	U16	1	N/A	N/A				
13	37012	BMS Error Code High	BMS高位故障代码	RO	U16	1	N/A	N/A				
14	37013	BMS Warning Code High	BMS高位警告代码	RO	U16	1	N/A	N/A				
15	37014	BMS Software Version	BMS软件版本	RO	U16	1	1	N/A				
16	37015	Battery Hardware Version	电池固件版本	RO	U16	1	1	N/A				
17	37016	Maximum Cell Temperature ID	最大电池温度ID	RO	U16	1	1	N/A			Battery module ID(1Byte) + Battery sample point(1Byte)	电池模块ID(1Byte) + 电池采样点(1Byte)
18	37017	Minimum Cell Temperature	最小电池温度	RO	U16	1	1	N/A				
19	37018	Maximum Cell Voltage ID	最大电池电压	RO	U16	1	1	N/A				
20	37019	Minimum Cell Voltage ID	最小电池电压	RO	U16	1	1	N/A				
21	37020	Maximum Cell Temperature	最大电池温度	RO	U16	1	10	°C				
22	37021	Minimum Cell Temperature	最小电池温度	RO	U16	1	10	°C				
23	37022	Maximum Cell Voltage	最大电池电压	RO	U16	1	1	mV				
24	37023	Minimum Cell Voltage	最小电池电压	RO	U16	1	1	mV				
25	37024	BMS1 Pass Infomation1	BMS1透传信息1	RO	U16	1	N/A	N/A			(GW BAT only)BAT system running status 0x00 : Init 0x04 : Sleep 0x01 : Idle 0x05 : Shutdown 0x02 : Standby 0x06 : Fault 0x03 : Run	(仅自研电池) 电池系统运行状态 0x00 : Init 初始化 0x04 : Sleep 休眠 0x01 : Idle 空闲 0x05 : Shutdown 关机 0x02 : Standby 准备就 0x06 : Fault 故障 0x03 : Run 运行 0x07 : Update 升级
26	37025	BMS1 Pass Infomation2	BMS1透传信息2	RO	U16	1	N/A	N/A			Fault slave group number	故障从簇编号
27	37026	BMS1 Pass Infomation3	BMS1透传信息3	RO	U16	1	N/A	N/A			Fault slave group alarm (0x24f data1-	故障从簇告警 (0x24f data1-2)

28	37027	BMS1 Pass Infomation4	BMS1透传信息 4	RO	U16	1	N/A	N/A			Fault slave group code high (0x24f	故障从簇故障码高 (0x24f data4-5)
29	37028	BMS1 Pass Infomation5	BMS1透传信息 5	RO	U16	1	N/A	N/A			Fault slave group code low (0x24f	故障从簇故障码低 (0x24f data6-7)
30	37029	BMS1 Pass Infomation6	BMS1透传信息 6	RO	U16	1	N/A	N/A				
31	37030	BMS1 Pass Infomation7	BMS1透传信息 7	RO	U16	1	N/A	N/A				
32	37031	BMS1 Pass Infomation8	BMS1透传信息 8	RO	U16	1	N/A	N/A				
33	37032	BMS1 Pass Infomation9	BMS1透传信息 9	RO	U16	1	N/A	N/A				
34	37033	BMS1 Pass Infomation10	BMS1透传信息 10	RO	U16	1	N/A	N/A				
35	37034	BMS1 Pass Infomation11	BMS1透传信息 11	RO	U16	1	N/A	N/A				
36	37035	BMS1 Pass Infomation12	BMS1透传信息 12	RO	U16	1	N/A	N/A				
37	37036	BMS1 Pass Infomation13	BMS1透传信息 13	RO	U16	1	N/A	N/A				
38	37037	BMS1 Pass Infomation14	BMS1透传信息 14	RO	U16	1	N/A	N/A				
39	37038	BMS1 Pass Infomation15	BMS1透传信息 15	RO	U16	1	N/A	N/A				
40	37039	BMS1 Pass Infomation16	BMS1透传信息 16	RO	U16	1	N/A	N/A				
41	37040	BMS1 Pass Infomation17	BMS1透传信息 17	RO	U16	1	N/A	N/A				
42	37041	BMS1 Pass Infomation18	BMS1透传信息 18	RO	U16	1	N/A	N/A				

43	37042	BMS1 Pass Infomation19	BMS1透传信息 19	RO	U16	1	N/A	N/A				
44	37043	BMS1 Pass Infomation20	BMS1透传信息 20	RO	U16	1	N/A	N/A				
45	37044	BMS1 Pass Infomation21	BMS1透传信息 21	RO	U16	1	N/A	N/A				
46	37045	BMS1 Pass Infomation22	BMS1透传信息 22	RO	U16	1	N/A	N/A				
47	37046	BMS1 Pass Infomation23	BMS1透传信息 23	RO	U16	1	N/A	N/A				
48	37047	BMS1 Pass Infomation24	BMS1透传信息 24	RO	U16	1	N/A	N/A				
49	37048	BMS1 Pass Infomation25	BMS1透传信息 25	RO	U16	1	N/A	N/A				
50	37049	BMS1 Pass Infomation26	BMS1透传信息 26	RO	U16	1	N/A	N/A				
51	37050	BMS1 Pass Infomation27	BMS1透传信息 27	RO	U16	1	N/A	N/A				
52	37051	BMS1 Pass Infomation28	BMS1透传信息 28	RO	U16	1	N/A	N/A				
53	37052	BMS1 Pass Infomation29	BMS1透传信息 29	RO	U16	1	N/A	N/A				
54	37053	BMS1 Pass Infomation30	BMS1透传信息 30	RO	U16	1	N/A	N/A				
55	37054	BMS1 Pass Infomation31	BMS1透传信息 31	RO	U16	1	N/A	N/A				
56	37055	BMS1 Pass Infomation32	BMS1透传信息 32	RO	U16	1	N/A	N/A				
57	37056	Battery Total Charge	电池总充电能	RO	U32	2	10	kwh				
58	37058	Battery Total Discharge Energy	电池总放电能 量	RO	U32	2	10	kwh				
59	37060	battery SN	电池1 SN	RO	STR	16	N/A	N/A				
60	37076	BMS1 Rated Capacity	BMS1 额定容	RO	U16	1	10	Ah				
61	37077	BMS1 Cluster NUM	BMS1 簇数	RO	U16	1	N/A	N/A				

62	37078	BMS1 Register Cluster number+BMS1 Online Cluster number	BMS1注册簇+功率在线簇数	RO	U16	1	N/A	N/A				
63	37079	BMS1 status monitor	BMS1 状态监	RO	U16	1	N/A	N/A				
64	37080	BMS1 Bat power	BMS1 电池功	RO	S32	2	N/A	w				
65	37082	BMS1 rated energy	BMS1 额定电	RO	U32	2	10	kwh				
66	37084	BMS1 Bat attribute (High/Low Vot)	BMS1 电池属性 (高低压)	RO	U16	1	N/A	N/A			0低压 ; 1高压 ; FF旧协议	
67	37085	The battery reestablishes the communication status	BMS1电池重新建立通信状态	RO	U16	1	N/A	N/A			The module is used to determine whether to refresh the slow-battery-read data.	模块判断是否需要刷新电池慢刷数据用
68	37086	Beta Version_BMS1	BMS1 SVN版本号	RO	U16	1	N/A	N/A				
69	37087	BAT1 need indicator	电池1是否需要维护	RO	U16	1	N/A	N/A			requires user maintenance (for charging calibration operation), 0: The battery is in good condition and does not require	1:电池需要用户维护(充满校准操作),0:电池良好不需要维护
70	37088	bat1cluster online status	电池1并簇在线状态	RO	U32	2	N/A	N/A				
71	37090	Max BMS1 Charge Power	BMS1最大充电功率	RO	U32	2	N/A	N/A				
72	37092	Max BMS1 Discharge Power	BMS1最大放电功率	RO	U32	2	N/A	N/A				

73	37094	BMS1 Support Function Enable Flag	BMS1功能使能标志	RO	U16	1	N/A	N/A	[0, 0xFFFF]	cluster information bit1: Upload of cell data bit2: DCDC module bit3: Fire information bit4: Liquid cooling information bit5: Air conditioning information bit6:rsvd bit7:rsvd bit8: Heating function bit9: Remote shutdown bit10: Breathing light bit11: bit12: Support two-stage heating time bit13:Support for battery self-adaption (1: Yes, 0: No) bit14: bit15: rsvd (If the above functions are supported, the BMS protocol will upload	bit0:单簇信息上传 bit1:电芯数据上传 bit2:DCDC模块 bit3:消防信息 bit4:液冷信息 bit5:空调信息 bit6:rsvd bit7:rsvd bit8:加热功能 bit9:远程关断 bit10:呼吸灯 bit11: bit12:支持两段加热时间 bit13:支持电池自适应(1支持，0不支持) bit14: bit15:rsvd (如果支持以上功能，BMS协议上传该bit强制0)
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BMS AC Data												
1	37151	Coil temperature	盘管温度(出风温度)	RO	U16	1	10	°C				
2	37152	Condensing temperature	冷凝温度	RO	U16	1	10	°C				
3	37153	Room temperature	柜内温度	RO	U16	1	10	°C				
4	37154	Room RH	柜内湿度	RO	U16	1	10	%				
5	37155	AC Voltage	交流电压	RO	U16	1	10	V				
6	37156	Frost temperature	容霜温度	RO	U16	1	10	°C				
7	37157	Internal Fan Speed	内风机转速	RO	U16	1	1	r/min				
8	37158	External Fan Speed	外风机转速	RO	U16	1	1	r/min				
9	37159	Air conditioner status	空调系统运行状态	RO	U16	1	NA	N/A			0x00 : Shutdown 0x01 : Blowing 0x02 : Cooling 0x03 : Warming 0x04 : Arefaction 0xFFFF : Unsupport	0x00 : Shutdown 关机 0x01 : Blowing 送风 0x02 : Cooling 制冷 0x03 : Warming 加热 0x04 : Arefaction除湿 0xFFFF : Unsupport
BMS1Opeartion Data Expansion												
1	37400	BMS1 ParallelBox SN	BMS1并机盒	RO	STR	16	N/A	N/A				
2	37416	BMS1 ParallelBox Version	BMS1并机盒版	RO	U16	1	N/A	N/A				
3	37417	BMS1 PackInfomationResponse	BMS1Pack信息答复	RO	STR	16	N/A	N/A				
BMS2 Operation Data												
1	39000	BMS2 Status	BMS2状态	RO	U16	1	N/A	N/A				
2	39001	BMS2 Package Temperature	BMS2电池包温度	RO	U16	1	10	°C				
3	39002	Max BMS2 Charge Current	BMS2充电最大电流	RO	U16	1	1	A				
4	39003	Max BMS2 Discharge Current	BMS2放电最大电流	RO	U16	1	1	A				
5	39004	BMS2 Error Code Low	BMS2低位故障代码	RO	U16	1	N/A	N/A				
6	39005	BMS2 SOC	BMS2电池剩余电量百分比	RO	U16	1	1	%				

7	39006	BMS2 SOH	BMS2电池健康度	RO	U16	1	1	%				
8	39007	BMS2 Battery Strings	BMS2电池节数	RO	U16	1	N/A	N/A				
9	39008	BMS2 Warning Code Low	BMS2低位警告代码	RO	U16	1	N/A	N/A				
10	39009	Battery Protocol	电池协议	RO	U16	1	N/A	N/A				
11	39010	BMS2 Error Code High	BMS2高位故障代码	RO	U16	1	N/A	N/A				
12	39011	BMS2 Warning Code High	BMS2高位警告代码	RO	U16	1	N/A	N/A				
13	39012	BMS2 Software Version	BMS2软件版本	RO	U16	1	1	N/A				
14	39013	Battery2 Hardware Version	电池组2固件版本	RO	U16	1	1	N/A				
15	39014	BMS2 Maximum Cell Temperature ID	BMS2最大电池温度ID	RO	U16	1	1	N/A			Battery module ID(1Byte) + Battery sample point(1Byte)	电池模块ID(1Byte) + 电池采样点(1Byte)
16	39015	BMS2 Minimum Cell Temperature ID	BMS2最小电池温度ID	RO	U16	1	1	N/A				
17	39016	BMS2 Maximum Cell Voltage ID	BMS2最大电池电压ID	RO	U16	1	1	N/A			Battery module ID(1Byte) + Battery number(1Byte)	电池模块ID(1Byte) + 电池编号(1Byte)
18	39017	BMS2 Minimum Cell Voltage ID	BMS2最小电池电压ID	RO	U16	1	1	N/A				
19	39018	BMS2 Maximum Cell Temperature	BMS2最大电池温度	RO	U16	1	10	°C				
20	39019	BMS2 Minimum Cell Temperature	BMS2最小电池温度	RO	U16	1	10	°C				
21	39020	BMS2 Maximum Cell Voltage	BMS2最大电池电压	RO	U16	1	1	mV				
22	39021	BMS2 Minimum Cell Voltage	BMS2最小电池电压	RO	U16	1	1	mV				

23	39022	BMS2 Pass Infomation1	BMS2 透传信息1	RO	U16	1	N/A	N/A			(GW BAT only)BAT system running status 0x00 : Init 0x04 : Sleep 0x01 : Idle 0x05 : Shutdown 0x02 : Standby 0x06 : Fault 0x03 : Run	(仅自研电池) 电池系统运行状态 0x00 : Init 初始化 0x04 : Sleep 休眠 0x01 : Idle 空闲 0x05 : Shutdown 关机 0x02 : Standby 准备就 0x06 : Fault 故障 0x03 : Run 运行 0x07 : Update 升级
24	39023	BMS2 Pass Infomation2	BMS2 透传信息2	RO	U16	1	N/A	N/A			Fault slave group number	故障从簇编号
25	39024	BMS2 Pass Infomation3	BMS2 透传信息3	RO	U16	1	N/A	N/A			Fault slave group alarm (0x24f data1-	故障从簇告警 (0x24f data1-2)
26	39025	BMS2 Pass Infomation4	BMS2 透传信息4	RO	U16	1	N/A	N/A			code high (0x24f	故障从簇故障码高 (0x24f data4-5)
27	39026	BMS2 Pass Infomation5	BMS2 透传信息5	RO	U16	1	N/A	N/A			code low (0x24f	故障从簇故障码低 (0x24f data6-7)
28	39027	BMS2 Pass Infomation6	BMS2 透传信息6	RO	U16	1	N/A	N/A				
29	39028	BMS2 Pass Infomation7	BMS2 透传信息7	RO	U16	1	N/A	N/A				
30	39029	BMS2 Pass Infomation8	BMS2 透传信息8	RO	U16	1	N/A	N/A				
31	39030	BMS2 Pass Infomation9	BMS2 透传信息9	RO	U16	1	N/A	N/A				
32	39031	BMS2 Pass Infomation10	BMS2 透传信息10	RO	U16	1	N/A	N/A				

33	39032	BMS2 Pass Infomation11	BMS2 透传信息11	RO	U16	1	N/A	N/A				
34	39033	BMS2 Pass Infomation12	BMS2 透传信息12	RO	U16	1	N/A	N/A				
35	39034	BMS2 Pass Infomation13	BMS2 透传信息13	RO	U16	1	N/A	N/A				
36	39035	BMS2 Pass Infomation14	BMS2 透传信息14	RO	U16	1	N/A	N/A				
37	39036	BMS2 Pass Infomation15	BMS2 透传信息15	RO	U16	1	N/A	N/A				
38	39037	BMS2 Pass Infomation16	BMS2 透传信息16	RO	U16	1	N/A	N/A				
39	39038	BMS2 Pass Infomation17	BMS2 透传信息17	RO	U16	1	N/A	N/A				
40	39039	BMS2 Pass Infomation18	BMS2 透传信息18	RO	U16	1	N/A	N/A				
41	39040	BMS2 Pass Infomation19	BMS2 透传信息19	RO	U16	1	N/A	N/A				
42	39041	BMS2 Pass Infomation20	BMS2 透传信息20	RO	U16	1	N/A	N/A				
43	39042	BMS2 Pass Infomation21	BMS2 透传信息21	RO	U16	1	N/A	N/A				
44	39043	BMS2 Pass Infomation22	BMS2 透传信息22	RO	U16	1	N/A	N/A				
45	39044	BMS2 Pass Infomation23	BMS2 透传信息23	RO	U16	1	N/A	N/A				
46	39045	BMS2 Pass Infomation24	BMS2 透传信息24	RO	U16	1	N/A	N/A				
47	39046	BMS2 Pass Infomation25	BMS2 透传信息25	RO	U16	1	N/A	N/A				
48	39047	BMS2 Pass Infomation26	BMS2 透传信息26	RO	U16	1	N/A	N/A				
49	39048	BMS2 Pass Infomation27	BMS2 透传信息27	RO	U16	1	N/A	N/A				

50	39049	BMS2 Pass Infomation28	BMS2 透传信息28	RO	U16	1	N/A	N/A				
51	39050	BMS2 Pass Infomation29	BMS2 透传信息29	RO	U16	1	N/A	N/A				
52	39051	BMS2 Pass Infomation30	BMS2 透传信息30	RO	U16	1	N/A	N/A				
53	39052	BMS2 Pass Infomation31	BMS2 透传信息31	RO	U16	1	N/A	N/A				
54	39053	BMS2 Pass Infomation32	BMS2 透传信息32	RO	U16	1	N/A	N/A				
55	39054	Battery Total Charge	电池总充电能	RO	U32	2	10	kwh				
56	39056	Battery Total Discharge Energy	电池总放电能量	RO	U32	2	10	kwh				
57	39058	battery SN	电池2 SN	RO	STR	16	N/A	N/A				
58	39074	BMS2 Rated Capacity	BMS2 额定容	RO	U16	1	10	kwh				
59	39075	BMS2 Cluster NUM	BMS2 簇数	RO	U16	1	N/A	N/A				
60	39076	BMS2 Register Cluster number+BMS2 Online Cluster number	BMS2注册簇+功率在线簇数	RO	U16	1	N/A	N/A				
61	39077	BMS2 status monitor	BMS2 状态监	RO	U16	1	N/A	N/A				
62	39078	BMS2 Bat power	BMS2 电池功	RO	S32	2	N/A	w				
63	39080	BMS2 rated energy	BMS2 额定电	RO	U32	2	10	kwh				
64	39082	BMS2 Bat attribute (High/Low Vot)	BMS2 电池属性 (高低压)	RO	U16	1	N/A	N/A			0低压；1高压；FF旧协议	模块识别版本用
65	39083	The battery reestablishes the communication status	BMS2电池重新建立通信状态	RO	U16	1	N/A	N/A			The module is used to determine whether to refresh the slow-battery-read data.	模块判断是否需要刷新电池慢刷数据用
66	39084	Beta Version_BMS2	BMS2 SVN版本号	RO	U16	1	N/A	N/A				
67	39085	BMS2 Type Index	电池2类型索引	RO	U16	1	N/A	N/A				

68	39086	BAT2 need indicator	电池2是否需要维护	RO	U16	1	N/A	N/A			requires user maintenance (for charging calibration operation), 0: The battery is in good condition and does not require	1:电池需要用户维护(充满校准操作),0:电池良好不需要维护
69	39087	bat2cluster online status	电池2并簇在线状态	RO	U32	2	N/A	N/A				
70	39089	Max BMS2 Charge Power	BMS2最大充电功率	RO	U32	2	N/A	N/A				
71	39091	Max BMS2 Discharge Power	BMS2最大放电功率	RO	U32	2	N/A	N/A				

72	39093	BMS1 Support Function Enable Flag	BMS2功能使能标志	RO	U16	1	N/A	N/A	[0, 0xFFFF]	cluster information bit1: Upload of cell data bit2: DCDC module bit3: Fire information bit4: Liquid cooling information bit5: Air conditioning information bit6: rsvd bit7: rsvd bit8: Heating function bit9: Remote shutdown bit10: Breathing light bit11: bit12: bit13: Support for battery self-adaption (1: Yes, 0: No) bit14: bit15: rsvd (If the above functions are supported, the BMS protocol will upload this bit with a value of	bit0: 单簇信息上传 bit1: 电芯数据上传 bit2: DCDC模块 bit3: 消防信息 bit4: 液冷信息 bit5: 空调信息 bit6: rsvd bit7: rsvd bit8: 加热功能 bit9: 远程关断 bit10: 呼吸灯 bit11: bit12: bit13: 支持电池自适应 (1支持, 0不支持) bit14: bit15: rsvd (如果支持以上功能, BMS协议上传该bit强制0)
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74	39095	BMS2 ParallelBox SN	BMS2并机盒	RO	STR	16	N/A	N/A				
75	39111	BMS2 ParallelBox Version	BMS2并机盒版	RO	U16	1	N/A	N/A				
76	39112	BMS2	BMS2Pack信	RO	STR	16	N/A	N/A				
LG HV BAT Log												
1	39499	LG battery type	LG电池类型	RO	U16	1	1	N/A			0: LV BAT 1:HV BAT	0 : 为低压电池 1 : 为高压电池
2	39500	LG Wakeup Data analyze Flag	LG电池唤醒数据解析标志	RO	U16	1	1	N/A				
3	39501	LG Running Data analyze Flag	LG电池运行数据解析标志	RO	U16	1	1	N/A				
4	39502	LG Fault Data analyze Flag	LG电池故障数据解析标志	RO	U16	1	1	N/A				
1	39503	BMS1 Battery state	BMS1电池状态	RO	U16	1	1	N/A				
2	39504	BMS1 DC Bus voltage	BMS1 DC BUS 电压	RO	U16	1	10	V				
3	39505	BMS1 Real-time power	BMS1 实时功	RO	S16	1	1	W				
4	39506	BMS1 Full pack energy available	BMS1 全包可用能量	RO	U16	1	1	Wh				
5	39507	BMS1 Energy remaining	BMS1 剩余能	RO	U16	1	1	Wh				
6	39508	BMS1 Lifetime energy charged	BMS1 终生充电能量	RO	U32	2	1	Wh				
8	39510	BMS1 Lifetime energy discharged	BMS1 终生放电能量	RO	U32	2	1	Wh				

[illegible]

[illegible]

1	39600	BMS1 Protocol version number	BMS1协议版本号	RO	U16	1	1	N/A				
2	39601	BMS1 Battery serial number	BMS1电池SN	RO	U32	2	1	N/A				
4	39603	BMS1 DC/DC FW version1	BMS1 DC/DC 固件版本1	RO	U16	1	1	N/A				
5	39604	BMS1 DC/DC FW version2	BMS1 DC/DC 固件版本2	RO	U16	1	1	N/A				
6	39605	BMS1 FW version	BMS1 固件版	RO	U32	2	1	N/A				
8	39607	BMS1 Inverter type	BMS1 逆变器类型	RO	U16	1	1	N/A				
9	39608	BMS1 Nameplate energy	BMS1 标称能	RO	U16	1	1	Wh				
1	39630	BMS2 Protocol version number	BMS2协议版本号	RO	U16	1	1	N/A				
2	39631	BMS2 Battery serial number	BMS2电池SN	RO	U32	2	1	N/A				
4	39633	BMS2 DC/DC FW version1	BMS2 DC/DC 固件版本1	RO	U16	1	1	N/A				
5	39634	BMS2 DC/DC FW version2	BMS2 DC/DC 固件版本2	RO	U16	1	1	N/A				
6	39635	BMS2 FW version	BMS2 固件版	RO	U32	2	1	N/A				
8	39637	BMS2 Inverter type	BMS2 逆变器类型	RO	U16	1	1	N/A				
9	39638	BMS2 Nameplate energy	BMS2标称能量	RO	U16	1	1	Wh				
1	39660	BMS1 Fault 0 Count	BMS1故障0计数	RO	U16	1	1	N/A				
2	39661	BMS1 Fault 0 BMS heartbeat	BMS1故障 0 BMS 心跳	RO	U32	2	1	N/A				
4	39663	BMS1 Fault 0 ID	BMS1故障0 ID	RO	U16	1	1	N/A				
5	39664	BMS1 Fault 0 Battery internal voltage	BMS1故障0电池内部电压	RO	U16	1	100	V				
6	39665	BMS1 Fault 0 Battery internal current	BMS1故障0电池内部电流	RO	S16	1	10	A				

7	39666	BMS1 Fault 0 Battery external voltage	BMS1故障0电池外部电压	RO	U16	1	10	V				
8	39667	BMS1 Fault 0 Battery external current	BMS1故障0电池外部电流	RO	S16	1	10	A				
9	39668	BMS1 Fault 0 Battery maximum temperature	BMS1故障0电池最大温度	RO	S16	1	10	°C				
10	39669	BMS1 Fault 0 BMS internal check	BMS1故障0 BMS内部检查	RO	U16	1	1	N/A				
11	39670	BMS1 Fault 1 Count	BMS1故障1计数	RO	U16	1	1	N/A				
12	39671	BMS1 Fault 1 BMS heartbeat	BMS1故障1 BMS心跳	RO	U32	2	1	N/A				
14	39673	BMS1 Fault 1 ID	BMS1故障1 ID	RO	U16	1	1	N/A				
15	39674	BMS1 Fault 1 Battery internal voltage	BMS1故障1电池内部电压	RO	U16	1	100	V				
16	39675	BMS1 Fault 1 Battery internal current	BMS1故障1电池内部电流	RO	S16	1	10	A				
17	39676	BMS1 Fault 1 Battery external voltage	BMS1故障1电池外部电压	RO	U16	1	10	V				
18	39677	BMS1 Fault 1 Battery external current	BMS1故障1电池外部电流	RO	S16	1	10	A				
19	39678	BMS1 Fault 1 Battery maximum temperature	BMS1故障1电池最大温度	RO	S16	1	10	°C				
20	39679	BMS1 Fault 1 BMS internal check	BMS1故障1 BMS内部检查	RO	U16	1	1	N/A				
21	39680	BMS1 Fault 2 Count	BMS1故障2计数	RO	U16	1	1	N/A				
22	39681	BMS1 Fault 2 BMS heartbeat	BMS2故障2 BMS心跳	RO	U32	2	1	N/A				
24	39683	BMS1 Fault 2 ID	BMS1故障2 ID	RO	U16	1	1	N/A				
25	39684	BMS1 Fault 2 Battery internal voltage	BMS1故障2电池内部电压	RO	U16	1	100	V				
26	39685	BMS1 Fault 2 Battery internal current	BMS1故障2电池内部电流	RO	S16	1	10	A				

27	39686	BMS1 Fault 2 Battery external voltage	BMS1故障2电池外部电压	RO	U16	1	10	V				
28	39687	BMS1 Fault 2 Battery external current	BMS1故障2电池外部电流	RO	S16	1	10	A				
29	39688	BMS1 Fault 2 Battery maximum temperature	BMS1故障2电池最大温度	RO	S16	1	10	°C				
30	39689	BMS1 Fault 2 BMS internal check	BMS1故障2BMS内部检查	RO	U16	1	1	N/A				
31	39690	BMS1 Fault 3 Count	BMS1故障3计数	RO	U16	1	1	N/A				
32	39691	BMS1 Fault 3 BMS heartbeat	BMS1故障3 BMS心跳	RO	U32	2	1	N/A				
34	39693	BMS1 Fault 3 ID	BMS1故障3 ID	RO	U16	1	1	N/A				
35	39694	BMS1 Fault 3 Battery internal voltage	BMS1故障3电池内部电压	RO	U16	1	100	V				
36	39695	BMS1 Fault 3 Battery internal current	BMS1故障3电池内部电流	RO	S16	1	10	A				
37	39696	BMS1 Fault 3 Battery external voltage	BMS1故障3电池外部电压	RO	U16	1	10	V				
38	39697	BMS1 Fault 3 Battery external current	BMS1故障3电池外部电流	RO	S16	1	10	A				
39	39698	BMS1 Fault 3 Battery maximum temperature	BMS1故障3电池最大温度	RO	S16	1	10	°C				
40	39699	BMS1 Fault 3 BMS internal check	BMS1故障3 BMS内部检查	RO	U16	1	1	N/A				
41	39700	BMS1 Fault 4 Count	BMS1故障4计数	RO	U16	1	1	N/A				
42	39701	BMS1 Fault 4 BMS heartbeat	BMS1故障4BMS心跳	RO	U32	2	1	N/A				
44	39703	BMS1 Fault 4 ID	BMS1故障4 ID	RO	U16	1	1	N/A				
45	39704	BMS1 Fault 4 Battery internal voltage	BMS1故障4电池内部电压	RO	U16	1	100	V				
46	39705	BMS1 Fault 4 Battery internal current	BMS1故障4电池内部电流	RO	S16	1	10	A				

47	39706	BMS1 Fault 4 Battery external voltage	BMS1故障4电池外部电压	RO	U16	1	10	V				
48	39707	BMS1 Fault 4 Battery external current	BMS1故障4电池外部电流	RO	S16	1	10	A				
49	39708	BMS1 Fault 4 Battery maximum temperature	BMS1故障4电池最大温度	RO	S16	1	10	°C				
50	39709	BMS1 Fault 4 BMS internal check	BMS1故障4 BMS内部检查	RO	U16	1	1	N/A				
51	39710	BMS1 Fault 5 Count	BMS1故障5计数	RO	U16	1	1	N/A				
52	39711	BMS1 Fault 5 BMS heartbeat	BMS1故障5 BMS心跳	RO	U32	2	1	N/A				
54	39713	BMS1 Fault 5 ID	BMS1故障5 ID	RO	U16	1	1	N/A				
55	39714	BMS1 Fault 5 Battery internal voltage	BMS1故障5电池内部电压	RO	U16	1	100	V				
56	39715	BMS1 Fault 5 Battery internal current	BMS1故障5电池内部电流	RO	S16	1	10	A				
57	39716	BMS1 Fault 5 Battery external voltage	BMS1故障5电池外部电压	RO	U16	1	10	V				
58	39717	BMS1 Fault 5 Battery external current	BMS1故障5电池外部电流	RO	S16	1	10	A				
59	39718	BMS1 Fault 5 Battery maximum temperature	BMS1故障5电池最大温度	RO	S16	1	10	°C				
60	39719	BMS1 Fault 5 BMS internal check	BMS1故障5 BMS内部检查	RO	U16	1	1	N/A				
61	39720	BMS1 Fault 6 Count	BMS1故障6计数	RO	U16	1	1	N/A				
62	39721	BMS1 Fault 6 BMS heartbeat	BMS1故障6 BMS心跳	RO	U32	2	1	N/A				
64	39723	BMS1 Fault 6 ID	BMS1故障6 ID	RO	U16	1	1	N/A				
65	39724	BMS1 Fault 6 Battery internal voltage	BMS1故障6电池内部电压	RO	U16	1	100	V				
66	39725	BMS1 Fault 6 Battery internal current	BMS1故障6电池内部电流	RO	S16	1	10	A				

67	39726	BMS1 Fault 6 Battery external voltage	BMS1故障6电池外部电压	RO	U16	1	10	V				
68	39727	BMS1 Fault 6 Battery external current	BMS1故障6电池外部电流	RO	S16	1	10	A				
69	39728	BMS1 Fault 6 Battery maximum temperature	BMS1故障6电池最大温度	RO	S16	1	10	°C				
70	39729	BMS1 Fault 6 BMS internal check	BMS1故障6 BMS内部检查	RO	U16	1	1	N/A				
71	39730	BMS1 Fault 7 Count	BMS1故障7计数	RO	U16	1	1	N/A				
72	39731	BMS1 Fault 7 BMS heartbeat	BMS1故障7 BMS 心跳	RO	U32	2	1	N/A				
74	39733	BMS1 Fault 7 ID	BMS1故障7 ID	RO	U16	1	1	N/A				
75	39734	BMS1 Fault 7 Battery internal voltage	BMS1故障7电池内部电压	RO	U16	1	100	V				
76	39735	BMS1 Fault 7 Battery internal current	BMS1故障7电池内部电流	RO	S16	1	10	A				
77	39736	BMS1 Fault 7 Battery external voltage	BMS1故障7电池外部电压	RO	U16	1	10	V				
78	39737	BMS1 Fault 7 Battery external current	BMS1故障7电池外部电流	RO	S16	1	10	A				
79	39738	BMS1 Fault 7 Battery maximum temperature	BMS1故障7电池最大温度	RO	S16	1	10	°C				
80	39739	BMS1 Fault 7 BMS internal check	BMS1故障7 BMS内部检查	RO	U16	1	1	N/A				
1	39760	BMS2 Fault 0 Count	BMS2故障0计数	RO	U16	1	1	N/A				
2	39761	BMS2 Fault 0 BMS heartbeat	BMS2故障 0 BMS 心跳	RO	U32	2	1	N/A				
4	39763	BMS2 Fault 0 ID	BMS2故障0 ID	RO	U16	1	1	N/A				
5	39764	BMS2 Fault 0 Battery internal voltage	BMS2故障0电池内部电压	RO	U16	1	100	V				

6	39765	BMS2 Fault 0 Battery internal current	BMS2故障0电池内部电流	RO	S16	1	10	A				
7	39766	BMS2 Fault 0 Battery external voltage	BMS2故障0电池外部电压	RO	U16	1	10	V				
8	39767	BMS2 Fault 0 Battery external current	BMS2故障0电池外部电流	RO	S16	1	10	A				
9	39768	BMS2 Fault 0 Battery maximum temperature	BMS2故障0电池最大温度	RO	S16	1	10	°C				
10	39769	BMS2 Fault 0 BMS internal check	BMS2故障0 BMS内部检查	RO	U16	1	1	N/A				
11	39770	BMS2 Fault 1 Count	BMS2故障1计数	RO	U16	1	1	N/A				
12	39771	BMS2 Fault 1 BMS heartbeat	BMS2故障1 BMS心跳	RO	U32	2	1	N/A				
14	39773	BMS2 Fault 1 ID	BMS2故障1 ID	RO	U16	1	1	N/A				
15	39774	BMS2 Fault 1 Battery internal voltage	BMS2故障1电池内部电压	RO	U16	1	100	V				
16	39775	BMS2 Fault 1 Battery internal current	BMS2故障1电池内部电流	RO	S16	1	10	A				
17	39776	BMS2 Fault 1 Battery external voltage	BMS2故障1电池外部电压	RO	U16	1	10	V				
18	39777	BMS2 Fault 1 Battery external current	BMS2故障1电池外部电流	RO	S16	1	10	A				
19	39778	BMS2 Fault 1 Battery maximum temperature	BMS2故障1电池最大温度	RO	S16	1	10	°C				
20	39779	BMS2 Fault 1 BMS internal check	BMS2故障1 BMS内部检查	RO	U16	1	1	N/A				
21	39780	BMS2 Fault 2 Count	BMS2故障2计数	RO	U16	1	1	N/A				
22	39781	BMS2 Fault 2 BMS heartbeat	BMS2故障2 BMS心跳	RO	U32	2	1	N/A				
24	39783	BMS2 Fault 2 ID	BMS2故障2 ID	RO	U16	1	1	N/A				
25	39784	BMS2 Fault 2 Battery internal voltage	BMS2故障2电池内部电压	RO	U16	1	100	V				

26	39785	BMS2 Fault 2 Battery internal current	BMS2故障2电池内部电流	RO	S16	1	10	A				
27	39786	BMS2 Fault 2 Battery external voltage	BMS2故障2电池外部电压	RO	U16	1	10	V				
28	39787	BMS2 Fault 2 Battery external current	BMS2故障2电池外部电流	RO	S16	1	10	A				
29	39788	BMS2 Fault 2 Battery maximum temperature	BMS2故障2电池最大温度	RO	S16	1	10	°C				
30	39789	BMS2 Fault 2 BMS internal check	BMS2故障2BMS内部检查	RO	U16	1	1	N/A				
31	39790	BMS2 Fault 3 Count	BMS2故障3计数	RO	U16	1	1	N/A				
32	39791	BMS2 Fault 3 BMS heartbeat	BMS2故障3 BMS心跳	RO	U32	2	1	N/A				
34	39793	BMS2 Fault 3 ID	BMS2故障3 ID	RO	U16	1	1	N/A				
35	39794	BMS2 Fault 3 Battery internal voltage	BMS2故障3电池内部电压	RO	U16	1	100	V				
36	39795	BMS2 Fault 3 Battery internal current	BMS2故障3电池内部电流	RO	S16	1	10	A				
37	39796	BMS2 Fault 3 Battery external voltage	BMS2故障3电池外部电压	RO	U16	1	10	V				
38	39797	BMS2 Fault 3 Battery external current	BMS2故障3电池外部电流	RO	S16	1	10	A				
39	39798	BMS2 Fault 3 Battery maximum temperature	BMS2故障3电池最大温度	RO	S16	1	10	°C				
40	39799	BMS2 Fault 3 BMS internal check	BMS2故障3 BMS内部检查	RO	U16	1	1	N/A				
41	39800	BMS2 Fault 4 Count	BMS2故障4计数	RO	U16	1	1	N/A				
42	39801	BMS2 Fault 4 BMS heartbeat	BMS2故障4BMS心跳	RO	U32	2	1	N/A				
44	39803	BMS2 Fault 4 ID	BMS2故障4 ID	RO	U16	1	1	N/A				
45	39804	BMS2 Fault 4 Battery internal voltage	BMS2故障4电池内部电压	RO	U16	1	100	V				

46	39805	BMS2 Fault 4 Battery internal current	BMS2故障4电池内部电流	RO	S16	1	10	A				
47	39806	BMS2 Fault 4 Battery external voltage	BMS2故障4电池外部电压	RO	U16	1	10	V				
48	39807	BMS2 Fault 4 Battery external current	BMS2故障4电池外部电流	RO	S16	1	10	A				
49	39808	BMS2 Fault 4 Battery maximum temperature	BMS2故障4电池最大温度	RO	S16	1	10	°C				
50	39809	BMS2 Fault 4 BMS internal check	BMS2故障4 BMS内部检查	RO	U16	1	1	N/A				
51	39810	BMS2 Fault 5 Count	BMS2故障5计数	RO	U16	1	1	N/A				
52	39811	BMS2 Fault 5 BMS heartbeat	BMS2故障5 BMS心跳	RO	U32	2	1	N/A				
54	39813	BMS2 Fault 5 ID	BMS2故障5 ID	RO	U16	1	1	N/A				
55	39814	BMS2 Fault 5 Battery internal voltage	BMS2故障5电池内部电压	RO	U16	1	100	V				
56	39815	BMS2 Fault 5 Battery internal current	BMS2故障5电池内部电流	RO	S16	1	10	A				
57	39816	BMS2 Fault 5 Battery external voltage	BMS2故障5电池外部电压	RO	U16	1	10	V				
58	39817	BMS2 Fault 5 Battery external current	BMS2故障5电池外部电流	RO	S16	1	10	A				
59	39818	BMS2 Fault 5 Battery maximum temperature	BMS2故障5电池最大温度	RO	S16	1	10	°C				
60	39819	BMS2 Fault 5 BMS internal check	BMS2故障5 BMS内部检查	RO	U16	1	1	N/A				
61	39820	BMS2 Fault 6 Count	BMS2故障6计数	RO	U16	1	1	N/A				
62	39821	BMS2 Fault 6 BMS heartbeat	BMS2故障6 BMS心跳	RO	U32	2	1	N/A				
64	39823	BMS2 Fault 6 ID	BMS2故障6 ID	RO	U16	1	1	N/A				
65	39824	BMS2 Fault 6 Battery internal voltage	BMS2故障6电池内部电压	RO	U16	1	100	V				

[illegible]

L G L V Data Log												
1	39870	Maximum charging voltage	最大充电电压	RO	U16	1	10	V				
2	39871	Maximum charging current	最大充电电流	RO	U16	1	10	A				
3	39872	Maximum Discharging	最大放电电流	RO	U16	1	10	A				
4	39873	Batttery Voltage	电池电压	RO	U16	1	10	V				
5	39874	Batttery Current	电池电流	RO	S16	1	10	A				
6	39875	Batttery Temperature	电池温度	RO	U16	1	10	°C				
7	39876	Cell Voltage #1	电芯1电压	RO	U16	1	1000	V				
8	39877	Cell Voltage #2	电芯2电压	RO	U16	1	1000	V				
9	39878	Cell Voltage #3	电芯3电压	RO	U16	1	1000	V				
10	39879	Cell Voltage #4	电芯4电压	RO	U16	1	1000	V				
11	39880	Cell Voltage #5	电芯5电压	RO	U16	1	1000	V				
12	39881	Cell Voltage #6	电芯6电压	RO	U16	1	1000	V				

13	39882	Cell Voltage #7	电芯7电压	RO	U16	1	1000	V				
14	39883	Cell Voltage #8	电芯8电压	RO	U16	1	1000	V				
15	39884	Cell Voltage #9	电芯9电压	RO	U16	1	1000	V				
16	39885	Cell Voltage #10	电芯10电压	RO	U16	1	1000	V				
17	39886	Cell Voltage #11	电芯11电压	RO	U16	1	1000	V				
18	39887	Cell Voltage #12	电芯12电压	RO	U16	1	1000	V				
19	39888	Cell Voltage #13	电芯13电压	RO	U16	1	1000	V				
20	39889	Cell Voltage #14	电芯14电压	RO	U16	1	1000	V				
21	39890	BMS Serial Number	电池序列号	RO	U32	2	1	N/A				
23	39892	BMS SW Version	电池软件版本	RO	U16	1	1	N/A				
24	39893	Battery series	电池系列	RO	U16	1	1	N/A				
25	39894	BMS Warning	电池告警信息	RO	U32	2	1	N/A			Refer to 8-14	释义见8-14
27	39896	BMS Alarm	电池故障信息	RO	U32	2	1	N/A			Refer to 8-14	释义见8-14
29	39898	SOC	电池剩余电量 百分比	RO	U16	1	1	%			Capacity of BAT	
30	39899	BMS SOH	电池健康度	RO	U16	1	1	%			BMS SOH	BMS电池健康度

#Address	English Name	Chinese Name	#R/W	#Type	#Size	#SF	#Units	Range	Flash Save	Note(English)	Note(Chinese)	
OEM Data												
1	40000	OEM SN	OEM设备序列号	RW	STR	8	1	N/A		Y	read and write inverter serial No. ASCII,16 bytes	读写逆变器串行编号 ASCII码，16个字节
2	40008	EMS Check Status	EMS自检状态	RO	U16	1	1	N/A		N	to read inverter operation status 0:checking 1:Normal 2:Fault	读取逆变器运行状态 0：检测中；1：正常； 2：故障
shelly smart device												
1	41000	Search for smart devices	搜索智能设备	RW	U16	1	1	N/A		N		
BTC/ETC												
1	42000	EMS Power Mode	能量管理模式	RW	U16	1	N/A	N/A		N	For BTC/ETC	BTC/ETC使用
2	42001	EMS Power Set	能量管理功率设置	RW	U32	2	N/A	N/A		N		
3	42003	Feed Power Enable	防逆流开关	RW	U16	1	N/A	N/A	[0,1]	Y	0:Disable 1:Enable Used together with register 42004	0:禁止 1:使能 和寄存器42004一起使用，BTC/ETC使用

4	42004	Feed Power Allowable On Grid Power	防逆流允许并网功率	RW	S32	2	N/A	W	[-50000,50000]	Y	as FeedPowerEnable is set as 1, then use this register to set the max export power allowed,For BTC/ETC If the maximum AC power of the energy storage machine is greater than 30K or when the machine is combined, the anti-counter-current power is set using this register	将防逆流开关设置为1，然后使用该寄存器设置允许的最大输出功率，BTC/ETC使用储能机单机最大AC功率大于30K或并机时，防逆流功率设置值使用该寄存器
5	42006	3 Phase Feed Power Enable	三相防逆流开关	RW	S32	1	N/A	N/A	[0,1]	Y	For BTC/ETC	BTC/ETC使用
6	42007	R Phase Feed Power Parameter	R相防逆流功率	RW	S32	2	N/A	W	[-20000,20000]	Y		
7	42009	S Phase Feed Power Parameter	S相防逆流功率	RW	S32	2	N/A	W	[-20000,20000]	Y		
8	42011	T Phase Feed Power Parameter	T相防逆流功率	RW	S32	2	N/A	W	[-20000,20000]	Y		

ETC Parameter												
2	42101	Remote Comm Loss Time	远程通讯异常时间	RW	U16	1	1	S	[0,65536]	Y	Config Remote Communication Loss Time 0: Disable Remote communication loss	配置远程通讯异常时间 0：禁止远程通讯异常检测功能
ETC Parameter												
1	42200	STS Module Installed	STS模块安装	RW	U16	1	1	N/A	[0,1]	Y	Config installation of STS module 0:STS module not installed 1:STS module	配置STS模块是否安装 0:STS模块未安装 1:STS模式安装

#Address	English Name	Chinese Name	#R/W	#Type	#Size	#SF	#Units	Range	Flash Save	Note(English)	Note(Chinese)	
OEM Data												
1	43500	Fix Q Enable	定Q控制使能	RW	U16	1	1	N/A	[0,1]		0: Disable 1: Enable (Added in 1547)	0: 关闭 1：使能 (1547新增)
2	43501	Fix PF Enable	定PF控制使能	RW	U16	1	1	N/A	[0,1]		0: Disable 1: Enable (Added in 1547)	0: 关闭 1：使能 (1547新增)
3	43502	Fix PF Lead/Lag	定PF控制功率因	RW	U16	1	1	N/A	[0,1]		0: Leading 1: Lagging (Added in Version 1547)	0: 超前 1：滞后(1547新增)
4	43503	Active power cor	有功调度模式	RO	U16	1	1	N/A	[0,5]	Y	Storage does not support high-frequency write operations 0: Not enabled 1: Active Power Value (Stored) 2: Active Power Percentage (Storage) 3: Active power value (not stored) 4: Active Power Percentage (Not Stored) 5: Fast Active Power Percentage (Not Saved)	存储，不支持高频写操作 0:不启用 1:有功功率值(存储) 2:有功百分比(存储) 3:有功功率值(不存储) 4:有功百分比(不存储) 5：快速有功百分比（不存储）
5	43504	Active power cor	有功功率值(存储)	RW	S32	2	1	W	[-400000,	Y	Does not support application scenarios with high-frequency read and write operations and frequent scheduling [Save]	不支持高频读写，频繁调度的应用场景 【存储】
7	43506	Active power cor	有功百分比(存储)	RW	S16	1	10	%Pn	[-1200,12	Y	Does not support application scenarios with high-frequency read and write operations and frequent scheduling [Save]	不支持高频读写，频繁调度的应用场景 【存储】

8	43507	Active power co	有功百分比(不存	WO	S16	1	10	%Pn	[-1200,1200]		that support high-frequency read and write operations and frequent scheduling, it is recommended to use this register [Not Saved]	支持高频读写，频繁调度的应用场景，推荐使用该寄存器 【不存储】
9	43508	Active power co	有功功率值(不存	WO	S32	2	1	W	[-400000,1000000]		that support high-frequency read and write operations and frequent scheduling, it is recommended to use this register [Not saved]	支持高频读写，频繁调度的应用场景，推荐使用该寄存器 【不存储】
11	43510	Reactive power	无功调度模式	RO	U16	1	1	N/A	[0,10]	Y	high-frequency write operations 0: Not enabled 1: Reactive Power Value (Stored) 2: Reactive Power Percentage (Stored) 3: Power Factor (Storage) 4: Reactive power value (not stored) 5: Reactive Power Percentage (Not Stored) 6: Power Factor (not stored) 7: Fast Reactive Power Percentage (not stored) 8: Q(U) Curve Activation 9: cos Phi (P) curve enabled	存储，不支持高频写操作 0:不启用 1:无功功率值（存储） 2:无功百分比（存储） 3:功率因数（存储） 4:无功功率值（不存储） 5:无功百分比（不存储） 6:功率因数（不存储） 7:快速无功百分比（不存储） 8：Q(U)曲线开启 9：cos Phi (P)曲线开启 10：Q(P)曲线开启"

12	43511	Reactive power	无功功率值 (存	RW	S32	2	1	Var	[-1000000]	Y	Perform reactive power compensation in the form of a fixed value Does not support application scenarios with high-frequency read and write operations and frequent scheduling [Save]	以固定值的方式进行无功功率补偿 不支持高频读写，频繁调度的应用场景 【存储】
14	43513	Power factor (S	功率因数 (存储	RW	S16	1	1000	N/A	[-1000,10]	Y	Reactive power compensation by power factor Does not support application scenarios with high-frequency read and write operations and frequent scheduling [Save].	以功率因数的方式进行无功功率补偿 不支持高频读写，频繁调度的应用场景 【存储】
15	43514	Reactive power	无功百分比 (存	RW	S16	1	10	%Pn	[-1250,12]	Y	compensation in percentage Does not support application scenarios with high-frequency read and write operations and frequent scheduling [Save]	以百分比的方式进行无功功率补偿 不支持高频读写，频繁调度的应用场景 【存储】
16	43515	Reactive power	无功功率值 (不	WO	S32	2	1	Var	[-1000000,1000000]		that support high-frequency read and write operations and frequent scheduling, it is recommended to use this register 【No Save】	支持高频读写，频繁调度的应用场景，推荐使用该寄存器 【不存储】

18	43517	Reactive power	无功百分比 (不	WO	S16	1	10	%Pn	[-1250,1250]		that support high-frequency read and write operations and frequent scheduling, it is recommended to use this register 【No Save】	支持高频读写，频繁调度的应用场景，推荐使用该寄存器 【不存储】
19	43518	Power factor (N	功率因数 (不存	WO	S16	1	1000	N/A	[-1000,1000]		that support high-frequency read and write operations and frequent scheduling, it is recommended to use this register 【No Save】	支持高频读写，频繁调度的应用场景，推荐使用该寄存器 【不存储】
20	43519	Shutdown reacti	关机无功调节速	RW	U16	1	10	%Pn/min	[0,65000]	Y	Storage does not support high-frequency write operations.	存储，不支持高频写操作
21	43520	Start-up reactive	开机无功调节速	RW	U16	1	10	%Pn/min	[0,65000]	Y	Storage does not support high-frequency write operations.	存储，不支持高频写操作
22	43521	Shutdown active	关机有功调节速	RW	U16	1	10	%Pn/min	[0,65000]	Y	Storage does not support high-frequency write operations.	存储，不支持高频写操作
23	43522	Start-up active p	开机有功调节速	RW	U16	1	10	%Pn/min	[0,65000]	Y	Storage does not support high-frequency write operations.	存储，不支持高频写操作

24	43523	Fast active power	快速有功百分比	WO	S16	1	10	%Pn	[-1200,1200]		Panoramic monitoring scenario rapid active power dispatch command, reaching the set value within 50ms [Not Saved] Applicable to CN	全景监控场景快速有功调度指令，50ms内达到设定值 【不存储】适用国内
25	43524	Fast reactive power	快速无功百分比	WO	S16	1	10	%Pn	[-1250,1250]		that support high-frequency read and write operations and frequent scheduling, it is recommended to use this register [Not Saved]	支持高频读写，频繁调度的应用场景，推荐使用该寄存器 【不存储】
26	43525	Reactive power	无功响应方式	RW	U16	1	1	N/A	[0,3]	Y	Mode of reactive power dispatch response. Slope mode or first-order low-pass filtering mode. Storage does not support high-frequency write operations 0: Not enabled 1: Slope Mode 2: First-order low-pass time constant 3: First-order low-pass response time	无功调度响应的方式。 斜率模式或一阶低通滤波模式。 存储，不支持高频写操作 0：不启用 1：斜率模式 2：一阶低通时间常数 3：一阶低通响应时间
27	43526	Reactive power response time constant/Reactive power response time	无功响应时间常数	RW	U32	2	1	ms	[0,360000]	Y	When 43525 is set to 2, it represents reactive power response time constant When 43525 is set to 3, it represents the reactive power response time	43525设置为2时，是无功响应时间常数 43525设置为3时，是无功响应时间

29	43528	Reactive power	无功响应斜率	RW	U32	2	10	%Pn/min	[1,600000]	Y	The setting will only take effect when 43525 is set to 1	43525设置为1时，设置才能生效
31	43530	Active power res	有功响应方式	RW	U16	1	1	N/A	[0,3]	Y	scheduling response. Slope mode or first-order low-pass filtering mode. Storage does not support high-frequency write operations. 0: Disabled 1: Slope mode 2: First-order low-pass time constant 3: First-order low-pass response time	有功调度响应的方式。 斜率模式或一阶低通滤波模式。 存储，不支持高频写操作 0：不启用 1：斜率模式 2：一阶低通时间常数 3：一阶低通响应时间
32	43531	Active power response time constant/Active power response time	有功响应时间常	RW	U32	2	1	ms	[0,360000]	Y	When 43530 is set to 2, it represents the active response time constant When 43530 is set to 3, it represents the active response time	43530设置为2时,是有功响应时间常数 43530设置为3时,是有功响应时间
34	43533	Active power (N	有功(不存储)响	RW	U32	2	10	%Pn/min	[1,600000]	Y	effect only when it is set to 1	43530设置为1时设置才能生效
36	43535	Load Power Lim	RCR限有功功率	RW	U16	1	1	%Pn	[0, 120]		Germany RCR limits active power	德国RCR限有功功率
37	43536	Active power co	功率调度禁能	RW	U16	1	1	N/A	[0,1]	Y	0: Accept instructions from a third party 1: Disable, do not accept instructions from a third party	0:接受第三方调度指令 1:禁能，不接收第三方调度指令

38	43537	Remote Active power	远程有功功率调	RW	U16	1	1	N/A	[0,1]	Y	0: Disable, do not receive remote scheduling instructions Accept remote scheduling instructions 1: Accept remote scheduling instructions	0:不接受远程调度指令 1:使能，接收远程调度指令
39	43538	Local Active power	本地有功功率调	RW	U16	1	1	N/A	[0,1]	Y	0: Disable, do not receive remote scheduling instructions Accept remote scheduling instructions 1: Accept local scheduling instructions	0:接受本地调度指令 1:使能，接收本地调度指令
40	43539	Active power Priority	有功优先级模式	RW	U16	1	1	N/A	[0,2]	Y	0: Take the smaller value 1: Remote active power takes precedence (4110, 4120 safety regulations) 2: Under-frequency loading takes precedence (4130 safety regulations)	0:有功取小 1:远程有功优先 (4110、4120安规) 2: 欠频加载优先 (4130安规)
Safety												
41	43560		新旧安规识别位	RO	U16	1	1	N/A	[0,0]		1: Use the new safety protocol address 0 or 0xFFFF: Use the old safety protocol address	1: 采用新的安规地址 0或者0xFFFF: 采用旧安规地址
42	43561	Safety software	安规软件版本号	RO	U16	1	1	N/A	[0,0]			
43	43562	Safety continent	安规洲区域	RO	U16	1	1	N/A	[0,0]		regulations parameter table	见安规参数表
44	43563	Safety code	安规代码	RW	U16	1	1	N/A	[0,65535]	Y	regulations parameter table	见安规参数表
45	43564	Safety version	安规版本号	RO	U16	1	1	N/A	[0,0]			
46	43565	grid voltage value	电网额定电压值	RW	U16	1	10	V	[0,8000]	Y	Rated phase voltage; It is recommended to enhance permission management	相电压额定值； 建议增加权限管理

47	43566	grid frequency v	电网额定频率值	RW	U16	1	100	Hz	[4500,650]	Y	enhance the authority management.	建议增加权限管理
48	43567	Phase voltage d	相电压检测使能	RW	U16	1	1	N/A	[0,1]	Y	0: Close 1: Open	0 : 关闭 1 : 打开
49	43568	Line voltage det	线电压检测使能	RW	U16	1	1	N/A	[0,1]	Y	0: Close 2: Open	0 : 关闭 1 : 打开
DCI												
61	43580	DCI protection m	DCI保护模式	RW	U16	1	1	N/A	[0,10]	Y	0: DCI Absolute Value (both primary and secondary levels are current absolute values) 1: DCI Percentage (both primary and secondary are percentages of rated current) 2: DCI Absolute Value && Percentage (Level 1 is current absolute value, Level 2 is rated current percentage)	0:DCI绝对值（一二级均为电流绝对值） 1:DCI百分比（一二级均为额定电流百分比） 2:DCI绝对值&&百分比（一级为电流绝对值，二级为额定电流百分比）
62	43581	DCI threshold st	DCI一级保护值	RW	U16	1	1	mA	[0,5000]	Y	DCI Level 1 Trip Corresponding Current Value	DCI一级跳脱对应电流值
63	43582	DCI percentage	DCI一级保护百分比	RW	U16	1	10	%In	[0,1000]	Y	Percentage of DCI Level 1 Trip Rated Current	DCI一级跳脱额定电流百分比
64	43583	DCI trip time sta	DCI一级保护时间	RW	U16	1	1	ms	[0,60000]	Y	DCI Level 1 Protection Trip Time	DCI一级保护跳脱时间
65	43584	DCI threshold st	DCI二级保护值	RW	U16	1	1	mA	[0,5000]	Y	DCI Secondary Trip Corresponding Current Value	DCI二级跳脱对应电流值
66	43585	DCI percentage	DCI二级保护百分比	RW	U16	1	10	%In	[0,1000]	Y	DCI Secondary Trip Rated Current Value	DCI二级跳脱额定电流百分比
67	43586	DCI trip time sta	DCI二阶保护时间	RW	U16	1	1	ms	[0,60000]	Y	DCI Level 2 Protection Trip Time	DCI二级保护跳脱时间

68	43587	DCI check Enab	DCI 检测使能开	RW	U16	1	1	N/A	[0,1]	Y		
69	43588	DCI check filter	DCI检测滤波开	RW	U16	1	1	N/A	[0,1]	Y		
78	43597	ROCOF withstar	频移耐受值	RW	U16	1	100	Hz/s	[0,1000]	Y	Frequency shift does not deviate from the rate of frequency change	频移不跳脱频率变化率
79	43598	ROCOF protecti	频移保护值	RW	U16	1	100	Hz/s	[0,1000]	Y	Frequency shift deviates from the rate of frequency change	频移跳脱频率变化率
80	43599	Frequency shift	频移保护阈值	RW	U16	1	100	Hz/s	[0,1000]	Y		
81	43600	ROCOF measurement window	频移检测窗口	RW	U16	1	1	ms	[10,10000]	Y	Frequency Shift Sliding Detection Width	频移滑动检测宽度
82	43601	ROCOF trip time	频移保护时间	RW	U16	1	1	ms	[10,65000]	Y	Frequency Shift Trigger Protection Trip Time	频移触发保护跳脱时间
Grid connection												
91	43610	Observation time	启机等待时间	RW	U16	1	1	s	[30, 3000]	Y	Machine startup self-check grid connection countdown time	机器启机自检并网倒计时时间
92	43611	Slope of power r	启机功率斜率	RW	U16	1	10	%Pn/min	[0,60000]	Y	Power ramp-up slope during machine startup and grid connection	机器启机并网功率上升斜率
93	43612	Slope of normal	启机充电功率加	RW	U16	1	10	%Pn/min	[0,60000]	Y	Set the loading rate for the initial power connection upon startup, with the charging power gradually increasing as the device starts up.	设置首次开机并网，设备启动时充电功率逐步上升的加载速率。
94	43613	Lower voltage of	启机电压下限	RW	U16	1	10	%Vn	[150,1000]	Y	Lower limit percentage of starting voltage	启机电压下限百分比
95	43614	Upper voltage of	启机电压上限	RW	U16	1	10	%Vn	[800,1400]	Y	Higher limit percentage of starting voltage	启机电压上限百分比
96	43615	Lower frequency	启机频率下限	RW	U16	1	100	Hz	[4500,6000]	Y	Lower limit percentage of starting frequency	启机频率下限百分比

97	43616	Upper frequency	启机频率上限	RW	U16	1	100	Hz	[5000,6500]	Y	Higher limit percentage of starting frequency	启机频率上限百分比
98	43617	Observation time	故障重连等待时间	RW	U16	1	1	s	[30, 3000]	Y	Countdown time for self-check and grid connection after machine fault reconnection	机器故障重连自检并网倒计时时间
99	43618	Power slope of reconnection	故障重连功率斜率	RW	U16	1	10	%Pn/min	[0,60000]	Y	during reconnection to the grid due to machine failure	机器故障重连并网功率上升斜率
100	43619	Slope of Fault charging power	重连充电功率加载斜率	RW	U16	1	10	%Pn/min	[0,60000]	Y	standards of certain countries or regions, the percentage of power increment that the inverter can output per minute when it is not the first time to connect to the grid. For example: If it is set to 10, it indicates that the reconnection charging loading slope is: 10% P/Srated%%.	根据某些国家或地区的标准要求，逆变器非首次并网时，每分钟可输出的功率增量百分比。例如：设置为10时，表示重连充电加载斜率为：10%P/Srated%%。
101	43620	Lower voltage of reconnection	故障重连电压下限	RW	U16	1	10	%Vn	[150,1000]	Y	of fault reconnection voltage	故障重连电压下限百分比
102	43621	Upper voltage of reconnection	故障重连电压上限	RW	U16	1	10	%Vn	[800,1400]	Y	of fault reconnection voltage	故障重连电压上限百分比
103	43622	Lower frequency of reconnection	故障重连频率下限	RW	U16	1	100	Hz	[4500,6000]	Y	fault reconnection frequency	故障重连频率下限百分比
104	43623	Upper frequency of reconnection	故障重连频率上限	RW	U16	1	100	Hz	[5000,6500]	Y	fault reconnection frequency	故障重连频率上限百分比
105	43624	Reconnection Restriction Mode	重连限制模式	RW	U16	1	1	N/A	[0,1]	Y	internal use	内部使用
QU Curve												
121	43640	Q(U) enable	Q(U)使能	RW	U16	1	1	N/A	[0,1]	Y	0: Off 1: On	0：关闭 1：打开

122	43641	Intentional delay	Q(U)初始延迟时间	RW	U16	1	10	s	[0,10000]	Y	First trigger Q(U) response delay time	首次触发Q(U)响应延迟时间
123	43642	Insensitivity voltage	Q(U)电压不灵敏	RW	U16	1	10	%Vn	[0,100]	Y	fluctuates within a certain range, the voltage input is not updated, and the output reactive power remains unchanged	Q(U)电压在一定范围波动，不更新电压输入，输出无功不变
124	43643	Power reference	Q(U)功率基准	RW	U16	1	1	N/A	[0,3]	Y	0: Rated power of Pn 1: Apparent power of Ps 2: Current power of Po 3: Maximum power of Pm	0：额定功率(Pn) 1：视在功率(Ps) 2：当前功率(Pm) 3：最大功率(Pmax)
125	43644	Enter the curve	Q(U)进入条件使能	RW	U16	1	1	N/A	[0,1]	Y	0: Close 1: Open	0：禁能 1：使能
126	43645	Lock in power of Q(U)	Q(U)进入曲线功率百分比	RW	U16	1	10	%Pn	[0,1100]	Y	QU entry curve power percentage.	Q(U)进入曲线功率百分比
127	43646	Lock out power of Q(U)	Q(U)退出曲线功率百分比	RW	U16	1	10	%Pn	[0,1100]	Y	QU exit curve power percentage.	Q(U)退出曲线功率百分比
128	43647	V1 of Q(U)	Q(U)电压值V1	RW	U16	1	10	%Vn	[0,1400]	Y	The voltage percentage corresponding to the QU curve.	QU曲线对应的电压百分比
129	43648	Q1 of Q(U)	Q(U)无功值Q1	RW	S16	1	10	%Pr	[-1000,10]	Y	reactive power corresponding to the QU curve.	QU曲线对应的无功功率百分比
130	43649	V2 of Q(U)	Q(U)电压值V2	RW	U16	1	10	%Vn	[0,1400]	Y	The voltage percentage corresponding to the QU curve.	QU曲线对应的电压百分比
131	43650	Q2 of Q(U)	Q(U)无功值Q2	RW	S16	1	10	%Pr	[-1000,10]	Y	reactive power corresponding to the QU curve.	QU曲线对应的无功功率百分比
132	43651	V3 of Q(U)	Q(U)电压值V3	RW	U16	1	10	%Vn	[0,1400]	Y	The voltage percentage corresponding to the QU curve.	QU曲线对应的电压百分比

133	43652	Q3 of Q(U)	Q(U)无功值Q3	RW	S16	1	10	%Pr	[-1000,10]	Y	reactive power corresponding to the QU curve.	QU曲线对应的无功功率百分比
134	43653	V4 of Q(U)	Q(U)电压值V4	RW	U16	1	10	%Vn	[0,1400]	Y	The voltage percentage corresponding to the QU curve.	QU曲线对应的电压百分比
135	43654	Q4 of Q(U)	Q(U)无功值Q4	RW	S16	1	10	%Pr	[-1000,10]	Y	reactive power corresponding to the QU curve.	QU曲线对应的无功功率百分比
136	43655	QU reference voltage	QU基准电压值	RW	U16	1	10	%Vn	[800,1200]	Y	(Added in 1547)	(1547新增)
137	43656	QU Reference Voltage	QU基准电压自动	RW	U16	1	1	N/A	[0,1]	Y	(Added in 1547)	(1547新增)
138	43657	Response type of Q(U)	Q(U)响应方式	RW	U16	1	1	N/A	[0,3]	Y	The way Q(U) outputs responses. Slope mode or first-order low-pass filtering mode. Storage does not support high-frequency write operations 0: Not enabled 1: Slope Mode 2: First-order low-pass time constant 3: First-order low-pass response time	Q(U)输出响应的方式。斜率模式或一阶低通滤波模式。 存储，不支持高频写操作 0：不启用 1：斜率模式 2：一阶低通时间常数 3：一阶低通响应时间
139	43658	Response time of Q(U)	Q(U)响应时间常数	RW	U32	2	1	ms	[0,3600000]	Y	When 43657 is set to 2, Q(U) response time constant When 43657 is set to 3, Q(U) response time	43657设置为2时，Q(U)响应时间常数 43657设置为3时，Q(U)响应时间
141	43660	Response slope of Q(U)	Q(U)响应斜率	RW	U16	1	10	%Pn/s	[0,65000]	Y	The setting takes effect only when 43657 is set to	43657设置为1时，设置才能生效
142	43661	Q(U) mode	Q(U)模式	RW	U16	1	1	N/A	[0,2]	Y	0: Reserved 1: Point Plotting Mode 2: Slope Mode	0：预留 1：描点模式 2：斜率模式

143	43662	Droop1 of Q(U)	Q(U)Droop1(slope)	RW	U16	1	10	%Qmax/ 1%Vn	[0,2000]	Y	Voltage value at point V1 in the equivalent tracing point mode, slope between V1 and V2	等同描点模式下电压值 V1点，V1到V2之间的斜 率
144	43663	Droop2 of Q(U)	Q(U)Droop2(slope)	RW	U16	1	10	%Qmax/ 1%Vn	[0,2000]	Y	Voltage value at point V4 and the slope between V3 and V4 in the equivalent tracing point mode	等同描点模式下电压值 V4点，V3到V4之间的斜 率
145	43664	Dead band of Q(U)	Q(U)电压死区宽度	RW	U16	1	10	%Vn	[0,100]	Y	Dead zone width in slope mode, equivalent to points V2 and V3 in point plotting mode	斜率模式下死区宽度， 等同描点模式下V2,V3点
146	43665		QU动态响应时间	RW	U32	2	1	ms	[0,360000]	Y	(Added in 1547)	(1547新增)
148	43667	Min PF of Q(U)	Q(U)最小PF使能	RW	U16	1	1	N/A	[0,1]	Y	0: Disable 1: Enable	0: 关闭 1：使能
149	43668	Min PF of Q(U)	Q(U)最小PF值	RW	S16	1	1000	N/A	[-1000,10]	Y	Minimum PF limit for Q(U); storage, does not support high-frequency write operations	Q(U)最小PF限值；存 储，不支持高频写操作
150	43669	K value of QU curve	QU曲线K值									
PU Curve												
171	43690	P(U) enable	P(U)使能	RW	U16	1	1	N/A	[0,1]	Y	0: Off 1: On	0：禁能 1：使能
172	43691	V1 of P(U)	P(U)电压值V1	RW	U16	1	10	%Vn	[0,1400]	Y	Voltage percentage corresponding to the P(U) curve	P(U)曲线对应的电压百 分比
173	43692	P1 of P(U)	P(U)有功值P1	RW	U16	1	10	%Pn	[0,1100]	Y	Active power percentage corresponding to the P(U) curve	P(U)曲线对应的有功功 率百分比
174	43693	V2 of P(U)	P(U)电压值V2	RW	U16	1	10	%Vn	[0,1400]	Y	Voltage percentage corresponding to the P(U) curve	P(U)曲线对应的电压百 分比

175	43694	P2 of P(U)	P(U)有功值P2	RW	U16	1	10	%Pn	[0,1100]	Y	Active power percentage corresponding to the P(U) curve	P(U)曲线对应的有功功率百分比
176	43695	V3 of P(U)	P(U)电压值V3	RW	U16	1	10	%Vn	[0,1400]	Y	Voltage percentage corresponding to the P(U) curve	P(U)曲线对应的电压百分比
177	43696	P3 of P(U)	P(U)有功值P3	RW	U16	1	10	%Pn	[0,1100]	Y	Active power percentage corresponding to the P(U) curve	P(U)曲线对应的有功功率百分比
178	43697	V4 of P(U)	P(U)电压值V4	RW	U16	1	10	%Vn	[0,1400]	Y	Voltage percentage corresponding to the P(U) curve	P(U)曲线对应的电压百分比
179	43698	P4 of P(U)	P(U)有功值P4	RW	S16	1	10	%Pn	[-1100,1100]	Y	Active power percentage corresponding to the P(U) curve	P(U)曲线对应的有功功率百分比
180	43699	Intentional delay	P(U)初始延迟时间	RW	U16	1	10	s	[0,10000]	Y	First trigger P(U) load shedding response delay time	首次触发P(U)降载响应延迟时间
181	43700	Response type	P(U)响应方式	RW	U16	1	1	N/A	[0,3]	Y	response mode. It is divided into slope mode and first-order response mode. 0: Off 1: Slope response 2: Time constant 3: Response time	斜率模式或一阶低通滤波模式。 存储，不支持高频写操作 0：不启用 1：斜率模式 2：一阶低通时间常数 3：一阶低通响应时间
182	43701	Response slope	P(U)响应斜率	RW	U16	1	10	%Pn/s	[0,65000]	Y	corresponding to the output response mode of the PU curve represents the slope of power change.	43700设置为1时，设置才能生效

183	43702	Response time	P(U)响应时间常	RW	U32	2	1	ms	[0,36000]	Y	The response time corresponding to the first-order response mode in the output response mode of the PU curve.	43700设置为2时，P(U)响应时间常数 43700设置为3时，P(U)响应时间
185	43704	Power reference	P(U)功率基准	RW	U16	1	1	N/A	[0,3]	Y	0: Rated power of Pn 1: Apparent power of Ps 2: Current power of Po 3: Maximum power of Pm	0：额定功率(Pn) 1：视在功率(Ps) 2：当前功率(Pm) 3：最大功率(Pmax)
<i>cos φ P Curve</i>												
201	43720	cos Phi (P) enable	cos Phi (P)使能	RW	U16	1	1	N/A	[0,1]	Y	Cos φ (P) curve enable switch. 0: Off 1: On	cos φ (P)曲线使能开关。 0：关闭 1：打开
202	43721	Intentional delay	cos Phi (P)初始	RW	U16	1	10	s	[0,10000]	Y	First trigger cos Phi (P) response delay time	首次触发cos Phi (P)响应延迟时间
203	43722	P1 of cos Phi (P)	cos Phi (P)有功	RW	U16	1	10	%Pn	[0,1100]	Y	Active power percentage corresponding to the cos Phi (P) curve	cos Phi (P)曲线对应的有功百分比
204	43723	P2 of cos Phi (P)	cos Phi (P)有功	RW	U16	1	10	%Pn	[0,1100]	Y	Active power percentage corresponding to the cos Phi (P) curve	cos Phi (P)曲线对应的有功百分比
205	43724	P3 of cos Phi (P)	cos Phi (P)有功	RW	U16	1	10	%Pn	[0,1100]	Y	Active power percentage corresponding to the cos Phi (P) curve	cos Phi (P)曲线对应的有功百分比
206	43725	P4 of cos Phi (P)	cos Phi (P)有功	RW	U16	1	10	%Pn	[0,1100]	Y	Active power percentage corresponding to the cos Phi (P) curve	cos Phi (P)曲线对应的有功百分比
207	43726	Cos Phi1 of cos	cos Phi (P)功率	RW	S16	1	1000	N/A	[-1000,-800] [800,1000]	Y	The corresponding value of cos φ for the cos φ (P) curve.	cos Phi (P)曲线对应的功率因数值

208	43727	Cos Phi2 of cos	cos Phi (P)功率	RW	S16	1	1000	N/A	[-1000,-800] [800,1000]	Y	The corresponding value of $\cos \varphi$ for the $\cos \varphi$ (P) curve.	cos Phi (P)曲线对应的功率因数
209	43728	Cos Phi3 of cos	cos Phi (P)功率	RW	S16	1	1000	N/A	[-1000,-800] [800,1000]	Y	The corresponding value of $\cos \varphi$ for the $\cos \varphi$ (P) curve.	cos Phi (P)曲线对应的功率因数
210	43729	Cos Phi4 of cos	cos Phi (P)功率	RW	S16	1	1000	N/A	[-1000,-800] [800,1000]	Y	The corresponding value of $\cos \varphi$ for the $\cos \varphi$ (P) curve.	cos Phi (P)曲线对应的功率因数
211	43730	cos Phi (P) mod	cos Phi (P)模式	RW	U16	1	1	N/A	[0,1]	Y	is processed according to RO. 0: Normal mode 1: Slope mode	0 : 描点模式 1 : 斜率模式
212	43731	Droop1 of cos P	cos Phi (P) Droop	RW	U16	1	10	Qmax/10	[0,2000]	Y	mode, equivalent to point P1 in point plotting mode, and the slope from P1 to P2	斜率模式下无功斜率，等同描点模式下P1点，P1到P2之前的斜率
213	43732	Droop2 of cos P	cos Phi (P) Droop	RW	U16	1	10	Qmax/10	[0,2000]	Y	Reactive slope in slope mode, equivalent to point P4 in point plotting mode, and the slope between P3 and P4	斜率模式下无功斜率，等同描点模式下P4点，P3到P4之间的斜率
214	43733	Lock in/out of co	cos Phi (P)进入	RW	U16	1	1	N/A	[0,1]	Y	0: Close 1: Open	0 : 禁能 1 : 使能
215	43734	Inductive Q lock	cos Phi (P)感性	RW	U16	1	10	%Vn	[0,1400]	Y	Reactive Power Entering Curve Voltage Percentage	cos Phi (P)感性无功进入曲线电压百分比
216	43735	Lock out voltage	cos Phi (P)退出	RW	U16	1	10	%Vn	[0,1400]	Y	cos Phi (P) Exit Curve Voltage Percentage	cos Phi (P)退出曲线电压百分比

[illegible]

261	43780	LFSM-O enable	P(F)过频使能	RW	U16	1	1	N/A	[0,1]	Y	0: Close 1: Open	0 : 禁能 1 : 使能
262	43781	LFSM-O mode	P(F)过频模式	RW	U16	1	1	N/A	[0,2]	Y	shedding curve determined based on over-frequency starting point and over-frequency slope) 1: Stop mode (load shedding curve determined based on overfrequency start point and overfrequency end	0 : slop模式(基于过频起点和过频斜率确定的降载曲线) 1 : stop模式 (基于过频起点和过频终点确定的降载曲线)
263	43782	Starting point of	P(F)过频起点	RW	U16	1	100	Hz	[5000,650]	Y	countries/regions, the standards require that when the grid frequency exceeds the over-frequency point, the active power output by the inverter should be	某些国家/地区的标准要求，电网频率超过过频起点时，逆变器输出的有功功率要进行降额
264	43783	Slope reference	P(F)过频斜率基	RW	U16	1	1	N/A	[0,3]	Y	the slope of the derating change in the active power output by the inverter when the grid frequency exceeds the over-frequency starting point in Slope mode 0: Rated Power (Pn) 1: Apparent Power (Ps) 2: Current Power (Pm) 3: Maximum Power	Slope模式下电网频率超过过频起点时，逆变器输出的有功功率降额变化的斜率的功率基准值 0 : 额定功率(Pn) 1 : 视在功率(Ps) 2 : 当前功率(Pm) 3 : 最大功率(Pmax)

265	43784	Slope of LFMSM-(P(F)过频斜率(sl	RW	U16	1	10	%Px/Hz	[0,10000]	Y	change in the active power output of the inverter when the grid frequency exceeds the over-frequency starting point in Slope mode	Slope模式下电网频率超过过频起点时，逆变器输出的有功功率降额变化的斜率
266	43785	Intentional delay P(F)过频初始延	RW	U16	1	10	s	[0,10000]	Y	First trigger load shedding response delay time	首次触发降载响应延迟时间
267	43786	Ending point of LP(F)过频终点(st	RW	U16	1	100	Hz	[5000,650]	Y	exceeds the over-frequency end point, the active power output by the inverter shall stop derating	电网频率超过过频终点后，逆变器输出的有功功率要停止降额
268	43787	Charge Pcmx eP(F)最大充电功	RW	U16	1	1	N/A	[0,0]	Y	endpoint corresponds to the actual maximum charging power (associated with the maximum charging power setting); otherwise, it is executed according to the rated power.	过频终点对应实际最大充电功率（和最大充电功率设置关联），否则按照额定功率执行
269	43788	Charge switch eP(F)允许转充电	RW	U16	1	1	N/A	[0,1]	Y	0: Close 1: Open	0：禁能 1：使能
270	43789	Charge switch FP(F)转充电转换	RW	U16	1	100	Hz	[5000,650]	Y	When the set frequency value is reached, the system switches from discharging to charging	当达到设置的频率值，系统由放电转为充电

271	43790	Response type	P(F)过频响应方	RW	U16	1	1	N/A	[0,3]	Y	Mode of reactive power dispatch response. Slope mode or first-order low-pass filtering mode. Storage does not support high-frequency write operations 0: Not enabled 1: Slope Mode 2: First-order low-pass time constant 3: First-order low-pass response time	无功调度响应的方式。 斜率模式或一阶低通滤波模式。 存储，不支持高频写操作 0：不启用 1：斜率模式 2：一阶低通时间常数 3：一阶低通响应时间
272	43791	Response slope	P(F)过频响应斜	RW	U16	1	10	%Pn/min	[0,65000]	Y	The setting will only take effect when 43790 is set to 1	43790设置为1时，设置才能生效
273	43792	Response time	P(F)过频响应时	RW	U32	2	1	ms	[0,360000]	Y	the over-frequency response time constant of P(F) When 43790 is set to 3, the P(F) over-frequency response time	43790设置为2时，P(F)过频响应时间常数 43790设置为3时，P(F)过频响应时间
275	43794	OF Into the pow	过频降载曲线进	RW	U16	1	10	%Pn	[0,1100]	Y		
276	43795	Upper Threshold	P(F)过频频率上	RW	U16	1	100	Hz	[5000,6500]	Y	If the frequency exceeds this frequency point, introduce curve calculation to maintain the frequency unchanged	频率如果超过此频率点，引入曲线计算频率维持其不变
277	43796	Output power lo	P(F)过频输出功	RW	U16	1	10	%Pn	[0,1100]	Y	output exceeds this power point, maintain its power constant	降载功率输出如果超过此功率点，维持其功率不变

278	43797	Deactivation threshold	P(F)过频滞回使	RW	U16	1	1	N/A	[0,1]	Y	After frequent load shedding, if the frequency decreases and is greater than the hysteresis point, the power remains unchanged 0: Disable 1: Enable	过频降载之后，如果频率变小且大于滞回点，功率维持不变 0：禁能 1：使能
279	43798	Deactivation threshold	P(F)过频滞回点	RW	U16	1	100	Hz	[4500,6500]	Y	After over-frequency load shedding, if the frequency decreases and is less than the hysteresis point for power recovery, it corresponds to the hysteresis point	过频降载之后，如果频率变小且小于滞回点功率恢复，对应滞回点
280	43799	Deactivation time	P(F)过频滞回等	RW	U16	1	1	s	[0,65000]	Y	After frequent load shedding, if the frequency decreases and is less than the hysteresis point, the power recovery response delay time 42654 takes effect after being enabled	过频降载之后，如果频率变小且小于滞回点，功率恢复响应延迟时间42654使能后生效

281	43800	Recover slope re	P(F)过频恢复斜	RW	U16	1	1	N/A	[0,4]	Y	shedding, if the frequency exits the curve, the power recovery corresponds to the recovery slope power reference 0: Rated Power (Pn) 1: Apparent Power (Ps) 2: Current Power (Pm) 3: Maximum Power (Pmax) 4: Power Difference (DeltaP)	过频降载之后，如果频率退出曲线、功率恢复对应恢复斜率功率基准 0：额定功率(Pn) 1：视在功率(Ps) 2：当前功率(Pm) 3：最大功率(Pmax) 4：功率差值 (DeltaP)
282	43801	Recover slope o	P(F)过频恢复斜	RW	U16	1	10	%Pn/min	[0,65000]	Y	After over-frequency load shedding, if the frequency exits the curve, the power recovery corresponds to the recovery slope	过频降载之后，如果频率退出曲线、功率恢复对应恢复斜率
283	43802	Quit of LFSM-O	P(F)过频退出点	RW	U16	1	100	Hz	[5000,6500]	Y	the hysteresis function is turned off, the frequency point at which the over-frequency load shedding is exited (it may not be the same point as the over-frequency starting point)	在滞回功能关闭的条件下，退出过频降载的频率点 (可能存在和过频起点不是一个点的情况)
284	43803	预留	预留			17						
301	43820	LFSM-U enable	P(F)欠频使能	RW	U16	1	1	N/A	[0,1]	Y	0: Off 1: On	0：禁能 1：使能

302	43821	LFSM-U mode	P(F)欠频模式	RW	U16	1	1	N/A	[0,2]	Y	increase curve determined based on under-frequency starting point and over-frequency slope) 1: Stop mode (load increase curve determined based on under-frequency starting point and under-	0 : slop模式(基于欠频起点和过频斜率确定的升载曲线) 1 : stop模式 (基于欠频起点和欠频终点确定的升载曲线)
303	43822	Starting of LFSM	P(F)欠频起点	RW	U16	1	100	Hz	[4500,600]	Y	countries/regions, the standards require that when the grid frequency exceeds the over-frequency point, the active power output by the inverter should be	某些国家/地区的标准要求，电网频率低压欠频起点时，逆变器输出的有功功率要进行加载
304	43823	Slope reference	P(F)欠频斜率基	RW	U16	1	1	N/A	[0,3]	Y	value of the slope rate of the reduction in active power output by the inverter when the grid frequency is below the under-frequency point in the Slope mode. 0: Pn - Rated power 1: Ps - Apparent power 2: Po - Current power 3: Pm - Maximum power	Slope模式下电网频率压欠频起点时，逆变器输出的有功功率升载变化的斜率的功率基准值 0 : 额定功率(Pn) 1 : 视在功率(Ps) 2 : 当前功率(Pm) 3 : 最大功率(Pmax)

305	43824	Slope of LFSM-1	P(F)欠频加载斜	RW	U16	1	10	%Px/Hz	[0,10000]	Y	rate of change in the reduction of the active power output by the inverter when the grid frequency is below the under-frequency point in the Slope mode.	Slope模式下电网频率压欠频起点时，逆变器输出的有功功率加载变化的斜率
306	43825	Intentional delay	P(F)欠频初始延	RW	U16	1	10	s	[0,10000]	Y	First trigger loading response delay time	首次触发加载响应延迟时间
307	43826	Ending of LFSM	P(F)欠频终点 (	RW	U16	1	100	Hz	[4500,600]	Y	is lower than the under-frequency starting point, the active power output by the inverter shall stop loading	电网频率低于欠频起点时，逆变器输出的有功功率要停止加载
308	43827	Discharge Pcma	P(F)最大放电功	RW	U16	1	1	N/A	[0,0]	Y	endpoint corresponds to the actual maximum discharge power (associated with the maximum discharge power setting); otherwise, it is executed according to the rated power.	欠频终点对应实际最大放电功率 (和最大放电功率设置关联)，否则按照额定功率执行
309	43828	Discharge switch	P(F)允许转放电	RW	U16	1	1	N/A	[0,1]	Y	0: Off 1: On	0：禁能 1：使能
310	43829	Discharge switch	P(F)转放电转换	RW	U16	1	100	Hz	[5000,650]	Y	When the set frequency value is reached, the system switches from buying electricity to selling electricity.	当达到设置的频率值，系统由充电转放电

311	43830	Response type	P(F)欠频响应方	RW	U16	1	1	N/A	[0,3]	Y	Modes of reactive power dispatch response. Slope mode or first-order low-pass filtering mode. Storage does not support high-frequency write operations 0: Not enabled 1: Slope Mode 2: First-order low-pass time constant 3: First-order low-pass response time	无功调度响应的方式。斜率模式或一阶低通滤波模式。 存储，不支持高频写操作 0：不启用 1：斜率模式 2：一阶低通时间常数 3：一阶低通响应时间
312	43831	Response slope	P(F)欠频响应斜	RW	U16	1	10	%Pn/min	[0,65000]	Y	only when 43830 is set to 1	43830设置为1时，设置才能生效
313	43832	Response time	P(F)欠频响应时	RW	U32	2	1	ms	[0,360000]	Y	When 43830 is set to 2, P(F) underfrequency response time constant When 43830 is set to 3, P(F) underfrequency response time	43830设置为2时，P(F)欠频响应时间常数 43830设置为3时，P(F)欠频响应时间
315	43834	UF Into the power	欠频加载曲线进	RW	U16	1	10	%Pn	[0,1100]	Y		
316	43835	Lower Threshold	P(F)欠频频率下	RW	U16	1	100	Hz	[5000,6500]	Y	If the frequency is lower than this frequency point, introduce curve calculation to maintain the frequency unchanged	频率如果低于此频率点，引入曲线计算频率维持其不变
317	43836	Output power up	P(F)欠频输出功	RW	U16	1	10	%Pn	[0,1100]	Y	exceeds this power point, maintain its power constant	加载功率输出如果超过此功率点，维持其功率不变

318	43837	Deactivation threshold	P(F)欠频滞回点	RW	U16	1	1	N/A	[0,1]	Y	loading, if the frequency increases and is less than the hysteresis point, the power remains unchanged 0: Disable 1: Enable	欠频加载之后，如果频率变大且小于滞回点，功率维持不变 0：禁能 1：使能
319	43838	Deactivation threshold	P(F)欠频滞回点	RW	U16	1	100	Hz	[4500,6500]	Y	loading, if the frequency increases and exceeds the hysteresis point for power recovery, it corresponds to the hysteresis point	欠频加载之后，如果频率变大且大于滞回点功率恢复，对应滞回点
320	43839	Deactivation time	P(F)欠频滞回点	RW	U16	1	1	s	[0,65000]	Y	After under-frequency loading, if the frequency increases and exceeds the hysteresis point, the power recovery response delay time	欠频加载之后，如果频率变大且大于滞回点，功率恢复响应延迟时间
321	43840	Recover slope reference	P(F)欠频恢复斜率	RW	U16	1	1	N/A	[0,4]	Y	loading, if the frequency exits the curve, the power recovery corresponds to the recovery slope power reference 0: Rated Power (Pn) 1: Apparent Power (Ps) 2: Current Power (Pm) 3: Maximum Power (Pmax) 4: Power Difference (DeltaP)	欠频加载之后，如果频率退出曲线、功率恢复对应恢复斜率功率基准 0：额定功率(Pn) 1：视在功率(Ps) 2：当前功率(Pm) 3：最大功率(Pmax) 4：功率差值 (DeltaP)

322	43841	Recover slope of P(F)	欠频恢复斜率	RW	U16	1	10	%Pn/min	[0,65000]	Y	After under-frequency loading, if the frequency exits the curve, the power recovery corresponds to the recovery slope	欠频加载之后，如果频率退出曲线、功率恢复对应恢复斜率
323	43842	Quit of LFSM-U	P(F)欠频退出点	RW	U16	1	100	Hz	[5000,65000]	Y	Under the condition that the hysteresis function is turned off, the frequency point at which the under-frequency loading is exited (it may not be the same point as the over-frequency starting point)	在滞回功能关闭的条件下，退出欠频加载的频率点（可能存在和过频起点不是一个点的情况）
Misc2Param												
341	43860	Frequency detection	频率检测周期值	RW	U16	1	1	N/A	[0,0]	Y		
342	43861	Reactive power	无功变化滤波参数	RW	U16	1	1000	s	[0,50000]	Y		
QP Curve												
351	43870	Q(P) enable	Q(P)使能	RW	U16	1	1	N/A	[0,1]	Y	QP enable switch. 0: Off 1: On	QP使能开关。 0：关闭 1：打开
352	43871	Charge P1 of Q(P)	Q(P)充电有功值	RW	S16	1	10	%Pn	[-1100,1100]	Y	Q(P) Charging Active Power Value Percentage	Q(P)充电有功值百分比
353	43872	Charge Q1 of Q(P)	Q(P)充电无功值	RW	S16	1	10	%Pn	[-1000,1000]	Y	Q(P) Charging Reactive Power Value Percentage	Q(P)充电无功值百分比
354	43873	Charge P2 of Q(P)	Q(P)充电功率值	RW	S16	1	10	%Pn	[-1100,1100]	Y	Q(P) Charging Power Value Percentage	Q(P)充电功率值百分比
355	43874	Charge Q2 of Q(P)	Q(P)充电无功值	RW	S16	1	10	%Pn	[-1000,1000]	Y	Power Value Percentage / Q(P) Reactive Power Value Percentage (Slope Mode)	Q(P)充电无功值百分比 / Q(P)无功值百分比(斜率模式)
356	43875	Charge P3 of Q(P)	Q(P)充电功率值	RW	S16	1	10	%Pn	[-1100,1100]	Y	Q(P) Charging Power Value Percentage / Q(P) Power Value Percentage (Slope Mode)	Q(P)充电功率值百分比 / Q(P)功率值百分比(斜率模式)

357	43876	Charge Q3 of Q	Q(P)充电无功点	RW	S16	1	10	%Pn	[-1000,10]	Y	Q(P) Charging Reactive Point Percentage / Q(P) Reactive Value Percentage (Slope Mode)	Q(P)充电无功点百分比 / Q(P)无功值百分比(斜率模式)
358	43877	Discharge P4 of	Q(P)放电功率点	RW	S16	1	10	%Pn	[-1100,11]	Y	Q(P) Discharge Power Point Percentage / Q(P) Power Value Percentage (Slope Mode)	Q(P)放电功率点百分比 / Q(P)功率值百分比(斜率模式)
359	43878	Discharge Q4 of	Q(P)放电无功点	RW	S16	1	10	%Pn	[-1000,10]	Y	Q(P) Discharge Reactive Point Percentage / Q(P) Reactive Point Percentage (Slope Mode)	Q(P)放电无功点百分比 / Q(P)无功点百分比(斜率模式)
360	43879	Discharge P5 of	Q(P)放电功率点	RW	S16	1	10	%Pn	[-1100,11]	Y	Percentage of Q(P) discharge power point	Q(P)放电功率点百分比
361	43880	Discharge Q5 of	Q(P)放电无功点	RW	S16	1	10	%Pn	[-1000,10]	Y	Q(P) Discharge Reactive Point Percentage / Q(P) Reactive Point Percentage (Slope Mode)	Q(P)放电无功点百分比 / Q(P)无功点百分比(斜率模式)
362	43881	Discharge P6 of	Q(P)放电功率点	RW	S16	1	10	%Pn	[-1100,11]	Y	Percentage of Q(P) discharge power point	Q(P)放电功率点百分比
363	43882	Discharge Q6 of	Q(P)放电无功点	RW	S16	1	10	%Pn	[-1000,10]	Y	Q(P) Discharge Reactive Point Percentage	Q(P)放电无功点百分比
364	43883	Reactive power	Q(P)无功功率基	RW	U16	1	1	N/A	[0,3]	Y	0: Rated power of Pn 1: Apparent power of Ps 2: Current power of Pm 3: Maximum power of Pm	0 : 额定功率(Pn) 1 : 视在功率(Ps) 2 : 当前功率(Pm) 3 : 最大功率(Pmax)
365	43884	Intentional delay	Q(P)初始延迟时	RW	U16	1	10	s	[0,10000]	Y	First trigger Q(P) response delay time	首次触发Q(P)响应延迟时间

366	43885	Response type	Q(P)响应方式	RW	U16	1	1	N/A	[0,3]	Y	The way Q(P) outputs responses. Slope mode or first-order low-pass filtering mode. Storage does not support high-frequency write operations 0: Not enabled 1: Slope Mode 2: First-order low-pass time constant 3: First-order low-pass response time	Q(P)输出响应的方式。斜率模式或一阶低通滤波模式。 存储，不支持高频写操作 0：不启用 1：斜率模式 2：一阶低通时间常数 3：一阶低通响应时间
367	43886	Response time	Q(P)响应时间常数	RW	U32	2	1	ms	[0,360000]	Y	When 43885 is set to 2, Q(P) response time constant When 43885 is set to 3, Q(P) response time	43885设置为2时，Q(P)响应时间常数 43885设置为3时，Q(P)响应时间
369	43888	Response slope	Q(P)响应斜率	RW	U16	1	10	%Pn/s	[0,65000]	Y	The setting will only take effect when 43885 is set to 1	43885设置为1时，设置才能生效
370	43889	Q(P) mode	Q(P)模式	RW	U16	1	1	N/A	[0,1]	Y	0: Point Plotting Mode 1: Slope Mode	0：描点模式 1：斜率模式
371	43890	Droop1 of Q(P)	Q(P) Droop1(slope)	RW	U16	1	10	Qmax/10	[0,2000]	Y	Q(P) Droop1(slope)	Q(P) Droop1(slope)
372	43891	Droop2 of Q(P)	Q(P) Droop2(slope)	RW	U16	1	10	Qmax/10	[0,2000]	Y	Q(P) Droop2(slope)	Q(P) Droop2(slope)
373	43892	Enter the curve	Q(P)进入条件使能	RW	U16	1	1	N/A	[0,1]	Y	0: Close 1: Open	0：禁能 1：使能
374	43893	Lock in voltage	Q(P)进入曲线电压百分比	RW	U16	1	10	%Vn	[500,1500]	Y	Q(P) Entering Curve Voltage Percentage	Q(P)进入曲线电压百分比
375	43894	Lock out voltage	Q(P)退出曲线电压百分比	RW	U16	1	10	%Vn	[500,1500]	Y	Q(P) Quitting Curve Voltage Percentage	Q(P)退出曲线电压百分比
Overvoltage and overfrequency												
391	43910	Under voltage threshold	一级欠压保护值	RW	U16	1	10	%Vn	[50,1000]	Y	Level 1 Undervoltage Percentage	一级欠压百分比

392	43911	Under voltage trip	一级欠压保护时	RW	U32	2	1	ms	[10,7200]	Y	Level 1 under-voltage protection tripping time	一级欠压保护跳脱时间
394	43913	Over voltage thr	一级过压保护值	RW	U16	1	10	%Vn	[800,1400]	Y	Level 1 Overvoltage Percentage	一级过压百分比
395	43914	Over voltage trip	一级过压保护时	RW	U32	2	1	ms	[10,7200]	Y	Level 1 over-voltage protection tripping time	一级过压保护跳脱时间
397	43916	Under voltage th	二级欠压保护值	RW	U16	1	10	%Vn	[0,1000]	Y	Level 2 Undervoltage Percentage	二级欠压百分比
398	43917	Under voltage tr	二级欠压保护时	RW	U32	2	1	ms	[10,7200]	Y	Level 2 under-voltage protection tripping time	二级欠压保护跳脱时间
400	43919	Over voltage thr	二级过压保护值	RW	U16	1	10	%Vn	[800,1400]	Y	Level 2 Overvoltage Percentage	二级过压百分比
401	43920	Over voltage trip	二级过压保护时	RW	U32	2	1	ms	[10,7200]	Y	Level 2 over-voltage protection tripping time	二级过压保护跳脱时间
403	43922	Under voltage th	三级欠压保护值	RW	U16	1	10	%Vn	[0,1000]	Y	Level 3 Undervoltage Percentage	三级欠压百分比
404	43923	Under voltage tr	三级欠压保护时	RW	U32	2	1	ms	[10,7200]	Y	Level 3 under-voltage protection tripping time	三级欠压保护跳脱时间
406	43925	Over voltage thr	三级过压保护值	RW	U16	1	10	%Vn	[800,1400]	Y	Level 3 Overvoltage Percentage	三级过压百分比
407	43926	Over voltage trip	三级过压保护时	RW	U32	2	1	ms	[10,7200]	Y	Level 3 over-voltage protection tripping time	三级过压保护跳脱时间
409	43928	Under voltage th	四级欠压保护值	RW	U16	1	10	%Vn	[0,1000]	Y	Level 4 Undervoltage Percentage	四级欠压百分比
410	43929	Under voltage tr	四级欠压保护时	RW	U32	2	1	ms	[10,7200]	Y	Level 4 under-voltage protection tripping time	四级欠压保护跳脱时间
412	43931	Over voltage thr	四级过压保护值	RW	U16	1	10	%Vn	[800,1400]	Y	Level 4 Overvoltage Percentage	四级过压百分比
413	43932	Over voltage trip	四级过压保护时	RW	U32	2	1	ms	[10,7200]	Y	Level 4 over-voltage protection tripping time	四级过压保护跳脱时间
415	43934	Under frequency	一级欠频保护值	RW	U16	1	100	Hz	[4500,600]	Y	Level 1 Under-frequency Percentage	一级欠频百分比
416	43935	Under frequency	一级欠频保护时	RW	U32	2	1	ms	[10,7200]	Y	Level 1 under-frequency protection tripping time	一级欠频保护跳脱时间

418	43937	Over frequency	一级过频保护值	RW	U16	1	100	Hz	[5000,6500]	Y	Level 1 Over-frequency Percentage	一级过频百分比
419	43938	Over frequency	一级过频保护时	RW	U32	2	1	ms	[10,72000]	Y	Level 1 over-frequency protection tripping time	一级过频保护跳脱时间
421	43940	Under frequency	二级欠频保护值	RW	U16	1	100	Hz	[4500,6000]	Y	Level 2 Under-frequency Percentage	二级欠频百分比
422	43941	Under frequency	二级欠频保护时	RW	U32	2	1	ms	[10,72000]	Y	Level 2 under-frequency protection tripping time	二级欠频保护跳脱时间
424	43943	Over frequency	二级过频保护值	RW	U16	1	100	Hz	[5000,6500]	Y	Level 2 over-frequency Percentage	二级过频百分比
425	43944	Over frequency	二级过频保护时	RW	U32	2	1	ms	[10,72000]	Y	Level 2 over-frequency protection tripping time	二级过频保护跳脱时间
427	43946	Under frequency	三级欠频保护值	RW	U16	1	100	Hz	[4500,6000]	Y	Level 3 Under-frequency Percentage	三级欠频百分比
428	43947	Under frequency	三级欠频保护时	RW	U32	2	1	ms	[10,72000]	Y	Level 3 under-frequency protection tripping time	三级欠频保护跳脱时间
430	43949	Over frequency	三级过频保护值	RW	U16	1	100	Hz	[5000,6500]	Y	Level 3 over-frequency Percentage	三级过频百分比
431	43950	Over frequency	三级过频保护时	RW	U32	2	1	ms	[10,72000]	Y	Level 3 over-frequency protection tripping time	三级过频保护跳脱时间
433	43952	Under frequency	四级欠频保护值	RW	U16	1	100	Hz	[4500,6000]	Y	Level 4 Under-frequency Percentage	四级欠频百分比
434	43953	Under frequency	四级欠频保护时	RW	U32	2	1	ms	[10,72000]	Y	Level 4 under-frequency protection tripping time	四级欠频保护跳脱时间
436	43955	Over frequency	四级过频保护值	RW	U16	1	100	Hz	[5000,6500]	Y	Level 4 over-frequency Percentage	四级过频百分比
437	43956	Over frequency	四级过频保护时	RW	U32	2	1	ms	[10,72000]	Y	Level 4 over-frequency protection tripping time	四级过频保护跳脱时间
439	43958	Over voltage thr	十分钟过压保护	RW	U16	1	10	%Vn	[800,1400]	Y	10-minute overpressure percentage	十分钟过压百分比
440	43959	Over voltage trip	十分钟过压保护	RW	U32	2	1	ms	[10,72000]	Y	10-minute overvoltage protection tripping time	十分钟过压保护跳脱时间
442	43961	Line voltage und	线电压欠压触发	RW	U16	1	10	%Vn	[50,1000]	Y		
443	43962	Line voltage ove	线电压过压触发	RW	U16	1	10	%Vn	[800,1400]	Y		

444	43963	Line voltage und	线电压欠压触发	RW	U16	1	10	%Vn	[0,1000]	Y	0: Shut down	0:关闭
445	43964	Line voltage ove	线电压过压触发	RW	U16	1	10	%Vn	[800,1400]	Y		
446	43965	Line voltage und	线电压欠压触发	RW	U16	1	10	%Vn	[0,1000]	Y	0: Shut down	0:关闭
447	43966	Line voltage ove	线电压过压触发	RW	U16	1	10	%Vn	[800,1400]	Y		
448	43967	Line voltage und	线电压欠压触发	RW	U16	1	10	%Vn	[0,1000]	Y	0: Shut down	0:关闭
449	43968	Line voltage ove	线电压过压触发	RW	U16	1	10	%Vn	[800,1400]	Y		
450	43969	First voltage pro	一级电压保护时	RW	S16	1	1	ms	[-10000,1]	Y		
451	43970	First frequency p	一级频率保护时	RW	S16	1	1	ms	[-10000,1]	Y		
452	43971	Second-third-fou	二三四级电压保	RW	S16	1	1	ms	[-10000,1]	Y		
453	43972	Second-third-fou	二三四级频率保	RW	S16	1	1	ms	[-10000,1]	Y		
High/low wear setting												
471	43990	VRT voltage mo	电压穿越电压模	RW	U16	1	1	N/A	[0,2]	Y	0: Fundamental positive sequence component Ud 1: Phase voltage reconstruction 2: Line voltage reconstruction	0:基波正序分量Ud 1:线电压重构 2:相电压重构
472	43991	Insensitivity rang	电压突变使能	RW	U16	1	1	N/A	[0,1]	Y	1: Enable Determine whether to enter the high and low crossing enable switch based on the voltage change	0 : 禁能 1 : 使能 根据电压变化量判断进入高低穿使能开关
473	43992	Threshold of ins	电压突变判断阈	RW	U16	1	10	%Vn	[0,1000]	Y	Voltage Ride-Through Voltage Variation Judgment Threshold	电压穿越电压变化量判断阈值
474	43993	RSV	RSV			1				Y		
475	43994	Q current inject	无功电流注入模	RW	U16	1	1	N/A	[0,1]	Y	injection, effective based on calculated reactive power 1: Relative value injection, increasing or decreasing reactive power based on the pre-crossing reactive	0 : 绝对值注入，根据计算无功生效 1 : 相对值注入，在穿越前无功基础上增加或减小无功

476	43995	Current mode	电流分配模式	RW	U16	1	1	N/A	[0,2]	Y	During the high-low crossing process, the maximum current is set according to I_{max}, reactive power should be fully generated as required by the calculation based on the K value, and insufficient active current is supplemented. 1. Active power priority: During the high-low crossing process, the active current equals I_{max}, and the insufficient reactive current is supplemented. 2. Constant current mode: Record the active and reactive current values before high and low voltage ride-through, and after triggering high and low voltage ride-through,	0、无功优先：高低穿过程中最大电流按照 I_{max}，无功依据K值计算要求应发尽发，不够的有功电流补充。 1、有功优先：高低穿过程中有功电流等于 I_{max}，不够的无功电流补充。 2、电流不变模式：记录高低穿前的有功和无功电流值，触发高低穿后按照记录的有功和无功电流值给定。
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477	43996	Response type	电压穿越有功恢	RW	U16	1	1	N/A	[0,3]	Y	VRT: Voltage Ride Through includes Low Voltage Ride Through (LVRT) and High Voltage Ride Through (HVRT) Storage does not support high-frequency write operations 0: Not enabled 1: Slope Mode 2: First-order low-pass time constant 3: First-order low-pass response time	VRT : 电压穿越包含低穿和高穿 存储, 不支持高频写操作 0 : 不启用 1 : 斜率模式 2 : 一阶低通时间常数 3 : 一阶低通响应时间
478	43997	Response slope	电压穿越有功恢	RW	U16	1	10	%ln/s	[0,65000]	Y	only when 43996 is set to 1	43996设置为1时, 设置才能生效
479	43998	Response type	电压穿越无功恢	RW	U16	1	1	N/A	[0,3]	Y	VRT: Voltage Ride Through includes Low Voltage Ride Through (LVRT) and High Voltage Ride Through (HVRT) Storage does not support high-frequency write operations 0: Not enabled 1: Slope Mode 2: First-order low-pass time constant 3: First-order low-pass response time	VRT : 电压穿越包含低穿和高穿 存储, 不支持高频写操作 0 : 不启用 1 : 斜率模式 2 : 一阶低通时间常数 3 : 一阶低通响应时间
480	43999	Response slope	电压穿越无功恢	RW	U16	1	10	%ln/s	[0,65000]	Y	The setting will only take effect when 43998 is set to 1	43998设置为1时, 设置才能生效

481	44000	Grid voltage trip	电压穿越电网电	RW	U16	1	1	N/A	[0,1]	Y	0: Disable, do not shield high and low penetration protection processing 1: Enable, shield high-low penetration protection processing	0：禁能，不屏蔽高低穿保护处理 1：使能，屏蔽高低穿保护处理
482	44001	RSV	RSV			2				Y		
484	44003	continuous ZVR	连续高低穿使能	RW	U16	1	1	N/A	[0,1]	Y	crossing is enabled by default, serving as a condition to meet the requirements of high-low crossing or multiple consecutive low crossings.	连续高低穿默认使能，作为满足高低穿交叉穿越或者低穿连续多次穿越的要求的条件。
485	44004	enter VRT silence	高低穿进入静默	RW	U16	1	1	10ms	[0,65500]	Y		
486	44005	exit VRT silence	高低穿退出静默	RW	U16	1	1	10ms	[0,65500]	Y		
487	44006	Response time d	电压穿越有功恢	RW	U32	2	1	ms	[0,360000]	Y	the active power recovery time constant for voltage ride-through When 43996 is set to 3, the active power recovery time during voltage ride-through	43996设置为2时，电压穿越有功恢复时间常数 43996设置为3时，电压穿越有功恢复时间
489	44008	Response time d	电压穿越无功恢	RW	U32	2	1	ms	[0,360000]	Y	the voltage ride-through reactive power recovery time constant When 43998 is set to 3, the voltage ride-through reactive power recovery time	43998设置为2时，电压穿越无功恢复时间常数 43998设置为3时，电压穿越无功恢复时间
491	44010	预留	预留			3				Y		
494	44013	LVRT enable	低穿使能	RW	U16	1	1	N/A	[0,1]	Y	0: Close 1: Open	0：禁能 1：使能

495	44014	Threshold of ent	低穿进入阈值	RW	U16	1	10	%Vn	[0,1000]	Y	Set the threshold for triggering low voltage ride through. The threshold setting must comply with the requirements of local grid standards	设置触发低电压穿越的阈值。阈值设置需要符合当地电网标准要求。
496	44015	Threshold of ext	低穿退出阈值	RW	U16	1	10	%Vn	[0,1000]	Y	Set the threshold for exiting low voltage ride through. The threshold setting must comply with the requirements of local grid standards.	设置退出低电压穿越的阈值。阈值设置需要符合当地电网标准要求。
497	44016	LVRT computati	低穿电压变化量	RW	U16	1	10	%Vn	[0,1000]	Y	Low-throughput reactive voltage reference, based on safety regulations requirements	低穿发无功的电压基准，基于安规要求
498	44017	Positive sequen	低穿正序有功调	RW	U16	1	1	N/A	[0,1]	Y	0: Close 1: Open	0：禁能 1：使能
499	44018	Positive sequen	低穿正序无功调	RW	U16	1	1	N/A	[0,1]	Y	0: Close 1: Open	0：禁能 1：使能
500	44019	Negative sequen	低穿负序无功调	RW	U16	1	1	N/A	[0,1]	Y	0: Close 1: Open	0：禁能 1：使能
501	44020	LVRT active pow	低穿有功电流限	RW	U16	1	10	%/In	[0,1100]	Y	current does not exceed this value	低穿有功电流不超过该值

502	44021	Positive sequen	低穿正序无功K	RW	U16	1	100	N/A	[0,1000]	Y	ride-through process, the inverter needs to inject positive sequence reactive power to support the power grid, and this parameter is used to set the magnitude of the positive sequence reactive power injected by the inverter	低电压穿越过程中，逆变器需要发出正序无功功率对电网进行支撑，该参数用于设置逆变器发出正序无功功率的大小
503	44022	LVRT Positive s	低穿正序无功静	RW	S16	1	10	%/ln	[-1000,10]	Y	Low-pass positive and negative reactive power accuracy deviation compensation amount	低穿正负无功精度偏差补偿量
504	44023	LVRT Positive s	低穿正序无功电	RW	U16	1	10	%/ln	[0,1100]	Y	the low-crossing positive-sequence reactive current shall not exceed this value	低穿正序无功电流最大不超过该值
505	44024	Negative Seque	低穿负序无功K	RW	U16	1	100	N/A	[0,1000]	Y	ride-through process, the inverter needs to inject negative sequence reactive power to support the power grid, and this parameter is used to set the magnitude of the negative sequence reactive power injected by the inverter.	低电压穿越过程中，逆变器需要发出负序无功功率对电网进行支撑，该参数用于设置逆变器发出负序无功功率的大小
506	44025	LVRT Negative s	低穿负序无功静	RW	S16	1	10	%/ln	[-1000,10]	Y	Low-pass negative sequence reactive power accuracy deviation compensation amount	低穿负序无功精度偏差补偿量

507	44026	LVRT Negative	低穿负序无功电	RW	U16	1	10	%/In	[0,1100]	Y	the low-crossing negative-sequence reactive current shall not exceed this value	低穿负序无功电流最大不超过该值
508	44027	0-current mode	低穿零电流模式	RW	U16	1	1	N/A	[0,1]	Y	1: Enable When "Zero Current Mode Enable" is enabled, during the LVRT process, if the grid voltage is less than the "LVRT Zero Current Mode Threshold", it will be executed according to the zero current mode; otherwise, it will be executed according to the mode set in the "Current Distribution Mode".	0 : 禁能 1 : 使能 当“零电流模式使能”使能时，在LVRT过程中，电网电压小于“LVRT零电流模式阈值”，按照零电流模式执行，否则按照“电流分配模式模式”设置的模式执行。
509	44028	Threshold of 0-c	低穿零电流模式	RW	U16	1	10	%/Vn	[0,1000]	Y	When the zero current mode enable switch is turned on and the voltage is below this threshold, the inverter does not output current.	零电流模式使能开关打开后，电压低于该阈值，则逆变器不输出电流。
510	44029	预留	预留			11				Y		
521	44040	LVRT Feature V	LVRT特征点V1	RW	U16	1	10	%Vn	[0,1000]	Y	percentage for the characteristic points 1 of the low-pass curve.	设置低穿曲线特征点1的电压百分比。
522	44041	LVRT Feature T	LVRT特征点T1	RW	U16	1	1	10ms	[0,65000]	Y	low-pass curve feature points 1.	设置低穿曲线特征点1的持续时间。
523	44042	LVRT Feature V	LVRT特征点V2	RW	U16	1	10	%Vn	[0,1000]	Y	percentage for the characteristic points 2 of the low-pass curve.	设置低穿曲线特征点2的电压百分比。

524	44043	LVRT Feature T	LVRT特征点T2	RW	U16	1	1	10ms	[0,65000]	Y	low-pass curve feature points 2.	设置低穿曲线特征点2的持续时间。
525	44044	LVRT Feature V	LVRT特征点V3	RW	U16	1	10	%Vn	[0,1000]	Y	percentage for the characteristic points 3 of the low-pass curve.	设置低穿曲线特征点3的电压百分比。
526	44045	LVRT Feature T	LVRT特征点T3	RW	U16	1	1	10ms	[0,65000]	Y	low-pass curve feature points 3.	设置低穿曲线特征点3的持续时间。
527	44046	LVRT Feature V	LVRT特征点V4	RW	U16	1	10	%Vn	[0,1000]	Y	percentage for the characteristic points 4 of the low-pass curve.	设置低穿曲线特征点4的电压百分比。
528	44047	LVRT Feature T	LVRT特征点T4	RW	U16	1	1	10ms	[0,65000]	Y	low-pass curve feature points 4.	设置低穿曲线特征点4的持续时间。
529	44048	LVRT Feature V	LVRT特征点V5	RW	U16	1	10	%Vn	[0,1000]	Y	percentage for the characteristic points 5 of the low-pass curve.	设置低穿曲线特征点5的电压百分比。
530	44049	LVRT Feature T	LVRT特征点T5	RW	U16	1	1	10ms	[0,65000]	Y	low-pass curve feature points 5.	设置低穿曲线特征点5的持续时间。
531	44050	LVRT Feature V	LVRT特征点V6	RW	U16	1	10	%Vn	[0,1000]	Y	percentage for the characteristic points 6 of the low-pass curve.	设置低穿曲线特征点6的电压百分比。
532	44051	LVRT Feature T	LVRT特征点T6	RW	U16	1	1	10ms	[0,65000]	Y	low-pass curve feature points 6.	设置低穿曲线特征点6的持续时间。
533	44052	LVRT Feature V	LVRT特征点V7	RW	U16	1	10	%Vn	[0,1000]	Y	percentage for the characteristic points 7 of the low-pass curve.	设置低穿曲线特征点7的电压百分比。
534	44053	LVRT Feature T	LVRT特征点T7	RW	U16	1	1	10ms	[0,65000]	Y	low-pass curve feature points 7.	设置低穿曲线特征点7的持续时间。
535	44054	预留	预留			11				Y		
546	44065	HVRT enable	高穿使能	RW	U16	1	1	N/A	[0,1]	Y	0: Disable 1: Enable	0 : 禁能 1 : 使能

547	44066	Threshold of ent	高穿进入阈值	RW	U16	1	10	%Vn	[1000,140	Y	triggering high voltage ride-through. The threshold setting must comply with the requirements of local grid standards.	设置触发高电压穿越的阈值。阈值设置需要符合当地电网标准要求。
548	44067	Threshold of ext	高穿退出阈值	RW	U16	1	10	%Vn	[1000,140	Y	Set the threshold for quitting high voltage ride-through. The threshold setting must comply with the requirements of local grid standards.	设置退出高电压穿越的阈值。阈值设置需要符合当地电网标准要求。
549	44068	HVRT computat	高穿电压变化量	RW	U16	1	10	%Vn	[1000,120	Y	High-throughput, non-functional voltage reference, based on safety requirements	高穿发无功的电压基准，基于安规要求
550	44069	Positive sequen	高穿正序有功调	RW	U16	1	1	N/A	[0,1]	Y	0: Disable 1: Enable	0：禁能 1：使能
551	44070	Positive sequen	高穿正序无功调	RW	U16	1	1	N/A	[0,1]	Y	0: Disable 1: Enable	0：禁能 1：使能
552	44071	Negative sequen	高穿负序无功调	RW	U16	1	1	N/A	[0,1]	Y	0: Disable 1: Enable	0：禁能 1：使能
553	44072	HVRT active pow	高穿有功电流限	RW	U16	1	10	%In	[0,1100]	Y	High penetration active current does not exceed this value	高穿有功电流不超过该值

554	44073	Positive sequen	高穿正序无功K	RW	U16	1	100	N/A	[0,1000]	Y	ride-through process, the inverter needs to output positive sequence reactive power to support the power grid, and this parameter is used to set the magnitude of the positive sequence reactive power output by the inverter	高电压穿越过程中，逆变器需要发出正序无功功率对电网进行支撑，该参数用于设置逆变器发出正序无功功率的大小
555	44074	HVRT Positive s	高穿正序无功静	RW	S16	1	10	%/ln	[-1000,10]	Y	High-penetration positive and negative reactive power accuracy deviation compensation amount	高穿正负无功精度偏差补偿量
556	44075	HVRT Positive s	高穿正序无功电	RW	U16	1	10	%ln	[0,1100]	Y	the high-penetration positive-sequence reactive current does not exceed this value	高穿正序无功电流最大不超过该值
557	44076	Negative Seque	高穿负序无功K	RW	U16	1	100	N/A	[0,1000]	Y	ride-through process, the inverter needs to inject negative sequence reactive power to support the power grid, and this parameter is used to set the magnitude of the negative sequence reactive power injected by the inverter	高电压穿越过程中，逆变器需要发出负序无功功率对电网进行支撑，该参数用于设置逆变器发出负序无功功率的大小
558	44077	HVRT Negative	高穿负序无功静	RW	S16	1	10	%/ln	[-1000,10]	Y	negative-sequence reactive power accuracy deviation compensation amount	高穿负序无功精度偏差补偿量

559	44078	HVRT Negative	高穿负序无功电	RW	U16	1	10	%In	[0,1100]	Y	the high-penetration negative-sequence reactive current shall not exceed this value	高穿负序无功电流最大不超过该值
560	44079	0-current mode	高穿零电流模式	RW	U16	1	1	N/A	[0,1]	Y	1: Enable When "Zero Current Mode Enable" is enabled, during the HVRT process, if the grid voltage is greater than the "HVRT Zero Current Mode Threshold", it shall be executed in accordance with the zero current mode; otherwise, it shall be executed in accordance with the mode set in the "Current	0: 禁能 1: 使能 当“零电流模式使能”使能时，在HVRT过程中，电网电压大于“HVRT零电流模式阈值”，按照零电流模式执行，否则按照“电流分配模式模式”设置的模式执行。
561	44080	Threshold of 0-c	高穿零电流模式	RW	U16	1	10	%Vn	[1000,140	Y	When the zero current mode enable switch is turned on and the voltage is higher than this threshold, the inverter does not output current.	零电流模式使能开关打开后，电压高于该阈值，则逆变器不输出电流。
562	44081	预留	预留			11				Y		
573	44092	HVRT Feature V	HVRT特征点V1	RW	U16	1	10	%Vn	[1000,140	Y	percentage for the high-pass curve feature point 1.	设置高穿曲线特征点1的电压百分比。
574	44093	HVRT Feature T	HVRT特征点T1	RW	U16	1	1	10ms	[0,65000]	Y	high-pass curve feature point 1.	设置高穿曲线特征点1的持续时间。
575	44094	HVRT Feature V	HVRT特征点V2	RW	U16	1	10	%Vn	[1000,140	Y	percentage for the high-pass curve feature point 2.	设置高穿曲线特征点2的电压百分比。

576	44095	HVRT Feature T	HVRT特征点T2	RW	U16	1	1	10ms	[0,65000]	Y	high-pass curve feature point 2.	设置高穿曲线特征点2的持续时间。
577	44096	HVRT Feature V	HVRT特征点V3	RW	U16	1	10	%Vn	[1000,140]	Y	percentage for the high-pass curve feature point 3.	设置高穿曲线特征点3的电压百分比。
578	44097	HVRT Feature T	HVRT特征点T3	RW	U16	1	1	10ms	[0,65000]	Y	high-pass curve feature point 3.	设置高穿曲线特征点3的持续时间。
579	44098	HVRT Feature V	HVRT特征点V4	RW	U16	1	10	%Vn	[1000,140]	Y	percentage for the high-pass curve feature point 4.	设置高穿曲线特征点4的电压百分比。
580	44099	HVRT Feature T	HVRT特征点T4	RW	U16	1	1	10ms	[0,65000]	Y	high-pass curve feature point 4.	设置高穿曲线特征点4的持续时间。
581	44100	HVRT Feature V	HVRT特征点V5	RW	U16	1	10	%Vn	[1000,140]	Y	percentage for the high-pass curve feature point 5.	设置高穿曲线特征点5的电压百分比。
582	44101	HVRT Feature T	HVRT特征点T5	RW	U16	1	1	10ms	[0,65000]	Y	high-pass curve feature point 5.	设置高穿曲线特征点5的持续时间。
583	44102	HVRT Feature V	HVRT特征点V6	RW	U16	1	10	%Vn	[1000,140]	Y	percentage for the high-pass curve feature point 6.	设置高穿曲线特征点6的电压百分比。
584	44103	HVRT Feature T	HVRT特征点T6	RW	U16	1	1	10ms	[0,65000]	Y	high-pass curve feature point 7.	设置高穿曲线特征点7的持续时间。
585	44104	HVRT Feature V	HVRT特征点V7	RW	U16	1	10	%Vn	[1000,140]	Y	percentage for the high-pass curve feature point 8.	设置高穿曲线特征点8的电压百分比。
586	44105	HVRT Feature T	HVRT特征点T7	RW	U16	1	1	10ms	[0,65000]	Y	high-pass curve feature point 8.	设置高穿曲线特征点8的持续时间。
primary frequency control												
601	44120	FSM enable	FSM使能	RW	U16	1	1	N/A	[0,1]	Y	0: Close 1: Open	0 : 禁能 1 : 使能
602	44121	Power reference	FSM功率基准	RW	U16	1	1	N/A	[0,3]	Y	1: Apparent power of Ps 2: Current power of Pm 3: Maximum power of Pmax	0 : 额定功率(Pn) 1 : 视在功率(Ps) 2 : 当前功率(Pm) 3 : 最大功率(Pmax)

603	44122	Frequency response	FSM频率死区	RW	U16	1	1	mHz	[0,500]	Y	triggered within the frequency deadband range	在频率死区范围内不触发功率变动
604	44123	Intentional delay	FSM初始延迟时间	RW	U16	1	1	ms	[0,65000]	Y	First Trigger FSM Response Delay Time	首次触发FSM响应延迟时间
605	44124	Power slope of FSM	FSM功率变化斜率	RW	U16	1	10	%P/Hz	[0,65000]	Y	Slope of the active power changes output by the inverter	逆变器输出的有功功率变化斜率
606	44125	Active power range	FSM最大功率变化量	RW	U16	1	10	%Pn	[0,1000]	Y	Active power range related to maximum capacity $ \Delta P1 / P_{max}$	FSM最大功率变化量
607	44126	Response type	FSM响应方式	RW	U16	1	1	N/A	[0,3]	Y	The way the FSM outputs responses. Slope mode or first-order low-pass filtering mode. Storage does not support high-frequency write operations 0: Not enabled 1: Slope Mode 2: First-order low-pass time constant 3: First-order low-pass response time	FSM输出响应的方式。斜率模式或一阶低通滤波模式。 存储，不支持高频写操作 0：不启用 1：斜率模式 2：一阶低通时间常数 3：一阶低通响应时间
608	44127	Response time	FSM响应时间常数	RW	U32	2	1	ms	[0,360000]	Y	When 44126 is set to 2, the FSM response time constant When 44126 is set to 3, the FSM response time	44126设置为2时，FSM响应时间常数 44126设置为3时，FSM响应时间
610	44129	Response slope	FSM响应斜率	RW	U16	1	10	%Pn/s	[0,65000]	Y	only when 44126 is set to 1	44126设置为1时，设置才能生效

611	44130	Frequency response	FSM频率不敏感	RW	U16	1	1	mHz	[0,50]	Y	When the frequency fluctuates within a certain range, the frequency input is not updated, and the output active power remains unchanged	频率在一定范围波动，不更新频率输入，输出有功不变
<i>primary frequency control</i>												
631	44150	FRT enable	频率穿越使能	RW	U16	1	1	N/A	[0,1]	Y	0: Disable 1: Enable FRT : (Frequency Ride Through).	0 : 禁能 1 : 使能 FRT : (Frequency Ride Through)
632	44151	UFRT Feature F	UFRT特征点F1	RW	U16	1	100	Hz	[4500,600]	Y	Frequency value of underfrequency crossing characteristic point 1	欠频穿越特征点1的频率值
633	44152	UFRT Feature T	UFRT特征点T1	RW	U32	2	1	ms	[20,7200]	Y	Duration of Underfrequency Crossing Characteristic Point 1	欠频穿越特征点1的持续时间
635	44154	UFRT Feature F	UFRT特征点F2	RW	U16	1	100	Hz	[4500,600]	Y	Frequency value of underfrequency crossing characteristic point 12	欠频穿越特征点2的频率值
636	44155	UFRT Feature T	UFRT特征点T2	RW	U32	2	1	ms	[20,7200]	Y	Duration of Underfrequency Crossing Characteristic Point 2	欠频穿越特征点2的持续时间
638	44157	UFRT Feature F	UFRT特征点F3	RW	U16	1	100	Hz	[4500,600]	Y	Frequency value of underfrequency crossing characteristic point 3	欠频穿越特征点3的频率值
639	44158	UFRT Feature T	UFRT特征点T3	RW	U32	2	1	ms	[20,7200]	Y	Duration of Underfrequency Crossing Characteristic Point 3	欠频穿越特征点3的持续时间
641	44160	OFRT Feature F	OFRT特征点F1	RW	U16	1	100	Hz	[5000,650]	Y	Frequency value of overfrequency crossing characteristic point 1	过频穿越特征点1的频率值

642	44161	OFRT Feature T	OFRT特征点T1	RW	U32	2	1	ms	[20,72000	Y	Duration of Overfrequency Crossing Characteristic Point 1	过频穿越特征点1的持续时间
644	44163	OFRT Feature F	OFRT特征点F2	RW	U16	1	100	Hz	[5000,650	Y	Frequency value of overfrequency crossing characteristic point 2	过频穿越特征点2的频率值
645	44164	OFRT Feature T	OFRT特征点T2	RW	U32	2	1	ms	[20,72000	Y	Duration of Overfrequency Crossing Characteristic Point 2	过频穿越特征点2的持续时间
647	44166	OFRT Feature F	OFRT特征点F3	RW	U16	1	100	Hz	[5000,650	Y	Frequency value of overfrequency crossing characteristic point 3	过频穿越特征点3的频率值
648	44167	OFRT Feature T	OFRT特征点T3	RW	U32	2	1	ms	[20,72000	Y	Duration of Overfrequency Crossing Characteristic Point 3	过频穿越特征点3的持续时间

#Address		English Name	Chinese Name	#R/W	#Type	#Size	#SF	#Units	Range	Flash Save	Note(English)	Note(Chinese)
Com Setting												
14	45127	Modbus Address	Modbus地址	RW	U16	1	1	N/A	[0,247]	Y	The deFault is 247. if multiple inverters are connected to the same controller, addr of each inverter must be different and 247 should not be used for any inverter	默认值为247。如果多台逆变器连接在同一个控制器上，每台逆变器的地址必须不同，且任何一台逆变器不能使用
16	45132	485 Modbus Baudrate	485 Modbus波特率	RW	U32	2	1	N/A	[1,5] [9600,115200]	Y	EM/ES/SBP series(ARM105) range[1,5] When the value is between 1 and 5, 1:9600; 2:19200; 3:38400; 4:57600; 5:115200 EH/ET/ETPLUS series(ARM205) range[9600,19200] other series(ARM205 and 745) range[9600,115200] If set wrong, EMS communicaiton fails	EM/ES/SBP系列 设置范围1-5，当值在1到5之间时,1:9600; 2:19200; 3:38400; 4:57600; 5:115200 EH/ET/ETPLUS系列 设置范围9600或19200，其他系列设置范围9600至115200 若果设置错误，则EMS通讯失败
19	45153	Modbus Protocol Type	modbus协议类型	RW	STR	1	1	N/A		Y	“0”goodwe “1” sunspec	“0”固德威 “1” SUNSPEC
25	45166	DHCP Enable	DHCP功能开启	RW	U16	1	1	N/A	[0,1]	Y	0 : close (Static) 1 : open (DHCP)	0 : 关闭 (静态) 1 : 开启 (DHCP)
26	45167	IPV4_IP1	IPV4的IP字段1	RW	U16	1	1	N/A	[0,255]	Y		
27	45168	IPV4_IP2	IPV4的IP字段2	RW	U16	1	1	N/A	[0,255]	Y		
28	45169	IPV4_IP3	IPV4的IP字段3	RW	U16	1	1	N/A	[0,255]	Y		
29	45170	IPV4_IP4	IPV4的IP字段4	RW	U16	1	1	N/A	[0,255]	Y		
30	45171	IPV4_MASK1	IPV4的IP掩码字段1	RW	U16	1	1	N/A	[0,255]	Y		
31	45172	IPV4_MASK2	IPV4的IP掩码字段2	RW	U16	1	1	N/A	[0,255]	Y		
32	45173	IPV4_MASK3	IPV4的IP掩码字段3	RW	U16	1	1	N/A	[0,255]	Y		
33	45174	IPV4_MASK4	IPV4的IP掩码字段4	RW	U16	1	1	N/A	[0,255]	Y		
34	45175	IPV4_GW1	IPV4的IP网关字段1	RW	U16	1	1	N/A	[0,255]	Y		
35	45176	IPV4_GW2	IPV4的IP网关字段2	RW	U16	1	1	N/A	[0,255]	Y		
36	45177	IPV4_GW3	IPV4的IP网关字段3	RW	U16	1	1	N/A	[0,255]	Y		
37	45178	IPV4_GW4	IPV4的IP网关字段4	RW	U16	1	1	N/A	[0,255]	Y		

Function Setting												
1	45200	RTC Setting	设备RTC时间	RW	U16	1	1	N/A	[13,99]-[1,12]	N	High Byte:Year/Low Byte:Month	高字节:年/低字节:月
2	45201	RTC Setting	设备RTC时间	RW	U16	1	1	N/A	[1,31]-[0,23]	N	High Byte:Day/Low Byte:Hour	高字节:日/低字节:时
3	45202	RTC Setting	设备RTC时间	RW	U16	1	1	N/A	[0,59]-[0,59]	N	High Byte:Minute/Low Byte:Second	高字节:分/低字节:秒
6	45216	Restore Factory Setting	恢复出厂设置	WO	U16	1	1	N/A	[1]	N	Choose "Warehouse" safety code first and then Set "1" to factory settings	先选择“Warehouse”安全码，然后将“1”设置为出厂设置
7	45217	Clear Data	清除数据	WO	U16	1	1	N/A	[1]	N	Reset inverter accumulated data like E-total, E-day, error log running data etc.	复位逆变器累计数据如E-total、E-day、错误日志运行数据等。
8	45218	On/Off(Allow to Grid/Forbid to Grid)	开/关机(允许并网/禁止并网)	RW	U16	1	1	N/A	[0 , 1]	N	0 Shutdown; 1 Startup. (Shutting down requires disconnecting the bypass relay	0关机；1开机。 (关机同时需要断开bypass继电器)
10	45220	Restart	重启	WO	U16	1	1	N/A	[1]	N	Inverter will recheck and reconnect to utility again. Inverter does not shutdown.	逆变器将再次检查并重新连接到实用程序。逆变器不停机
12	45222	PV Energy-Total	PV 总电量	RW	U32	2	10	1KW.Hr		Y	To read or write the total PV production energy from the installation date.	读取或写入自安装日起的光伏总发电量
13	45224	PV Energy-Day	PV 日电量	RW	U32	2	10	1KW.Hr		Y	To read or write the total PV production energy of the day.	读取或写入每天的光伏发电量

14	45226	Energy-Total-Sell	总卖电量	RW	U32	2	10	1KW.Hr		Y	To read or write the accumulated exporting energy to Grid from the installation date.	读取或写入自安装之日起累计输至电网的电量
15	45228	Hour-Total	累计工作时间	RW	U32	2	1	H		Y	To read or write the accumulated operation hours from the installation date.	读取或写入自安装之日起累计工作时间
16	45230	Energy-Day-Sell	日卖电量	RW	U16	1	10	1KW.Hr		Y	To read or write the accumulated exporting energy to Grid of the day.	读取或写入自每天输至电网的电量
17	45231	Energy-Total-Buy	总买电量	RW	U32	2	10	1KW.Hr		Y	To read or write the accumulated energy imported from Grid from the installation date.	读取或写入自安装之日起累计从电网输入的电量
18	45233	Energy-Day-Buy	日买电量	RW	U16	1	10	1KW.Hr		Y	To read or write the accumulated energy imported from Grid of the day.	读取或写入每天从电网输入的电量
19	45234	Energy-Total-Load	总负载电量	RW	U32	2	10	1KW.Hr		Y	To read or write the accumulated load consumption energy from the installation date, not include backup load.	读取或写入自安装之日起累计的负载消耗电量，不包括backup负载。

20	45236	Energy-Load-Day	日负载电量	RW	U16	1	10	1KW.Hr		Y	To read or write the accumulated load consumption energy of the day Not include backup loads.	读取或写入每天的负载消耗电量，不包括backup负载。
21	45237	Energy-Battery Charge	总电池充电量	RW	U32	2	10	1KW.Hr		Y	To read or write the accumulated energy charged to Battery from the installation date,not from BMS.	读取或写入从安装之日起累计向电池充电的电量，不是从BMS
22	45239	Energy-Charge-Day	日电池充电量	RW	U16	1	10	1KW.Hr		Y	To read or write the accumulated energy charged to Battery of the day,not from BMS.	读取或写入每天向电池充电的电量，不是从BMS
23	45240	Energy-Battery Discharge	总电池放电量	RW	U32	2	10	1KW.Hr		Y	To read or write the accumulated energy Battery discharged, from the installation date,not from BMS.	读取或写入从安装之日起累计向电池放电的电量，不是从BMS
24	45242	Energy-Discharge-Day	日电池放电量	RW	U16	1	10	1KW.Hr		Y	To read or write the accumulated energy Battery discharged, of the day,not from BMS.	读取或写入每天向电池放电的电量，不是从BMS
26	45244	Safety Country	安规国家	RW	U16	1	1	N/A	[0,65535]	N	To set safety code for inverter or read the preset safety code for the	设置逆变器安全码或读取预设的逆变器安全码
27	45245	ISO Limit	ISO限值	RW	U16	1	1	10kΩ	[0,1000]	Y	deFault 100kΩ, to read or set Isolation protection threshold for	默认100kΩ，读取或设置逆变器的“隔离保护
28	45246	LVRT Enable	低电压穿越使能	RW	U16	1	1	N/A	[0,1]	Y	as deFault is deactivated, set "1" to activate LVRT function, Set "2" to activate HVRT The same as 45499	默认激活，设置“1”为开启LVRT功能，设置“2”为开启HVRT功能和45499一样
32	45250	PV Start Voltage	PV 启动电压	RW	U16	1	10	V	[1800,8500]	Y	To write or read the start up PV Voltage of the inverter.Please refer to the user manual.	写入或读取逆变器的启动PV电压。请参考使用手册

33	45251	Enable MPPT4 Shadow	阴影扫描使能	RW	U16	1	1	N/A	[0,1]	Y	as deFault is deactivated, set "1" to activate "Shadow Scan" functtion	默认禁用，设置"1"激活"阴影扫描"功能
34	45252	BackUp Enable	backup使能	RW	U16	1	1	N/A	[0,1]	Y	as deFault is activated, set "0" to deactivate "Backup" functtion	默认激活，设置"0"禁用"Backup"功能
35	45253	Auto Start Backup	backup自启动	RW	U16	1	1	N/A	[0,1]	Y	Off-Grid Auto startup, as deFault is deactivated, set "1" to activate "Shadow Scan" functtion.	离网自动启动，默认激活，设置"1"激活"阴影扫描"功能
36	45254	Grid Wave Check Level	电网波形检测	RW	U16	1	1	N/A	[0,2]	Y	0 : full (High quality) wave check ; 1 : half (low quality) wave check ; 2 : close wave check	0 : 全波(高质量)检测;1:半波(低质量)检测;2:关闭.默认为"1"半
37	45255	Rapid Cut Off	快速切断	RW	U16	1	1	N/A		N	Reapid shutdown,to cut Grid connection, System turn to off-Grid operation.	快速关闭，切换电网连接系统至离网运行。
38	45256	Backup Start Delay	backup启动延时	RW	U16	1	1	N/A		N	Default: 1500 (30s)	默认1500(30s)
39	45257	UPS Standard Voltage Type	不间断电源标准电压类型	RW	U16	1	N/A	N/A	[0,3]	Y	0:208V, 1:20V, 2:240V, 3:230V	0:208V, 1:220V, 2:240V, 3:230V
45	45263	Derate Rate(VDE)	安规降额	RW	U16	1	N/A	N/A	[0,100]	Y	decrease the load,only can set 70, only for German	降载70%,只能设置70,且仅用于德国
46	45264	Three Phase Unbalanced Output	三相不平衡输出	RW	U16	1	N/A	N/A	[0,1]	Y	this function is deactivated as deFault, set "1" to activate. After activated, All power needs to be turned off and restarted	该功能默认禁用,设置"1"激活.激活后,所有的电源都需要关闭并重新启动
48	45266	High Impedance Mode	高阻抗模式	RW	U16	1	N/A	N/A		Y	For weak Grid area	用于弱电区域
53	45271	ARC Self Check	拉弧自检	WO	U16	1	N/A	1	[1450]	N	only for inverters with AFCI function	仅用于带电弧分段保护功能的逆变器
54	45272	ARC Fault Remove	拉弧故障手动清除	WO	U16	1	N/A	1	[1290]	N		
57	45275	ISO Check Mode	ISO检测模式	RW	U16	1	N/A	1	[0,1]	Y	0:Normal mode 1:cancel ISO test when offGrid to onGrid	0:常规模式 1:当离网转并网时取消ISO测试
58	45276	Off Grid To On Grid Delay	离网转并网延迟	RW	U16	1	N/A	sec	[10,7200]	Y	The delay time when Grid is available	电网可用时的延时时间
59	45277	Off Grid Under Voltage Protect Ccoefficient	离网输出欠压保护系数	RW	U16	1	N/A	%	[50,90]	Y	If set 80%, when offGrid output Voltage less than 230*80%=184V, inverter will have the error.DeFault setting is 80%.	如果设置为80%，当离网输出电压小于230*80%=184V,逆变器会发生故障。默认设置为80%。

60	45278	Battery Mode PV Charge Enable	PV给电池模式充电使能	RW	U16	1	1	1	[0,1]	Y	When offGrid and the Battery SOC is low, PV charge the Battery first.	当离网和电池剩余电量过低, PV优先给电池充
61	45279	DCV Check Coefficient	离网DCV检测系数	RW	U16	1	N/A	N/A	[1,20]	Y	Default: 1	默认设置为1
62	45280	Force MircoGrid Run	微网强制启动充电	RW	U16	1	N/A	N/A	[0,1]	Y	Only for MircoGrid Function inverter	仅用于具有微网功能的逆变器 0:常规模式 1:无视电池DOD, 进入微网充电
63	45281	Battery PreCharge Function	电池预充电功能	RW	U16	1	N/A	1	[0 , 1]	Y	For configuring whether the Battery comes with a pre-charge function	用于配置电池是否自带预充电功能
68	45286	GL Operation Mode Control Duration	GL工作模式控制有效时长	RW	U16	1	N/A	sec	[0,65535]	Y		
70	45288	PE Relay Switch	接地继电器开关	RW	U16	1	N/A	N/A	[0,1]	Y	1 close 0 open	1:闭合 0:断开
71	45289	GFCI fault remove	GFCI故障手动清除	WO	U16	1	N/A	N/A	[0 , 0x050A]	N		写入0x050A清除GFCI故障
73	45291	PV Connect Mode	PV 接入模式设置	RW	U16	1	N/A	N/A	[0 , 255]	N	PV Connect Mode , Set 0 for each MPPT , Set 1 for 12,34MPPT,Set 2 for 1MPPT	PV接入模式, 设置0, 4路单独MPPT, 设置12路并联MPPT, 34路并联MPPT, 设置2, 4路并联MPPT
74	45292	GFCI Disable Check	GFCI禁止检测	RW	U16	1	N/A	N/A	[0,1]	Y	0:Normal mode 1:cancel GFCI check	0:常规模式 1:产线校正站并网不进行GFCI检
75	45293	SPD enable	防雷报警使能	RW	U16	1	N/A	N/A	[0,1]	Y		1: 使能防雷报警功能
76	45294	Force MircoGrid Frequence Adjust	强制启动微网频率控制	RW	U16	1	N/A	N/A	[0 , 1]	Y	Only for MircoGrid Function inverter	仅用于具有微网功能的逆变器
77	45295	Shadow scan cycle setting	阴影扫描周期设定	RW	U16	1	N/A	Min	[5 , 300]	Y		

78	45296	PV only UPS Auto En	单独PV供电离网自启动使能	RW	U16	1	N/A	N/A	[0,1]	Y	1:Enable	1：使能单PV供电离网自启动使能
79	45297	Ac Phase Order Self-Adaption	AC相序自适应开关	RW	U16	1	N/A	N/A	[0,1]	Y	1:Enable	1：使能AC相序自适应开关
80	45298	AFCI Fault Reconnect Time	AFCI故障自动重连时间	RW	U16	1	N/A	s	[10,3600]	Y	Default: 300s	设置AFCI故障出现后故障自动清除的时间（默认300s）
81	45299	AFCI Fault Reconnect Count Limit	AFCI故障重连次数限制（单日）	RW	U16	1	N/A	N/A	[0,1000]	Y	Default: 5 times	设置单日内允许自动清除AFCI故障的最大次数（默认5次）
82	45300	Port multiplexing mode	端口复用模式选择	RW	U16	1	1	N/A	[0,2]	Y	ESC Port multiplexing mode (default: 0) 0: generator mode 1: Large load mode 2: indicates the normal BACKUP mode	ESC端口复用模式（默认为0） 0：发电机模式 1：大负载模式 2：普通BACKUP模式
83	45301	PV charge enable	PV充电使能	RW	U16	1	1	N/A	[0,1]	Y		
84	45302	Battery SPS SW_TurnOff Enable	电池sps软件关机使能	RW	U16	1	1	N/A	[0,1]	Y	ES G2	ESG2/ES Uniq适用
85	45303	Battery SPS TurnOff Value	电池sps关机点	RW	U16	1	10	V	[300,480]	Y	ES G2	ESG3/ES Uniq适用
86	45304	PX Curve Enable	PX曲线开关	RW	U16	1	1	N/A	[0,1]	Y	Three-phase inverter	三相机适用
87	45305	PX Curve Voltage	PX电压设定阈值	RW	U16	1	10	V	[2300, 3000]	Y	Three-phase inverter	三相机适用
88	45306	external Trigger emergency stop signal	外部触发设备急停标志	RW	U16	1	1	N/A	[0,1]	Y		
89	45307	Micro Grid Frequency vs. SOC slope	微网频率对SOC斜率	RW	U16	1	100	Hz / %	[0,50]	Y		
90	45308	Single Island Check Enable	单相孤岛检测使能	RW	U16	1	1	N/A	[0,1]	Y	Default: OFF (three-phase inverter)	默认关闭（三相机适用）
91	45309	Maximum input current limit on the grid side	网侧最大输入电流限制	RW	U16	1	1	A	[0,60000]	Y	Used for users to set their own circuit breaker size to prevent	用于用户设置自己的空开大小防止跳闸
92	45310	N-PE Fault Limit	N-PE报错阈值	RW	U16	1	10	V	[200,2300]	Y	N-PE voltage overvoltage fault threshold. Default: 50V	N-PE电压过压故障阈值。 默认50V
93	45311	IslandDetectReactiveFactor	孤岛检测频率无功系数	RW	U16	1	1	N/A	[0,1000]	Y		
94	45312	Meter Connect Check Flag	电表检测使能	RW	U16	1	N/A	N/A	[0,1]	N	0:Disable 1:Enable	0:禁能 1:使能

97	45316	Total electricity consumption of Static Transfer Switch Smart Port	切换柜智能端口总用电量	RW	U32	2	10	1KW.Hr	[0,0]	Y	To read or write the accumulated exporting energy to SP from the installation date.	读取或写入自安装之日起累计输至智能端口(SP)的电量
98	45318	Total power generation of Static Transfer Switch Smart Port	切换柜智能端口总发电量	RW	U32	2	10	1KW.Hr	[0,0]	Y	To read or write the accumulated energy imported from SP from the installation date.	读取或写入自安装之日起累计从智能端口(SP)输入的电量
99	45320	"Incorrect operation of the bypass switch" Fault clear instruction	"旁路开关误动作"故障清除指令	RW	U16	1	N/A	N/A	[0,1]	N	"Incorrect operation of the bypass switch" Fault clear instruction from APP	APP指令，清除“旁路开关误动作”故障
100	45321	Output Type	输出类型	RW	U16	1	1	N/A	[0,4]	Y	0:single phase 1:three phase four wire system 2:three phase three wire system 3:two phase 4:split phase	0:单相 1:三相四线制 2:三相三线制 3:两相(120°) 4:裂相(180°)
101	45322	Three-Phase Phase Setting	组三相相位设置	RW	U16	1	N/A	N/A	[0,3]	Y	0 : Invalid 1 : Phase R 2 : Phase S 3 : Phase T	0 : 无效 1 : R相 2 : S相 3 : T相
102	45323	Ac-side connect Test(Parallel system)	交流侧接线检测(并机)	RW	U16	1	N/A	N/A	[0,1]	Y	0 : UnTest 1 : Tested	0 : 未检测 1 : 已检测
103	45324	Meter High/Low Brush Type Settings	高刷电表模式刷新模式	RW	U16	1	1	N/A	[0,1]	Y	0: Low refresh rate, 1: High refresh rate.	0 : 低刷，1高刷
DSP BAT setting												
1	45350	Lead Battery Capacity	电池容量	RW	U16	1	1	AH	[25,2000]	Y		
2	45351	Battery Strings	电池节数	RW	U16	1	1	N/A	[4,16]	Y		

3	45352	Battery Charge Volaget Max	电池最大充电电压	RW	U16	1	10	V	[400,7200]	Y	these registers is to set the protection Parameters on Battery charge/discharge operation ON INVERTER SIDE. The real operation will still follow Battery BMSlimitations (or registers 47900~47916) if it is not out of the range. Eg. Set BattChargeCurrMax (45353) as 25A, but Battery BMSlimit the max charge Current as 20A, then the Battery charge at max 20A. but if Battery BMSlimit max charge Current as 50A, then the real charge Current of the	这些寄存器在逆变器侧 电池充电/放电运行处 设置保护参数。如果没有超出范围，实际运行 仍将遵循电池BMS限制 (或寄存器 47900~47916)。 例如：设置电池最大充 电电流(45353)为25A, 但电池BMS限制最大充 电电流为20A,电池会以 最大20A进行充电。但 如果电池BMS将最大充 电电流限制为50A，那 么电池的实际充电电流
4	45353	Battery Charge Current Max	电池最大充电电流	RW	U16	1	10	A	[0,3000]	Y		
5	45354	Battery Voltage Under Min	电池电压下限	RW	U16	1	10	V	[0,5760]	Y		
6	45355	Battery Discharge Current Max	电池最大放电电流	RW	U16	1	10	A	[0,3000]	Y		
7	45356	Battery SOC Under	电池剩余电量下限	RW	U16	1	1	%	[0,100]	Y		
8	45357	Battery Offline Voltage Under Min	电池离网电压下限	RW	U16	1	10	V	[0,5760]	Y		
9	45358	Battery Offline SOC Under Min	电池离网剩余电量下限	RW	U16	1	N/A	%	[0,100]	Y		
25	45374	Battery2 enable	电池2使能	RW	U16	1	1	N/A	[0,1]	Y		
26	45375	Lead Battery2 Capacity	电池2容量	RW	U16	1	1	AH	[25,65534]	Y		
27	45376	Battery2 Strings	电池2节数	RW	U16	1	1	N/A	[1,16]	Y		

28	45377	Battery2 Charge Voltage Max	电池2最大充电电压	RW	U16	1	10	V	[400,8500]	Y		
29	45378	Battery2 Charge Current Max	电池2最大充电电流	RW	U16	1	10	A	[0,60000]	Y		
30	45379	Battery2 Voltage Under Min	电池2电压下限	RW	U16	1	10	V	[0,5760]	Y		
31	45380	Battery2 Discharge Current Max	电池2最大放电电流	RW	U16	1	10	A	[0,60000]	Y		
32	45381	Battery2 SOC Under	电池2剩余电量下限	RW	U16	1	1	%	[0,100]	Y		
33	45382	Battery2 Offline Voltage Under Min	电池2离网电压下限	RW	U16	1	10	V	[0,5760]	Y		
34	45383	Battery2 Offline SOC Under Min	电池2离网剩余电量下限	RW	U16	1	N/A	%	[0,100]	Y		
40	45389	Separate Battery Mode	电池接入方式	RW	U16	1	1	N/A	[0,1]	Y		
41	45390	off-grid enable Switch	离网使能开关	RW	U16	1	1	N/A	[1,2]	Y	Uniq Use 2 : 45252 is invalid 1 : 45252 is valid	Uniq使用 2 : 45252不生效 1 : 45252生效
Safety Parameter Setting												
1	45400	Grid Voltage High S1	过压触发一阶值	RW	U16	1	10	V	[600,3000]	Y		
2	45401	Grid Voltage High S1 Time	过压触发一阶跳脱时间	RW	U16	1	1	periods	[1,65535]	Y		
3	45402	Grid Voltage Low S1	欠压触发一阶值	RW	U16	1	10	V	[600,3000]	Y		
4	45403	Grid Voltage Low S1 Time	欠压触发一阶跳脱时间	RW	U16	1	1	periods	[1,65535]	Y		
5	45404	Grid Voltage High S2	过压触发二阶值	RW	U16	1	10	V	[600,3000]	Y		
6	45405	Grid Voltage High S2 Time	过压触发二阶跳脱时间	RW	U16	1	1	periods	[1,65535]	Y		
7	45406	Grid Voltage Low S2	欠压触发二阶值	RW	U16	1	10	V	[600,3000]	Y		
8	45407	Grid Voltage Low S2 Time	欠压触发二阶跳脱时间	RW	U16	1	1	periods	[1,65535]	Y		
9	45408	Grid Voltage Quality	10min过压触发值	RW	U16	1	10	V	[600,3000]	Y		
10	45409	Grid Frequency High	过频触发一阶值	RW	U16	1	100	Hz	[3000,8000]	Y		
11	45410	Grid Frequency High S1 Time	过频触发一阶跳脱时间	RW	U16	1	1	periods	[1,65535]	Y		
12	45411	Grid Frequency Low	欠频触发一阶值	RW	U16	1	100	Hz	[3000,8000]	Y		
13	45412	Grid Frequency Low S1 Time	欠频触发一阶跳脱时间	RW	U16	1	1	periods	[1,65635]	Y		

14	45413	Grid Frequency High	过频触发二阶值	RW	U16	1	100	Hz	[3000,8000]	Y		
15	45414	Grid Frequency High S2 Time	过频触发二阶跳脱时间	RW	U16	1	1	periods	[1,65635]	Y		
16	45415	Grid Frequency Low	欠频触发二阶值	RW	U16	1	100	Hz	[3000,8000]	Y		
17	45416	Grid Frequency Low S2 Time	欠频触发二阶跳脱时间	RW	U16	1	1	periods	[1,65635]	Y		
18	45417	Grid Voltage High	连接电压上限	RW	U16	1	10	V	[600,3000]	Y		
19	45418	Grid Voltage Low	连接电压下限	RW	U16	1	10	V	[600,3000]	Y		
20	45419	Grid Frequency High	连接频率上限	RW	U16	1	100	Hz	[3000,8000]	Y		
21	45420	Grid Frequency Low	连接频率下限	RW	U16	1	100	Hz	[3000,8000]	Y		
22	45421	Waiting Time of On	并网等待时间	RW	U16	1	1	s	[1,1200]	Y		
23	45422	Grid Voltage Recover High of Fault Condition	故障条件连接电压上限	RW	U16	1	10	V	[600,3000]	Y		
24	45423	Grid Voltage Recover Low High of Fault Condition	故障条件连接电压下限	RW	U16	1	10	V	[600,3000]	Y		
25	45424	Grid Frequency Recover High High of Fault Condition	故障条件连接频率上限	RW	U16	1	100	Hz	[3000,8000]	Y		
26	45425	Grid Frequency Recover Low High of Fault Condition	故障条件连接频率下限	RW	U16	1	100	Hz	[3000,8000]	Y		
27	45426	Waiting Time of On Grid of Voltage Fault Condition	电压故障条件并网等待 时间	RW	U16	1	1	s	[1,1200]	Y		
28	45427	Waiting Time of On Grid of Frequency Fault Condition	频率故障条件并网等待 时间	RW	U16	1	1	s	[1,1200]	Y		
29	45428	On Grid Power Slope	并网功率斜率	RW	U16	1	1	s	[0,1200]	Y		
30	45429	On Grid Power Slope of Fault Condition	故障条件并网功率斜率	RW	U16	1	1	s	[0,1200]	Y		
31	45430	Power Decrease Slope	功率减少斜率	RW	U16	1	100	N/A	[0,1200]	Y		
32	45431	On Grid Protect Switch	并网保护开关	RW	U16	1	1	N/A	[0,1]	Y		
33	45432	On Grid Slope Switch	并网斜率开关	RW	U16	1	1	N/A	[0,1]	Y		
34	45433	Enable Curve	cos φ (P) 曲线	RW	U16	1	1	N/A	[0,1]	Y		
35	45434	A Point Power	A点功率	RW	S16	1	1	‰	[-1000, 1000]	Y		

36	45435	A Point cos φ	A点cos φ 值	RW	S16	1	1	N/A	[-100, 100]	Y		
37	45436	B Point Power	B点功率	RW	S16	1	1	‰	[-1000, 1000]	Y		
38	45437	B Point cos φ	B点cos φ 值	RW	S16	1	1	N/A	[-100, 100]	Y		
39	45438	C Point Power	C点功率	RW	S16	1	1	‰	[-1000, 1000]	Y		
40	45439	C Point cos φ	C点cos φ 值	RW	S16	1	1	N/A	[-100, 100]	Y		
41	45440	Lock In Curve Voltage	进入曲线电压	RW	U16	1	10	V	[600, 3000]	Y		
42	45441	Lock Out Curve	退出曲线电压	RW	U16	1	10	V	[600, 3000]	Y		
43	45442	Lock Out Curve Power	退出曲线功率	RW	S16	1	1	‰	[-1000, 1000]	Y		
44	45443	Over Frequency Decrease Load Curve	过频降载曲线	RW	U16	1	1	N/A	[0,1]	Y	bit0: 0:off,1:on bit1:response mode 1:fstop,0: slope	bit0: 0:关闭,1:打开 bit1:响应模式 1:fstop,0: 0:斜坡
45	45444	Ffrozen-Discharge (Frequency of Pm)	放电模式过频点	RW	U16	1	100	Hz	[3000,8000]	Y		
46	45445	Ffrozen-Charge (Frequency of Pm)	充电模式欠频点	RW	U16	1	100	Hz	[3000,8000]	Y		
47	45446	fstop-Discharge	放电模式过频终点	RW	U16	1	100	Hz	[3000,8000]	Y		
48	45447	fstop-Charge	充电模式欠频终点	RW	U16	1	100	Hz	[3000,8000]	Y		
49	45448	Over Frequency Recovery Waiting Time	过频恢复等待时间	RW	U16	1	100	s	[0,1200]	Y		
50	45449	Recovery Frequency1	恢复频率上限	RW	U16	1	100	Hz	[3000,8000]	Y		
51	45450	Recovery Frequency2	恢复频率下限	RW	U16	1	100	Hz	[3000,8000]	Y		
52	45451	Over Frequency Recovery Slope	过频恢复斜率	RW	U16	1	1	N/A	[0,1200]	Y		

53	45452	Frequency Power Curve Configuration	频率功率曲线配置寄存器	RW	U16	1	1	N/A	[0,65535]	Y		
54	45453	Over Frequency Decrease Load Slope	过频降载斜率	RW	S16	1	1	%/Hz	[-1000, 1000]	Y		
55	45454	Over Frequency Increase Load Slope	欠频加载斜率	RW	S16	1	1	%/Hz	[-1000, 1000]	Y		
56	45455	Over Frequency Recover Rate	过频恢复速率	RW	S16	1	1	%/Min	[-1000, 2000]	Y		
57	45456	QU Curve	QU曲线	RW	U16	1	1	N/A	[0,1]	Y		
58	45457	Lock In Curve Power	进入曲线功率	RW	S16	1	1	%	[-1000, 1000]	Y		
59	45458	Lock Out Curve Power	退出曲线功率	RW	S16	1	1	%	[-1000, 1000]	Y		
60	45459	V1 Voltage	V1电压值	RW	U16	1	10	V	[600, 3000]	Y		
61	45460	V1 Value (var % Rated VA)	V1无功值	RW	S16	1	1	%	[-1000, 1000]	Y		
62	45461	V2 Voltage	V2电压值	RW	U16	1	10	V	[600, 3000]	Y		
63	45462	V2 Value (var % Rated VA)	V2无功值	RW	S16	1	1	%	[-1000, 1000]	Y		
64	45463	V3 Voltage	V3电压值	RW	U16	1	10	V	[600, 3000]	Y		
65	45464	V3 Value (var % rated VA)	V3无功值	RW	S16	1	1	%	[-1000, 1000]	Y		
66	45465	V4 Voltage	V4电压值	RW	U16	1	10	V	[600, 3000]	Y		
67	45466	V4 Value (var % rated VA)	V4无功值	RW	S16	1	1	%	[-1000, 1000]	Y		
68	45467	K Value	K值	RW	U16	1	1	N/A	[0,65535]	Y		
69	45468	Filter Time Constant	滤波时间常数	RW	U16	1	1	N/A	[0,4096]	Y		
70	45469	Miscellanea	杂项	RW	U16	1	1	N/A	[0,65535]	Y		
71	45470	Rated Voltage(Korea)	额定电压(韩国)	RW	U16	1	1	N/A	[0,65535]	Y		
72	45471	Response Time(Korea)	响应时间(韩国)	RW	U16	1	1	N/A	[0,65535]	Y		
73	45472	PU Curve	PU曲线	RW	U16	1	1	N/A	[0,1]	Y		
74	45473	Power Change Rate	功率变化速率	RW	U16	1	1	s	[0,1200]	Y		
75	45474	V1 Voltage	V1电压值	RW	U16	1	10	V	[600,3000]	Y		
76	45475	V1 Value ((P/Pn)%)	V1有功值	RW	S16	1	1	%	[-1000,1000]	Y		
77	45476	V2 Voltage	V2电压值	RW	U16	1	10	V	[600,3000]	Y		
78	45477	V2 Value ((P/Pn)%)	V2有功值	RW	S16	1	1	%	[-1000,1000]	Y		
79	45478	V3 Voltage	V3电压值	RW	U16	1	10	V	[600,3000]	Y		
80	45479	V3 Value ((P/Pn)%)	V3有功值	RW	S16	1	1	%	[-1000,1000]	Y		
81	45480	V4 Voltage	V4电压值	RW	U16	1	10	V	[600,3000]	Y		
82	45481	V4 Value ((P/Pn)%)	V4有功值	RW	S16	1	1	%	[-1000,1000]	Y		

83	45482	Fixed Power Factor	固定功率因数	RW	U16	1	1	N/A	[0,20] [80,100]	Y	80=0.8Pf , 20= -0.8Pf	80=0.8Pf , 20= -0.8Pf
84	45483	Fixed Reactive Power	固定无功功率	RW	S16	1	1	‰	[-600, 600]	Y	Set the percentage of rated power of the inverter	设置逆变器额定功率百分比
85	45484	Fixed Active Power	固定有功功率	RW	U16	1	1	‰	[0, 1100]	Y		
86	45485	Grid Limit By Voltage Start Voltage	电网限制按电压启动电压	RW	U16	1	1	N/A	[0,65535]	Y		
87	45486	Grid Limit By Voltage Start Percent	按电压启动百分比的电网限制	RW	U16	1	1	N/A	[0,65535]	Y	Only for Japan	仅用于日本
88	45487	Grid Limit By Voltage Slope	按电压启动斜率的电网限制	RW	U16	1	1	N/A	[0,65535]	Y		
92	45491	All Power Curve Disable	禁用所有功率曲线	RW	U16	1	1	N/A	[0,1]	Y	this must be turned off to do Meter test . "1" means Off	进行仪表测试时必须关闭, "1"为关闭
93	45492	R Phase Fixed Active Power	R相固定有功功率	RW	U16	1	1	‰	[0,1000]	Y	If it is 1-phase inverter, then use only R phase. Unbalance output function must be turned on to set different values for R/S/T phases.	如果是单相逆变器则只用R相。必须打开不平衡输出功能, 以便为R/S/T相设置不同的值
94	45493	S Phase Fixed Active Power	S相固定有功功率	RW	U16	1	1	‰	[0,1000]	Y		
95	45494	T Phase Fixed Active Power	T相固定有功功率	RW	U16	1	1	‰	[0,1000]	Y		
96	45495	Grid Voltage High S3	过压触发三阶值	RW	U16	1	10	V	[2200,2992]	Y	Only for countries where it needs 3-stage Grid Voltage protection, Eg. Czech Republic.	仅适用于需要3级电网电压保护的国家, 如捷克共和国。
97	45496	Grid Voltage High S3 Time	过压触发三阶跳脱时间	RW	U16	1	1	periods	[1,65535]	Y		
98	45497	Grid Voltage Low S3	欠压触发三阶值	RW	U16	1	10	V	[33,220]	Y		
99	45498	Grid Voltage Low S3 Time	欠压触发三阶跳脱时间	RW	U16	1	1	periods	[1,65535]	Y		
100	45499	Zvrt Configuration	高低穿寄存器	RW	U16	1	1	N/A	[0,3]	Y	0:Disable 1:Only Lvrt 2:Only Hvrt 3: Both Lvrt&Hvrt	0:关闭 1:仅打开低穿 2:仅打开高穿 3:高低穿都打开
101	45500	Lvrt Start Voltage	低穿起点电压	RW	U16	1	10	V	[0,2300]	Y		

102	45501	Lvrt End Voltage	低穿终点电压	RW	U16	1	10	V	[0,2300]	Y		
103	45502	Lvrt Start Trip Time	低穿起点跳脱时间	RW	U16	1	1	periods	[1,65535]	Y		
104	45503	Lvrt End Trip Time	低穿终点跳脱时间	RW	U16	1	1	periods	[1,65535]	Y		
105	45504	Lvrt Trip Limit Voltage	低穿跳脱阈值	RW	U16	1	10	V	[0,2300]	Y		
106	45505	Hvrt Start Voltage	高穿起点电压	RW	U16	1	10	V	[1000,3000]	Y		
107	45506	Hvrt End Voltage	高穿终点电压	RW	U16	1	10	V	[1000,3000]	Y		
108	45507	Hvrt Start Trip Time	高穿起点跳脱时间	RW	U16	1	1	periods	[1,65535]	Y		
109	45508	Hvrt End Trip Time	高穿终点跳脱时间	RW	U16	1	1	periods	[1,65535]	Y		
110	45509	Hvrt Trip Limit Voltage	高穿跳脱阈值	RW	U16	1	10	V	[2300,3000]	Y		
111	45510	PF Time Constant	PF滤波时间常数	RW	U16	1	1	N/A	[0, 4096]	Y		
112	45511	Power Frequency Filter Time Constant	频率曲线滤波时间常数	RW	U16	1	NA	N/A	[0, 4096]	Y		
113	45512	PU Curve Filter Time Constant	PU曲线滤波时间常数	RW	U16	1	1	N/A	[0, 4096]	Y	0:Disable 1:Only Lvrt 2:Only Hvrt 3: Both Lvrt&Hvrt	0:关闭 1:仅打开低穿 2:仅打开高穿 3:高低穿都打开
114	45513	D Point Power	D点功率	RW	S16	1	1	‰	[-1000, 1000]	Y		
115	45514	D Point cosφ	D点cos φ值	RW	S16	1	1	N/A	[-100, 100]	Y		
116	45515	Under Frequency Recovery Waiting Time	欠频恢复等待时间	RW	U16	1	1	s	[0,1200]	Y		
117	45516	Under Frequency Recovery Slope	欠频恢复斜率	RW	U16	1	1	s	[0,1200]	Y		
118	45517	Under Frequency Power Rate	欠频恢复速率	RW	U16	1	1	‰/Min	[-1000, 2000]	Y		
119	45518	On Grid Charging Power Slope	并网充电功率斜率	RW	U16	1	1	s	[0,1200]	Y		
120	45519	On Grid Charging Power Slope of Fault	故障条件并网充电功率斜率	RW	U16	1	1	s	[0,1200]	Y		
121	45520	Under Frequency Stop Charging Frequency	欠频停止充电频率	RW	U16	1	100	Hz	[3000,8000]	Y		

122	45521	Over Frequency Stop Charging Frequency	过频停止放电频率	RW	U16	1	100	Hz	[3000,8000]	Y		
123	45522	Over/Under Frequency Two Step Flag	过/欠频曲线两段标志	RW	U16	1	1	N/A	[0,1]	Y		
124	45523	Frequency Extension Function Bit	频率曲线拓展功能标志位	RW	U16	1	1	N/A	[0,2]	Y	Holu special requirements: 0:No effect 1:Enable only over frequency curve	Holu特殊需求: 0:无作用 1:仅使能过频曲线 2:仅
125	45524	Protect parameter uint	安规PF设置方式	RW	U16	1	1	N/A	[0,2]	Y	1 : Setting method is -0.8~+0.8 Not 1 : Setting method is 1-20、80-100	1 : 正负0.8的设置方式 非1 : 1-20、80-100的设置方式
1	45526	PQ Curve Enable	PQ曲线使能	RW	U16	1	1	N/A	[0,1]	Y		
2	45527	Point1 Active Power	P1点有功功率	RW	S16	1	1	‰	[-1000,1000]	Y		
3	45528	Point1 Reactive Power	P1点无功功率	RW	S16	1	1	‰	[-1000,1000]	Y		
4	45529	Point2 Active Power	P2点有功功率	RW	S16	1	1	‰	[-1000,1000]	Y		
5	45530	Point2 Reactive Power	P2点无功功率	RW	S16	1	1	‰	[-1000,1000]	Y		
6	45531	Point3 Active Power	P3点有功功率	RW	S16	1	1	‰	[-1000,1000]	Y		
7	45532	Point3 Reactive Power	P3点无功功率	RW	S16	1	1	‰	[-1000,1000]	Y		
8	45533	Point4 Active Power	P4点有功功率	RW	S16	1	1	‰	[-1000,1000]	Y		
9	45534	Point4 Reactive Power	P4点无功功率	RW	S16	1	1	‰	[-1000,1000]	Y		
10	45535	Point5 Active Power	P5点有功功率	RW	S16	1	1	‰	[-1000,1000]	Y		
11	45536	Point5 Reactive Power	P5点无功功率	RW	S16	1	1	‰	[-1000,1000]	Y		
12	45537	Point6 Active Power	P6点有功功率	RW	S16	1	1	‰	[-1000,1000]	Y		

13	45538	Point6 Reactive Power	P6点无功功率	RW	S16	1	1	‰	[-1000,1000]	Y		
14	45539	Fixed Power Factor	固定PF使能标志	RW	U16	1	1	N/A	[0,1]	Y	1547-1 Communication	1547-1通讯需求
15	45540	Power Factor	固定PF功率因数	RW	U16	1	100	N/A	[0,1000]	Y		
16	45541	Fixed PF Power Factor Over\Under	固定PF功率因数超前\滞后	RW	U16	1	1	N/A	[0,1]	Y	1 : Lead ; 0 : Lag	1 : 超前 ; 0 : 滞后
17	45542	Fixed Q Power Flag	固定Q使能标志	RW	U16	1	1	N/A	[0,1]	Y		
18	45543	Fixed P Power Flag	固定P使能标志	RW	U16	1	1	N/A	[0,1]	Y		
19	45544	QU Reference Voltage Adjust Flag	QU基准电压校准标志位	RW	U16	1	1	N/A	[0,1]	Y		
20	45545	QU Reference Voltage Adjust Time	QU基准电压校准调节时间	RW	U16	1	1	s	[300,5000]	Y		
21	45546	QU Reference Voltage Adjust Value	QU基准电压校准值	RW	U16	1	10	V	[2000,2560]	Y		
22	45547	Frequency Droop Dead Beat	功率频率曲线过频死区	RW	U16	1	100	Hz	[0,200]	Y		
23	45548	Frequency Droop Dead Beat	功率频率曲线欠频死区	RW	U16	1	100	Hz	[0,200]	Y		
24	45549	Power Frequency Curve Slope of Over Frequency and	功率频率曲线过频降载斜率	RW	U16	1	100	Hz/‰	[200,500]	Y		
25	45550	Power Frequency Curve Slope of Over Frequency and	功率频率曲线过频加载斜率	RW	U16	1	100	Hz/‰	[200,500]	Y		
26	45551	Power Frequency Curve Reponse Time	功率频率曲线响应时间	RW	U16	1	100	s	[0,10]	Y		

27	45552	FCAS Ready Enable	FCAS准备使能	RW	U16	1	1	N/A	[0,1]	Y	FCAS Function Requirements	FCAS功能需求
28	45553	Under Frequency Start	欠频起始点	RW	U16	1	100	Hz	[3000,8000]	Y		
29	45554	Over Frequency Start	过频起始点	RW	U16	1	100	Hz	[3000,8000]	Y		
30	45555	Under Frequency End	欠频终点	RW	U16	1	100	Hz	[3000,8000]	Y		
31	45556	Over Frequency End	过频终点	RW	U16	1	100	Hz	[3000,8000]	Y		
32	45557	Frequency Response Resolution Ratio	频率响应分辨率	RW	U16	1	100	Hz	[1,1000]	Y		
33	45558	Min Discharge SOC	放电最小剩余电量	RW	U16	1	1	%	[0 , 100]	Y		
34	45559	Max Charge SOC	充电最大剩余电量	RW	U16	1	1	%	[0 , 100]	Y		
35	45560	Discharge Duration	放电持续时间	RW	U16	1	1	sec	[0,65535]	Y		
36	45561	Discharge Max Power Delta	放电最大功率偏移量	RW	U16	1	1	W	0-4600	Y		
37	45562	Charge Duration	充电持续时间	RW	U16	1	1	sec	[0,65535]	Y		
38	45563	Charge Max Power	充电最大功率偏移量	RW	U16	1	1	W	0-4600	Y		
39	45564	Battery Power Charge/Discharge Enable	充放电使能	RW	U16	1	1	N/A	[0,1]	Y		
40	45565	Battery Charge Limit	电池充电功率上限	RW	U16	1	1	W	0-4600	Y		
41	45566	Battery Discharge Limit	电池放电功率上限	RW	U16	1	1	W	0-4600	Y		
42	45567	Inverter AC Input/Output Limit	逆变器买卖电上限	RW	U16	1	1	W	0-4600	Y		

43	45568	Fixed Q Filter Time Constant	定无功响应滤波时间常数	RW	U16	1	10	s	[0,60000]	Y		
44	45569	uwFrtK_Value	中压低穿无功K值	RW	U16	1	N/A	N/A	0-10	Y		
45	45570	Fixed Q Over\Under	固定无功超前\滞后	RW	U16	1	1	N/A	[0,1]	Y	1 : Lead ; 0 : Lag	1 : 超前 ; 0 : 滞后
46	45571	QU curve operation mode	QU曲线模式	RW	U16	1	1	N/A	[0,1]	Y	0 : Basic Mode 1 : Slope Mode	0 : 基础模式 ; 1 : 斜率模式
47	45572	QU curve Slope1	V1点对应斜率	RW	S16	1	10	%Qmax/1% Vn	[-2000,2000]	Y		
48	45573	QU curve Slope2	V4点对应斜率	RW	S16	1	10	%Qmax/1% Vn	[-2000,2000]	Y		
49	45574	QU curve Volt Dead	电压死区宽度	RW	U16	1	10	%Vn	[0,100]	Y		
52	45577	AS477 Parameters	安规状态标志位	RW	U16	1	1	N/A	[0,65535]	Y	Japanese safety Parameters, corresponding to the old version 0x056D (1:on 0:off) Bit14:Effective power Bit13:Reverse power prevention Bit12:OVR detection function Bit11:EMS Loss Check Bit10:Enable manual recovery Bit9:Reactive power oscillation suppression Bit8:Voltage imbalance detection Bit7:FRT. Bit6:Passive islanding Bit5:Active anti-islanding Bit4:Step reactive power injection Bit3:Frequency feedback reactive power injection Bit2:Reserved Bit1:Reserved Bit0:Invalid power	日本安规参数，对应旧 版0x056D (1:on 0:off) Bit14:有效电力 Bit13: 逆电力防止 Bit12:OVR检出机能 Bit11:EMS Loss Check Bit10:开启手动恢复 Bit9:无功震荡抑制 Bit8:电压不平衡检测 Bit7:FRT ; Bit6:被动孤岛 Bit5:主 动式反孤岛 Bit4:步进无功注入 Bit3:频率反馈无功注入 Bit2:保留 Bit1:保留 Bit0:无效电力
53	45578	Voltage Rise Suppression V1	电压上升抑制V1	RW	U16	1	10	N/A	[0,65535]	Y	Japanese safety Parameters, corresponding to the old version 0x0591	日本安规参数，对应旧 版0x0591
54	45579	Voltage Rise Suppression Limit Power	电压上升抑制功率	RW	U16	1	1	N/A	[0,65535]	Y	Japanese safety Parameters, corresponding to the old version 0x0592	日本安规参数，对应旧 版0x0592

55	45580	Anti-Island(Passive) Angle	被动孤岛检测阈值	RW	U16	1	1	N/A	[0,65535]	Y	Japanese safety Parameters, corresponding to the old version 0x0593	日本安规参数，对应旧 版0x0593
56	45581	Volt Rise Suppression V2	电压上升抑制V2	RW	U16	1	10	N/A	[0,65535]	Y	Japanese safety Parameters, corresponding to the old version 0x0594	日本安规参数，对应旧 版0x0594
57	45582	Anti-Island Status	主动孤岛状态设置	RW	U16	1	1	N/A	[0,65535]	Y	Japanese safety Parameters, corresponding to the old version 0x7150 0:regular 1:standby	日本安规参数，对应旧 版0x7150 0:常规 1:备用
58	45583	On Grid Waiting Manual Recover Set	并网等待手动恢复	RW	U16	1	1	N/A	[0,65535]	Y	Japanese safety Parameters, corresponding to the old version 0x7151 Bit0:Recovery manually	日本安规参数，对应旧 版0x7151 Bit0:手动恢复
59	45584	Frequency Feedback Inject Efficiency	频率反馈注入效率	RW	U16	1	1	N/A	[0,65535]	Y	Japanese safety Parameters, corresponding to the old version 0x7152 0.25p.u/0.001Hz	日本安规参数，对应旧 版0x7152 0.25p.u/0.001Hz
60	45585	Action Signal Pin Setting	动作信号引脚设置	RW	U16	1	1	N/A	[0,65535]	Y	Japanese safety Parameters, corresponding to the old version 0x7153 0:MODE0 1:MODE1 2:MODE2	日本安规参数，对应旧 版0x7153 0:模式0 1:模式1 2:模 式2
61	45586	Output Control Time Set	出力制御时间设置	RW	S16	1	1	min	[0,10]	Y	Japanese safety Parameters 0:Disable 5-10:5min-10min	日本安规参数 0:取消线性增加/减少功 率 5-10 : 5min-10min
62	45587	Output Control Power Set	出力制御功率设置	RW	S16	1	1	‰	[0,1000]	Y	Japanese safety Parameters	日本安规参数
63	45588	Number of Standby Transfer Set	待机转移次数设置	RW	U16	1	1	N/A	[0,65535]	Y	Japanese safety Parameters	日本安规参数
64	45589	Recovery Cycle Setting When the Number of Standby Transfer is Less Than 2	待机转移次数小于等于 2时的回复周期设置	RW	U16	1	1	periods	[0,65535]	Y	Japanese safety Parameters	日本安规参数
65	45590	Recovery Cycle Setting When the Number of Standby Transfer is Less Than 3	待机转移次数等于3时 的回复周期设置	RW	U16	1	1	periods	[0,65535]	Y	Japanese safety Parameters	日本安规参数

66	45591	Lost Communication Flag	通信丢失标志位	RW	U16	1	1	N/A	[0,65535]	Y	Japanese safety Parameters	日本安规参数
67	45592	Frequency Feedback Reactive Power Infection Standard Unit	频率反馈无功注入标么值	RW	U16	1	1	0.0001p.u	[0,2500]	Y	Japanese safety Parameters	日本安规参数
68	45593	Upper limit of the generator operating voltage range	发电机工作电压范围上限	RW	U16	1	1	V	[80, 300]	Y	default 280v	默认280v
69	45594	Lower limit of the generator operating voltage range	发电机工作电压范围下限	RW	U16	1	1	V	[80, 300]	Y	default 180v	默认180v
70	45595	Upper limit of the generator operating frequency range	发电机工作频率范围上限	RW	U16	1	100	HZ	[4000, 6000]	Y	default 550	默认5500
71	45596	Lower limit of the generator operating frequency range	发电机工作频率范围下限	RW	U16	1	100	HZ	[4000, 6000]	Y	default 450	默认4500
72	45597	Delay before load	投载前延时间	RW	U16	1	1	S	[10, 300]	Y	default 10s	默认10s
73	45598	Generator safety trip time	安规跳脱时间	RW	U16	1	1	S	[1,65535]	Y	Generator's Safety Parameter	发电机安规参数
74	45599	reserved	预留			1						
75	45600	uwPFUCurveFlag	PFU曲线使能	RW	U16	1	1	N/A	[0 1]	Y	default 0	默认0
76	45601	uwPFUV1Volt	V1点电压	RW	U16	1	10	V	[600 3000]	Y	default 2162	默认2162
77	45602	wPFUCurvePointAPf	A点功率因数	RW	S16	1	100	N/A	[-100 100]	Y	default 90	默认90
78	45603	uwPFUV2Volt	V2点电压	RW	U16	1	10	V	[600 3000]	Y	default 2231	默认2231
79	45604	wPFUCurvePointBPf	B点功率因数	RW	S16	1	100	N/A	[-100 100]	Y	default 100	默认100
80	45605	uwPFUV3Volt	V3点电压	RW	U16	1	10	V	[600 3000]	Y	default 2415	默认2415
81	45606	wPFUCurvePointCPf	C点功率因数	RW	S16	1	100	N/A	[-100 100]	Y	default 100	默认100
82	45607	uwPFUV4Volt	V4点电压	RW	U16	1	10	V	[600 3000]	Y	default 2484	默认2484
83	45608	wPFUCurvePointDPf	D点功率因数	RW	S16	1	100	N/A	[-100 100]	Y	default -90	默认-90
84	45609	wPFUTimeConstant	PFU曲线离散系数	RW	U16	1	10	s	[0,60000]	Y	default 0	默认0
85	45610	wPFULockInPower	进入曲线有功功率	RW	S16	1	10	%Pn	[-1000 1000]	Y	default 200	默认200
86	45611	wPFULockOutPower	退出曲线有功功率	RW	S16	1	10	%Pn	[-1000 1000]	Y	default 100	默认100
87	45612	Output power control setting	出力制御功率设置	RW	U16	1	1	W	[0,65533]	Y	Japanese safety Parameters	日本安规参数
88	45613	FIT contract type	FIT 模式 类型	RW	U16	1	1	N/A	[0,1]	Y	Japanese safety Parameters	日本安规参数
89	45614	Conversion coefficient	转换系数	RW	U16	1	1	N/A	[0,100]	Y	Japanese safety Parameters	日本安规参数

90	45615	Output power restraint status	出力制御状态	RW	U16	1	1	N/A	[65,69]	Y	Japanese safety Parameters	日本安规参数
91	45616	System interconnection status	系统互联状态	RW	U16	1	1	N/A	[0,3]	N	Japanese safety Parameters	日本安规参数
92	45617	reserved				1						
93	45618	reserved				1						
94	45619	reserved				1						
95	45620	cos φ P Mode	cos φ P曲线模式	RW	U16	1	1	N/A	[0,65535]	Y	0 : Basic Mode 1 : Slope Mode	0 : 基础模式 1 : 斜率模式
96	45621	Lock In Curve LowVoltage	容性进入曲线电压	RW	U16	1	10	V	[600, 3000]	Y	The specific logic of Italian safety regulations, the requirements for reactive power generation at low voltage according to the cos φ (P) curve	意大利安规特有的逻辑, cos φ (P)曲线在低压的发无功的要求
97	45622	QP Curve Mode	QP功能模式	RW	U16	1	1	N/A	[0,65535]	Y	0 : Basic Mode 1 : Slope Mode	0 : 基础模式 1 : 斜率模式
98	45623	QP Curve Slope1	QP曲线P1对应斜率	RW	S16	1	10	Qmax/10 %Pn	[-2000,2000]	Y		
99	45624	QP Curve Slope2	QP曲线P4对应斜率	RW	S16	1	10	Qmax/10 %Pn	[-2000,2000]	Y		
100	45625	Int QP Curve Grid Volt	QP曲线检入电压	RW	U16	1	10	%Vn	[500,1500]	Y		
101	45626	Out QP Curve Grid	QP曲线检出电压	RW	U16	1	10	%Vn	[500,1500]	Y		
102	45627	QP Enter the curve condition enabling	QP曲线进入条件使能开关	RW	U16	1	1	N/A	[0,1]	Y	0 : OFF 1 : ON	0 : 关闭 1 : 打开
103	45628	QP Curve Rective Power Refer	QP无功功率参考值	RW	U16	1	1	N/A	[0,4]	Y	0 : Pn Rated Power 1 : Ps Apparent Power 2 : Po Current Power 3 : Pm Max Power 4 : Qm Max Reactive Power	0 : Pn 额定功率 1 : Ps 视在功率 2 : Po 当前功率 3 : Pm 最大功率 4 : Qm 最大无功
104	45629	SoftWareVersionbySoft yCountry	软件版本号	RW	U32	2	1	N/A	[0,0xFFFFFFFF]	Y	From high bit to Low bit : Master DSP、Slave DSP、ARM、Reserved	由高位到低位 : 主DSP版本、副DSP版本、arm版本、预留
113	45638	DCI parameter type setting	DCI参数类型	RW	U16	1	1	N/A	[0,10]		0: Absolute value of DCI 1: Percentage of DCI Others: Absolute value &	0 : DCI绝对值 1 : DCI百分比 其他 : 绝对值&百分比
114	45639	DCI first order tripping value	DCI一阶限幅	RW	U16	1	1	mA	[0,5000]			

115	45640	DCI first order tripping value	DCI一阶百分比(0.1%)	RW	U16	1	10	%In	[0,1000]			
116	45641	DCI first order tripping time	DCI一阶限幅保护时间	RW	U16	1	1	ms	[0,60000]			
117	45642	DCI second order tripping value	DCI二阶限幅	RW	U16	1	1	mA	[0,5000]			
118	45643	DCI second order tripping value	DCI二阶百分比(0.1%)	RW	U16	1	10	%In	[0,1000]			
119	45644	DCI second order tripping time	DCI二阶限幅保护时间	RW	U16	1	1	ms	[0,60000]			
120	45645	DCI check Enable	DCI 检测使能开关	RW	U16	1	1	N/A	[0,1]	Y		
121	45646	DCI check filter	DCI检测滤波开关	RW	U16	1	1	N/A	[0,1]	Y		
122	45647	Grid Frequency High	过频触发三阶值	RW	U16	1	100	Hz	[3000,8000]	Y		
123	45648	Grid Frequency Low	欠频触发三阶值	RW	U16	1	100	Hz	[3000,8000]	Y		
124	45649	Grid Frequency High	过频触发四阶值	RW	U16	1	100	Hz	[3000,8000]	Y		
125	45650	Grid Frequency Low	欠频触发四阶值	RW	U16	1	100	Hz	[3000,8000]	Y		
1	45651	VFmsEnFlag	安规时间支持ms设置标志	RO	U16	1	1	N/A	[0,1]	Y		
2	45652	Undervoltage trigger first order value(0.1%)	欠压触发一阶值(0.1%)	RW	U16	1	10	%Vn	[150,1000]	Y		
3	45653	Undervoltage trigger first-order trip time	欠压触发一阶跳脱时间	RW	U32	2	1	ms	[10,7200000]	Y		
5	45655	Overvoltage trigger first order value(0.1%)	过压触发一阶值(0.1%)	RW	U16	1	10	%Vn	[800,1400]	Y		
6	45656	Overvoltage trigger first-order trip time	过压触发一阶跳脱时间	RW	U32	2	1	ms	[10,7200000]	Y		
8	45658	Undervoltage trigger second order value(0.1%)	欠压触发二阶值(0.1%)	RW	U16	1	10	%Vn	[150,1000]	Y		
9	45659	Undervoltage trigger second-order trip time	欠压触发二阶跳脱时间	RW	U32	2	1	ms	[10,7200000]	Y		
11	45661	Overvoltage trigger second order value(0.1%)	过压触发二阶值(0.1%)	RW	U16	1	10	%Vn	[800,1400]	Y		

12	45662	Overvoltage trigger second-order trip time	过压触发二阶跳脱时间	RW	U32	2	1	ms	[10,7200000]	Y		
14	45664	Undervoltage trigger third order value(0.1%)	欠压触发三阶值(0.1%)	RW	U16	1	10	%Vn	[150,1000]	Y	0(Default): Not being used If 0, data will not be displayed.	不使用默认值为0 ; APP读取如果为0,则不
15	45665	Undervoltage trigger third-order trip time	欠压触发三阶跳脱时间	RW	U32	2	1	ms	[10,7200000]	Y		
17	45667	Overvoltage trigger third order value(0.1%)	过压触发三阶值(0.1%)	RW	U16	1	10	%Vn	[800,1400]	Y		
18	45668	Overvoltage trigger third-order trip time	过压触发三阶跳脱时间	RW	U32	2	1	ms	[10,7200000]	Y		
20	45670	Undervoltage trigger forth order value(0.1%)	欠压触发四阶值(0.1%)	RW	U16	1	10	%Vn	[150,1000]	Y		
21	45671	Undervoltage trigger forth-order trip time	欠压触发四阶跳脱时间	RW	U32	2	1	ms	[10,7200000]	Y		
23	45673	Overvoltage trigger forth order value(0.1%)	过压触发四阶值(0.1%)	RW	U16	1	10	%Vn	[800,1400]	Y		
24	45674	Overvoltage trigger forth-order trip time	过压触发四阶跳脱时间	RW	U32	2	1	ms	[10,7200000]	Y		
26	45676	Underfrequency trigger first-order trip time	欠频触发一阶跳脱时间	RW	U32	2	1	ms	[10,7200000]	Y		
28	45678	Overfrequency trigger first-order trip time	过频触发一阶跳脱时间	RW	U32	2	1	ms	[10,7200000]	Y		
30	45680	Underfrequency trigger second-order trip time	欠频触发二阶跳脱时间	RW	U32	2	1	ms	[10,7200000]	Y		
32	45682	Overfrequency trigger second-order trip time	过频触发二阶跳脱时间	RW	U32	2	1	ms	[10,7200000]	Y		
34	45684	Underfrequency trigger third-order trip time	欠频触发三阶跳脱时间	RW	U32	2	1	ms	[10,7200000]	Y		

36	45686	Overfrequency trigger third-order trip time	过频触发三阶跳脱时间	RW	U32	2	1	ms	[10,7200000]	Y		
38	45688	Underfrequency trigger forth-order trip time	欠频触发四阶跳脱时间	RW	U32	2	1	ms	[10,7200000]	Y		
40	45690	Overfrequency trigger forth-order trip time	过频触发四阶跳脱时间	RW	U32	2	1	ms	[10,7200000]	Y		
42	45692	10min overvoltage trigger value (0.1%)	10min过压触发值 (0.1%)	RW	U16	1	10	%Vn	[800,1400]	Y		
43	45693	10min trigger trip time	10min过压跳脱时间	RW	U32	2	1	ms	[10,7200000]	Y		
45	45695	QU Reference Voltage Regulation Time	QU曲线响应时间参数	RW	U16	1	10	s	[0,600000]	Y		
46	45696	PU Curve OutPut Regulation Time	PU曲线响应时间参数	RW	U16	1	10	s	[0,600000]	Y		
47	45697	cos ϕ (P) curve response time	cos ϕ (P)曲线响应时间参数	RW	U16	1	10	s	[0,600000]	Y		
48	45698	PQ Reactive power variation filter	PQ曲线响应时间参数	RW	U16	1	10	s	[0,600000]	Y		
49	45699	Power Frequency Filter Time Constant	频率曲线滤波时间常数	RW	U16	1	10	s	[0,600000]	Y		

QU Curve												
50	45700	QU curve ref	QU曲线功率基准	RW	U16	1	1	N/A	[0,3]	Y	0 : Pn Rated Power 1 : Ps Apparent Power 2 : Po Current Power 3 : Pm Max Power	0 : Pn 额定功率 1 : Ps 视在功率 2 : Po 当前功率 3 : Pm 最大功率
51	45701	QU Voltage1(0.1%)	V1电压值(0.1%)	RW	U16	1	10	%Vn	[0,1400]	Y		
52	45702	QU Voltage2(0.1%)	V2电压值(0.1%)	RW	U16	1	10	%Vn	[0,1400]	Y		
53	45703	QU Voltage3(0.1%)	V3电压值(0.1%)	RW	U16	1	10	%Vn	[0,1400]	Y		
54	45704	QU Voltage4(0.1%)	V4电压值(0.1%)	RW	U16	1	10	%Vn	[0,1400]	Y		
55	45705	QU reference voltage value	QU基准电压值	RW	U16	1	10	%Vn	[800,1200]	Y		
56	45706	QU Reference Voltage Autoscale Enable	QU基准电压自动调节使能	RW	U16	1	1	N/A	[0,1]	Y		
57	45707	QU curve response slope	QU曲线响应无功变化斜率	RW	U16	1	10	%Pn/s	[0,65000]	Y		
58	45708	QU curve output regulation mode	QU曲线输出响应方式	RW	U16	1	1	N/A	[0,2]	Y	0 : OFF 1 : Slope Response 2 : Low-pass filter Response	0 : 关闭 1 : 斜率响应 2 : 低通滤波模式响应
59	45709	QU Enter the curve condition enabling	QU曲线进入条件使能开关	RW	U16	1	1	N/A	[0,1]	Y	0 : OFF 1 : ON	0 : 关闭 1 : 打开
60	45710	QU min PF	QU曲线最小PF	RW	S16	1	1000	N/A	[-1000,1000]	Y		
61	45711	Voltage1(0.1%)	电压不灵敏度值(0.1%)	RW	U16	1	10	%Vn	[0,100]	Y		
62	45712	QU curve Slope1	V1点对应斜率	RW	U16	1	10	%Qmax/1% Vn	[0,2000]	Y		
63	45713	QU curve Slope2	V4点对应斜率	RW	U16	1	10	%Qmax/1% Vn	[0,2000]	Y		

Others												
64	45714	Normal condition upper limit of connection voltage(0.1%)	启机并网电压下限	RW	U16	1	10	%Vn	[150,1000]	Y		
65	45715	Normal condition lowerer limit of connection	启机并网电压上限	RW	U16	1	10	%Vn	[800,1400]	Y		
66	45716	Fault condition upper limit of connection	重连并网电压下限	RW	U16	1	10	%Vn	[150,1000]	Y		
67	45717	Fault condition lowerer limit of connection voltage	重连并网电压上限	RW	U16	1	10	%Vn	[800,1400]	Y		
PU Curve												
70	45720	PU Voltage1(0.1%)	V1电压值(0.1%)	RW	U16	1	10	%Vn	[0,1400]	Y		
71	45721	PU Voltage2(0.1%)	V2电压值(0.1%)	RW	U16	1	10	%Vn	[0,1400]	Y		
72	45722	PU Voltage3(0.1%)	V3电压值(0.1%)	RW	U16	1	10	%Vn	[0,1400]	Y		
73	45723	PU Voltage4(0.1%)	V4电压值(0.1%)	RW	U16	1	10	%Vn	[0,1400]	Y		
74	45724	PU Curve OutPut Regulation Mode	PU曲线输出响应模式	RW	U16	1	1	N/A	[0,2]	Y	0 : OFF 1 : Slope Response 2 : Low-pass filter Response	0 : 关闭 1 : 斜率响应 2 : 低通滤波模式响应
75	45725	Slope of PU Curve power slope	PU曲线输出功率变化速率	RW	U16	1	10	%Pn/s	[0,65000]	Y		
76	45726	PU Curve operation mode	PU曲线模式	RW	U16	1	1	N/A	[0,65535]	Y		
77	45727	PU_Pref	PU曲线功率基准	RW	U16	1	1	N/A	[0,3]		0 : Pn 额定功率 1 : Ps 视在功率 2 : Po 当前功率 3 : Pm 最大功率	
78	45728	start response time delay	PU曲线开始响应延时时间	RW	U16	1	10	s	[0,10000]	Y		

cosφ P Curve												
83	45733	voltage ofEnter the curve (0.1%)	进入曲线电压(0.1%)	RW	U16	1	10	%Vn	[0,1400]	Y		
84	45734	voltage of quit the curve(0.1%)	退出曲线电压(0.1%)	RW	U16	1	10	%Vn	[0,1400]	Y		
85	45735	cos φ(P) curve response Slope	cosφ(P)曲线输出响应斜率	RW	U16	1	10	%Pn/s	[0,65000]	Y		
86	45736	cos φ(P) curve response Model	cosφ(P)曲线响应时间方式	RW	U16	1	1	N/A	[0,2]	Y	0 : OFF 1 : Slope Response 2 : Low-pass filter Response	0 : 关闭 1 : 斜率响应 2 : 低通滤波模式响应
87	45737	cos φ A(pf,0.001)	A点cos φ值(pf,0.001)	RW	S16	1	1000	N/A	[-1000,-800] [800,1000]	Y		
88	45738	cos φ B(pf,0.001)	B点cos φ值(pf,0.001)	RW	S16	1	1000	N/A	[-1000,-800] [800,1000]	Y		
89	45739	cos φ C(pf,0.001)	C点cos φ值(pf,0.001)	RW	S16	1	1000	N/A	[-1000,-800] [800,1000]	Y		
90	45740	cos φ D(pf,0.001)	D点cos φ值(pf,0.001)	RW	S16	1	1000	N/A	[-1000,-800] [800,1000]	Y		
91	45741	Power E(0.1%)	E点功率(0.1%)	RW	S16	1	10	%Pn	[0,1100]	Y		
92	45742	cos φ E(pf,0.001)	E点cos φ值(pf,0.001)	RW	S16	1	1000	N/A	[-1000,-800] [800,1000]	Y		
93	45743	cos φ(P) Enter the curve condition	cos φ P曲线进入条件使能开关	RW	U16	1	1	N/A	[0,1]	Y		
94	45744	cos φ(P) curve P1 Slope	P1对应斜率	RW	U16	1	10	Qmax/10 %Pn	[0,2000]	Y		
95	45745	cos φ(P) curve P4 Slope	P4对应斜率	RW	U16	1	10	Qmax/10 %Pn	[0,2000]	Y		
Frequency Curve												
101	45751	P(F) Curve Eable	FP曲线过频使能位	RW	U16	1	1	N/A	[0,1]		0 : OFF 1 : ON	0 : 关闭 1 : 打开
102	45752	P(F) Curve Mode	FP曲线过频模式	RO	U16	1	1	N/A	[0,2]	Y	0 : slop mode 1 : stop mode 2 : Stop_V mode(AU)	0 : slop模式 1 : stop模式 2 : Stop_V模式 (储能澳洲模式)
103	45753	P(F) Curve Transition Mode	FP曲线过频过渡模式	RW	U16	1	1	N/A	[0,1]	Y	0 : OFF 1 : ON	0 : 关闭 1 : 打开
104	45754	silence time	进入过频曲线静默时间	RW	U16	1	10	s	[0,10000]	Y		

105	45755	OF dereating power benchmark	过频降载功率基准	RW	U16	1	1	N/A	[0,3]	Y	0 : Pn Rated Power 1 : Ps Apparent Power 2 : Po Current Power 3 : Pm Max Power	0 : Pn 额定功率 1 : Ps 视在功率 2 : Po 当前功率 3 : Pm 最大功率
106	45756	P(F) power slope (Slope)	过频曲线斜率	RW	U16	1	10	%Px/Hz	[0,10000]			
107	45757	Hysteresis eable	过频降载滞回开关	RW	U16	1	1	N/A	[0,1]	Y	0 : OFF 1 : ON	0 : 关闭 1 : 打开
108	45758	Hysteresis frequency Point	过频降载滞回频率	RW	U16	1	100	Hz	[5000,6500]	Y		
109	45759	Deactivation observation time	停用观察时间	RW	U16	1	10	s	[0,10000]	Y		
110	45760	Power response mode	功率响应方式	RW	U16	1	1	N/A	[0,2]	Y	0 : OFF 1 : Slope Response 2 : Low-pass filter Response	0 : 关闭 1 : 斜率响应 2 : 低通滤波模式响应
111	45761	Recover power slope	过频曲线功率恢复速率	RW	U16	1	10	%Pn/min	[0,65000]	Y		
112	45762	Into the power point	进入功率点	RW	U16	1	10	%Pn	[0,1100]	Y		
113	45763	Upper Threshold frequency	上限阈值频率	RW	U16	1	100	Hz	[5000,6500]	Y		
114	45764	Output power lower	输出功率下限	RW	U16	1	10	%Pn	[0,1100]	Y		
115	45765	uwOFRecoverPwrRef	过频降载恢复功率参考	RW	U16	1	1	N/A	[0,4]	Y	0 : Pn Rated Power 1 : Ps Apparent Power 2 : Po Current Power 3 : Pm Max Power 4: Pd Power Difference	0 : Pn 额定功率 1 : Ps 视在功率 2 : Po 当前功率 3 : Pm 最大功率 4 : Pd 功率差值
116	45766	Response slope	过频降载响应斜率	RW	U16	1	10	%Pn/min	[0,65000]	Y		
117	45767	Overfrequency curve Charge Enable	过频降载允许充电使能	RW	U16	1	1	N/A	[0,1]	Y		
118	45768	reserved				1						

119	45769	Overfrequency P _{cm} ax flag	过频降载终点P _{cm} ax标志	RW	U16	1	1	N/A	[0,1]	Y		
120	45770	Frequency-Watt curve overfrequency dead zone	功率频率曲线过频死区	RW	U16	1	100	Hz	[0,200]	Y		
121	45771	Over frequency quit Point	过频退出点	RW	U16	1	100	Hz	[5000,6500]	Y		
122	45772	Power response filtering time	功率响应滤波时间	RW	U32	2	1	ms	[0,36000000]	Y		
1	45776	Under frequency Point	FP曲线欠频使能位	RW	U16	1	1	N/A	[0,1]	Y	0 : OFF 1 : ON	0 : 关闭 1 : 打开
2	45777	FP UF curve transition mode	FP曲线欠频模式	RO	U16	1	1	N/A	[0,2]	Y	0 : slop mode 1 : stop mode 2 : Stop_V mode(AU)	0 : slop模式 1 : stop模式 2 : Stop_V模式 (储能澳洲模式)
3	45778	silence time	进入欠频加载静默时间	RW	U16	1	10	s	[0,10000]	Y		
4	45779	Power reference (Slope)	欠频加载功率基准	RW	U16	1	1	N/A	[0,4]	Y	0 : P _n Rated Power 1 : P _s Apparent Power 2 : P _o Current Power 3 : P _m Max Power 4 : P _d Power Difference	0 : P _n 额定功率 1 : P _s 视在功率 2 : P _o 当前功率 3 : P _m 最大功率 4 : P _d 功率差值
5	45780	P(F) power slope (Slope)	欠频功率斜率	RW	U16	1	10	%P _x /Hz	[0,10000]	Y		
6	45781	Hysteresis eable	欠频加载滞回开关	RW	U16	1	1	N/A	[0,1]	Y		
7	45782	Hysteresis frequency Point	欠频加载滞回频率	RW	U16	1	100	Hz	[4500,6000]	Y		
8	45783	Deactivation observation time	退出欠频曲线静默时间	RW	U16	1	10	s	[0,10000]	Y		
9	45784	Power response mode	功率响应方式	RW	U16	1	1	N/A	[0,2]	Y	0 : OFF 1 : Slope Response 2 : Low-pass filter Response	0 : 关闭 1 : 斜率响应 2 : 低通滤波模式响应
10	45785	Recover power slope	恢复功率斜率	RW	U16	1	10	%P _n /min	[0,65000]	Y		
11	45786	Into the power point	进入功率点	RW	U16	1	10	%P _n	[0,1100]	Y		

12	45787	Lower Threshold frequency	下限阈值频率	RW	U16	1	100	Hz	[4500,6000]	Y		
13	45788	Output power upper	输出功率上限	RW	U16	1	10	%Pn	[0,1100]	Y		
14	45789	Recover power reference	恢复功率参考	RW	U16	1	1	N/A	[0,4]	Y	0 : Pn Rated Power 1 : Ps Apparent Power 2 : Po Current Power 3 : Pm Max Power 4 : Pd Power Difference	0 : Pn 额定功率 1 : Ps 视在功率 2 : Po 当前功率 3 : Pm 最大功率 4 : Pd 功率差值
15	45790	Response slope	响应斜率	RW	U16	1	10	%Pn/min	[0,65000]	Y		
16	45791	UF curve Charge Enable	欠频加载允许放电使能	RW	U16	1	1	N/A	[0,1]	Y		
17	45792	reserved				1						
18	45793	UF Psmax flag	欠频加载终点Psmax标志	RW	U16	1	1	N/A	[0,1]	Y		
19	45794	Frequency-Watt curve underfrequency dead zone	功率频率曲线欠频死区	RW	U16	1	100	Hz	[0,200]	Y		
20	45795	Under frequency quit Point	欠频退出点	RW	U16	1	100	Hz	[4500,6000]	Y		
21	45796	Power response filtering time	功率响应滤波时间	RW	U32	2	1	ms	[0,36000000]	Y		
25	45800	Generate enable	起机使能位	RW	U16	1	1	N/A	[0,1]	Y		
26	45801	Reconnect enable	重连使能位	RW	U16	1	1	N/A	[0,1]	Y		
27	45802	Decrease enable	降载使能位	RW	U16	1	1	N/A	[0,1]	Y		
28	45803	ACT power change	功率调度响应斜率	RW	U32	2	1	%Pn/min	[0,FFFFFFFF]	Y		
QP Curve												
38	45813	QP Reactive power variation filter	QP曲线输出响应斜率	RW	U16	1	10	%Pn/s	[0,65000]	Y		
39	45814	QP Reactive output Model	QP曲线输出响应方式	RW	U16	1	1	N/A	[0,2]	Y	0 : OFF 1 : Slope Response 2 : Low Filter Mode Response	0 : 关闭 1 : 斜率响应 2 : 低通滤波模式响应
42	45817	Slop of frequency shift	不跳脱对应频率变化率	RW	U16	1	100	Hz/s	[0,1000]	Y		
43	45818	Slop of frequency shift	跳脱对应频率变化率	RW	U16	1	100	Hz/s	[0,1000]	Y		
44	45819	Frequency shift detection time	频移检测时间宽度	RW	U16	1	1	ms	[10,10000]	Y		
45	45820	Frequency shift protect time	频移保护时间	RW	U16	1	1	ms	[10,65000]	Y		

HVRT&HVRT												
46	45821	Fault ride-through processing volume mode selection	故障穿越加工量模式选择	RW	U16	1	1	N/A	[0,2]	Y	0: Fundamental positive sequence component Ud 1: Phase voltage reconstruction 2: Line voltage reconstruction	0 : 基波正序分量Ud 1: 相电压重构 2 : 线电压重构
47	45822	Fault ride through variation machining mode enable	Δ高低穿模式使能	RW	U16	1	1	N/A	[0,1]	Y	0 : OFF 1 : ON	0 : 关闭 1 : 打开
48	45823	Variation Threshold	Δ高低穿判断阈值	RW	U16	1	10	%Vn	[0,1000]	Y		
49	45824	Reactive Current Injection Mode	无功电流注入模式	RW	U16	1	1	N/A	[0,1]	Y	0: Absolute injection 1: relative injection	0 : 绝对注入 1 : 相对注入
50	45825	Current distribution mode	电流分配模式	RW	U16	1	1	N/A	[0,2]	Y	0: Reactive power priority 1: Active power priority 2: Constant current mode	0 : 无功优先 1 : 有功优先 2: 恒电流模式
51	45826	Ride through end active power recover Model	穿越结束有功恢复模式	RW	U16	1	1	N/A	[0,2]	Y	0 : OFF 1 : Slope Response 2 : Low Filter Mode Response	0 : 关闭 1 : 斜率响应 2 : 低通滤波模式响应
52	45827	Ride through end active power recover	穿越结束有功恢复速率	RW	U16	1	10	%In/s	[0,65000]	Y		
53	45828	Ride through end reactive power recover Model	穿越结束无功恢复模式	RW	U16	1	1	N/A	[0,2]	Y	0 : OFF 1 : Slope Response 2 : Low Filter Mode Response	0 : 关闭 1 : 斜率响应 2 : 低通滤波模式响应
54	45829	Ride through end reactive power recover speed	穿越结束无功恢复速率	RW	U16	1	10	%In/s	[0,65000]	Y		
55	45830	VRT grid voltage protection shield	VRT时电网电压保护屏蔽	RW	U16	1	1	N/A	[0,1]	Y		
56	45831	continuous ZVRT enable	连续高低穿使能位	RW	U16	1	1	N/A	[0,1]	Y	0 : OFF 1 : ON	0 : 关闭 1 : 打开
57	45832	enter VRT silence time	高低穿进入静默时间	RW	U16	1	1	10ms	[0,65500]	Y		
58	45833	exit VRT silence time	高低穿退出静默时间	RW	U16	1	1	10ms	[0,65500]	Y		
59	45834	Active power recovery slope at the end of high and low wear (first-order low-pass filtering)	穿越结束有功恢复一阶低通滤波	RW	U32	2	1	ms	[0,36000000]	Y		

61	45836	Reactive power recovery slope at the end of high and low wear (first-order low-	穿越结束无功恢复一阶低通滤波	RW	U32	2	1	ms	[0,36000000]	Y		
63	45838	LVRT enable	低穿使能位	RW	U16	1	1	N/A	[0,1]	Y	0 : OFF 1 : ON	0 : 关闭 1 : 打开
64	45839	The judgment threshold of entering	进入低穿的判断阈值	RW	U16	1	10	%Vn	[0,1000]	Y		
65	45840	The judgment threshold of quitting	退出低穿的判断阈值	RW	U16	1	10	%Vn	[0,1000]	Y		
66	45841	LVRT computational benchmark	低穿电压变化量计算基准电压	RW	U16	1	10	%Vn	[0,1000]	Y		
67	45842	LVRT positive active power current adjust enable	低穿正序有功电流调节使能位	RW	U16	1	1	N/A	[0,1]	Y	0 : OFF 1 : ON	0 : 关闭 1 : 打开
68	45843	LVRT positive reactive power current adjust enable	低穿正序无功电流调节使能位	RW	U16	1	1	N/A	[0,1]	Y	0 : OFF 1 : ON	0 : 关闭 1 : 打开
69	45844	LVRT negative active power current adjust enable	低穿负序无功电流调节使能位	RW	U16	1	1	N/A	[0,1]	Y	0 : OFF 1 : ON	0 : 关闭 1 : 打开
70	45845	LVRT active power current limit percent	低穿有功电流限幅百分比	RW	U16	1	10	%/In	[0,1100]	Y		
71	45846	LVRT positive sequence reactive K	低穿正序无功K值	RW	U16	1	100	N/A	[0,1000]	Y		
72	45847	LVRT Positive sequence reactive power static error	低穿正序无功静差偏移量	RW	S16	1	10	%/In	[-1000,1000]	Y		
73	45848	LVRT positive reactive power current limit percent	低穿正序无功电流限幅百分比	RW	U16	1	10	%/In	[0,1100]	Y		
74	45849	LVRT negative sequence reactive K	低穿负序无功K值	RW	U16	1	100	N/A	[0,1000]	Y		
75	45850	LVRT negative sequence reactive power static error	低穿负序无功静差偏移量	RW	S16	1	10	%/In	[-1000,1000]	Y		

76	45851	LVRT negative reactive power current limit percent	低穿负序无功电流限幅百分比	RW	U16	1	10	%/In	[0,1100]	Y		
77	45852	LVRT null-current mode enable	低穿零电流模式使能位	RW	U16	1	1	N/A	[0,1]	Y	0 : OFF 1 : ON	0 : 关闭 1 : 打开
78	45853	LVRT null-current mode enter voltage	低穿零电流模式进入电压阈值	RW	U16	1	10	%/Vn	[0,1000]	Y		
79	45854	LVRT depth 1	低穿深度1	RW	U16	1	10	%Vn	[0,1000]	Y		
80	45855	maintenance time 1	维持时间1	RW	U16	1	1	10ms	[0,65000]	Y		
81	45856	LVRT depth 2	低穿深度2	RW	U16	1	10	%Vn	[0,1000]	Y		
82	45857	maintenance time 2	维持时间2	RW	U16	1	1	10ms	[0,65000]	Y		
83	45858	LVRT depth 3	低穿深度3	RW	U16	1	10	%Vn	[0,1000]	Y		
84	45859	maintenance time 3	维持时间3	RW	U16	1	1	10ms	[0,65000]	Y		
85	45860	LVRT depth 4	低穿深度4	RW	U16	1	10	%Vn	[0,1000]	Y		
86	45861	maintenance time 4	维持时间4	RW	U16	1	1	10ms	[0,65000]	Y		
87	45862	LVRT depth 5	低穿深度5	RW	U16	1	10	%Vn	[0,1000]	Y		
88	45863	maintenance time 5	维持时间5	RW	U16	1	1	10ms	[0,65000]	Y		
89	45864	LVRT depth 6	低穿深度6	RW	U16	1	10	%Vn	[0,1000]	Y		
90	45865	maintenance time 6	维持时间6	RW	U16	1	1	10ms	[0,65000]	Y		
91	45866	LVRT depth 7	低穿深度7	RW	U16	1	10	%Vn	[0,1000]	Y		
92	45867	maintenance time 7	维持时间7	RW	U16	1	1	10ms	[0,65000]	Y		
93	45868	reserved				3						
96	45871	HVRT enable	高穿使能位	RW	U16	1	1	N/A	[0,1]	Y	0 : OFF 1 : ON	0 : 关闭 1 : 打开

97	45872	The judgment threshold of entering	进入高穿的判断阈值	RW	U16	1	10	%Vn	[1000,1400]	Y		
98	45873	The judgment threshold of quitting	退出高穿的判断阈值	RW	U16	1	10	%Vn	[1000,1400]	Y		
99	45874	HVRT computational benchmark	高穿电压变化量计算基准	RW	U16	1	10	%Vn	[1000,1200]	Y		
100	45875	HVRT positive active power adjust enable	高穿正序有功调节使能	RW	U16	1	1	N/A	[0,1]	Y	0 : OFF 1 : ON	0 : 关闭 1 : 打开
101	45876	HVRT positive reactive power adjust enable	高穿正序无功调节使能	RW	U16	1	1	N/A	[0,1]	Y	0 : OFF 1 : ON	0 : 关闭 1 : 打开
102	45877	HVRT negative reactive power adjust	高穿负序无功调节使能	RW	U16	1	1	N/A	[0,1]	Y	0 : OFF 1 : ON	0 : 关闭 1 : 打开
103	45878	HVRT active power current limit percent	高穿有功电流限制百分比	RW	U16	1	10	%In	[0,1100]	Y		
104	45879	HVRT positive sequence reactive K	高穿正序无功K值	RW	U16	1	100	N/A	[0,1000]	Y		
105	45880	HVRT Positive sequence reactive power static error	高穿正序无功静差偏移量	RW	S16	1	10	%/In	[-1000,1000]	Y		
106	45881	HVRT positive reactive power current limit percent	高穿正序无功电流限制百分比	RW	U16	1	10	%In	[0,1100]	Y		
107	45882	HVRT negative sequence reactive K value	高穿负序无功K值	RW	U16	1	100	N/A	[0,1000]	Y		
108	45883	HVRT negative sequence reactive power static error	高穿负序无功静差偏移量	RW	S16	1	10	%/In	[-1000,1000]	Y		
109	45884	HVRT negative reactive power current limit percent	高穿负序无功电流限制百分比	RW	U16	1	10	%In	[0,1100]	Y		

110	45885	HVRT null-current mode enable	高穿零电流模式使能	RW	U16	1	1	N/A	[0,1]	Y	0 : OFF 1 : ON	0 : 关闭 1 : 打开
111	45886	HVRT null-current mode enter voltage	高穿零电流模式进入电压阈值	RW	U16	1	10	%Vn	[1000,1400]	Y		
112	45887	HVRT depth 1	高穿深度1	RW	U16	1	10	%Vn	[1000,1400]	Y		
113	45888	maintenance time 1	维持时间1	RW	U16	1	1	10ms	[0,65000]	Y		
114	45889	HVRT depth 2	高穿深度2	RW	U16	1	10	%Vn	[1000,1400]	Y		
115	45890	maintenance time 2	维持时间2	RW	U16	1	1	10ms	[0,65000]	Y		
116	45891	HVRT depth 3	高穿深度3	RW	U16	1	10	%Vn	[1000,1400]	Y		
117	45892	maintenance time 3	维持时间3	RW	U16	1	1	10ms	[0,65000]	Y		
118	45893	HVRT depth 4	高穿深度4	RW	U16	1	10	%Vn	[1000,1400]	Y		
119	45894	maintenance time 4	维持时间4	RW	U16	1	1	10ms	[0,65000]	Y		
120	45895	HVRT depth 5	高穿深度5	RW	U16	1	10	%Vn	[1000,1400]	Y		
121	45896	maintenance time 5	维持时间5	RW	U16	1	1	10ms	[0,65000]	Y		
122	45897	HVRT depth 6	高穿深度6	RW	U16	1	10	%Vn	[1000,1400]	Y		
123	45898	maintenance time 6	维持时间6	RW	U16	1	1	10ms	[0,65000]	Y		
124	45899	HVRT depth 7	高穿深度7	RW	U16	1	10	%Vn	[1000,1400]	Y		
125	45900	maintenance time 7	维持时间7	RW	U16	1	1	10ms	[0,65000]	Y		
1	45901	FRT Enable	频率穿越使能位	RW	U16	1	1	N/A	[0,1]	Y		
2	45902	FRT UF1	一阶欠频穿越频率点_UF1	RW	U16	1	100	Hz	[4500,6000]	Y		
3	45903	FRT UT1	一阶欠频穿越时间_UT1	RW	U32	2	1	ms	[20,7200000]	Y		
5	45905	FRT UF2	二阶欠频穿越频率点_UF2	RW	U16	1	100	Hz	[4500,6000]	Y		

6	45906	FRT UT2	二阶欠频穿越时间_UT2	RW	U32	2	1	ms	[20,7200000]	Y		
8	45908	FRT UF3	三阶欠频穿越频率点_UF3	RW	U16	1	100	Hz	[4500,6000]	Y		
9	45909	FRT UT3	三阶欠频穿越时间_UT3	RW	U32	2	1	ms	[20,7200000]	Y		
11	45911	FRT OF1	一阶过频穿越频率点_OF1	RW	U16	1	100	Hz	[5000,6500]	Y		
12	45912	FRT OT1	一阶过频穿越时间	RW	U32	2	1	ms	[20,7200000]	Y		
14	45914	FRT OF2	二阶过频穿越频率点_OF2	RW	U16	1	100	Hz	[5000,6500]	Y		
15	45915	FRT OT2	二阶过频穿越时间	RW	U32	2	1	ms	[20,7200000]	Y		
17	45917	FRT OF3	三阶过频穿越频率点_OF3	RW	U16	1	100	Hz	[5000,6500]	Y		
18	45918	FRT OT3	三阶过频穿越时间	RW	U32	2	1	ms	[20,7200000]	Y		
20	45920	LvrtVolt1	低穿电压点1	RW	U16	1	10	V	[0,2300]	Y	用于1.0安规模式的储能机适配澳洲5点式低穿 注意：需要关联28550寄存器的bit3	
21	45921	LvrtTime1	低穿时间点1	RW	U16	1	1	period	[0,65535]	Y	Note that the value of register 1 represents 0.02 seconds.	注意寄存器值1代表0.02秒
22	45922	LvrtVolt2	低穿电压点2	RW	U16	1	10	V	[0,2300]	Y		
23	45923	LvrtTime2	低穿时间点2	RW	U16	1	1	period	[0,65535]	Y		
24	45924	LvrtVolt3	低穿电压点3	RW	U16	1	10	V	[0,2300]	Y		
25	45925	LvrtTime3	低穿时间点3	RW	U16	1	1	period	[0,65535]	Y		
26	45926	LvrtVolt4	低穿电压点4	RW	U16	1	10	V	[0,2300]	Y		
27	45927	LvrtTime4	低穿时间点4	RW	U16	1	1	period	[0,65535]	Y		
28	45928	LvrtVolt5	低穿电压点5	RW	U16	1	10	V	[0,2300]	Y		
29	45929	LvrtTime5	低穿时间点5	RW	U16	1	1	period	[0,65535]	Y		
DSP Function Switch												
1	46700	island mode	孤岛功能开关	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0：关闭，1：使能
4	46703	LVRT	低穿功能位开关	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0：关闭，1：使能
5	46704	HVRT	高穿功能位开关	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0：关闭，1：使能
9	46708	Anti-backflow function	防逆流开关	RW	U16	1	1	N/A	[0,1]	Y	Anti-backflow function enable switch. 0: Off, 1: Enable	防逆流功能使能开关。 0：关闭，1：使能

10	46709	Three-phase anti-backflow mode selection	三相防逆流方式选择	RW	U16	1	1	N/A	[0,0]	Y	The anti-backflow mode selection is either the single-phase anti-backflow method or the sum of three-phase anti-backflow method. 0: The total power of three phases cannot be reversed. 1: Any one phase cannot be	防逆流模式选择单相防逆流方式或三相之和和防逆流方式。 0: 三相总功率不能逆流 1: 任意一相不能逆流
11	46710	Hardware anti-backflow function	硬件防逆流开关	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0 : 关闭 , 1 : 使能
12	46711	Q(U) Cruve	Q(U)曲线	RW	U16	1	1	N/A	[0,1]	Y	Used solely for display	只用于显示
13	46712	P(U) Cruve	P(U)曲线	RW	U16	1	1	N/A	[0,1]	Y	Used solely for display	只用于显示
14	46713	cos φ (P) Cruve	cos φ (P)曲线	RW	U16	1	1	N/A	[0,1]	Y	Used solely for display	只用于显示
15	46714	P(F) Cruve	P(F)曲线	RW	U16	1	1	N/A	[0,1]	Y	Used solely for display	只用于显示
16	46715	Q(P) Cruve	Q(P)曲线	RW	U16	1	1	N/A	[0,1]	Y	Used solely for display	只用于显示
23	46722	shadow mppt function 1	阴影mppt功能开关1	RW	U16	1	1	N/A	[0,1]	Y	When the inverter is applied to photovoltaic string arrays with significant shading, if this function is enabled, the inverter will perform a global MPPT scan at regular intervals to find the maximum	逆变器应用于光伏组串有明显遮挡的场景时, 使能该功能, 则逆变器会每隔一段时间进行一次全局MPPT扫描, 找到功率最大值。
24	46723	shadow mppt function	阴影mppt功能开关2	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0 : 关闭 , 1 : 使能
25	46724	shadow mppt function	阴影mppt功能开关3	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0 : 关闭 , 1 : 使能
40	46739	Night reactive mode	夜间无功功能开关	RW	U16	1	1	N/A	[0,1]	Y	Switch for the night reactive power function. 1 - On, 0 - Off	夜间无功功能的开关。 1-打开, 0-关闭
43	46742	Output method (grid type)	输出方式(电网类型)	RW	U16	1	1	N/A	[0,1]	Y	Used solely for display	只用于显示
44	46743	Phase sequence adaptation of three-phase grid	三相电网相序自适应	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0 : 关闭 , 1 : 使能
45	46744	Fault recover settings	故障恢复设置	RW	U16	1	1	N/A	[0,1]	Y	0: Automatic recovery after failure 1: Manual recovery after failure	0 : 故障后自动恢复 1 : 故障后手动恢复
46	46745	GFCI Device Self-checking Switch	GFCI设备自检检测开关	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0 : 关闭 , 1 : 使能
47	46746	THD Estimation Switch	电网谐波预估开关	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0 : 关闭 , 1 : 使能
48	46747	Fan detector switch	风扇检测开关	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0 : 关闭 , 1 : 使能

49	46748	Communication disconnect switch	通讯断链开关	RW	U16	1	1	N/A	[0,1]	Y	0: Communication disconnection without turning off the device 1: Communication disconnection with device turned off	0: 通讯断链不关机 1: 通讯断链关机
50	46749	Communication disconnection recovery settings	通讯折断恢复设置	RW	U16	1	1	N/A	[0,1]	Y	0: Automatic Recovery 1: Manual Recovery	0: 自动恢复 1: 手动恢复
51	46750	Output control communication abnormal shutdown	出力制御通讯异常关机设置	RW	U16	1	1	N/A	[0,1]	Y	0: Normal startup 1: Abnormal shutdown	0: 正常开机 1: 异常关机
52	46751	NPE Overvoltage fault detection switch	NPE电压过压故障检测开关	RW	U16	1	1	N/A	[0,1]	Y	0: Prohibit PN voltage overvoltage fault 1: Enable PN voltage overvoltage fault	0: 禁止PN电压过压故障 1: 使能PN电压过压故障
53	46752	OVGR setting	OVGR设置	RW	U16	1	1	N/A	[0,1]	Y	0: Enable OVGR 1: Disable OVGR	0: 使能OVGR 1: 禁能OVGR
54	46753	PV Reverse connection detection switch	PV反接检测开关	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0: 关闭, 1: 使能
55	46754	Fuse blown detection switch	熔丝熔断检测开关	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0: 关闭, 1: 使能
56	46755	Quick power-off protection detection switch	快速断电保护检测开关	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0: 关闭, 1: 使能
57	46756	ISO detection switch	ISO检测开关	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0: 关闭, 1: 使能
58	46757	Lightning protection module detection	防雷模块检测开关	RW	U16	1	1	N/A	[0,1]	Y	The enable switch for the lightning protection function. 0: Off, 1: Enable	防雷功能的使能开关。 0: 关闭, 1: 使能
59	46758	GFCI detection switch	GFCI检测开关	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0: 关闭, 1: 使能
60	46759	AFCI detection switch	AFCI检测开关	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0: 关闭, 1: 使能
61	46760	DCI detection switch	DCI检测开关	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0: 关闭, 1: 使能
62	46761	Anti-PID function enable	PID预防功能使能	RW	U16	1	1	N/A	[0,1]	Y	0: Disable 1: Enable	0: 禁能 1: 使能
63	46762	Ac Sps	Ac Sps功能开关	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0: 关闭, 1: 使能
64	46763	Grid unbalance detection switch	电网电压不平衡检测开关	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0: 关闭, 1: 使能
65	46764	Grid current unbalance detection switch	电网电流不平衡检测开关	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0: 关闭, 1: 使能
66	46765	Inv Over Current Limit Switch	逐波限流开关	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0: 关闭, 1: 使能

67	46766	HVrt Augmented	高穿增强开关	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0 : 关闭 , 1 : 使能
68	46767	PID repair function switch	PID修复功能开关	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0 : 关闭 , 1 : 使能
69	46768	PID Night protection function switch	PID夜间保护功能开关	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0 : 关闭 , 1 : 使能
70	46769	Phase Angle offset protection	相角偏移保护	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0 : 关闭 , 1 : 使能
71	46770	power off command	指令关机掉电保持	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0 : 关闭 , 1 : 使能
72	46771	pf(U) Cruve	pf(U)曲线	RW	U16	1	1	N/A	[0,1]	Y	The pf and U curve enable switch is only used for display purposes.	pf与U曲线使能开关 , 只用于显示
73	46772	start up time enable	启动并网时间使能	RW	U16	1	1	N/A	[0,1]	Y	Enable the grid connection start time. 0 means off, 1 means start. This address is associated with 40371 for starting the grid	使能启动并网时间 , 0 为关闭 , 1为开始。此地址关联40371启动并网时间。
74	46773	Grid On Off Mode	并离网模式	RW	U16	1	1	N/A	[0,1]	Y	0: Grid-connected 1: Off-grid	0 : 并网 1 : 离网
75	46774	VSG mode	VSG模式	RW	U16	1	1	N/A	[0,1]	Y	0: Disable 1: Enable	0 : 禁能 1 : 使能
76	46775	master-slave mode	主从模式	RW	U16	1	1	N/A	[0,1]	Y	0: Master 1: Slave	0 : 主机 1 : 从机
77	46776	Iso Morning Fault Enable	早上九点ISO检测故障使能开关	RW	U16	1	1	N/A	[0,1]	Y	0: Close 1: Open	0 : 关闭 1 : 打开
78	46777	ISO on-line check Enable	在线ISO检测使能	RW	U16	1	1	N/A	[0,1]	Y	0: Close 1: Open	0 : 关闭 1 : 打开
79	46778	shadow mppt function	阴影mppt功能开关4	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0 : 关闭 , 1 : 使能
80	46779	shadow mppt function	阴影mppt功能开关5	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0 : 关闭 , 1 : 使能
81	46780	shadow mppt function	阴影mppt功能开关6	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0 : 关闭 , 1 : 使能
82	46781	shadow mppt function	阴影mppt功能开关7	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0 : 关闭 , 1 : 使能
83	46782	shadow mppt function	阴影mppt功能开关8	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0 : 关闭 , 1 : 使能
84	46783	shadow mppt function	阴影mppt功能开关9	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0 : 关闭 , 1 : 使能
85	46784	shadow mppt function	阴影mppt功能开关10	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0 : 关闭 , 1 : 使能
86	46785	Zero power block PWM enable	零功率封波使能开关	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable After enabling and setting the power scheduling to 0, the PWM module stops the pulse width modulation, reducing losses, but there will be an indication of reactive power from the grid-side	0 : 关闭 , 1 : 使能 使能后功率调度均为0后 , PWM模块封波 , 减少损耗 , 但会有网侧电容无功功率显示。

87	46786	Topspeed Enable	极速调度使能开关	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable The energy storage device is used. After enabling, it allows for rapid scheduling of charge and discharge conversions (completed within 50 milliseconds).	0 : 关闭 , 1 : 使能 储能机使用 , 使能后允 许充放电转换极速调度 (50ms内完成)
79	46787	RPCOptimize	微逆轻载无功补偿模式	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0 : 关闭 , 1 : 使能
80	46788	Fan Manual Test	风扇手动自检开关	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0 : 关闭 , 1 : 使能
81	46789	Fan Reverse Enable	风扇反转功能开关	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0 : 关闭 , 1 : 使能
82	46790	HVRT Certifi Enable	VRT认证模式使能	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0 : 关闭 , 1 : 使能
83	46791	PV Input Mode Protect Enable	PV接入模式保护使能	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0 : 关闭 , 1 : 使能
85	46793	Backup Mode Parallel Enable	离网并网使能	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0 : 关闭 , 1 : 使能
86	46794	AFCI Fault Enable	AFCI故障保护开关	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0 : 关闭 , 1 : 使能
87	46799	RSVD	预留			101						

Optional function Setting												
7	46906	BUS balance bridge Enable	BUS母线正负平衡电路使能	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0 : 关闭 , 1 : 使能
8	46907	Smart Port Parallel Connection Mode	Smart口并机接线模式	RW	U16	1	1	N/A	[1,2]	Y	1: 485 Parallel Smart Port Parallel Connection 2: 485 parallel Smart port is not paralleled (default)	1 : 485并机Smart口并联 2 : 485并机Smart口不并联(默认)
ARM Setting												
1	47000	App Work Mode Index	APP工作模式索引	RW	U16	1	N/A	N/A		Y	The same function as that for Operation Mode on PV Master App 0:selfuse mode 1:off gird mode 2:backup mode 3:economic mode 4:peakshaving 5:Advanced self-use	和在PV主程序上运行模式的功能相同 0:自发自用 1:离网模式 2:备用模式 3:经济模式 4:调峰 5:高级自发自用
2	47001	Meter Check Value	电表检测结果	RO	U16	1	N/A	N/A	[0, 60000]	N	1: Correct connection 2: Connection reserved (CT) 4: Incorrect connection 8: Wrong phase for CT and voltage sampling 0: Not detected For example: 0X0124 means R phase connection is incorrect, the T phase connection is retained, and the S phase connection is	1:连接正确 2:连接保留(CT) 4:连接不正确 8 : CT和电压采样同时错相 0:未检测 例如 : 0X0124就是R相连接不正确, T相连接保留, S相连接正确
3	47002	Meter Connect Check Flag	电表检测使能	RW	U16	1	N/A	N/A	[0,2]	N	to read the Meter Test status 0:Not open test/end test 1:Being test	读电表检测状态 0:未开启检测/检测结束 1:正在检测 2:等待检测
6	47005	Log Data Enable	断点续传使能位	RW	U16	1	N/A	N/A	[0,1]	Y	Breakpoint Resume for Data transferring. Activated as deFault, time interval 5 minutes.	断点恢复数据传输。默认激活, 时间间隔5分钟
7	47006	Data Send Interval	数据传输间隔	RW	U16	1	N/A	*5s	[0,256]	Y	Time interval for data send to cloud or EMS,deFault is 1 minute.	数据发送到云或EMS的时间间隔, 默认为1分

8	47007	DRED command	命令响应设备指令	RW	U16	1	N/A	N/A	[0,255]	N	DRED0:0x00FF DRED1:0x0001 DRED2:0x0002 DRED3:0x0004 DRED4:0x0008 DRED5:0x0010 DRED6:0x0020 DRED7:0x0040 DRED8:0x0080 Only for Australia, Refer to Table 8-22	DRED0:0x00FF DRED1:0x0001 DRED2:0x0002 DRED3:0x0004 DRED4:0x0008 DRED5:0x0010 DRED6:0x0020 DRED7:0x0040 DRED8:0x0080 仅用于澳大利亚,见 表8-22
10	47009	WiFi or Lan Switch	wifi或lan模式	WO	U16	1	N/A	N/A	[4,5]	Y	4:wifi 5:lan For wifi+Lan module, to switch to LAN or WiFi communicaiton mode	4:wifi 5:lan wifi+Lan模块, 切换到 LAN或WiFi通信模式。
13	47012	Led Blink Time	Led灯闪烁时间	RW	U16	1	N/A	s		Y		
14	47013	WiFi Led State	wifi指示灯状态	RW	U16	1	N/A	N/A	[1,5]	N	1:off 2:on 3:flash1x 4:flash2x 5:flash4x	1:关闭 2:打开 3:flash1x 4:flash2x
15	47014	Communication Led State	通信灯状态	RW	U16	1	N/A	N/A	[1,5]	N		
16	47015	Meter CT1 Reverse Enable	电表CT1预留使能	RW	U16	1	N/A	N/A	[1,5]	Y	1:on 0:off ony for single phase Smart meter	1:打开 0:关闭 仅用于单相智能电表
17	47016	Error Log Read Page	故障记录读取页	RW	U16	1	N/A	N/A	[0,255]	N		
18	47017	Modbus TCP Without Internet	ModbusTCP 不联网应用	RW	U16	1	N/A	N/A	[0,255]	Y	1:on 0:off If not connect to Internet, please set 1.	1:打开 0:关闭 如果不能联网, 就设置1
19	47018	Backup Led	backup Led指示灯	RW	U16	1	N/A	N/A	[1,5]	N	1:off 2:on 3:flash1x 4:flash2x 5:flash4x	1:关闭 2:打开 3:flash1x 4:flash2x
20	47019	Grid Led	电网指示灯	RW	U16	1	N/A	N/A	[1,5]	N		
21	47020	SOC Led 1	剩余电量指示灯1	RW	U16	1	N/A	N/A	[1,5]	N		
22	47021	SOC Led 2	剩余电量指示灯2	RW	U16	1	N/A	N/A	[1,5]	N		
23	47022	SOC Led 3	剩余电量指示灯3	RW	U16	1	N/A	N/A	[1,5]	N		
24	47023	SOC Led 4	剩余电量指示灯4	RW	U16	1	N/A	N/A	[1,5]	N		
25	47024	Battery Led	电池指示灯	RW	U16	1	N/A	N/A	[1,5]	N		
26	47025	System Led	系统指示灯	RW	U16	1	N/A	N/A	[1,5]	N		
27	47026	Fault Led	故障指示灯	RW	U16	1	N/A	N/A	[1,5]	N		

28	47027	Energy Led	能量指示灯	RW	U16	1	N/A	N/A	[1,5]	N		
29	47028	Led External Control	Led灯外部控制	RW	U16	1	N/A	N/A	[42343]	N	0xA567	0xA567
36	47038	Off Grid Set Save	离网设置存储	RW	U16	1	1	N/A	[0,1]	Y	1 Enable, After restart the inverter, setting saved	1使能, 重启逆变器后, 保存设置
38	47040	WIFI Modbus TCP Enable	使能modbusTCP功能	RW	U16	1	1	N/A	[0,1]	Y		
43	47117	API Remote TimeOut Enable	API超时判断使能	RW	U16	1	1	N/A	[0,1]	N		
44	47118	API time out	API超时时间	RW	U16	1	1	min	[1,60]	Y		
45	47119	EMS 485 Port Use Type	EMS 485口使用类型	RW	U16	1	1	N/A	[0, 65535]	Y	bit0: SAPN FeedPower send Grid-machine SW bit2: Schedule On-Grid Inverter to Control Micro Grid Function bit3: Charge Pile Function	bit0: sapn防逆流计划发送并网机开关(储能机作为485主机, 将远程设置的sapn防逆流计划通过485转发给并网机) bit1: 调度并网机控制微网功能
46	47120	Meter target power Offset	电表目标功率偏置	RW	S16	1	1	W	[-32767,32767]	Y	Positive value: partial selling electricity Negative value: partial power	正值: 偏卖电 负值: 偏买电
47	47121	Blower Open Temperature	外部风扇开启温度设置值	RW	U16	1	1	N/A	0, 1, [40, 80]	Y	ATS Mode : 0: Force Close, 1: Force Open(Only used in produce) User Mode : When not set, the default open temperaturure is 65°C. [40,80]°C set open temperaturure, close the temperature is open temperature is -10°C autoly	ATS模式下: 0强制关闭, 1强制开启(仅在生产时使用) 用户: 不设置时默认开启温度为65°C [40,80]°C设置开启温度, 关闭温度自动为开启温度-10°C
48	47122	Micro Grid Feed Power Offset	微网防逆流偏置	RW	S16	1	1	W	[-1000,0]	Y	Negative value: The anti-backflow value has decreased.	负值: 防逆流值减小
49	47123	Remote ShutDown Battery	远程关断电池	RW	U16	1	1	N/A	[0, 1]	N		
50	47124	Remote On Off Battery	远程开关机电池	RW	U16	1	1	N/A	[0, 1]	N		

51	47125	Grid Port Connect Device Type	Grid端口接入设备类型	RW	U16	1	1	N/A		N	0(Value is 0): Not connect device bit0: Connect Generator bit1: Connect On-Grid Inverter bit2: Connect Large Load	0(值是0): 未接入设备 bit0: 接入发电机 bit1: 接入并网机 bit2: 接入大负载
52	47126	Backup Port Connect Device Type	Backup端口接入设备类型	RW	U16	1	1	N/A		N	0(Value is 0): Not connect device bit0: N/A bit1: Connect On-Grid Inverter bit2: Connect Large Load	0(值是0): 未接入设备 bit0: N/A bit1: 接入并网机 bit2: 接入大负载
53	47127	Smart Port Connect Device Type	智能端口接入设备类型	RW	U16	1	1	N/A			0(Value is 0): Not connect device bit0: Connect Generator bit1: Connect On-Grid Inverter bit2: Connect Large Load	0(值是0): 未接入设备 bit0: 接入发电机 bit1: 接入并网机 bit2: 接入大负载
54	47128	Dry Contact Large Load Control Mode	干接点大负载控制模式	RW	U16	1	1	N/A		Y	0: Disable 1: Switching Mode 2: Time Manage Mode 3: SOC Mode	0: 关闭 1: 开关模式 2: 时间管理模式 3: SOC模式
55	47129	Dry Contact LargeLoad Switch	干接点大负载切换开关	RW	U16	1	1	N/A		Y	0: Open Dry Contact 1: Close Dry Contact	0: 断开干接点 1: 闭合干接点
56	47130	Dry Contact LargeLoad Backup Switch SOC Min	干接点大负载Backup切换最小剩余电量	RW	U16	1	1	%	[0, 100]	Y	For load control function, if the controlLed load on Backup side, use this to switch the load off when Battery reaches the SOC set here	对于负载控制功能，如果被控制的负载在Backup侧，当电池达到设定的SOC时，使用此功能来关闭负载。
58	47148	Generator Peakshaving Max	柴发peakshaving最大功率点	RW	U16	1	1	N/A		N		
60	47154	Start Time_15	启动时间_15	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y	hh:mm	hh:mm
61	47155	End Time_15	结束时间_15	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
62	47156	Work Week_15	工作周_15	RW	U16	1	N/A	N/A		Y	0xFE: Standard Load Mode 0xFD: Smart Load Mode	0xFE: 标准负载模式 0xFD: 智能负载模式
63	47157	Parameter15_1	参数15_1	RW	S16	1	1	W	[-32768, 32767]	Y	Smart Load Mode Peak Sell Power	智能负载模式下卖电功率峰值
64	47158	Parameter15_2	参数15_2	RW	U16	1	1	min	[0,100]/NA/[0,1440]/[0,100]	Y	Smart Load Mode Continue Open Time	智能负载模式下开启持续时间
65	47159	Parameter15_3	参数15_3	RW	U16	1	N/A	N/A		Y		
66	47160	Start Time_16	启动时间_16	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y	hh:mm	hh:mm
67	47161	End Time_16	结束时间_16	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
68	47162	Work Week_16	工作周_16	RW	U16	1	N/A	N/A		Y	0xFE: Standard Load Mode 0xFD: Smart Load Mode	0xFE: 标准负载模式 0xFD: 智能负载模式

69	47163	Parameter16_1	参数16_1	RW	S16	1	1	W	[-32768, 32767]	Y	Smart Load Mode Peak Sell Power	智能负载模式下卖电功率峰值
70	47164	Parameter16_2	参数16_2	RW	U16	1	1	min	[0,100]/NA/[0,1440]/[0,100]	Y	Smart Load Mode Continue Open Time	智能负载模式下开启持续时间
71	47165	Parameter16_3	参数16_3	RW	U16	1	N/A	N/A		Y		
72	47166	Start Time_17	启动时间_17	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y	hh:mm	hh:mm
73	47167	End Time_17	结束时间_17	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
74	47168	Work Week_17	工作周_17	RW	U16	1	N/A	N/A		Y	0xFE: Standard Load Mode 0xFD: Smart Load Mode	0xFE: 标准负载模式 0xFD: 智能负载模式
75	47169	Parameter17_1	参数17_1	RW	S16	1	1	W	[-32768, 32767]	Y	Smart Load Mode Peak Sell Power	智能负载模式下卖电功率峰值
76	47170	Parameter17_2	参数17_2	RW	U16	1	1	min	[0,100]/NA/[0,1440]/[0,100]	Y	Smart Load Mode Continue Open Time	智能负载模式下开启持续时间
77	47171	Parameter17_3	参数17_3	RW	U16	1	N/A	N/A		Y		
78	47172	Start Time_18	启动时间_18	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y	hh:mm	hh:mm
79	47173	End Time_18	结束时间_18	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
80	47174	Work Week_18	工作周_18	RW	U16	1	N/A	N/A		Y	0xFE: Standard Load Mode 0xFD: Smart Load Mode	0xFE: 标准负载模式 0xFD: 智能负载模式
81	47175	Parameter18_1	参数18_1	RW	S16	1	1	W	[-32768, 32767]	Y	Smart Load Mode Peak Sell Power	智能负载模式下卖电功率峰值
82	47176	Parameter18_2	参数18_2	RW	U16	1	1	min	[0,100]/NA/[0,1440]/[0,100]	Y	Smart Load Mode Continue Open Time	智能负载模式下开启持续时间
83	47177	Parameter18_3	参数18_3	RW	U16	1	N/A	N/A		Y		
22	47464	Extern Meter Enable Flag	外置电表使能标志	RW	U16	1	1	N/A	[0,4]	Y	0: Built-in electricity meter, default 0 1: External electricity meter, 2: Mixed use of built-in and external electricity meters, 3: Dual external electricity meters, 4: No electricity meter and CT	0: 内置电表, 默认0 1: 外置电表, 2: 内置电表外置电表混用, 3: 双外置电表, 4: 无电表和CT
24	47466	CT Correct mode	CT纠正方式	RW	U16	1	1	N/A		Y	0: Automatic correction, 1: Manual correction, default 0	0: 自动纠正, 1: 手动纠正 默认0
25	47467	Correct Complete Flag	纠正执行完成标志	RO	U16	1	1	N/A		N	0: Automatic correction not executed, 1: One automatic correction completed, Default value is 0, used in conjunction with CT correction	0: 未执行自动纠正, 1: 完成了一次自动纠正, 默认0, 配合CT的纠正结果使用

26	47468	CT Correct Status	内置CT纠正状态	RO	U16	1	1	NA		Y	0: No correction made 1: Correction in progress 2: The results of the electricity meter inspection cannot be corrected 3: Attempted to correct but failed 4: Correction successful	0 : 未进行纠正 1 : 正在进行纠正 2 : 电表检测的结果无法进行纠正 3 : 尝试纠正但失败了 4 : 纠正成功
27	47469	CT1 Correct Result	CT1的纠正结果	RW	U16	1	1	NA		Y	[15:12]: Indicates whether the corresponding CT is reverse-connected, with 1 representing	[15:12] : 表示为对应的CT是否反接, 1为反接 [11:8]:L3, [7:4]:L2,
28	47470	CT2 Correct Result	CT2的纠正结果	RW	U16	1	1	NA		Y		
29	47471	CT3 Correct Result	CT3的纠正结果	RW	U16	1	1	NA		Y		
30	47472	Battery discharge reset setting value	电池放电回置设置	RW	U16	1	1	NA	[0-5]	Y	After the battery reaches the charging upper limit of 47760, it restores the next charging value, and the current SOC is greater than or equal to 47760 - 47472	电池达到充电上限47760后, 恢复下一次充电值, 当前SOC大于等于47760-47472
31	47473	High Refresh Meter Refresh Mode Settings	高刷电表刷新模式设置	RW	U16	1	1	N/A		Y	Fast Brush Mode: 0xAA55, Slow Brush Mode: 0xBBCC,	快刷模式: 0xAA55, 慢刷模式: 0xBBCC,
32	47474	Scene switching	场景切换	RW	U16	1	1	N/A	[0,1]	Y	Balcony micro-storage can be used for household micro energy storage and can also be taken outdoors for use as a portable power source. 0 - Balcony Micro Energy Storage Scenario; 1 - Outdoor Power	阳台微储可用于户用微型储能, 也可带到户外做便携式电源使用。 0-阳台微储场景;1-户外电源场景;
33	47475	EPO polarity configuration	急停极性配置	RW	U16	1	1	N/A	[0,1]	Y	0: Normally closed. Open circuit indicates an emergency stop fault; 1: Normally open. Short circuit indicates an emergency stop fault;	0 : 常闭, 开路为急停故障; 1 : 常开, 短路为急停故障;
34	47476	Energy saving mode switch	节能模式开关	RW	U16	1	1	N/A	[0,1]	Y	0 - Off; 1 - On;	0-关闭; 1-开启;
35	47477	Lamp Ring Control	环形灯光熄灭时间控制	RW	U16	1	1	NA	[0,6]	Y	0 - Always On; 1-1min; 2-2min; 3-3min; 4-4min; 5-5min; 6 - Always off;	0-常亮; 1-1min; 2-2min; 3-3min; 4-4min; 5-5min; 6-常灭;

36	47478	Configuration of paralleling CAN communication matching resistors	并机DSPCAN通讯的匹配电阻控制	RW	U16	1	1	NA	[0,1]	Y	0 - Disability; 1 - Enablement	0-失能; 1-使能
ARM BMS&EMS parameter												
6	47505	Manufacturer Code	制造商类型	RW	U16	1	N/A	N/A		Y	If using EMS, must set to "2"	如果使用EMS模式，必须设置"2"
10	47509	Feed Power Enable	防逆流开关	RW	U16	1	N/A	N/A	[0,1]	Y	1:total power anti-counter Current Use FeedPowerEnable (47509) to activate export power limit function. And EMSPowerSet (47510) to set the max allowed export power to Grid. The 42004-42005 register is used to set the maximum allowable output power to the power grid when the AC power of a single inverter is greater than 30K or	1:总功率防逆流使用防逆流开关(47509)激活导出功率限制功能。使用EMS功率设置(47510)来设置输电网的最大允许输出功率 逆变器单机AC功率大于30K或并机时使用42004-42005寄存器设置输电网的最大允许

11	47510	Feed Power Parameter	防逆流允许并网功率	RW	S16	1	N/A	N/A	[0,10000]	Y		
12	47511	EMS Power Mode	能量管理模式	RW	U16	1	N/A	N/A		N		
13	47512	EMS Power Set	能量管理功率设置	RW	U16	1	N/A	N/A	[0,10000]	N		
15	47514	Battery Protocol	电池协议	RW	U16	1	N/A	N/A	[0,511]	Y	Refer to 8-33	见表8-33
16	47515	Start Time_1	启动时间_1	RW	U16	1	N/A	N/A	[0,23]-[0,59]	Y	hh:mm	hh:mm
17	47516	End Time_1	结束时间_1	RW	U16	1	N/A	N/A	[0,23]-[0,59]	Y		
18	47517	Bat Power Percent_1	电池功率百分比_1	RW	S16	1	N/A	%	[-100,100]	Y		
19	47518	Work Week_1	工作周_1	RW	U16	1	N/A	N/A		Y	High byte FF means enable, low byte 0~6bit means Sunday~Saturday a whole week	高字节FF表示使能，低字节0~6bit表示周日~周六一整个星期
20	47519	Start Time_2	启动时间_2	RW	U16	1	N/A	N/A	[0,23]-[0,59]	Y	hh:mm	hh:mm
21	47520	End Time_2	结束时间_2	RW	U16	1	N/A	N/A	[0,23]-[0,59]	Y		
22	47521	Battery Power	电池功率百分比_2	RW	S16	1	N/A	%	[-100,100]	Y		
23	47522	Work Week_2	工作周_2	RW	U16	1	N/A	N/A		Y	High byte FF means enable, low byte 0~6bit means Sunday~Saturday a whole week	高字节FF表示使能，低字节0~6bit表示周日~周六一整个星期
24	47523	Start Time_3	启动时间_3	RW	U16	1	N/A	N/A	[0,23]-[0,59]	Y	hh:mm	hh:mm
25	47524	End Time_3	结束时间_3	RW	U16	1	N/A	N/A	[0,23]-[0,59]	Y		

26	47525	Battery Power	电池功率百分比_3	RW	S16	1	N/A	%	[-100,100]	Y		
27	47526	Work Week_3	工作周_3	RW	U16	1	N/A	N/A		Y	High byte FF means enable, low byte 0~6bit means Sunday~Saturday a whole week	高字节FF表示使能，低字节0~6bit表示周日~周六一整个星期
28	47527	Start Time_4	启动时间_4	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y	hh:mm	hh:mm
29	47528	End Time_4	结束时间_4	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
30	47529	Battery Power	电池功率百分比_4	RW	S16	1	N/A	%	[-100,100]	Y		
31	47530	Work Week_4	工作周_4	RW	U16	1	N/A	N/A		Y	High byte FF means enable, low byte 0~6bit means Sunday~Saturday a whole week	高字节FF表示使能，低字节0~6bit表示周日~周六一整个星期
32	47531	Start Charge SOC	强充起始剩余电量	RW	U16	1	10	%		Y	To set the SOC Level to start/stop Battery force charge. (this is not the command from BMS, but the protection on inverter side. Eg. StartchgSOC (47531) is set as 5%, but the Battery BMS gives a force charge signal at SOC 6%, then Battery will start force charge at 6% SOC; if BMS does not send force charge command at 5% SOC, then Battery will still start force charge at 5% SOC.) Note: the deFault setting is 5% SOC to start and 10% to stop. force charge power is 1000W from PV or Grid as well.	"设置SOC水平来启动/停止电池强制充电。(这不是BMS的指令，而是对逆变器侧的保护。例如，StartchgSOC(47531)设置为5%，但电池BMS在SOC为6%时发出强制充电信号，那么电池将在SOC为6%时开始强制充电;如果BMS没有在SOC为5%时发出强制充电指令，那么电池仍将在SOC为5%时开始强制充电。) 注意：默认设置是SOC为5%时开始，10%停止。强行充电的功率是1000W，来自光伏或电

33	47532	Stop Charge SOC	强充停止剩余电量	RW	U16	1	10	%		Y		
34	47533	Clear ECO Time	清除电池时间设置使能位	WO	U16	1	N/A	N/A		N	to clear all economical mode settings (47515~47530), enter Self-Use Mode	清除所有的经济模式设置(47515~47530), 进入自发自用模式化
40	47539	Wifi Reset	复位wifi模块	WO	U16	1	N/A	N/A	[0,1]	N		
42	47541	Wifi Reload	wifi模块恢复出厂	WO	U16	1	N/A	N/A	[0,1]	N		
43	47542	Peak Shaving Power Limit	削峰功率限制	RW	U32	2	N/A	W		Y	to set the threshold of importing power, where peak-shaving acts. Eg. If set peak-shaving power as 20kW, then Battery will only discharge when imported power from Grid exceed 20kW to make sure the importing power keeps below 20kW	设定输入功率的阈值, 在该值下削峰。例如, 如果设定削峰功率为20kW, 则只有当从电网输入的功率超过20kW时, 电池才会放电, 以确保输入的功率保持在20kW以下。
44	47544	Peak Shaving SOC	削峰电池剩余电量	RW	U16	1	N/A	%		Y	to set the mimum Battery SOC reserved to respind peak-shaving use only.	设置最小电池SOC, 预留给重塑峰值使用。
45	47545	Fast Charge Enable	快速充电使能	RW	U16	1	N/A	1	[0,3]	N	0: Disable 1:Enable	0: 禁用 1:使能
46	47546	Fast Charge Stop SOC	快速充电SOC上限	RW	U16	1	N/A	1	[1,100]	N		
47	47547	Start Time_1	启动时间_1	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y	hh:mm	hh:mm
48	47548	End Time_1	结束时间_1	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		

49	47549	Work Week_1	工作周_1	RW	U16	1	N/A	N/A		Y	Table 8-34	
50	47550	Parameter1_1	参数1_1	RW	U16	1	1/NA/ 100/1 00/1	%/N/A/KW/K W/N/A/%/‰	[- 100,100]/NA/[0,1 0000]/[0,50000]/[0,100]/[- 1000,1000]	Y	0xFF:Battery power percentage 0xFE:N/A 0xFD:Rated Power 0xF9:Battery power permillage	0xFF:电池功率百分比 0xFE:N/A 0xFD:额定功率 0xF9:电池功率千分比
51	47551	Parameter1_2	参数1_2	RW	U16	1	1/NA/ 1/1	%/N/A/min/ %	[0,100]/NA/[0,144 0]/[0,100]	Y	Parameter Group Charging, SOC Upper Limit for Economic Mode Charging Parameter Group Discharge, Economic Mode Discharge SOC Lower Limit 0xFE:N/A 0xFD: Minimum operating time	参数组充电, 经济模式 充电SOC上限 参数组放电, 经济模式 放电SOC下限限 0xFE:N/A 0xFD:最小工作时间
52	47552	Parameter1_3	参数1_3	RW	U16	1	N/A	N/A		Y	0xFF:each month	0xFF:每月
53	47553	Start Time_2	启动时间_2	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y	hh:mm	hh:mm
54	47554	End Time_2	结束时间_2	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
55	47555	Work Week_2	工作周_2	RW	U16	1	N/A	N/A		Y	Table 8-34	
56	47556	Parameter2_1	参数2_1	RW	S16	1	1/NA/ 100/1 00/1	%/N/A/KW/K W/N/A/%/‰	[- 100,100]/NA/[0,1 0000]/[0,50000]/[0,100]/[- 1000,1000]	Y	0xFF:Battery power percentage 0xFE:N/A 0xFD:Rated Power 0xF9:Battery power permillage	0xFF:电池功率百分比 0xFE:N/A 0xFD:额定功率 0xF9:电池功率千分比
57	47557	Parameter2_2	参数2_2	RW	U16	1	1/NA/ 1/1	%/N/A/min/ %	[0,100]/NA/[0,144 0]/[0,100]	Y	Parameter Group Charging, SOC Upper Limit for Charging in Economy Mode Parameter Group Discharge, Economic Mode Discharge SOC Lower Limit 0xFE:N/A 0xFD: Minimum operating time	参数组充电, 经济模式 充电SOC上限 参数组放电, 经济模式 放电SOC下限限 0xFE:N/A 0xFD:最小工作时间
58	47558	Parameter2_3	参数2_3	RW	U16	1	N/A	N/A		Y	0xFF:each month	0xFF:每月
59	47559	Start Time_3	启动时间_3	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y	hh:mm	hh:mm
60	47560	End Time_3	结束时间_3	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
61	47561	Work Week_3	工作周_3	RW	U16	1	N/A	N/A		Y	Table 8-34	
62	47562	Parameter3_1	参数3_1	RW	S16	1	1/NA/ 100/1 00/1	%/N/A/KW/K W/N/A/%/‰	[- 100,100]/NA/[0,1 0000]/[0,50000]/[0,100]/[- 1000,1000]	Y	0xFF:Battery power percentage 0xFE:N/A 0xFD:Rated Power 0xF9:Battery power permillage	0xFF:电池功率百分比 0xFE:N/A 0xFD:额定功率 0xF9:电池功率千分比

63	47563	Parameter3_2	参数3_2	RW	U16	1	1/NA/ 1/1	%/N/A/min/ %	[0,100]/NA/[0,144 0]/[0,100]	Y	0xFF:Economy mode charging SOC limit 0xFE:N/A 0xFD:Minimum operating time	0xFF:经济模式充电 SOC上限 0xFE:N/A 0xFD:最小工作时间
64	47564	Parameter3_3	参数3_3	RW	U16	1	N/A	N/A		Y	0xFF:each month	0xFF:每月
65	47565	Start Time_4	启动时间_4	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y	hh:mm	hh:mm
66	47566	End Time_4	结束时间_4	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
67	47567	Work Week_4	工作周_4	RW	U16	1	N/A	N/A		Y	Table 8-34	
68	47568	Parameter4_1	参数4_1	RW	U16	1	1/NA/ 100/1 00/1	%/N/A/KW/K W/N/A/%/‰	[- 100,100]/NA/[0,1 0000]/[0,50000]/[0,100]/[- 1000,1000]	Y	0xFF:Battery power percentage 0xFE:N/A 0xFD:Rated Power 0xF9:Battery power permillage	0xFF:电池功率百分比 0xFE:N/A 0xFD:额定功率 0xF9:电池功率千分比
69	47569	Parameter4_2	参数4_2	RW	U16	1	1/NA/ 1/1	%/N/A/min/ %	[0,100]/NA/[0,144 0]/[0,100]	Y	Parameter Group Charging, SOC Upper Limit for Charging in Economy Mode Parameter group discharge, economic mode discharge SOC lower limit 0xFE:N/A 0xFD: Minimum operating time	参数组充电, 经济模式 充电SOC上限 参数组放电, 经济模式 放电SOC下下限 0xFE:N/A 0xFD:最小工作时间
70	47570	Parameter4_3	参数4_3	RW	U16	1	N/A	N/A		Y	0xFF:each month	0xFF:每月
71	47571	Start Time_5	启动时间_5	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y	hh:mm	hh:mm
72	47572	End Time_5	结束时间_5	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
73	47573	Work Week_5	工作周_5	RW	U16	1	N/A	N/A		Y	Table 8-34	
74	47574	Parameter5_1	参数5_1	RW	S16	1	1/NA/ 100/1 00/1	%/N/A/KW/K W/N/A/%/‰	[- 100,100]/NA/[0,1 0000]/[0,50000]/[0,100]/[- 1000,1000]	Y	0xFF:Battery power percentage 0xFE:N/A 0xFD:Rated Power 0xF9:Battery power permillage	0xFF:电池功率百分比 0xFE:N/A 0xFD:额定功率 0xF9:电池功率千分比
75	47575	Parameter5_2	参数5_2	RW	U16	1	1/NA/ 1/1	%/N/A/min/ %	[0,100]/NA/[0,144 0]/[0,100]	Y	Parameter Group Charging, SOC Upper Limit for Charging in Economy Mode Parameter Group Discharge, Economic Mode Discharge SOC Lower Limit 0xFE:N/A 0xFD: Minimum operating time	参数组充电, 经济模式 充电SOC上限 参数组放电, 经济模式 放电SOC下下限 0xFE:N/A 0xFD:最小工作时间
76	47576	Parameter5_3	参数5_3	RW	U16	1	N/A	N/A		Y	0xFF:each month	0xFF:每月
77	47577	Start Time_6	启动时间_6	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y	hh:mm	hh:mm
78	47578	End Time_6	结束时间_6	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		

79	47579	Work Week_6	工作周_6	RW	U16	1	N/A	N/A		Y	Table 8-34	
80	47580	Parameter6_1	参数6_1	RW	S16	1	1/NA/ 100/1 00/1	%/N/A/KW/K W/N/A/%/‰	[- 100,100]/NA/[0,1 0000]/[0,50000]/[0,100]/[- 1000,1000]	Y	0xFF:Battery power percentage 0xFE:N/A 0xFD:Rated Power 0xF9:Battery power permillage	0xFF:电池功率百分比 0xFE:N/A 0xFD:额定功率 0xF9:电池功率千分比
81	47581	Parameter6_2	参数6_2	RW	U16	1	1/NA/ 1/1	%/N/A/min/ %	[0,100]/NA/[0,144 0]/[0,100]	Y	Parameter Group Charging, SOC Upper Limit for Charging in Economy Mode Parameter Group Discharge, Economic Mode Discharge SOC Lower Limit 0xFE:N/A 0xFD: Minimum operating time	参数组充电, 经济模式 充电SOC上限 参数组放电, 经济模式 放电SOC下限限 0xFE:N/A 0xFD:最小工作时间
82	47582	Parameter6_3	参数6_3	RW	U16	1	N/A	N/A		Y	0xFF:each month	0xFF:每月
83	47583	Start Time_7	启动时间_7	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y	hh:mm	hh:mm
84	47584	End Time_7	结束时间_7	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
85	47585	Work Week_7	工作周_7	RW	U16	1	N/A	N/A		Y	Table 8-34	
86	47586	Parameter7_1	参数7_1	RW	S16	1	1/NA/ 100/1 00/1	%/N/A/KW/K W/N/A/%/‰	[- 100,100]/NA/[0,1 0000]/[0,50000]/[0,100]/[- 1000,1000]	Y	0xFF:Battery power percentage 0xFE:N/A 0xFD:Rated Power 0xF9:Battery power permillage	0xFF:电池功率百分比 0xFE:N/A 0xFD:额定功率 0xF9:电池功率千分比
87	47587	Parameter7_2	参数7_2	RW	U16	1	1/NA/ 1/1	%/N/A/min/ %	[0,100]/NA/[0,144 0]/[0,100]	Y	Parameter Group Charging, SOC Upper Limit for Charging in Economy Mode Parameter Group Discharge, Economic Mode Discharge SOC Lower Limit 0xFE:N/A 0xFD: Minimum operating time	参数组充电, 经济模式 充电SOC上限 参数组放电, 经济模式 放电SOC下限限 0xFE:N/A 0xFD:最小工作时间
88	47588	Parameter7_3	参数7_3	RW	U16	1	N/A	N/A		Y	0xFF:each month	0xFF:每月
89	47589	Start Time_8	启动时间_8	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y	hh:mm	hh:mm
90	47590	End Time_8	结束时间_8	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
91	47591	Work Week_8	工作周_8	RW	U16	1	N/A	N/A		Y	Table 8-35	
92	47592	Parameter8_1	参数8_1	RW	U16	1	1/NA/ 100/1 00/1	%/N/A/KW/K W/N/A/%/‰	[- 100,100]/NA/[0,1 0000]/[0,50000]/[0,1000]	Y	0xFC:Peak shaving power limit 0xFA:Limit permillage	0xFC:削峰功率限制 0xFA:限制千分比

93	47593	Parameter8_2	参数8_2	RW	U16	1	1/NA/ 1/1	%/N/A/min/ %	[0,100]/NA/[0,144 0]/[0,100]	Y	Parameter Group Charging, SOC Upper Limit for Charging in Economy Mode Parameter Group Discharge, Lower Limit of SOC for Discharge in Economy Mode 0xFC: Peak Charging SOC 0xFA: max SOC	参数组充电, 经济模式 充电SOC上限 参数组放电, 经济模式 放电SOC下限 0xFC:峰值充电SOC 0xFA: max SOC
94	47594	Parameter8_3	参数8_3	RW	U16	1	N/A	N/A		Y	0xFA:each month	0xFA:每月
95	47595	Load Regulation Index	负载规范索引	RW	U16	1	N/A	N/A	[0,3]	Y	0:Disable 1:switching mode 2:Time manage mode 3:off-Grid load mode Only for inverter with ARM version >18 To select Load control mode	0:关闭 1:开关模式 2:时间管理模式 3:离网 负载模式 仅用于ARM芯片高于 18版本的逆变器 , 用于选择负载控制模 式
96	47596	Load Switch Status	负载切换状态	RW	U16	1	N/A	N/A	[0,1]	N	Only for inverter with ARM version >18, to read the load control status	仅用于ARM芯片高于 18版本的逆变器, 用于 读取负载控制状态
97	47597	Backup Switch SOC Min	Backup切换最小剩余电 量	RW	U16	1	N/A	N/A	[0,100]	Y	For load control function, if the controlLed load on Backup side, use this to switch the load off when Battery reaches the SOC set here	对于负载控制功能, 如 果被控制的负载在 Backup侧, 当电池达 到设定的SOC时, 使用 此功能来关闭负载。
99	47599	Hardware Feed Power Disable	硬防逆流	RW	U16	1	N/A	N/A	[0,1]	Y		
100	47600	PCS Powersave Mode	PCS功率节约模式	RW	U16	1	N/A	N/A	[0,1]	Y	When enabLed,discharge mode turn to wait when the SPI Battery disables discharge	使能后, SPI电池禁止 放电时, 放电模式下转 wait
101	47601	Old Meter Protocol	旧电表协议	RW	U16	1	N/A	N/A	[0,1]	Y		
102	47602	DOD Holding Enable	DOD保持使能	RW	U16	1	N/A	N/A	[0,1]	Y		
104	47604	Load Regulation or Generator Flag	负载安规或发电机标志	RW	U16	1	N/A	N/A		Y	Dry node function configuration, deFault load control, version≥ 18 0:generator 1:load safety	干节点功能配置, 默认 负载控制, 18版本以上 0:发电机 1:负载安规
105	47605	Backup Mode Enable	backup模式使能	RW	U16	1	N/A	N/A	1Enable	Y		
106	47606	Max Charge Power from Gird	来自电网的最大充电功 率	RW	U16	1	N/A	‰	[0,1000]	Y		
107	47607	Reserved	预留	RW	U32	2	N/A	N/A		N		

108	47609	Smart Charging Mode Enable	智能充电模式使能	RW	U16	1	N/A	N/A	1Enable	Y		
109	47610	SelfUse Mode Enable	自发自用模式使能	RW	U16	1	N/A	N/A		N		
110	47611	PeakShaving Mode Enable	需量管理模式使能	RW	U16	1	N/A	N/A		N		
111	47612	ECO Mode Enable	经济模式使能	RW	U16	1	N/A	N/A		Y		
112	47613	PV sell first	PV卖电优先	RW	U16	1	N/A	N/A		Y		
113	47614	Bat FeedPower offset	耦合机电池防逆流偏移	RW	U16	1	1	W		Y		
114	47615	Battery Current Coff	电池电流系数	RW	U16	1	1	%	[0,100]	Y		
115	47616	Parallel Strong Charge Power permillage	并簇强充可调功率千分比	RW	U16	1	1	‰	[0,1000]	Y		
117	47618	Battery2 Protocol	电池2协议	RW	U16	1	N/A	N/A	[0,511]	Y	Refer to 8-33	见表8-33
118	47619	Feed Power Start Time_1	防逆流起始时间1	RW	U32	2	N/A	S		N		
119	47621	Feed Power limit_1	防逆流限值1	RW	S32	2	N/A	W		N		
120	47623	Feed Power Period_1	防逆流持续时间1	RW	U16	1	N/A	S		N		
121	47624	Feed Power Start Time_2	防逆流起始时间2	RW	U32	2	N/A	S		N		
122	47626	Feed Power limit_2	防逆流限值2	RW	S32	2	N/A	W		N		
123	47628	Feed Power Period_2	防逆流持续时间2	RW	U16	1	N/A	S		N		
124	47629	Feed Power Start Time_3	防逆流起始时间3	RW	U32	2	N/A	S		N		
125	47631	Feed Power limit_3	防逆流限值3	RW	S32	2	N/A	W		N		
126	47633	Feed Power Period_3	防逆流持续时间3	RW	U16	1	N/A	S		N		
127	47634	Feed Power Start Time_4	防逆流起始时间4	RW	U32	2	N/A	S		N		
128	47636	Feed Power limit_4	防逆流限值4	RW	S32	2	N/A	W		N		
129	47638	Feed Power Period_4	防逆流持续时间4	RW	U16	1	N/A	S		N		
130	47639	Feed Power Start Time_5	防逆流起始时间5	RW	U32	2	N/A	S		N		
131	47641	Feed Power limit_5	防逆流限值5	RW	S32	2	N/A	W		N		
132	47643	Feed Power Period_5	防逆流持续时间5	RW	U16	1	N/A	S		N		
133	47644	Feed Power Start Time_6	防逆流起始时间6	RW	U32	2	N/A	S		N		
134	47646	Feed Power limit_6	防逆流限值6	RW	S32	2	N/A	W		N		
135	47648	Feed Power Period_6	防逆流持续时间6	RW	U16	1	N/A	S		N		

136	47649	Feed Power Start Time_7	防逆流起始时间7	RW	U32	2	N/A	S		N		
137	47651	Feed Power limit_7	防逆流限值7	RW	S32	2	N/A	W		N		
138	47653	Feed Power Period_7	防逆流持续时间7	RW	U16	1	N/A	S		N		
139	47654	Feed Power Start Time_8	防逆流起始时间8	RW	U32	2	N/A	S		N		
140	47656	Feed Power limit_8	防逆流限值8	RW	S32	2	N/A	W		N		
141	47658	Feed Power Period_8	防逆流持续时间8	RW	U16	1	N/A	S		N		
142	47659	Feed Power Start Time_9	防逆流起始时间9	RW	U32	2	N/A	S		N		
143	47661	Feed Power limit_9	防逆流限值9	RW	S32	2	N/A	W		N		
144	47663	Feed Power Period_9	防逆流持续时间9	RW	U16	1	N/A	S		N		
145	47664	Feed Power Start Time_10	防逆流起始时间10	RW	U32	2	N/A	S		N		
146	47666	Feed Power limit_10	防逆流限值10	RW	S32	2	N/A	W		N		
147	47668	Feed Power Period_10	防逆流持续时间10	RW	U16	1	N/A	S		N		
148	47669	Feed Power Start Time_11	防逆流起始时间11	RW	U32	2	N/A	S		N		
149	47671	Feed Power limit_11	防逆流限值11	RW	S32	2	N/A	W		N		
150	47673	Feed Power Period_11	防逆流持续时间11	RW	U16	1	N/A	S		N		
151	47674	Feed Power Start Time_12	防逆流起始时间12	RW	U32	2	N/A	S		N		
152	47676	Feed Power limit_12	防逆流限值12	RW	S32	2	N/A	W		N		
153	47678	Feed Power Period_12	防逆流持续时间12	RW	U16	1	N/A	S		N		
154	47679	Feed Power Start Time_13	防逆流起始时间13	RW	U32	2	N/A	S		N		
155	47681	Feed Power limit_13	防逆流限值13	RW	S32	2	N/A	W		N		
156	47683	Feed Power Period_13	防逆流持续时间13	RW	U16	1	N/A	S		N		
157	47684	Feed Power Start Time_14	防逆流起始时间14	RW	U32	2	N/A	S		N		
158	47686	Feed Power limit_14	防逆流限值14	RW	S32	2	N/A	W		N		
159	47688	Feed Power Period_14	防逆流持续时间14	RW	U16	1	N/A	S		N		
160	47689	Feed Power Start Time_15	防逆流起始时间15	RW	U32	2	N/A	S		N		
161	47691	Feed Power limit_15	防逆流限值15	RW	S32	2	N/A	W		N		
162	47693	Feed Power Period_15	防逆流持续时间15	RW	U16	1	N/A	S		N		
163	47694	Feed Power Start Time_16	防逆流起始时间16	RW	U32	2	N/A	S		N		

164	47696	Feed Power limit_16	防逆流限值16	RW	S32	2	N/A	W		N		
165	47698	Feed Power Period_16	防逆流持续时间16	RW	U16	1	N/A	S		N		
166	47699	Feed Power Start Time_17	防逆流起始时间17	RW	U32	2	N/A	S		N		
167	47701	Feed Power limit_17	防逆流限值17	RW	S32	2	N/A	W		N		
168	47703	Feed Power Period_17	防逆流持续时间17	RW	U16	1	N/A	S		N		
169	47704	Feed Power Start Time_18	防逆流起始时间18	RW	U32	2	N/A	S		N		
170	47706	Feed Power limit_18	防逆流限值18	RW	S32	2	N/A	W		N		
171	47708	Feed Power Period_18	防逆流持续时间18	RW	U16	1	N/A	S		N		
172	47709	Feed Power Start Time_19	防逆流起始时间19	RW	U32	2	N/A	S		N		
173	47711	Feed Power limit_19	防逆流限值19	RW	S32	2	N/A	W		N		
174	47713	Feed Power Period_19	防逆流持续时间19	RW	U16	1	N/A	S		N		
175	47714	Feed Power Start Time_20	防逆流起始时间20	RW	U32	2	N/A	S		N		
176	47716	Feed Power limit_20	防逆流限值20	RW	S32	2	N/A	W		N		
177	47718	Feed Power Period_20	防逆流持续时间20	RW	U16	1	N/A	S		N		
178	47719	Feed Power Start Time_21	防逆流起始时间21	RW	U32	2	N/A	S		N		
179	47721	Feed Power limit_21	防逆流限值21	RW	S32	2	N/A	W		N		
180	47723	Feed Power Period_21	防逆流持续时间21	RW	U16	1	N/A	S		N		
181	47724	Feed Power Start Time_22	防逆流起始时间22	RW	U32	2	N/A	S		N		
182	47726	Feed Power limit_22	防逆流限值22	RW	S32	2	N/A	W		N		
183	47728	Feed Power Period_22	防逆流持续时间22	RW	U16	1	N/A	S		N		
184	47729	Feed Power Start Time_23	防逆流起始时间23	RW	U32	2	N/A	S		N		
185	47731	Feed Power limit_23	防逆流限值23	RW	S32	2	N/A	W		N		
186	47733	Feed Power Period_23	防逆流持续时间23	RW	U16	1	N/A	S		N		
187	47734	Feed Power Start Time_24	防逆流起始时间24	RW	U32	2	N/A	S		N		
188	47736	Feed Power limit_24	防逆流限值24	RW	S32	2	N/A	W		N		
189	47738	Feed Power Period_24	防逆流持续时间24	RW	U16	1	N/A	S		N		
190	47739	SAPN UP Rate	输入输出上升斜率	RW	U16	1	100	%Pn/min		Y		

191	47740	SAPN Down Rate	输入输出下降斜率	RW	U16	1	100	%Pn/min		Y		
192	47741	SAPN Feed Power Preset	SAPN预设防逆流	RW	S32	2	N/A	W		Y		
193	47743	Single Battery Paral Enable	单电池并机使能位	RW	U16	1	N/A	N/A		Y		
194	47744	Battery Busbar Mode	电池汇流排接入模式使能位	RW	U16	1	N/A	N/A		Y		
195	47745	Generator start mode selection	发电机启动方式选择	RW	U16	1	N/A	N/A		Y	0 : Automatic , 1 : Manual , 2(Default): Without Generator	0 : 自动 , 1 : 手动 , 2 : 未安装发电机 , 默
196	47746	One-click enable	一键启动使能	RW	U16	1	N/A	N/A		Y	0 : Disabled , 1 : Enabled	0 : Disabled , 1 : Enabled
197	47747	Generator charge to the battery permillage	发电机给电池充电额定功率千分比	RW	U16	1	1	‰	[0,1000]	Y	0 : Disabled , Other:Rated charging power permillage	0 : Disabled , 其它 : 额定充电功率千分比
198	47748	reserved	预留			1						
199	47749	Week1	工作周1	RW	U16	1	N/A	N/A		Y	Higher 8 bits are flag bits, 0xFF: enabled, 0x00: disabled, 0x55: not set bit0-Sunday bit1-Monday bit2-Tuesday bit3-Wednesday bit4-Thuesday bit5-Friday bit6-Saturday	高8位为标志位 , 0xFF : 使能 , 0x00 : 未使能 , 0x55 : 未设置 bit0-Sunday bit1-Monday bit2-Tuesday bit3-Wednesday bit4-Thuesday bit5-Friday bit6-Saturday
200	47750	Prohibit the start time of work1	禁止工作起始时间1	RW	U16	1	N/A	N/A	[0,23]-[0,59]	Y	Start time	时间段1开始时刻
201	47751	Prohibit work end	禁止工作结束时间1	RW	U16	1	N/A	N/A	[0,23]-[0,59]	Y	End time	时间段1结束时刻
202	47752	Week2	工作周2	RW	U16	1	N/A	N/A		Y	高8位为标志位 , 0xFF : 使能 , 0x00 : 未使能 , 0x55 : 未设置 bit0-Sunday bit1-Monday bit2-Tuesday bit3-Wednesday bit4-Thuesday bit5-Friday bit6-Saturday	高8位为标志位 , 0xFF : 使能 , 0x00 : 未使能 , 0x55 : 未设置 bit0-Sunday bit1-Monday bit2-Tuesday bit3-Wednesday bit4-Thuesday bit5-Friday bit6-Saturday

203	47753	Prohibit the start time of work2	禁止工作起始时间2	RW	U16	1	N/A	N/A	[0,23]-[0,59]	Y	Start time	时间段2开始时刻
204	47754	Prohibit work end	禁止工作结束时间2	RW	U16	1	N/A	N/A	[0,23]-[0,59]	Y	End time	时间段2结束时刻
205	47755	reserved	预留			1						
206	47756	open soC or open voltage	开启SOC或开启电压	RW	U16	1	1	N/A	[20, 90]%, [40, 55]V	Y	SOC 40% by default; The default voltage corresponds to the battery type	SOC默认40%；电压默认值与电池类型相对应
207	47757	close soC or close voltage	关闭SOC或关闭电压	RW	U16	1	1	N/A	[40, 95]%, [45, 60]V	Y	SOC 90% by default; The default voltage corresponds to the battery type	SOC默认90%；电压默认值与电池类型相对应
208	47758	Generator run time	发电机运行时间	RW	U16	1	10	h	[0, 240]	Y	Generator max operating time, default 8h	发电机最大运行时间，默认8h
209	47759	Generator rated power	发电机额定功率	RW	U16	1	100	KW		Y		
210	47760	reserved	预留			1						
211	47761	reserved	预留			1						
212	47762	reserved	预留			1						
213	47763	SAPN Enable	SAPN使能标志	RW	U16	1	1	N/A	[0, 1]	Y	SPAN, 1:ON , 0: OFF	SPAN功能开关，1为开，0为关
214	47764	lead_Acid Battery Float Charge Voltage	铅酸电池浮充电压	RW	U16	1	10	v	[0,65535]	Y		
215	47765	lead_Acid Battery Absorp Charge	铅酸电池恒充电压	RW	U16	1	10	v	[0,65535]	Y		
216	47766	lead_Acid Battery Equal Charge Voltage	铅酸电池均充电压	RW	U16	1	10	v	[0,65535]	Y		
217	47767	lead_Acid Battery InterRes	铅酸电池内阻	RW	U16	1	1	mΩ	[0,255]	Y		
218	47768	lead_Acid Battery Equal Charge Enable	铅酸电池均充使能	RW	U16	1	1	N/A	[0,1]	Y	0:OFF , 1 : ON	0:关闭，1：使能
219	47769	lead_Acid Battery Charge Coefficient	铅酸电池充电系数	RW	U16	1	100	%	[0,100]	Y		
220	47770	lead_Acid Battery Max Float Charge Current	铅酸电池浮充最大电流	RW	U16	1	10	A	[0,100]	Y	Max current of lead-acid battery during floating charge	铅酸电池浮充阶段最大电流
221	47771	lead_Acid Battery Equal Charge cycle	铅酸电池均充周期	RW	U16	1	1	day	[0,365]	Y		
222	47772	lead_Acid Battery Current To Float ChrgSts	铅酸电池进入浮充阶段电流阈值	RW	U16	1	10	A	[0,255]	Y		

223	47773	lead_Acid BattleTempcCompensation	铅酸电池温度影响系数	RW	S16	1	1	‰	[-200,200]	Y		
224	47774	lead_Acid BattleToFloatStsTime	铅酸电池进入浮充时间 阈值	RW	U16	1	1	S	[0,65535)	Y		
225	47775	LG VPP Control	LG VPP控制使能位	RW	U16	1	1	NA	[0,1]	N	1 : Open , 0 : Close	1 : 开 , 0 : 关
226	47776	LG VPP Control Start Time	LG VPP控制开始时间	RW	U32	2	1	S		N	Local Timestamp	本地时间戳
227	47778	LG VPP Control End Time	LG VPP控制结束时间	RW	U32	2	1	S		N	Local Timestamp	本地时间戳
228	47780	LG VPP Control Battery Mode	LG VPP控制电池模式	RW	U16	1	1	NA		N	2 : Charge , 3 : Discharge	2 : 充电 , 3 : 放电
229	47781	LG VPP Control Battery Power	LG VPP控制电池功率	RW	S32	2	1	W		N		
231	47785	Negative Electric Price Enable	负电价功能使能开关	RW	U16	1	1	N/A	[0,1]	N	0: Off, 1: On	0 : 关闭 , 1 : 开启
232	47786	RTC Time 1	RTC时间组1 (月日)	RW	U16	1	1	N/A	bit[15:8]月:[1,12] bit[0:7]日:[1,31]	N	Today RTC Time [15:8]month , [7:0]day	当日的RTC时间 [15:8]月 , [7:0]日
233	47787	Sell Electric Price Plan 1-1	卖电价格计划组1-1	RW	U16	1	1	N/A	[0,0]	N	96bits, ench bit means the sell electric price in 15min, bit equals 1 means the price is negative, 0 means positive.	共96个bit位 , 每个bit位 代表15分钟的卖电价格 计划, bit位置1表示价 格为负, 置0表示价格
234	47788	Sell Electric Price Plan 1-2	卖电价格计划组1-2	RW	U16	1	1	N/A	[0,0]	N		
235	47789	Sell Electric Price Plan 1-3	卖电价格计划组1-3	RW	U16	1	1	N/A	[0,0]	N		
236	47790	Sell Electric Price Plan 1-4	卖电价格计划组1-4	RW	U16	1	1	N/A	[0,0]	N		
237	47791	Sell Electric Price Plan 1-5	卖电价格计划组1-5	RW	U16	1	1	N/A	[0,0]	N		
238	47792	Sell Electric Price Plan 1-6	卖电价格计划组1-6	RW	U16	1	1	N/A	[0,0]	N		
239	47793	RTC Time 2	RTC时间组2 (月日)	RW	U16	1	1	N/A	bit[15:8]月:[1,12] bit[0:7]日:[1,31]	N	Tomorrow RTC Time [15:8]month , [7:0]day	次日的RTC时间 [15:8]月 , [7:0]日
240	47794	Sell Electric Price Plan 2-1	卖电价格计划组2-1	RW	U16	1	1	N/A	[0,0]	N	96bits, ench bit means the sell electric price in 15min, bit equals 1 means the price is negative, 0 means positive.	共96个bit位 , 每个bit位 代表15分钟的卖电价格 计划, bit位置1表示价 格为负, 置0表示价格

241	47795	Sell Electric Price Plan 2-2	卖电价格计划组2-2	RW	U16	1	1	N/A	[0,0]	N		
242	47796	Sell Electric Price Plan 2-3	卖电价格计划组2-3	RW	U16	1	1	N/A	[0,0]	N		
243	47797	Sell Electric Price Plan 2-4	卖电价格计划组2-4	RW	U16	1	1	N/A	[0,0]	N		
244	47798	Sell Electric Price Plan 2-5	卖电价格计划组2-5	RW	U16	1	1	N/A	[0,0]	N		
245	47799	Sell Electric Price Plan 2-6	卖电价格计划组2-6	RW	U16	1	1	N/A	[0,0]	N		
246	47800	Negative Electric Price Function Buy Power Switch	负电价功能买电切换开关	RW	U16	1	1	N/A	[0,2]	N	0 : No operation, set to zero feed power 1 : Buy power to charge battery 2 : Buy power to charge battery+Turn on the load control switch(Only for 745 ARM)	0 : 不操作，设置为零防逆流 1 : 买电给电池充电 2 : 买电给电池充电+打开负载控制开关(仅限于745ARM平台)
247	47801	Buy Electric Price Plan 1-1	买电价格计划组1-1	RW	U16	1	1	N/A	[0,0]	N	96bits, each bit means the buy electric price in 15min, bit equals 1 means the price is negative, 0 means positive.	共96个bit位，每个bit位代表15分钟的卖电价格计划，bit位置1表示价格为负，置0表示价格
248	47802	Buy Electric Price Plan 1-2	买电价格计划组1-2	RW	U16	1	1	N/A	[0,0]	N		
249	47803	Buy Electric Price Plan 1-3	买电价格计划组1-3	RW	U16	1	1	N/A	[0,0]	N		
250	47804	Buy Electric Price Plan 1-4	买电价格计划组1-4	RW	U16	1	1	N/A	[0,0]	N		
251	47805	Buy Electric Price Plan 1-5	买电价格计划组1-5	RW	U16	1	1	N/A	[0,0]	N		
252	47806	Buy Electric Price Plan 1-6	买电价格计划组1-6	RW	U16	1	1	N/A	[0,0]	N		
253	47807	Buy Electric Price Plan 2-1	买电价格计划组2-1	RW	U16	1	1	N/A	[0,0]	N	96bits, each bit means the buy electric price in 15min, bit equals 1 means the price is negative, 0 means positive.	共96个bit位，每个bit位代表15分钟的卖电价格计划，bit位置1表示价格为负，置0表示价格
254	47808	Buy Electric Price Plan 2-2	买电价格计划组2-2	RW	U16	1	1	N/A	[0,0]	N		
255	47809	Buy Electric Price Plan 2-3	买电价格计划组2-3	RW	U16	1	1	N/A	[0,0]	N		

256	47810	Buy Electric Price Plan 2-4	买电价格计划组2-4	RW	U16	1	1	N/A	[0,0]	N		
257	47811	Buy Electric Price Plan 2-5	买电价格计划组2-5	RW	U16	1	1	N/A	[0,0]	N		
258	47812	Buy Electric Price Plan 2-6	买电价格计划组2-6	RW	U16	1	1	N/A	[0,0]	N		
259	47813	Reserved	预留给副电价	RW	U16	17	1	N/A		N		
262	47832	Battery heating switch	电池加热开关	RW	U16	1	N/A	N/A	[0,1]	N	0 : Close , 1 : Open	0 : 关闭 ; 1 : 打开
263	47833	Battery heating mode	电池加热模式	RW	U16	1	N/A	N/A	[0,2]	N	0 : Low Power Mode , 1 : Middle Power Mode , 2 : High Power Mode ,	0 : 低功率模式 ; 1 : 中功率模式 ; 2 : 高功率模式 ;
264	47834	Start Time_9	启动时间_9	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y	hh:mm	hh:mm
265	47835	End Time_9	结束时间_9	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
266	47836	Work Week_9	工作周_9	RW	U16	1	N/A	N/A		Y	Table 8-35	
267	47837	Parameter9_1	参数9_1	RW	U16	1	1/NA/100/100/1	%/N/A/KW/KW/N/A/%	[-100,100]/NA/[0,10000]/[0,50000]/[0,1000]	Y	0xFC:Peak shaving power limit 0xFA:Limit permillage	0xFC:削峰功率限制 0xFA:限制千分比
268	47838	Parameter9_2	参数9_2	RW	U16	1	1/NA/1/1	%/N/A/min/%	[0,100]/NA/[0,1440]/[0,100]	Y	Parameter 1 is set to charging, and Parameter 2 corresponds to the upper limit of charging SOC. Parameter 1 is set to discharge, and Parameter 2 corresponds to the lower limit of discharge SOC, 0xFC: Peak Charging SOC 0xFA: max SOC	参数1设置是充电, 参数2设置对应是充电SOC上限, 参数1设置是放电, 参数2设置对应是放电SOC下限, 0xFC:峰值充电SOC 0xFA: max SOC
269	47839	Parameter9_3	参数9_3	RW	U16	1	N/A	N/A		Y	0xFA:each month	0xFA:每月
270	47840	Start Time_10	启动时间_10	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y	hh:mm	hh:mm
271	47841	End Time_10	结束时间_10	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
272	47842	Work Week_10	工作周_10	RW	U16	1	N/A	N/A		Y	Table 8-34	
273	47843	Parameter10_1	参数10_1	RW	S16	1	1/NA/100/100/1	%/N/A/KW/KW/N/A/%	[-100,100]/NA/[0,10000]/[0,50000]/[0,100]/[-1000,1000]	Y	0xFF:Battery power percentage 0xFE:N/A 0xFD:Rated Power 0xF9:Battery power permillage	0xFF:电池功率百分比 0xFE:N/A 0xFD:额定功率 0xF9:电池功率千分比

274	47844	Parameter10_2	参数10_2	RW	U16	1	1/NA/ 1/1	%/N/A/min/ %	[0,100]/NA/[0,144 0]/[0,100]	Y	Parameter 1 is set to charging, and Parameter 2 corresponds to the upper limit of charging SOC. Parameter 1 is set to discharge, and Parameter 2 corresponds to the lower limit of discharge SOC, 0xFE:N/A 0xFD: Minimum operating time	参数1设置是充电, 参 数2设置对应是充电 SOC上限, 参数1设置是放电, 参 数2设置对应是放电 SOC下限, 0xFE:N/A 0xFD:最小工作时间
275	47845	Parameter10_3	参数10_3	RW	U16	1	N/A	N/A		Y	0xFF:each month	0xFF:每月
276	47846	Start Time_11	启动时间_11	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y	hh:mm	hh:mm
277	47847	End Time_11	结束时间_11	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
278	47848	Work Week_11	工作周_11	RW	U16	1	N/A	N/A		Y	Table 8-34	
279	47849	Parameter11_1	参数11_1	RW	S16	1	1/NA/ 100/1 00/1	%/N/A/KW/K W/N/A/%/‰	[- 100,100]/NA/[0,1 0000]/[0,50000]/[0,100]/[- 1000,1000]	Y	0xFF:Battery power percentage 0xFE:N/A 0xFD:Rated Power 0xF9:Battery power permillage	0xFF:电池功率百分比 0xFE:N/A 0xFD:额定功率 0xF9:电池功率千分比
280	47850	Parameter11_2	参数11_2	RW	U16	1	1/NA/ 1/1	%/N/A/min/ %	[0,100]/NA/[0,144 0]/[0,100]	Y	0xFF:Economy mode charging SOC limit 0xFE:N/A 0xFD:Minimum operating time	0xFF:经济模式充电 SOC上限 0xFE:N/A 0xFD:最小工作时间
281	47851	Parameter11_3	参数11_3	RW	U16	1	N/A	N/A		Y	0xFF:each month	0xFF:每月
282	47852	Start Time_12	启动时间_12	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y	hh:mm	hh:mm
283	47853	End Time_12	结束时间_12	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
284	47854	Work Week_12	工作周_12	RW	U16	1	N/A	N/A		Y	Table 8-34	
285	47855	Parameter12_1	参数12_1	RW	S16	1	1/NA/ 100/1 00/1	%/N/A/KW/K W/N/A/%/‰	[- 100,100]/NA/[0,1 0000]/[0,50000]/[0,100]/[- 1000,1000]	Y	0xFF:Battery power percentage 0xFE:N/A 0xFD:Rated Power 0xF9:Battery power permillage	0xFF:电池功率百分比 0xFE:N/A 0xFD:额定功率 0xF9:电池功率千分比
286	47856	Parameter12_2	参数12_2	RW	U16	1	1/NA/ 1/1	%/N/A/min/ %	[0,100]/NA/[0,144 0]/[0,100]	Y	0xFF:Economy mode charging SOC limit 0xFE:N/A 0xFD:Minimum operating time	0xFF:经济模式充电 SOC上限 0xFE:N/A 0xFD:最小工作时间
287	47857	Parameter12_3	参数12_3	RW	U16	1	N/A	N/A		Y	0xFF:each month	0xFF:每月
288	47858	Start Time_13	启动时间_13	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y	hh:mm	hh:mm
289	47859	End Time_13	结束时间_13	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
290	47860	Work Week_13	工作周_13	RW	U16	1	N/A	N/A		Y	Table 8-34	

291	47861	Parameter13_1	参数13_1	RW	S16	1	1/NA/ 100/1 00/1	%/N/A/KW/K W/N/A/%/‰	[- 100,100]/NA/[0,1 0000]/[0,50000]/[0,100]/[- 1000,1000]	Y	0xFF:Battery power percentage 0xFE:N/A 0xFD:Rated Power 0xF9:Battery power permillage	0xFF:电池功率百分比 0xFE:N/A 0xFD:额定功率 0xF9:电池功率千分比
292	47862	Parameter13_2	参数13_2	RW	U16	1	1/NA/ 1/1	%/N/A/min/ %	[0,100]/NA/[0,144 0]/[0,100]	Y	0xFF:Economy mode charging SOC limit 0xFE:N/A 0xFD:Minimum operating time	0xFF:经济模式充电 SOC上限 0xFE:N/A 0xFD:最小工作时间
293	47863	Parameter13_3	参数13_3	RW	U16	1	N/A	N/A		Y	0xFF:each month	0xFF:每月
294	47864	Start Time_14	启动时间_14	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y	hh:mm	hh:mm
295	47865	End Time_14	结束时间_14	RW	U16	1	N/A	N/A	[0,23],[0,59]	Y		
296	47866	Work Week_14	工作周_14	RW	U16	1	N/A	N/A		Y	Table 8-34	
297	47867	Parameter14_1	参数14_1	RW	S16	1	1/NA/ 100/1 00/1	%/N/A/KW/K W/N/A/%/‰	[- 100,100]/NA/[0,1 0000]/[0,50000]/[0,100]/[- 1000,1000]	Y	0xFF:Battery power percentage 0xFE:N/A 0xFD:Rated Power 0xF9:Battery power permillage	0xFF:电池功率百分比 0xFE:N/A 0xFD:额定功率 0xF9:电池功率千分比
298	47868	Parameter14_2	参数14_2	RW	U16	1	1/NA/ 1/1	%/N/A/min/ %	[0,100]/NA/[0,144 0]/[0,100]	Y	0xFF:Economy mode charging SOC limit 0xFE:N/A 0xFD:Minimum operating time	0xFF:经济模式充电 SOC上限 0xFE:N/A 0xFD:最小工作时间
299	47869	Parameter14_3	参数14_3	RW	U16	1	N/A	N/A		Y	0xFF:each month	0xFF:每月
300	47870	Standby mode buy power charging enable flag	备用模式买电充电使能 标志	RW	U16	1	1	1	[0, 1]	Y	0: Shut down 1: Start	0:关闭 1: 开启
303	47873	Force MircoGrid Run On Hour Enable	微网整点唤醒使能	RW	U16	1	1	1	[0, 1]	Y	Uniq is applicable. Set 1 to activate the microgrid hourly wake-up	Uniq适用，置1开启微 网整点唤醒功能
304	47874	RCR Mode	RCR模式选择开关	RW	U16	1	1	1	[0, 1]	Y	0: 4-port RCR function (default) 1: 3-port RCR and 14A function	0: 4端口RCR功能 (默认) 1: 3端口RCR和14A功 能
305	47875	14A mode Buy Power Limit	14A功能买电功率限制	RW	U32	2	1	W	[0,4200]	Y	German 14A regulations on power limit for electricity purchase	德国14A法规买电功率 限制
306	47877	RCR2 Power Limit Percent	RCR2功率限制百分比	RW	U16	1	1	%	[0,100]	Y	RCR2 electricity selling power limit percentage	RCR2卖电功率限制百 分比
307	47878	RCR3 Power Limit Percent	RCR3功率限制百分比	RW	U16	1	1	%	[0,100]	Y	RCR3 electricity selling power limit percentage	RCR3卖电功率限制百 分比

308	47879	BAT breathing light SW	控制电池呼吸灯开关	RW	U16	1	1	NA	[0,2]	N	0: Default (lights for three minutes then turns off); 1: Always off; 2: Always on	0 : default (亮三分钟 灭) ; 1 : 常灭 2 : 常亮
309	47880	BAT Force Charge Power permillage	电池强充可调功率千分比	RW	U16	1	1	‰	[10,1000]	Y		
310	47881	Lead-acid Battery Float Charge Voltage2	第二路铅酸电池浮充电压	RW	U16	1	10	V	[400,600]	Y	The upper limit of the charging voltage in the float charging mode	浮充模式的充电电压上限
311	47882	lead_Acid Battery Absorp Charge	第二路铅酸电池恒充电压	RW	U16	1	10	V	[400,600]	Y	The upper limit of the charging voltage in the constant charge	恒充模式的充电电压上限
312	47883	lead_Acid Battery Equal Charge Voltage	第二路铅酸电池均充电压	RW	U16	1	10	V	[400,600]	Y	The upper limit of the charging voltage in the equalization charging	均充模式的充电电压上限
313	47884	lead_Acid Battery InterRes	第二路铅酸电池内阻	RW	U16	1	1	mΩ	[1,255]	Y	The total internal resistance of a lead-acid battery pack is not the internal resistance of each	铅酸电池组的总内阻，并不是每一节的内阻
314	47885	lead_Acid Battery Equal Charge Enable2	第二路铅酸电池均充使能	RW	U16	1	1	N/A	[0,1]	Y	0: Off, 1: Enable	0:关闭，1：使能
315	47886	lead_Acid Battery Charge Coefficient 2	第二路铅酸电池充电系数	RW	U16	1	100	0.01C	[0,20]	Y	C represents the rated capacity of the battery: 45350/45375. If the capacity is 200AH, each 0.01C represents 2A. If the register value is 20, the maximum charging current of the battery is 40A; the discharge current limit is fixed at 1C, and	C指电池额定容量：45350/45375，若容量为200AH，每0.01C代表2A 若寄存器值为20，则电池最大充电电流为40A；放电电流固定为1C，无寄存器
316	47887	lead_Acid Battery Max Float Charge Current 2	第二路铅酸电池浮充最大电流	RW	U16	1	10	A	[0,255]	Y	The current during the float charging stage must not exceed	浮充阶段的电流不能超过此值
317	47888	lead_Acid Battery Equal Charge cycle2	第二路铅酸电池均充周期	RW	U16	1	1	day	[0,180]	Y		
318	47889	lead_Acid Battery Current To Float ChrgSts2	第二路铅酸电池转浮充的电流阈值	RW	U16	1	10	A	[0,255]	Y	Below this current and for a period of time longer than that, it will then switch to float charging.	低于此电流，且持续一定时间以上，才会转为浮充
319	47890	lead_Acid BattleTempcCompensatio2	第二路铅酸电池温度影响系数	RW	S16	1	1	-1mV/°C	[-200,200]	Y	3 represents -3 mV/°C	3代表-3mV/°C
320	47891	lead_Acid BattleToFloatStsTime2	第二路铅酸电池进入浮充时间阈值	RW	U16	1	1	S	[1, 65535]	Y		

321	47892	PV priority selling	PV优先卖电模式	RW	U16	1	1	NA	[0,1)	Y	0: Off, 1: Enable	0:关闭, 1: 使能
322	47893	Function configuration word 1 of PCS module	PCS模块功能配置字1	RW	U16	1	1	1	[0,65535]	Y	Bit0: 1 - Ignore BMS power limitation; 0 - Respond to BMS power limitation. Bit1: 1 - Fault shutdown when both EMS and BMS communications are lost; 0 - Fault shutdown when either EMS or BMS communication is lost. Bit15 ~ Bit2: Reserved.(PCS Module Function Configuration	"bit0:1-不响应BMS限功率, 0-响应BMS限功率; bit1:1-EMS通信与BMS通信同时丢失故障停机, 0-单一EMS通信丢失或BMS通信丢失故障停机 Bit15~bit2:rsv;" PCS模块功能配置字1
323	47894	RCR output power control mode	RCR输出功率控制模式	RW	U16	1	1	NA	[0,1]	Y	0: RCR controls the output power on the AC side. 1: RCR controls the output power on the metering side.	0:RCR控制的是AC侧的输出功率 1: RCR控制的是电表侧输出功率
324	47895	DG control mode	柴发控制方式	RW	U16	1	1	1	[0,2]	Y	0: Generator not connected 1: DO control 2: Communication control	0: 未接入发电机 1: DO控制 2: 通讯控制
324	47896	wetility timer	wetility计时器	RW	U16	1	1	h	[0,8760]	Y	The wetility timer function limits the battery power and backup output after the module loses communication and reaches the	wetility计时器功能, 模块断通信到设定时间后限制电池功率和backup输出
326	47897	Backup Switch SOC Max	Backup切换最大剩余电量	RW	U16	1	1	%	[0,100]	Y	For the load control function, if the controlled load is on the Backup side, when the battery reaches the set SOC level, this function is used to turn on the load. (Used in conjunction with 47597, when bit0 of 28501 is 1, it takes effect.)	对于负载控制功能, 如果被控制的负载在Backup侧, 当电池达到设定的SOC时, 使用此功能来打开负载。 (与47597配合使用, 28501的bit0=1时生
328	47899	remote shutdown contact definition	远程关断触点定义	RW	U16	1	1	NA	[0,1]	Y	0: Turn off triggered by low voltage 1: Turn off triggered by high	0: 低电平时触发关断 1: 高电平时触发关断
ARM BMS Passthrough												
1	47900	BMS Version	BMS版本	RW	U16	1	N/A	N/A		N		
2	47901	Battery Strings	电池串2	RW	U16	1	N/A	N/A		N		
3	47902	Max BMS Battery Charge Voltage	BMS电池最大充电电压	RW	U16	1	N/A	V		N	Real-time max charge voltage limit for BAT BMS	电池BMS的实时最大充电电压限制

4	47903	Max BMS Battery Charge Current	BMS电池最大充电电流	RW	U16	1	N/A	A		N	Real-time max charge current limit for BAT BMS	电池BMS的实时最大充电电流限制
5	47904	Min BMS Battery Discharge Voltage	BMS电池最小放电电压	RW	U16	1	N/A	V		N	Real-time min charge voltage limit for BAT BMS	电池BMS的实时最小放电电压限制
6	47905	Min BMS Battery Discharge Current	BMS电池最小放电电流	RW	U16	1	N/A	A		N	Real-time min charge current limit for BAT BMS	电池BMS的实时最小放电电流限制
7	47906	BMS Battery Voltage	BMS电池电压	RW	U16	1	N/A	V		N	Real-time BAT voltage(BMS Detection)	实时电池电压(电池BMS检测)
8	47907	BMS Battery Current	BMS电池电流	RW	U16	1	N/A	A		N	Real-time BAT current(BMS Detection)	实时电池充放电电流(电池BMS检测)
9	47908	BMS Battery SOC	BMS电池剩余电量百分比	RW	U16	1	N/A	%		N		
10	47909	BMS Battery SOH	BMS电池健康度	RW	U16	1	N/A	N/A		N		
11	47910	BMS Battery Temperature	BMS电池温度	RW	U16	1	10	°C		N		
12	47911	BMS Warning Code	BMS警告代码	RW	U32	2	N/A	N/A		N		
13	47913	BMS Alarm Code	BMS报警代码	RW	U32	2	N/A	N/A		N		
14	47915	BMS Status	BMS状态	RW	U16	1	N/A	N/A		N		
15	47916	BMS Communication Loss Disable	BMS通信损失禁用	RW	U16	1	N/A	N/A		N		

16	47917	BMS Battery String Rate Voltage	BMS电池串额定电压	RW	U16	1	10	V	[200,2000]	N		
17	47918	BMS Version2	BMS版本2	RW	U16	1	N/A	N/A		N		
18	47919	Battery Strings2	电池串2	RW	U16	1	N/A	N/A		N		
19	47920	Max BMS Battery2 Charge Voltage	BMS电池2最大充电电压	RW	U16	1	N/A	V		N		
20	47921	Max BMS Battery2 Charge Current	BMS电池2最大充电电流	RW	U16	1	N/A	A		N		
21	47922	Min BMS Battery2 Discharge Voltage	BMS电池2最小放电电压	RW	U16	1	N/A	V		N		
22	47923	Min BMS Battery2 Discharge Current	BMS电池2最小放电电流	RW	U16	1	N/A	A		N		
23	47924	BMS Battery2 Voltage	BMS2电池电压	RW	U16	1	N/A	V		N		
24	47925	BMS Battery2 Current	BMS2电池电流	RW	U16	1	N/A	A		N		
25	47926	BMS Battery2 SOC	BMS电池2剩余电量百分比	RW	U16	1	N/A	%		N		
26	47927	BMS Battery2 SOH	BMS2电池健康度	RW	U16	1	N/A	N/A		N		
27	47928	BMS Battery2 Temperature	BMS2电池温度	RW	U16	1	10	°C		N		
28	47929	BMS2 Warning Code	BMS2警告代码	RW	U32	2	N/A	N/A		N		
29	47931	BMS2 Alarm Code	BMS2报警代码	RW	U32	2	N/A	N/A		N		
30	47933	BMS2 Status	BMS2状态	RW	U16	1	N/A	N/A		N		
31	47934	BMS2 Communication Loss Disable	BMS2通信损失禁用	RW	U16	1	N/A	N/A		N		

32	47935	BMS Battery String2 RateVoltage	BMS电池串2额定电压	RW	U16	1	10	V	[200,2000]	N		
33	47936	Max BMS Battery Discharge Current Offline	离网放电BMS限流值	RW	U16	1	10	A		N		
34	47937	Max BMS Battery Discharge Current Offline2	离网放电BMS限流值2	RW	U16	1	10	A		N		
35	47938	BMS1 Heating Start Time	电池1加热起始时间	RW	U16	1	N/A	N/A	[0,0]	Y	High byte h: Low byte min	高字节h : 低字节min
36	47939	BMS1 Heating End	电池1加热结束时间	RW	U16	1	N/A	N/A	[0,0]	Y	High byte h: Low byte min	高字节h : 低字节min
37	47940	BMS2 Heating Start Time	电池2加热起始时间	RW	U16	1	N/A	N/A	[0,0]	Y	High byte h: Low byte min	高字节h : 低字节min
38	47941	BMS2 Heating End	电池2加热结束时间	RW	U16	1	N/A	N/A	[0,0]	Y	High byte h: Low byte min	高字节h : 低字节min
39	47942	BMS1 Heating Start Time2	电池1加热起始时间2	RW	U16	1	N/A	N/A	[0,0]	Y	High byte h: Low byte min	高字节h : 低字节min
40	47943	BMS1 Heating End Time2	电池1加热结束时间2	RW	U16	1	N/A	N/A	[0,0]	Y	High byte h: Low byte min	高字节h : 低字节min
41	47944	BMS2 Heating Start Time2	电池2加热起始时间2	RW	U16	1	N/A	N/A	[0,0]	Y	High byte h: Low byte min	高字节h : 低字节min
42	47945	BMS2 Heating End Time2	电池2加热结束时间2	RW	U16	1	N/A	N/A	[0,0]	Y	High byte h: Low byte min	高字节h : 低字节min
ARM controls parameters of DSP												
60	48080	Charge Power of Charging Pile of DSP	DSP的充电桩充电功率	RW	U32	2	1	N/A		Y	ARM给到DSP的充电桩充电功率	Charge Power of Charging Pile from ARM to DSP

Table 8-1 Work Mode

Mode	Code	Description
Wait	0x00	cut off all the connection to Inverter
On-Grid	0x01	PV inputs to Inverter, Inverter outputs to Grid
Off-Grid	0x02	PV inputs to Inverter(1st),Battery inputs to Inverter(2nd),Inverter work as AC source
Fault	0x03	Fault ,fault mode, something is in fault mode
Flash	0x04	Inverter upgrade
Check	0x05	Power on self-check of inverter

Table 8-2 Error Message

Bit NO	Error message	Description
Bit31	Internal Communication Failure	Communication between microcontrollers is failure
Bit30	EEPROM R/W Failure	EEPROM cannot be read or written
Bit29	Fac Failure	The grid frequency is out of tolerable range
Bit28	DSP communication failure	Communication between ARM and DSP is failure
Bit27	PhaseAngleFailure	Phase angle out of range (110°~140°)
Bit26	TBD	NA
Bit25	Relay Check Failure	Relay check is failure
Bit24	TBD	NA

Bit23	Vac Consistency Failure	Different value between Master and Slave for grid voltage
Bit22	Fac Consistency Failure	Different value between Master and Slave for grid frequency
Bit21	TBD	NA
Bit20	Back-Up Over Load	NA
Bit19	DC Injection High	The DC injection to grid is too high
Bit18	Isolation Failure	Isolation resistance of PV-plant too low
Bit17	Vac Failure	Grid voltage out of tolerable range
Bit16	External Fan Failure	The external fan failure
Bit15	PV Over Voltage	Pv input voltage is over the tolerable maximum value
Bit14	Utility Phase Failure	Utility Phase Failure
Bit13	Over Temperature	Temperature is too high
Bit12	InternalFan Failure	The fan in case failure
Bit11	DC Bus High	Dc bus is too high
Bit10	Ground I Failure	Ground current is too high
Bit9	Utility Loss	Utility is unavailable
Bit8	AC HCT Failure	AC HCT check failure 3 times
Bit7	Relay Device Failure	Relay check failure 3 times
Bit6	GFCI Device Failure	GFCI check failure 3 times
Bit5	TBD	NA
Bit4	GFCI Consistency	Different GFCI values on Master &Slave
Bit3	DCI Consistency	Different DCI value on Master and Slave
Bit2	TBD	NA
Bit1	AC HCT Check Failure	The output current sensor is abnormal
Bit0	GFCI Device Check	The GFCI detecting circuit is abnormal

Table 8-3 PV Mode

Byte	Description
0	PV1 mode, refer to table 8-4
1	PV2mode, refer to table 8-4
2	PV3 mode, refer to table 8-4
3	PV4 mode, refer to table 8-4

Table 8-4 PV Mode Code

Mode Code	Description
0x00	NO PV, inverter disconnects to PV
0x01	Standby, PV does not output power
0x02	Work, PV output power

Table 8-7 BMS Alarm Code

Bit31	Bit30	Bit29	Bit28	Bit27
Reserved	Reserved	Reserved	Reserved	Reserved
Bit15	Bit14	Bit13	Bit12	Bit11
Charging over-voltage ³	Discharge	Cell High	Communication failure ²	Charging circuit Failure
Bit7	Bit6	Bit5	Bit4	Bit3
DC bus fault	Precharge fault	Discharging	Charging overcurrent ²	Cell Low temperature ²

Table 8-8 BMS Warning Code

Bit31~Bit15	Bit14	Bit13	Bit12	Bit11
Reserved				System High temperature
Bit7	Bit6	Bit5	Bit4	Bit3
System Reboot	communication	Discharge over-	Charge over-current ¹	Cell Low temperature ¹

Table 8-9 Battery Status

Mode Code	Description
0x00	No Battery, or battery disconnected
0x01	Standby, no discharging or charging
0x02	Discharging
0x03	Charging

Table 8-10 Grid Status

Mode Code	Description
0x00	Loss, inverter disconnects to Grid
0x01	OK, inverter connects to Grid
0x02	Fault, something is wrong

Table 8-11 Backup Status

Mode Code	Description
0x00	ON, inverter connects to Load
0x01	OFF, inverter disconnects to Load

Table 8-12 Operation Mode

Mode Code	Description
0x01	Cut off all the connection to Inverter(wait mode)
0x02	PV intputs to Inverter, Inverter outputs to Grid(online mode)
0x04	PV inputs to Inverter(First),Battery inputs to Inverter(Second),Inverter work as AC source(battery mode)
0x10	Fault, fault mode, something is in fault mode(fault mode)

Table 8-13 Diagnostic Status

Code	Diagnose Info	Expalantion
0x0000000	Battery Precharge Relay Off	Battery Precharge Relay Off
0x0000000	ByPass Relay Stick	Bypass relay is sticking
0x0000000 4	SPD Fault Feed Status	SPD Fault Feed Status
0x0000000	PV Power limit status	PV功率受限标识
0x0000001 0	PV Derate current for temperature	PV电流过温降额
0x0000002 0	Bat Derate current for temperature	电池电流过温降额
0x0000004 0	Inv Derate current for temperature	逆变电流过温降额
0x0000008 0	Ac Port Phase NegOrder	AC接入负相序状态
0x0000010 0	PV Derate By AFCI	PV由于AFCI检测降额
0x1000000 0	Meter voltage sample fault	Meter voltage sample fault
0x2000000	External Stop Mode Enable	DRED or ESD stop the inverter
0x4000000	Battery Offgrid DOD	Battery SOC less than Offgrid DOD

0x8000000	Battery SOC Adjust Enable	Only for BYD, adjust the SOC
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Table 8-14 Diagnostic Status

Bit	Diagnose Info	Explanation	Type
0	BatteryVoltLow	1:Battery not discharge caused by low battery voltage	Affect discharge
1	BatterySOCLow	1:Battery not discharge caused by low SOC	
2	BatterySOCInBack	1:Battery SOC not recover to allow-discharge level	
3	BMSDischargeDisable	1:BMS not allow discharge	
4	DischargeTimeOn	Discharge time is set, 1: On, 0: OFF	
5	ChargeTimeOn	Charge time is set, 1: On, 0: OFF	
6	DischargeDriveOn	1:Discharge driver is turned on	
7	BMSDischgCurrentLow	1:BMS discharge current limit is too low	
8	DischargeCurrentLow	1:Discharge current limit is too low (from App)	
9	MeterCommLoss	1:Smart Meter communication failure	
10	MeterConnectReverse	1:Smart Meter connection reversed	
11	SelfUseLoadLight	1:Low load power, cannot activate battery discharge	
12	EMSDischargeIZero	1:Discharge current limit 0A from EMS	
13	DischargeBUSHigh	1:Battery not discharge caused by over high PV voltage	Affect charging
14	BatteryDisconnect	1:Battery disconnected	
15	BatteryOvercharge	1:Battery overcharged	
16	BMSOverTemperature	1:Lithium battery over temperature	
17	BMSOvercharge	1:Lithium battery overcharged or an individual cell voltage is higher	
18	BMSChargeDisable	1:BMS does not allow charge	Affect
19	SelfUseOff	1:Self-use mode turned off	
20	SOCDeltaOverRange	1:SOC Jumps abnormally	Other
21	BatterySelfDischarge	1:Battery discharge at low current for long time, continuously over 30% of battery	
22	OffgridSOCLow	1:SOC is low under off-grid statues	
23	GridWaveUnstable	1:Grid wave is bad, switch to back-up mode frequently	

24	FeedPowerLimit	1:Export power limit is set	
25	PFValueSet	1:PF value is set	
26	RealPowerLimit	1:Active power value is set	
28	SOCProtectOff	1:SOC protect Off	

Table 8-15 DRM Status

Value	DRMx	Description
0	DRM0	Operate the disconnection device
1	DRM1	Do not consume power
2	DRM2	Do not consume at more than 50% of rated power
3	DRM3	Do not consume at more than 75% of rated power and source reactive power if capable
4	DRM4	Increase power consumption(Subjects to constraints from other active DRMs)
5	DRM5	Do not generate power
6	DRM6	Do not generate at more than 50% of rated power
7	DRM7	Do not generate at more than 50% of rated power and sink reactive power if capable
8	DRM8	Increase power generation(Subjects to constraints from other active DRMs)
0xFF	No command	Disable

Table 8-16 EMS Power Mode

Application scenarios	MODE	COMMAND		PV	Grid	Battery
		EMSPowerMode	EMSPowerSet	Power priority (Green is the control object)		
System shutdown	Stopped	0x00FF		NA		
	Note :Stop working and switch to wait mode					
Self-use	Auto	0x0001		NA		
	Note: PBattery =PInv - Pmeter – Ppv (Discharge/Charge)					
	The battery power is controlled by the meter power when the meter communication is normal.					
Control the battery to keep charging	Charge-PV	0x0002		Xmax ^[2]	High	Low ^[1] Energy In
	Note :PBattery =Xmax + PV (Charge)					
	Xmax is to allow the power to be taken from the grid, and PV power is preferred. When set to 0, only PV power is used.					
Control the battery to keep discharging	Dischg+PV	0x0003		Xmax	High	Energy Out Low
	Note : PBattery = Xmax (Discharge)					
	Xmax is the allowable discharge power of the battery. When the power fed into the grid is limited, PV power will be used					
The inverter is used as a unit for power grid energy scheduling	Import-AC	0x0004		Xset ^[3]	Low	High Energy In
	Note :PBattery = Xset + PV (Charge)					
	Xset refers to the power purchased from the power grid. The power purchased from the grid is preferred. If the PV power					
	Export-AC	0x0005		Xset	High	Energy Out Low
	Note : PBattery = Xset (Discharge)					
Xset is to sell power to the grid. PV power is preferred. When PV energy is insufficient, the battery will discharge.PV						
Off-grid reservation mode	Conserve	0x0006		NA		
	Note : PBattery = PV (Charge)					
	In on-grid mode, the battery is continuously charged, and only PV power (AC Couple model takes 10% of the rated power					

Off-Grid Mode	Off-Grid	0x0007	NA	
	Note : PBattery =Pbackup – Ppv (Charge/Discharge)			
	Forced off-grid operation (Disconnect from grid)			

No battery mode for hybrid inverter	Battery standby	0x0008	NA	
	Note : PBattery =0 (Standby)			
	The battery does not charge and discharge			

Regional energy management	Buy Power	0x0009	Xset	Low	High	Energy In/Out
	Note :PBattery = PInv – (Pmeter + Xset)– Ppv (Charge/Discharge)					
	When the meter communication is normal, the power purchased from the power grid is controlled as Xset. When the PV					
	Sell Power	0x000A	Xset	High	Energy Out	Low
	Note : PBattery = PInv – (Pmeter – Xset) – Ppv (Charge/Discharge)					
	When the communication of electricity meter is normal, the power sold from the power grid is controlled as Xset, PV					

Force the battery to work at set power value	Charge-BAT	0x000B	Xset	High	Low	Energy In
	Note : PBattery = Xset (Charge)					
	Xset is the charging power of the battery. PV power is preferred. When PV power is insufficient, it will buy power from the					
	Discharge-BAT	0x000C	Xset	Low	Energy In	High
	Note : PBattery = Xset (Discharge)					
	Xset is the discharge power of the battery, and the battery discharge has priority. If the PV power is too large, MPPT will be limited. Discharge power is also affected by discharge current limit.					

Note:

[1] for low-priority energy sources, when the battery charging power is limited or the rated output power of the inverter is limited, the load shall be

[2] Xmax represents the upper limit of the power control value, and the actual power will be adjusted according to the working condition.

[3] Xset represents the target value of power control, and the actual power must reach the set value.

Table 8-17 CPLD Warning Code

VALUE	Error message
1	PV1 Over Current HW
2	PV2 Over Current HW
3	Battery Over Current HW
4	Bus Over Voltage HW
5	R InvOverCurr HW
6	S InvOverCurr HW
7	T InvOverCurr HW
8	BatRelayFail

Table 8-18 Power Factor

Data	Description
1	0.99 lagging
2	0.98 lagging
3	0.97 lagging
4	0.96 lagging
5	0.95 lagging
6	0.94 lagging
7	0.93 lagging
8	0.92 lagging
9	0.91 lagging
10	0.90 lagging
11	0.89 lagging
12	0.88 lagging
13	0.87 lagging
14	0.86 lagging

15	0.85 lagging
16	0.84 lagging
17	0.83 lagging
18	0.82 lagging
19	0.81 lagging
20	0.80 lagging
80	0.80 leading
81	0.81 leading
82	0.82 leading
83	0.83 leading
84	0.84 leading
85	0.85 leading
86	0.86 leading
87	0.87 leading
88	0.88 leading
89	0.89 leading
90	0.90 leading
91	0.91 leading
92	0.92 leading
93	0.93 leading
94	0.94 leading
95	0.95 leading
96	0.96 leading
97	0.97 leading
98	0.98 leading
99	0.99 leading
100	1

Table 8-20 Weekly schedule

	Bit NO	Definition
High byte	Bit15~8	0xFF :enable
		0x00 : disable
Low byte	Bit7	NA
	Bit6	Saturday
	Bit5	Friday
	Bit4	Thursday
	Bit3	Wednesday
	Bit2	Tuesday
	Bit1	Monday
	Bit0	Sunday

Table 8-21

Code	Description	Grid connection standards
0x00	Italy	ENEL (Un: 230Vac)
0x01	Czech	EN50438(CZ) (Un: 230Vac)
0x02	Germany	VDE-AR-N 4105(Un: 230Vac)
0x03	Spain	RD1699(Un: 230Vac)
0x04	GreeceMainland	EN50438(GR) (Un: 230Vac)
0x05	Danmark	EN50438(DK)
0x06	Belgium	C10/C11(Un: 230Vac)
0x07	Romania	(Un: 230Vac)
0x08	G98	G83/2 G59/3 (Un: 230Vac)
0x09	Australia	AS/NZS 4777.2
0x0A	France	VDE0126
0x0B	China	NB-T
0x0C	60Hz Grid Default	CSA
0x0D	Poland	EN50438

0x0E	South Africa	(Un: 230Vac)
0x0F	AustraliaL	AS/NZS 4777.2
0x10	Brazil	(Un: 220Vac)
0x11	Thailand MEA	MEA
0x12	Thailand PEA	PEA
0x13	Mauritius	(Un: 230Vac)
0x14	Holland	EN50438
0x15	G99	Northern Ireland
0x16	ChinaHigher	NB-T
0x17	French 50Hz	(Un: 230Vac)
0x18	French 60Hz	(Un: 230Vac)
0x19	Australia Ergon	AS/NZS 4777.2
0x1A	Australia Energex	AS/NZS 4777.2
0x1B	Holland 16/20A	EN50438
0x1C	Korea	(Un: 220Vac)
0x1D	China Station	NB-T
0x1E	Austria	(Un: 230Vac)
0x1F	India	IEC61727
0x20	50Hz Grid Default	Default
0x21	Warehouse	Warehouse
0x22	Philippines	Philippines
0x23	Ireland	EN50438 Ireland
0x24	Taiwan	(Un: 230Vac)
0x25	Bulgaria	EN50438
0x26	Barbados	(Un: 230Vac)
0x27	ChinaHighest	NB-T
0x28	G99 reserve	G59/3 (Un: 230Vac)
0x29	Sweden	EN50438 (Un: 230Vac)
0x2A	Chile	BISI 4.0 (Un: 220Vac)

0x2B	Brazil LV	(Un: 220Vac)
0x2C	NewZealand	AS/NZS 4777.2
0x2D	IEEE1547 208Vac	IEEE1547 (Un: 120/208Vac)
0x2E	IEEE1547 220Vac	IEEE1547 (Un: 127/220Vac)
0x2F	IEEE1547 240Vac	IEEE1547 (Un: 138.6/240Vac)
0x30	60Hz LV Default	60Hz LV Default
0x31	50Hz LV Default	50Hz LV Default
0x32	Australia Western	AS/NZS 4777.2
0x33	Australia MicroGrid	AS/NZS 4777.2
0x34	JP_50Hz	JP_50Hz
0x35	JP_60Hz	JP_60Hz
0x36	India Higher	IEC61727 (Un: 230Vac))
0x37	DEWA LV	DEWA (Un:230Vac)
0x38	DEWA MV	DEWA (Un:230Vac)
0x39	Slovakia	EN50438(SV) (Un: 230Vac)
0X3A	GreenGrid	AS/NZS 4777.2
0x3B	Hungary	(Un: 230Vac)
0x3C	SriLanka	(Un: 230Vac)
0x3D	SpainIslands	RD1699 (Un: 230Vac)
0x3E	Ergon30K	(Un: 230Vac)
0x3F	Energe30K	(Un: 230Vac)
0x40	IEEE1547_230VAC	IEEE1547 (Un: 230/400Vac)
0x41	IEC61727_60Hz	IEC61727 (Un: 230Vac)
0x42	Switzerland	VDE-AR-N 4105 (Un: 230Vac)

0x43	CEI_016	CEI-016 (Un: 230Vac)
0x44	Australia Horizon	AS/NZS 4777.2
0x45	Cyprus	(Un: 230Vac)
0x46	Australia SAPN	AS/NZS 4777.2
0x47	Australia Ausgrid	AS/NZS 4777.2
0x48	Australia Essential	AS/NZS 4777.2
0x49	Australia	AS/NZS 4777.2
0x4A	China Hongkong	
0x4B	Poland MV	
0x4C	Holland MV	
0x4D	Sweden MV	
0x4E	VDE4110	
0x4F	Germany	
0x50	Spain MV	
0x51	Australia Endeavour	
0x52	Argentina	
0x53	AustralianB	
0x54	AustralianC	

Table 8-22

DRED0	DRED1	DRED2	DRED3	DRED4
0x00FF	0x0001	0x0002	0x0004	0x0008

Table 8-30

Bit NO	Grid detailed fault	Description
Bit0	GridZeroLossErr	电网停电/Power outage
Bit1	GridVoltLowErrSt1	电网欠压一级故障/Grid undervoltage first level failure
Bit2	GridVoltLowErrSt2	电网欠压二级故障/Grid undervoltage second level fault
Bit3	GridVoltLowErrSt3	电网欠压三级故障/Grid undervoltage third level fault
Bit4	GridVoltHighErrSt1	电网过压一级故障/Grid overvoltage first level failure
Bit5	GridVoltHighErrSt2	电网过压二级故障/Grid overvoltage second level fault
Bit6	GridVoltHighErrSt3	电网过压三级故障/Grid overvoltage third level fault
Bit7	Grid10minAvgVoltErr	电网平均电压高故障/Grid average voltage high fault
Bit8	GridFreqLowErrSt1	电网欠频一级故障/Grid underfrequency first level failure
Bit9	GridFreqLowErrSt2	电网欠频二级故障/Grid underfrequency second level fault
Bit10	GridIslandFreqLowErr	孤岛保护欠频故障/Islanding protection underfrequency fault
Bit11	GridFreqHighErrSt1	电网过频一级故障/Grid overfrequency first level failure
Bit12	GridFreqHighErrSt2	电网过频二级故障/Grid overfrequency second level fault
Bit13	GridIslandFreqHighErr	孤岛保护过频故障/Islanding protection overfrequency fault
Bit14	GridFreqShiftChkErr	电网频移故障/Grid frequency shift fault
Bit15	GridWaveCheckErr	电网波形检测故障/Grid waveform check fault
Bit16	GridLLVoltErrFlag	电网线电压故障标志/Grid line voltage fault flag
Bit17	GridLvrtErr	电网低电压穿越故障/Grid low voltage ride-through fault
Bit18	GridHvrtErr	电网高电压穿越故障/Grid high voltage ride-through fault
Bit19	GridVoltSampOverErr	电网电压超出采样上限/ Grid voltage exceeds the upper sampling limit
Bit20	GridConnVoltHighErr	电网连接电压高/Grid connection voltage high
Bit21	GridConnVoltLowErr	电网连接电压低/Grid connection voltage low
Bit22	GridConnFreqHighErr	电网连接频率高/Grid connection Frequency high
Bit23	GridConnFreqLowErr	电网连接频率低/Grid connection Frequency low
Bit...		
Bit63		

Table 8-31

Bit NO	Inverter detailed error	Description
Bit0	BattLLCHardOCErr	LLC硬件过流/LLC hardware overcurrent
Bit1	BattBoostHardOCErr	电池boost硬件过流/Battery boost hardware overcurrent
Bit2	BattBoostSoftOCErr	电池boost软件过流/ Battery boost software overcurrent
Bit3	BattBMSFaultErr	电池BMS故障/Battery BMS fault
Bit4	BattBMSDischgDisErr	电池BMS禁止放电/Battery BMS discharge disable
Bit5	BattCurrRmsOCErr	电池电流有效值过流/Battery current rms overcurrent
Bit6	OffgridBmsCurrLimitErr	离网模式超出BMS限流/Off-grid mode exceeds BMS current limit
Bit7	BusSoftStartFailedErr	Bus电压软启动失败/Bus voltage soft start failed
Bit8	BusVoltTooLowErr	Bus电压过低/Bus voltage is too low
Bit9	BusSampVoltTooHigh	Bus采样电压过高/ Bus voltage is too High
Bit10	InvHardOCErr	逆变硬件过流/Inverter hardware overcurrent
Bit11	InvCurrSoftOCErr	逆变软件过流/Inverter software overcurrent
Bit12	PvBoostHardOCErr	PV boost硬件过流/PV boost hardware overcurrent
Bit13	PvBoostSoftOCErr	PV boost软件过流/PV boost software overcurrent
Bit14	GridBackflowErr	电网倒灌/Grid backflow
Bit15	OffgridBattVoltLowErr	离网电池电压低/Off-grid mode battery voltage is low
Bit16	OffgridUpsVoltHighErr	离网AC电压过高/Off-grid mode AC voltage is too low
Bit17	OffgridUpsVoltLowErr	离网AC电压过低/ Off-grid mode AC voltage is too high
Bit18	UpsOverLoadErr	Backup overload
Bit19	OffGridZeroLossErr	离网过零错误/OffGridZero Error
Bit20	PowerFastRetrackErr	功率快速重追错误/Power fast retrack Error
Bit21	BypassRelaySwErr	Backup旁路继电器切换错误/Bypass Relay Switch Error
Bit22	LoadRelaySwErr	Backup负载继电器闭合错误/Backup load Relay switch Error
Bit23		
Bit...		
Bit63		

Table 8-32

Bit NO	Inverter detailed	Description
Bit0	SafetyOverFreqCurveF	进入过频曲线/Over-frequency curve running
Bit1	SafetyUnderFreqCurve	进入欠频曲线/Under frequency curve running
Bit2	SafetyFreqRecoCurve	频率曲线退出恢复中/Frequency curve exiting recovery
Bit3	SafetyPUCurveOVFlag	进入PU过压曲线/PU overvoltage curve running
Bit4	SafetyPUCurveUVFlag	进入PU欠压曲线/PU undervoltage curve running
Bit5	SafetyQUCurveFlag	进入QU曲线/QU curve running
Bit6	SafetyPFCurveFlag	进入PF曲线/PF curve running
Bit7	FixedPFSettingFlag	固定PF已设定/Fixed PF is set
Bit8	FixedQSettingFlag	固定无功已设定/Fixed reactive power is set
Bit9	InvOverTempFlag	机器过温降载/Inverter over-temp. derating curve operation
Bit10	DREDSellPowerLimitFl	澳洲DRED卖电/Australian DRED electricity sale status
Bit11	DREDBuyPowerLimitFl	澳洲DRED买电/Australian DRED purchase status
Bit12	ActivePowerSettingFla	有功功率限制已设定/Active power limit is set
Bit13	GeDratePowerFlag	德国70%降额打开70% derating (Germany) has been opened
Bit14	AutoTestEnableFlag	CEI021 selftest running
Bit15	GridVoltSt1DrateFlag	一级电压保护前降载/Inverter first level overvoltage derate
Bit16	ForceOffGridFlag	外部强制离网标志/Force OffGrid Flag
Bit17	ForceStopModeFlag	外部强制停机标志/Force StopMode Flag
Bit18	OffGridMpptChgUpsOff	离网充电关backup输出标志/PV charge, Off backup output Flag
Bit19	SafetyQUCurveOVFlag	QU曲线过压状态/QUCurveOverVoltageFlag
Bit20	SafetyQUCurveUVFlag	QU曲线欠压状态/QUCurveUnderVoltageFlag
Bit...		

Table 8-33

Battery manufacture	Battery series	Code	comments
GoodWe	SECU-S/LX S-H/LX F-H	0x122	
PYLONTECH	Powercube H1/Force H1/Force	0x101	
BYD	BYD-Box H	0x102	
BYD	BYD-Box Premium HVS	0x106	
BYD	BYD-Box Premium HVM/HVL	0x105	
LG	RESU_HV_Type-R	0x104	
OLOID	LBS	0x11E	
DYNESS	Tower	0x11E	
Soluna	HV BATTERY	0x11E	
EMS Use	EMS Battery	0x11F	Used when there is no direct communication between inverter and battery

Table 8-34

H-byte	Not set	0x55
	ECO mode	0xFF-Enable
		0x00-Disable
	Dry contact load mode	0xFE-Enable
		0x01-Disable
	Dry contact smart load mode	0xFD-Enable
		0x02-Disable
	peakshaving function	0xFC-Enable
		0x03-Disable
	Back-up mode	0xFB-Enable
		0x04-Disable
L-byte	Day select	bit0-Sunday
		bit1-Monday
		bit2-Tuesday
		bit3-Wednesday
		bit4-Thursday
		bit5-Friday
		bit6-Saturday


```
0x78, 0xB8, 0xB9, 0x79, 0xBB, 0x7B, 0x7A, 0xBA, 0xBE, 0x7E, 0x7F, 0xBF, 0x7D, 0xBD, 0xBC, 0x7C, 0xB4, 0x74, 0x75, 0xB5,
0x77, 0xB7, 0xB6, 0x76, 0x72, 0xB2, 0xB3, 0x73, 0xB1, 0x71, 0x70, 0xB0, 0x50, 0x90, 0x91, 0x51, 0x93, 0x53, 0x52, 0x92,
0x96, 0x56, 0x57, 0x97, 0x55, 0x95, 0x94, 0x54, 0x9C, 0x5C, 0x5D, 0x9D, 0x5F, 0x9F, 0x9E, 0x5E, 0x5A, 0x9A, 0x9B, 0x5B,
0x99, 0x59, 0x58, 0x98, 0x88, 0x48, 0x49, 0x89, 0x4B, 0x8B, 0x8A, 0x4A, 0x4E, 0x8E, 0x8F, 0x4F, 0x8D, 0x4D, 0x4C, 0x8C,
0x44, 0x84, 0x85, 0x45, 0x87, 0x47, 0x46, 0x86, 0x82, 0x42, 0x43, 0x83, 0x41, 0x81, 0x80, 0x40
};
```

```
INT16U sCRC16(INT8U *puchMsg, INT16U usDataLen)
{
    INT8U uchCRCHi = 0xFF ;
    INT8U uchCRCLo = 0xFF ;
    INT8U ulIndex ;
    while (usDataLen--)
    {
        ulIndex = uchCRCHi ^ *puchMsg++ ;
        uchCRCHi = uchCRCLo ^ uchCRCHi[ulIndex] ;
        uchCRCLo = uchCRCLo[ulIndex] ;
    }
    return ((INT16U)uchCRCHi << 8 | uchCRCLo) ;
}
```